

**SingleRAN**

# **IPsec Feature Parameter Description**

|              |            |
|--------------|------------|
| <b>Issue</b> | Draft B    |
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# 1 Change History

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This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes  
Changes in functions and their corresponding parameters
- Editorial changes  
Improvements or revisions to the documentation

## 1.1 SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

### Technical Changes

None

### Editorial Changes

Revised the descriptions of IPv4 VRF-based isolation and tunnel-interface-based configuration mode. For details, see [4.1.4.2.2 IPv4 VRF-based Isolation](#) and [Tunnel-Interface-based Configuration Mode](#).

## 1.2 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 05 (2025-08-30).

## Technical Changes

| Change Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Parameter Change                                                                                                                                                                                                                                                                                           | Base Station Model                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Added the function of using ESNs defined in the ESP protocol. For details, see: <ul style="list-style-type: none"><li>• <a href="#">4.1.2.1.4 Extended Serial Number</a></li><li>• <a href="#">4.3.2.1.2 Mutually Exclusive Functions</a></li><li>• <a href="#">4.3.2.2.2 Mutually Exclusive Functions</a></li><li>• <a href="#">4.3.2.3.2 Mutually Exclusive Functions</a></li><li>• <a href="#">4.3.2.4.2 Mutually Exclusive Functions</a></li><li>• <a href="#">4.3.2.5.2 Mutually Exclusive Functions</a></li><li>• <a href="#">4.3.2.6.2 Mutually Exclusive Functions</a></li><li>• <a href="#">4.3.3 Hardware</a></li><li>• <a href="#">4.4.2.1 Required Information</a></li><li>• <a href="#">4.4.2.2.1 Ethernet-Interface-based Configuration Mode</a></li></ul> | None                                                                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"><li>• 3900 and 5900 series base stations (macro base stations)</li><li>• DBS3900 LampSite and DBS5900 LampSite</li></ul> |
| Added the <b>Support Forcible Execution</b> parameter to commands that modify ACL rules for IPsec tunnel services. For details, see: <ul style="list-style-type: none"><li>• <a href="#">4.1.2.2.1 ACL</a></li><li>• <a href="#">4.4.2.2.2 Tunnel-Interface-based Configuration Mode</a></li><li>• <a href="#">5.4.1.1 Data Preparation</a></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                    | Modified parameters: <ul style="list-style-type: none"><li>• Added the <b>ACLRULE.FORCEEXEC UTE (5G gNodeB, LTE eNodeB)</b> parameter to the <b>MOD ACLRULE</b> command.</li><li>• Added the <b>ACLRULE6.FORCEEXEC UTE (5G gNodeB, LTE eNodeB)</b> parameter to the <b>MOD ACLRULE6</b> command.</li></ul> | <ul style="list-style-type: none"><li>• 3900 and 5900 series base stations (macro base stations)</li><li>• DBS3900 LampSite and DBS5900 LampSite</li></ul> |

## Editorial Changes

Added descriptions of source IP address verification for ACL rules. For details, see [4.1.2.2.1 ACL](#).

Added descriptions of hardware requirements for DH groups. For details, see [4.3.3 Hardware](#).

Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

# 2 About This Document

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## 2.1 General Statements

### Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

#### NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

### Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

## 2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and New Radio (NR).

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

## 2.3 Features in This Document

This document describes the following features.

| RAT     | Feature ID    | Feature Name                        | Chapter/Section              |
|---------|---------------|-------------------------------------|------------------------------|
| GSM     | GBFD-113524   | BTS Integrated IPsec                | <b>4 IPv4 IPsec</b>          |
| UMTS    | WRFD-140209   | NodeB Integrated IPsec              |                              |
| LTE FDD | LOFD-003009   | IPsec                               |                              |
| LTE TDD | TDLOFD-003009 | IPsec                               |                              |
| NB-IoT  | MLOFD-003009  | IPsec                               |                              |
| NR      | FOFD-010080   | IPsec (IPsec)                       |                              |
| GSM     | MRFD-121116   | Multi-mode BS Common IPsec(GSM)     |                              |
| UMTS    | MRFD-121126   | Multi-mode BS Common IPsec(UMTS)    |                              |
| LTE FDD | MRFD-121136   | Multi-mode BS Common IPsec(LTE)     |                              |
| LTE TDD | MRFD-121146   | Multi-mode BS Common IPsec(LTE TDD) |                              |
| NB-IoT  | MRFD-121156   | Multi-mode BS Common IPsec(NB-IoT)  |                              |
| NR      | MRFD-151165   | Multi-mode BS Common IPsec(NR)      |                              |
| LTE FDD | LOFD-003019   | IPsec Tunnel Backup                 | <b>6 IPsec Tunnel Backup</b> |
| NB-IoT  | MLOFD-003019  | IPsec Tunnel Backup                 |                              |
| LTE TDD | TDLOFD-131212 | IPsec Tunnel Backup                 |                              |
| NR      | FOFD-010080   | IPsec (IPsec Tunnel Backup)         |                              |



| RAT     | Feature ID    | Feature Name                                  | Chapter/Section                                |
|---------|---------------|-----------------------------------------------|------------------------------------------------|
| GSM     | GBFD-160209   | IPsec redundancy among multiple SeGWs         | <b>7 IPsec Redundancy Among Multiple SeGWs</b> |
| UMTS    | WRFD-160274   | IPsec redundancy among multiple SeGWs         |                                                |
| LTE FDD | LOFD-070211   | IPsec redundancy among multiple SeGWs         |                                                |
| NB-IoT  | MLOFD-070211  | IPsec redundancy among multiple SeGWs         |                                                |
| LTE TDD | TDLOFD-070211 | IPsec redundancy among multiple SeGWs         |                                                |
| NR      | FOFD-010080   | IPsec (IPsec redundancy among multiple SeGWs) |                                                |
| GSM     | GBFD-171206   | BTS Supporting IPsec Redirection              | <b>8 NE Supporting IPsec Redirection</b>       |
| UMTS    | WRFD-171221   | NodeB Supporting IPsec Redirection            |                                                |
| LTE FDD | LOFD-081281   | eNodeB Supporting IPsec Redirection           |                                                |
| NB-IoT  | MLOFD-081281  | eNodeB Supporting IPsec Redirection           |                                                |
| LTE TDD | TDLOFD-081211 | eNodeB Supporting IPsec Redirection           |                                                |
| NR      | FOFD-010080   | IPsec (IPsec Redirection)                     |                                                |
| LTE FDD | LOFD-003024   | IPsec for IPv6                                | <b>5 IPv6 IPsec</b>                            |
| LTE TDD | TDLOFD-003024 | IPsec for IPv6                                |                                                |
| NB-IoT  | MLOFD-003024  | IPsec for IPv6                                |                                                |
| NR      | FOFD-010080   | IPsec (IPsec for IPv6)                        |                                                |

# 3 Overview

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The move to IP-based networks has improved network performance and reduced network deployment costs. However, there are various threats and vulnerabilities specific to IP networks.

Before IP Security (IPsec) was introduced, base station IP-layer data was transmitted in plaintext, which could be easily intercepted or tampered with if the base station was deployed in an insecure network. Huawei base stations use IPsec tunnels to protect the transmitted data.

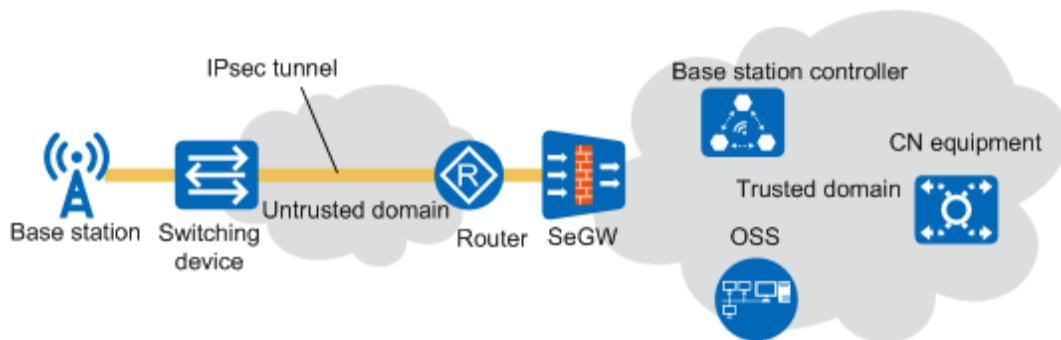
The IPsec protocol is defined by the Internet Engineering Task Force (IETF). It is implemented at the IP layer and uses three different protocols: Authentication Header (AH), Encapsulating Security Payload (ESP), and Internet Key Exchange (IKE). IPsec provides end-to-end security services for IP network communications, protecting them against cyber-attacks, such as eavesdropping or tampering.

With IPsec, two communicating peers (also known as IPsec peers) encrypt packets and authenticate the data sources to ensure the following:

- Confidentiality: Data transmissions are encrypted, preventing unauthorized users from obtaining the transmission contents.
- Integrity: Received data is verified to ensure that it has not been tampered with.
- Authenticity: The data source is authenticated.
- Anti-replay: Packets that are intercepted and repeatedly sent by malicious users are identified and rejected.

By establishing an IPsec tunnel between a base station and a security gateway (SeGW), base station data transmitted in an insecure network will be protected, as shown in [Figure 3-1](#).

**Figure 3-1** Secure networking



Based on the IP version, IPsec can be classified as follows:

- IPv4 IPsec. For details, see [4 IPv4 IPsec](#).  
To improve IPsec tunnel reliability, IPv4 IPsec supports the functions described in the following sections:
  - [6 IPsec Tunnel Backup](#)
  - [7 IPsec Redundancy Among Multiple SeGWs](#)
  - [8 NE Supporting IPsec Redirection](#)
- IPv6 IPsec. For details, see [5 IPv6 IPsec](#).  
To improve IPsec tunnel reliability, IPv6 IPsec supports the function described in [7 IPsec Redundancy Among Multiple SeGWs](#).

IPv4 IPsec protects only data transmitted over an Ethernet port or Ethernet cascading interface (CI) port or in an Ethernet link aggregation group. IPv6 IPsec protects only data transmitted over a tunnel interface. IPsec cannot protect data transmitted over other ports on a base station, such as data transmitted over the backplane tunnel between boards or data transmitted between the BBU and RF modules.

# 4 IPv4 IPsec

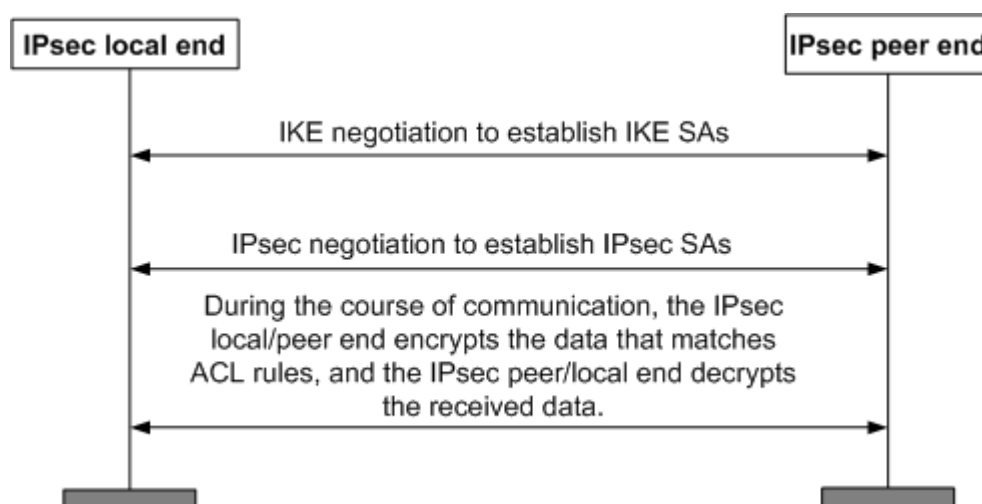
## 4.1 Principles

A security association (SA) must be established to define security policies which will be negotiated between communicating peers before IPsec tunnels are established.

There are two types of SAs in the IPsec framework: IKE SAs and IPsec SAs.

[Figure 4-1](#) shows the IPsec service procedure.

**Figure 4-1** IPsec service procedure



### NOTE

If IKE or IPsec negotiation fails, ALM-25891 IKE Negotiation Failure is reported.

An IKE SA is an agreement negotiated between IKE peers. An IKE SA defines the IKE SA lifecycle as well as the encryption algorithms, authentication algorithms, authentication method, and pseudorandom function (PRF) algorithms used between IKE peers. For details, see [4.1.1 IKE SA](#). IPsec SAs are negotiated under

the protection of IKE SAs. Each IPsec tunnel corresponds to a single IKE SA, but each IKE SA may correspond to multiple IPsec SAs. For details, see [4.1.2 IPsec SA](#).

For details about typical application of IPsec, see [4.1.3 IPsec Application](#).

## 4.1.1 IKE SA

Internet Key Exchange (IKE) is a security mechanism based on the Internet Security Association and Key Management Protocol (ISAKMP) framework. IKE provides encryption and authentication algorithms and key negotiation services for communicating peers. IKE can be used to automatically establish SAs, simplifying implementation and management of IPsec. Currently, IPsec SAs can only be automatically established for base stations using IKE.

IKE provides a security mechanism that allows for securely distributing keys, authenticating identities, and establishing IPsec SAs on insecure networks. The details are as follows:

- **IKE SA establishment**  
An ISAKMP SA (also known as an IKE SA) is established based on the first stage of IKE negotiation. The IKE SA provides an authenticated, secure channel for data exchange. IPsec SAs are negotiated under the protection of IKE SAs.  
IKE policy negotiation involves the IKE protocol version and IKE proposal. For details about the IKE proposal, see [4.1.1.1 IKE Proposal](#).
- **Session key generation**  
Communicating peers use a Diffie-Hellman (DH) algorithm to exchange session key materials and generate multiple session keys. These session keys are then used for IKE encryption, IKE data integrity check, IKE identity authentication, and IPsec encryption separately.
- **Identity authentication**  
Communicating peers exchange information to identify each other. This information includes authentication methods agreed upon during IKE negotiation and keys generated by DH exchange.

For details about the negotiation process, see [4.1.1.2 IKE Negotiation](#).

The dead peer detection (DPD) function is introduced to ensure the connectivity of IKE SAs. For details, see [4.1.1.3 IKE DPD](#).

For details about IKE, see IETF RFC 4301, IETF RFC 2409, IETF RFC 4306, IETF RFC 4754, IETF RFC 7427, and IETF RFC 7296.

### 4.1.1.1 IKE Proposal

During an IKE negotiation, the IKE local end uses its IKE proposal to negotiate with the IKE peer end and establish an IKE SA, which provides security services for IPsec SA negotiation.

An IKE proposal includes the encryption algorithms, authentication algorithms, authentication method, DH group, PRF algorithms, IKE SA lifecycle, and IKE reauthentication (supported only by IKEv2).

#### 4.1.1.1.1 Encryption and Authentication Algorithms

##### Encryption Algorithms

The encryption algorithm used by an IKE proposal is specified by the following parameters:

- **IKEPROPOSAL.ENCALG (LTE eNodeB, 5G gNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB)

The parameter value can be **DES**, **3DES**, **AES128**, **AES192**, **AES256**, **AES\_GCM\_128** (supported only by IKEv2), or **AES\_GCM\_256** (supported only by IKEv2). DES, 3DES, and AES stand for Data Encryption Standard, Triple Data Encryption Standard, and Advanced Encryption Standard, respectively.

##### NOTE

The DES algorithm has been cracked, and the 3DES algorithm is considered insufficiently secure in the industry. If these algorithms of low security levels are used, encrypted data may be cracked by attackers. It is recommended that the AES\_GCM\_128 or AES\_GCM\_256 encryption algorithm be used to ensure IKE SA security.

You can run **SET SECPY** to adjust the configuration security level to prohibit the parameter from being set to **DES** or **3DES**.

- **BTSIKEPROPOSAL.ENCALG** (for the GBTS)

The parameter value can be **DES**, **3DES**, **AES128**, **AES192**, or **AES256**.

##### Authentication Algorithms

The authentication algorithm used by an IKE proposal is specified by the following parameters:

- **IKEPROPOSAL.AUTHALG (5G gNodeB, LTE eNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB)

The parameter value can be **MD5**, **SHA1**, **AES-XCBC-96** (supported only by IKEv2), **SHA256** (supported only by IKEv2), or **SHA384** (supported only by IKEv2).

##### NOTE

The MD5, SHA1, and AES-XCBC-96 algorithms are considered insufficiently secure in the industry. If these algorithms of low security levels are used, the transmitted data may be forged by attackers. It is recommended that the SHA256 or SHA384 algorithm for authentication be used to ensure IKE SA security.

You can run **SET SECPY** to adjust the configuration security level to prohibit the parameter from being set to **MD5**, **SHA1**, or **AES-XCBC-MAC-96**.

- **BTSIKEPROPOSAL.AUTHALG** (for the GBTS)

The parameter value can be **AES-XCBC-96** (supported only by IKEv2).

For details about SHA, see IETF RFC 2404. For details about AES-XCBC-96, see IETF RFC 3566.

#### 4.1.1.1.2 Authentication Methods

IKE authentication methods refer to the methods used to authenticate IPsec peers during IKE communications. IKE authentication methods include PSK-based

authentication and digital-certificate-based authentication. PSK means pre-shared key.

- PSK-based authentication: Communicating parties use the same key to authenticate each other. After PSK message encryption, the sending party sends the encrypted message to the receiving party. The receiving party decrypts the message with an identical PSK. If the message is decrypted successfully, the authentication is successful.
- Digital-certificate-based authentication: Communicating parties authenticate each other using digital certificates.
- Certificates are difficult to counterfeit and are managed with a complete mechanism. For example, certificates have validity periods and can be revoked. Therefore, digital-certificate-based authentication is more reliable than PSK-based authentication. A public key infrastructure (PKI) system manages digital certificates for network equipment. Base stations support Rivest-Shamir-Adleman (RSA) and Elliptic Curve Digital Signature Algorithm (ECDSA) certificates. For details, see *PKI*.

#### NOTE

When an ECDSA certificate is used for IPsec, only IKEv2 negotiation is supported, and IKEv1 negotiation will fail.

The base station checks for the KeyUsage field of the IKE peer certificate. If the KeyUsage field exists, it must contain the digitalSignature or nonRepudiation bit. Otherwise, the base station reports ALM-26834 Peer Certificate Invalid.

When IPsec is enabled, the base station by default uses all its trust certificates to verify the device certificates of the peer end. If the value of the **IKEPEER.TRUSTGROUPID (LTE eNodeB, 5G gNodeB)** parameter is changed, only the trust certificates in the specified IPsec trust group are used for verification.

When IPsec is enabled on the base station, the **IKEPEER.OCSPSW (LTE eNodeB, 5G gNodeB)** parameter can be modified to specify whether to use IKE Online Certificate Status Protocol (OCSP) to verify the certificate revocation status of the peer end. IKE OCSP supports only IKEv2 negotiation. For details about OCSP, see RFC 4806.

In IPsec networking scenarios, the LiteCA cannot be used.

- During authentication method modification, if the IKE negotiation fails, the base station will retain the original SA for five minutes to ensure that services will not be interrupted and the authentication method on the peer will be changed to the same as that on the base station. Five minutes later, the base station deletes the original SA.

The following parameters specify the IKE authentication method:

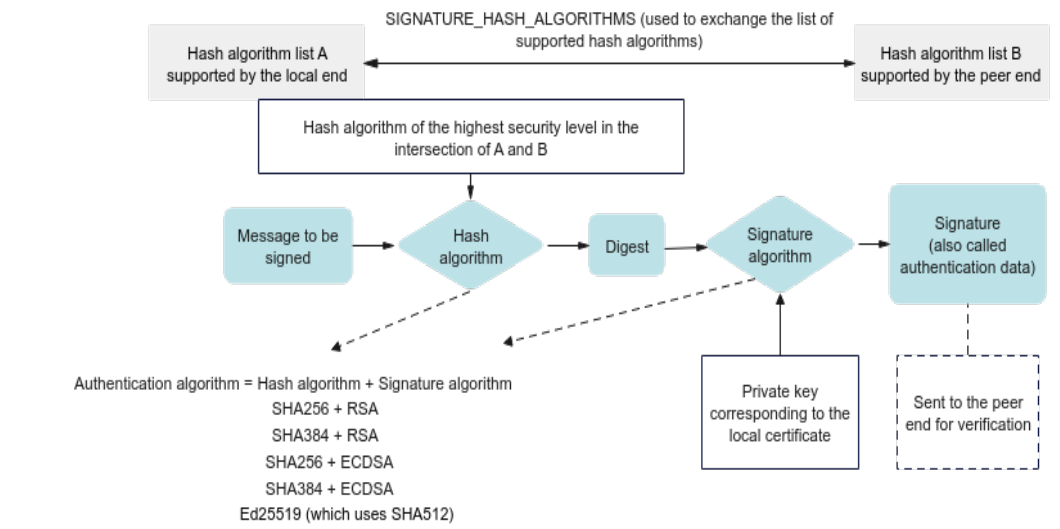
- **IKEPROPOSAL.AUTHMETH (LTE eNodeB, 5G gNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB)
- **BTSIKEPROPOSAL.AUTHMETH** (for the GBTS)

## Signature Hash Algorithm Negotiation

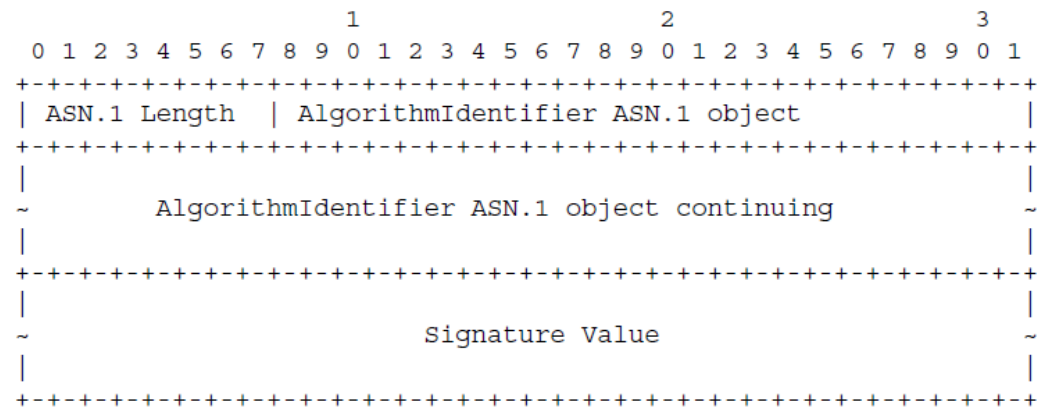
If the base station uses digital certificates for authentication (that is, the **IKEPROPOSAL.AUTHMETH (5G gNodeB, LTE eNodeB)** parameter is set to **IKE\_CERT\_SIG**), it uses a new digital signature authentication method (see RFC 7427) and sends an IKE\_SA\_INIT packet containing the payload SIGNATURE\_HASH\_ALGORITHMS to the peer to negotiate the signature hash algorithm. If the peer responds to this packet, the base station sets Auth Method

to 14, and uses a hash algorithm of the highest security level supported by the peer and the authentication algorithm corresponding to the certificate key type to generate authentication data, as shown in [Figure 4-2](#). The authentication data format complies with RFC 7427, as shown in [Figure 4-3](#).

**Figure 4-2** Generation of authentication data with hash algorithm negotiation supported



**Figure 4-3** New authentication data format with Auth Method 14



When the base station uses an RSA certificate, RSA padding can be configured using the **IKEPROPOSAL.RSAPADDING (5G gNodeB, LTE eNodeB)** parameter.



 NOTE

When digital-certificate-based authentication is used, the hash algorithm of the authentication payload can be SHA1, SHA256, or SHA384. If both ends support RFC 7427, the base station uses the most secure hash algorithm negotiated by the two ends. If the peer end does not support RFC 7427 and the local end uses an RSA certificate, the base station uses the hash algorithm specified by the **IKEPROPOSAL.RSASIGHASHALG (LTE eNodeB, 5G gNodeB)** parameter (the default value of this parameter is **SHA256**).

You can run **SET SECPLY** to adjust the configuration security level to prohibit the parameter from being set to **PKCS1V15**.

### 4.1.1.1.3 DH Group and PRF Algorithm

IKE enables both communicating peers to exchange data and use the DH algorithm and PRF as the core technologies of IKE to calculate and determine session keys without key transmission. Even if a third party intercepts all the exchanged data, the correct key cannot be calculated.

## DH Group

A DH group determines the length of the material for key generation. A longer length indicates a higher security level of the key.

The following parameters specify a DH group:

- **IKEPROPOSAL.DHGRP (LTE eNodeB, 5G gNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB)

The parameter value can be **DH\_GROUP1**, **DH\_GROUP2**, **DH\_GROUP14**, **DH\_GROUP15**, **DH\_GROUP19**, **DH\_GROUP20**, or **DH\_GROUP31**.

The CPU usage temporarily increases during IKE negotiation. If a large number of calculations involving **DH\_GROUP19** and **DH\_GROUP20** are performed, the base station will start flow control, which increases the IKE negotiation duration.

 NOTE

The **DH\_GROUP1** and **DH\_GROUP2** algorithms have been cracked in the industry. If these algorithms of low security levels are used, the negotiated keys may be cracked by attackers. It is recommended that the **DH\_GROUP14**, **DH\_GROUP15**, **DH\_GROUP19**, **DH\_GROUP20**, or **DH\_GROUP31** algorithm be used to ensure IKE SA security.

You can run **SET SECPLY** to adjust the configuration security level to prohibit the parameter from being set to **DH\_GROUP1** or **DH\_GROUP2**.

- **BTSIKEPROPOSAL.DHGRP** (for the GBTS)

The parameter value can be **DH\_GROUP1**, **DH\_GROUP2**, **DH\_GROUP14**, or **DH\_GROUP15**.

 NOTE

The **DH\_GROUP1** and **DH\_GROUP2** algorithms have been cracked in the industry. If these algorithms of low security levels are used, the negotiated keys may be cracked by attackers. It is recommended that the **DH\_GROUP14** or **DH\_GROUP15** algorithm be used to ensure IKE SA security.

## PRF Algorithm

PRF is a highly reliable unidirectional function that generates keys. After exchanging key materials using a DH algorithm, communicating peers use these materials as an input to PRF to generate a key.

The following parameters specify a PRF algorithm:

- **IKEPROPOSAL.PRFBALG (LTE eNodeB, 5G gNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB)

The parameter value can be **HMAC\_MD5**, **HMAC\_SHA1**, **AES128\_XCBC**, **HMAC\_SHA256**, or **HMAC\_SHA384**.

### NOTE

The HMAC\_MD5 algorithm has been cracked. As this algorithm is weak in security, the pseudorandom numbers generated using this algorithm may be guessed by attackers. It is recommended that the HMAC\_SHA256 or HMAC\_SHA384 algorithm be used to ensure IKE SA security.

You can run **SET SECPY** to adjust the configuration security level to prohibit the parameter from being set to **HMAC\_MD5**, **HMAC\_SHA1**, or **AES128\_XCBC**.

- **BTSIKEPROPOSAL.PRFBALG** (for the GBTS)

The parameter value can be **AES128\_XCBC**.

For details about PRF, see IETF RFC 4306 and IETF RFC 7296.

### 4.1.1.1.4 IKE SA Lifecycle

An IKE SA has a lifecycle, and it will be automatically updated before the lifecycle expires.

An excessively long lifecycle may cause keys to be cracked, while an excessively short lifecycle will trigger frequent IKE negotiations, which may interrupt ongoing IPsec sessions. IKE negotiation takes a long period of time as DH exchange and session key calculation are involved. A lifecycle longer than 3600 seconds is recommended.

The following parameters specify an IKE SA lifecycle:

- **IKEPROPOSAL.DURATION (5G gNodeB, LTE eNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB)
- **BTSIKEPROPOSAL.DURATION** (for the GBTS)

### NOTE

In several lifecycles after a base station is reset, IKE SA renegotiations are performed around the same period of time. The values of counters related to IKE SA re-key in these lifecycles may be different from those before the base station reset.

After these lifecycles, renegotiations are performed at different periods of time due to the network fluctuation. Therefore, the values of the counters related to IKE SA re-key tend to be consistent with those before the base station reset.

### 4.1.1.1.5 IKE Reauthentication

Only IKEv2 supports IKE reauthentication.

IKE authentication is performed only when the first IKE SA is established, that is, in the IKE\_AUTH exchange (the third and fourth messages shown in [Figure 4-4](#)) of

IKEv2 negotiation. Before the IKE SA lifecycle expires, the IKE SA is updated through the CREATE\_CHILD\_SA exchange of IKEv2 negotiation (see [Figure 4-5](#)). However, this exchange does not authenticate the identities of communicating peers, raising security risks.

To address this issue, IKE reauthentication is adopted without adding new procedures to IKEv2 negotiation. Before the IKE reauthentication period begins, reauthentication is initiated through IKE\_INIT (the first and second messages shown in [Figure 4-5](#)) and IKE\_AUTH (the third and fourth messages shown in [Figure 4-5](#)) messages. When 70% of the reauthentication period has elapsed, the base station starts to negotiate a new IKE SA with the peer. After the IKE reauthentication, the IKE SA and all IPsec SAs under the IKE SA are updated. The IKE SA will be updated when the IKE SA lifecycle or the IKE reauthentication period expires. Therefore, it is recommended that the IKE reauthentication period be longer than the IKE SA lifecycle. Otherwise, IKE SA updates will never be triggered by the expiration of the IKE SA lifecycle.

Base stations perform IKE reauthentication in either active mode or passive mode. The passive mode is enabled by default for base stations.

- In active mode, IKE reauthentication is performed based on the local reauthentication policy. The local end sends an IKE\_INIT/IKE\_AUTH message to initiate IKE reauthentication and does not send an AUTH\_LIFETIME notification to the peer (for example, an SeGW).
- In passive mode, the local end does not proactively send an IKE\_INIT/IKE\_AUTH message to initiate IKE reauthentication. Instead, the local end responds to the AUTH\_LIFETIME notification with an IKE\_INIT/IKE\_AUTH message to initiate IKE reauthentication. The passive mode is generally used with the Extensible Authentication Protocol (EAP). For details, see IETF RFC 4478 "Repeated Authentication in Internet Key Exchange (IKEv2) Protocol."

The **IKEPROPOSAL.REAUTHSW (LTE eNodeB, 5G gNodeB)** and **IKEPROPOSAL.REAUTHLT (LTE eNodeB, 5G gNodeB)** parameters specify whether to enable the active mode and the reauthentication period, respectively. The following describes two IKE reauthentication scenarios for base stations:

- Base stations support only passive mode (supported by default).  
In this scenario, the **IKEPROPOSAL.REAUTHSW (LTE eNodeB, 5G gNodeB)** parameter is set to **OFF**. To enable IKE reauthentication, the peer must support the active IKE reauthentication mode or must be able to function as a responder and send an AUTH\_LIFETIME notification to the local end. Otherwise, IKE reauthentication cannot be initiated.
- Base stations support both the active and passive modes.  
In this scenario, the **IKEPROPOSAL.REAUTHSW (LTE eNodeB, 5G gNodeB)** parameter is set to **ON**. The peer must support IKEv2 negotiation.
  - If the peer supports active IKE reauthentication or can send an AUTH\_LIFETIME notification to the base station, the IKE reauthentication period is the smaller one of the IKE reauthentication periods at two ends.
  - If the peer does not support active IKE reauthentication and cannot send an AUTH\_LIFETIME notification to the base station, the IKE reauthentication period is that configured at the local end. Generally, there is either active IKE reauthentication or an AUTH\_LIFETIME notification sent by the peer, but not both.

To support IKEv2 reauthentication, base stations have the following hardware requirements:

- The GBTS must be configured with GTMUc+UMPT\_L.
- The eGBTS, NodeB, eNodeB, gNodeB, and MBTS must be configured with a UMPT, UMDU, UTRPc, or MDUC.

 **NOTE**

The MDUC supports only GSM, UMTS, or GSM and UMTS dual-mode.

#### 4.1.1.2 IKE Negotiation

There are two IKE versions: IKEv1 and IKEv2. The two versions have different negotiation processes. The following parameters specify the IKE version:

- **IKEPEER.IKEVERSION (LTE eNodeB, 5G gNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB)
- **BTSIKEPEER.IKEVERSION** (for the GBTS)

 **NOTE**

If the **IKEPEER.IKEVERSION (LTE eNodeB, 5G gNodeB)** parameter is set to **IKE\_V1** and the **IKEPEER.IDTYPE (LTE eNodeB, 5G gNodeB)** parameter is set to **DN** for a base station, an IPsec tunnel will fail to be established between the base station and SeGW.

There is a low probability that both ends of an IKE negotiation (IKEv1/IKEv2) initiate the negotiation at the same time. In such a case, the base station and peer end may use different policies to avoid duplicate IKE SAs, causing negotiation exceptions. Therefore, it is recommended that the SeGW be configured to work in response-only mode.

When the **IKEPEER.IKEVERSION (5G gNodeB, LTE eNodeB)** parameter is set to **IKE\_V1** for a base station, only the default routing domain (VRF index is 0) is supported in IPv4.

An IKE negotiation requires the following parameters:

- **IKEPEER.LOCALIP (LTE eNodeB, 5G gNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB) or **BTSIKEPEER.LOCALIP** (for the GBTS): indicates the IP address of the IKE local end.
- **IKEPEER.REMOTEIP (LTE eNodeB, 5G gNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB) or **BTSIKEPEER.REMOTEIP** (for the GBTS): indicates the IP address of the IKE peer.

After an IKE negotiation, seven keys are generated, of which:

- Two are used for encryption of subsequent messages.
- Two are used for integrity protection of subsequent messages.
- Two serve as the cipher keys for identity authentication.
- One is used to derive the integrity protection key and cipher key for the IPsec SAs.

Subsequent messages are sent during IKE negotiation after key exchange.

For details about how keys are generated, see [4.1.1.3 DH Group and PRF Algorithm](#).

The base station checks the IKE negotiation status every 7 minutes. If an IKE negotiation fails and digital-certificate-based authentication is used, the base station checks the digital certificates used for the IKE negotiation. If a digital

certificate is abnormal (for example, the digital certificate has expired or been revoked or the certificate file does not exist), an automatic certificate obtaining procedure is triggered. For details, see *PKI*.

## IKEv2 Negotiation

An IKEv2 negotiation process consists of an initial exchange and a CREATE\_CHILD\_SA exchange. The initial exchange involves interactions of four messages, as shown in [Figure 4-4](#). The former two messages are used to negotiate IKE SA parameters, and the latter two are used to establish an IKE SA and a pair of IPsec SAs. If more than one pair of IPsec SAs needs to be established, one CREATE\_CHILD\_SA exchange is required for each additional pair of IPsec SAs, as shown in [Figure 4-5](#).

Figure 4-4 IKEv2 initial exchange

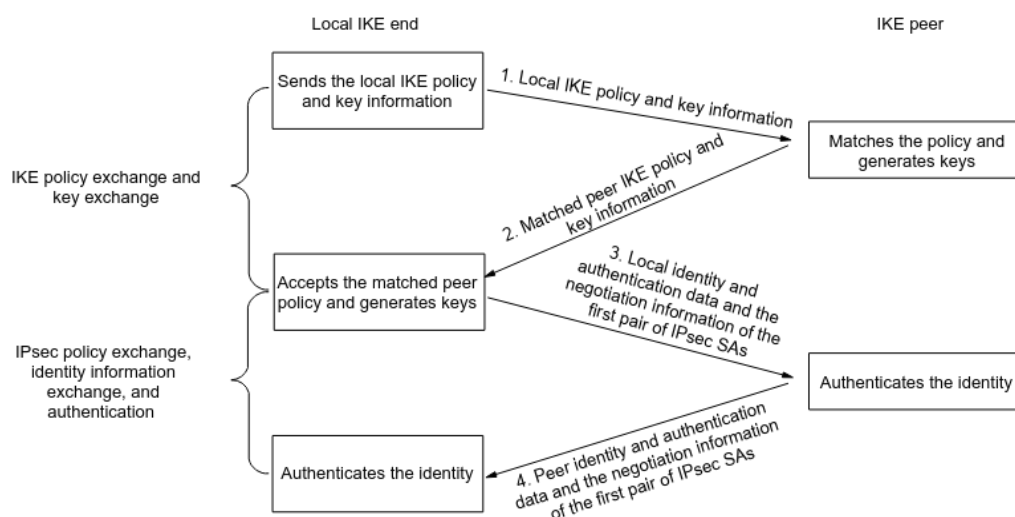
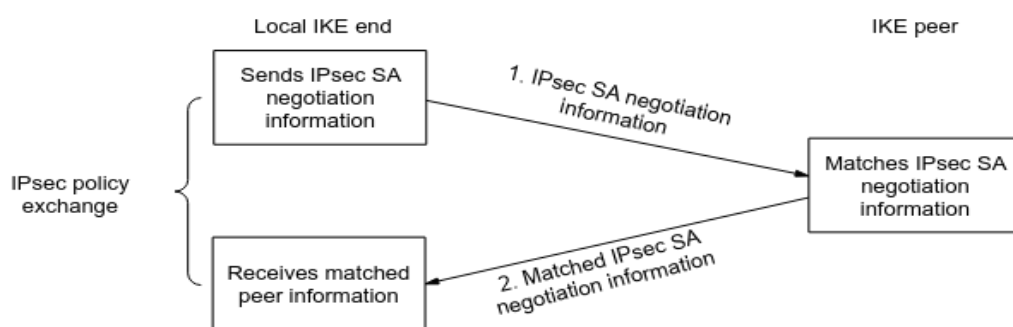


Figure 4-5 IKEv2 CREATE\_CHILD\_SA exchange



### 4.1.1.3 IKE DPD

IP and IPsec are unidirectional, connectionless protocols. If one end does not respond due to reasons such as system failures, the peer end will not necessarily know that anything is wrong. If this occurs and the sender continues to send IPsec service flows, the service flows will be lost. Dead peer detection (DPD) has been introduced to resolve this issue for base stations. DPD is enabled by setting the

**IKEPEER.DPD (LTE eNodeB, 5G gNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB) or **BTSIKEPEER.DPD** (for the GBTS) parameter to **PERIODIC**.

The local end starts DPD if it does not receive IPsec packets from the peer end for a period of time specified by the **IKEPEER.DPDIDLETIME (LTE eNodeB, 5G gNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB) or **BTSIKEPEER.DPDIDLETIME** (for the GBTS) parameter. The local end also starts DPD if it needs to send IPsec packets to the peer end.

If the local end receives an acknowledgement from the peer end after sending a DPD message, it considers the peer end online and normal. If the local end does not receive any acknowledgement from the peer end after retransmitting the DPD message multiple times, it considers the peer end unresponsive. If this happens, the local end re-initiates IKE negotiation. The number of DPD message retransmissions is specified by the **IKEPEER.DPDRETRN (LTE eNodeB, 5G gNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB) or **BTSIKEPEER.DPDRETRN** (for the GBTS) parameter. The local end retransmits DPD messages at an interval specified by the **IKEPEER.DPDRETRI (LTE eNodeB, 5G gNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB) or **BTSIKEPEER.DPDRETRI** (for the GBTS) parameter.

For details about DPD, see IETF RFC 3706.

#### 4.1.1.4 IKEv2 Packet Fragmentation

IKE packets that contain excessive payload, such as certificates, may be fragmented at the IP layer. To address this problem, IETF RFC 7383 introduces a fragmentation mechanism for IKEv2 packets. IKEv2 packets that exceed a specified size are fragmented in advance with the IKEv2 packet header contained after fragmentation. An IKEv2 packet fragment, under proper configurations, will have a size less than the transmission path MTU (PMTU) and do not need to be fragmented at the IP layer. This function is supported by both IPv4 IPsec and IPv6 IPsec.

The **IKEPEER.IKEV2FRGSW (LTE eNodeB, 5G gNodeB)** parameter specifies whether to enable IKEv2 packet fragmentation negotiation at the local base station. The default value is **OFF**. The base station fragments IKEv2 packets and reassembles IKEv2 fragments only when the negotiation function is enabled on both the local base station and the peer SeGW. For details about how to enable IKEv2 packet fragmentation negotiation on the SeGW side, see the SeGW product documentation.

The size of IKEv2 packet fragments sent by the base station is specified by the **IKEPEER.IKEV2FRGMTU (LTE eNodeB, 5G gNodeB)** parameter, which is set to **1280** (in the unit of bytes) by default. The **IKE\_AUTH** and **CREATE\_CHILD\_SA** packets that exceed the parameter value will be fragmented.

The IKEv2 fragmentation negotiation result is indicated by the **IKEv2 Fragmentation Negotiation Result** parameter, which can be queried by running the **DSP IKESA** command with **Display Mode** set to **VERBOSE** and **User Display Phase** set to **PHASE1**.

The IKEv2 fragment sequence numbers can be obtained by using the IKE tracing function on the LMT.

### 4.1.1.5 IKEv2 OCS

As specified in RFC 4806, OCS can be used to check the peer certificate status during IKEv2 negotiation.

The **IKEPEER.OCS** (*LTE eNodeB, 5G gNodeB*) parameter can be set on the local base station to determine whether the local IKEv2 end uses OCS to check the certificate revocation status of the peer end. When this switch is turned on, IKE messages sent by the local end contain OCS requests, requiring the IKE messages sent by the peer end to contain OCS responses. This switch applies only to IKEv2 and takes effect only when the **IKERESPONSE.AUTHMETH** (*LTE eNodeB, 5G gNodeB*) parameter is set to **IKE\_CERT\_SIG**. The effective status of this switch can be queried by running the **DSP IKESA** command.

If the IKE messages sent by the peer end (SeGW or the peer base station) contain OCS requests, the base station includes the OCS status it has obtained in the IKE messages returned to the peer end. For details about how the base station obtains the OCS status, see *PKI*.

The IKEv2 OCS-based check policy and CRL-based check policy of a base station can be configured by the **IKPEER.CRLPOLICY** (*LTE eNodeB, 5G gNodeB*) parameter.

- If this parameter is set to **GLOBAL**, the CRL-based policy used for the base station in IKE scenarios is the same as that in the **SET CRLPOLICY** command. For details about how to set the check policy through the **SET CRLPOLICY** command, see *PKI*.
- If this parameter is set to **ALARM**, the base station performs CRL- and OCS-based certificate validity checks on the peer NE, and reports ALM-26834 Peer Certificate Invalid if the peer certificate is invalid.
- If this parameter is set to **DISCONNECT**, the base station performs CRL- and OCS-based certificate validity checks on the peer NE, and reports ALM-26834 Peer Certificate Invalid and disconnects from the NE if the peer certificate is invalid.

#### NOTE

In the case that the base station needs to determine whether the peer certificate is invalid based on CRL and OCS, it determines that the peer certificate is invalid and reports ALM-26834 Peer Certificate Invalid if any of the following conditions is met:

- The peer certificate's OCS status is revoked or unknown.
- Verification of the peer certificate's OCS status fails.
- Acquisition of the peer certificate's OCS status fails.
- The CRL contains the revocation record of the certificate in question.

---

#### CAUTION

Before configuring the **IKPEER.OCS** (*LTE eNodeB, 5G gNodeB*) parameter, ensure that the peer device supports OCS. If the switch is turned on but the peer device does not support OCS, IKE links will be disconnected.

After IKEv2 OCS is enabled, the size of IKE packets may exceed the interface maximum transmission unit (MTU), causing packet fragmentation. To prevent this, the transmission MTU must be properly planned.

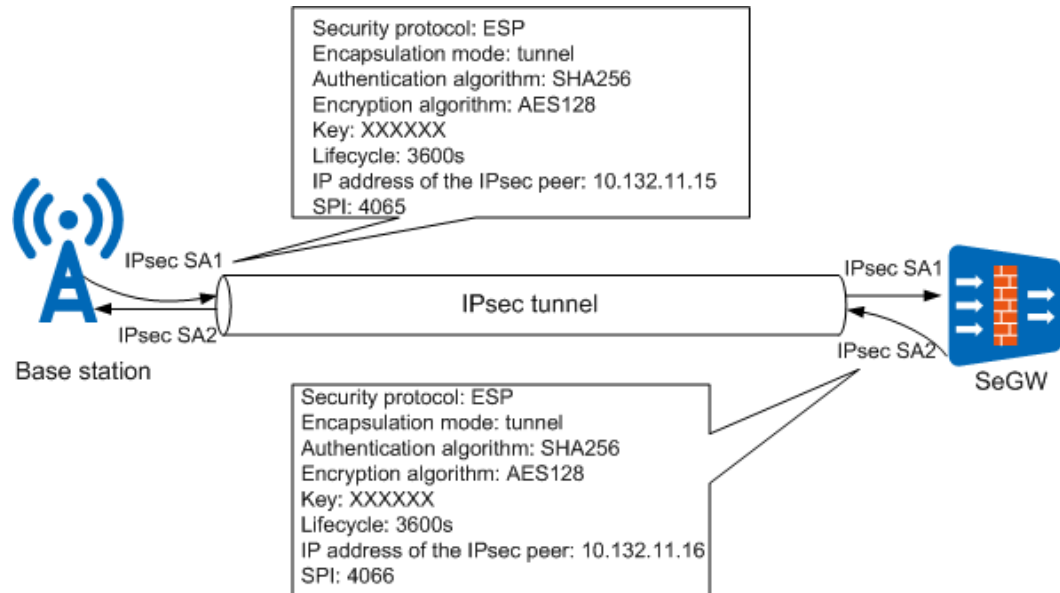
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## 4.1.2 IPsec SA

IPsec SAs are unidirectional. At least two IPsec SAs are required to protect two-way communications, one for each direction. [Figure 4-6](#) shows an example of an IPsec SA.

**Figure 4-6** Example of an IPsec SA



### NOTE

The security parameter index (SPI) is used to identify an IPsec SA. Each IPsec SA has a unique SPI.

As shown in [Figure 4-6](#), IPsec SAs define security measures for packets exchanged between communicating peers. For details, see:

- [4.1.2.1 IPsec Proposal](#)
- [4.1.2.2 IPsec Policies](#)

If the size of a packet in plaintext is greater than or equal to the maximum transmission unit (MTU) of the access network, the packet will be fragmented. For details, see [4.1.2.3 IPsec Packet Fragmentation](#).

For details about IPsec SAs, see IETF RFC 4301.

### 4.1.2.1 IPsec Proposal

An IPsec proposal defines the security protocol, encapsulation mode, encryption algorithm, and authentication algorithm for encapsulating packets.

In practical applications, security protocols, encapsulation modes, and encryption and authentication algorithms are negotiated between IPsec peers.

For details about security protocols, see [4.1.2.1.1 Security Protocols](#). For details about packet encapsulation modes, see [4.1.2.1.2 Encapsulation Modes](#). For details about encryption and authentication algorithms used by encryption and integrity protection, see [4.1.2.1.3 Encryption and Authentication Algorithms](#).



Algorithm keys are generated based on IKE negotiation between communicating peers. For details, see [4.1.1 IKE SA](#).

4.1.2.1.1 Security Protocols

IPsec uses two security protocols: AH and ESP, which are described in [Table 4-1](#).

Table 4-1 AH and ESP

| Security Protocol | Function                                          | Verification Scope                                            | Application Scenario  |
|-------------------|---------------------------------------------------|---------------------------------------------------------------|-----------------------|
| AH                | Integrity protection and anti-replay              | Entire IP packet (including the IP packet header and payload) | Non-confidential data |
| ESP               | Integrity protection, anti-replay, and encryption | IP payload only                                               | Confidential data     |

AH and ESP can be used separately or jointly. When they are used jointly, ESP and AH are used in sequence for packet encapsulation. If both AH and ESP are used, each IPsec peer requires two IPsec SAs: one for AH and the other for ESP. The following parameters specify whether to use AH or ESP:

- **IPSECPROPOSAL.TRANMODE (LTE eNodeB, 5G gNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB)
- **BTSIPSECPROPOSAL.TRANMODE** (for the GBTS)

 NOTE

It is recommended that only the ESP protocol be used on the base station side.

For details about AH, see IETF RFC 4302. For details about ESP, see IETF RFC 4303.

4.1.2.1.2 Encapsulation Modes

IPsec tunnels protect IP packets by packet encapsulation. Two IPsec packet encapsulation modes are supported: transport mode and tunnel mode.

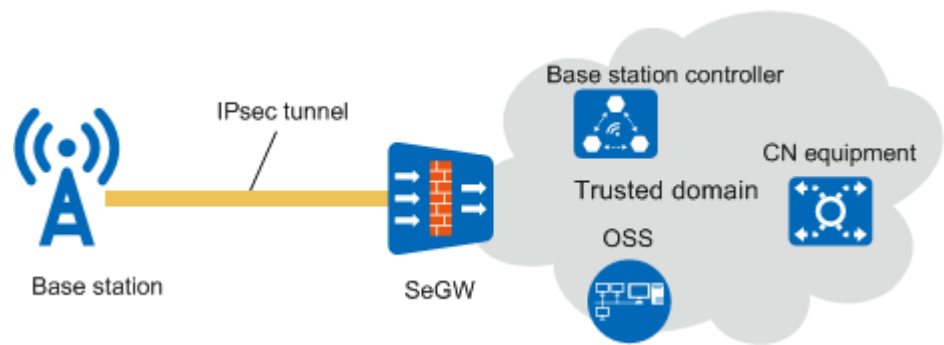
- In transport mode, only the upper-layer data of the IP packets is encrypted. This encapsulation mode features higher performance and is generally used for end-to-end security protection. The two devices that encrypt and decrypt packets must be the original sender and final receiver, respectively, of the packets. For example, this mode applies to data communication between hosts.
- In tunnel mode, IPsec protects entire IP packets. This encapsulation mode features higher security and can be applied to end-to-end security protection or the protection of a certain segment in a tunnel. That is, tunnel mode can be used when the device that encrypts or decrypts packets is not the original sender or final receiver of the packets. Generally, IPsec peers communicate with each other in tunnel mode. [Figure 4-7](#) shows an example of using the

tunnel mode over the IPsec tunnel between a Huawei base station and an SeGW.

The following parameters specify the packet encapsulation mode:

- **IPSECPROPOSAL.ENCAPMODE (5G gNodeB, LTE eNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB)
- **BTSIPSECPROPOSAL.ENCAPMODE** (for the GBTS)

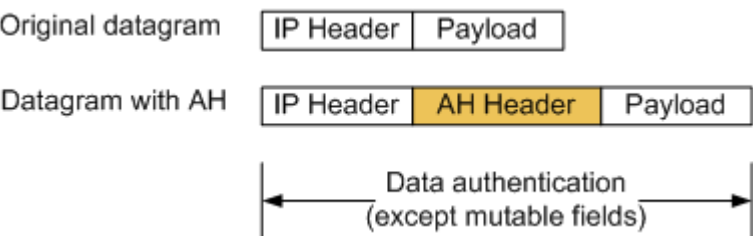
Figure 4-7 Tunnel mode example



Transport Mode

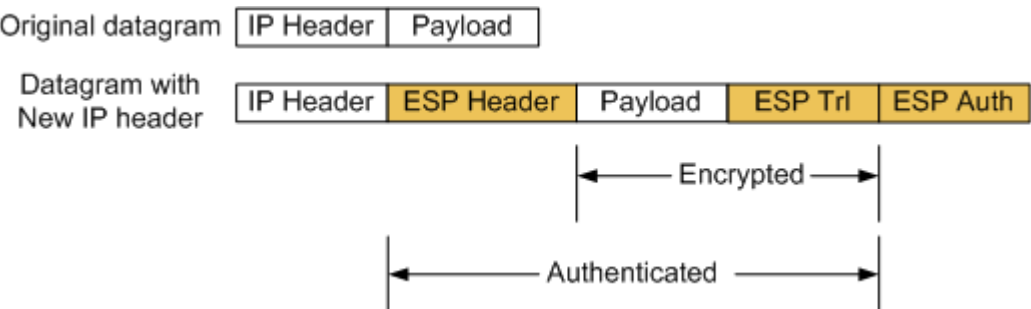
An AH header is inserted after the IP header of the original packet, but before the payload of the original packet, as shown in Figure 4-8.

Figure 4-8 AH packet encapsulation format used in transport mode



An ESP header is inserted after the IP header of the original packet, but before the payload of the original packet. An ESP trailer and an ESP authenticator are attached to the rear of the original packet, as shown in Figure 4-9.

Figure 4-9 ESP packet encapsulation format used in transport mode

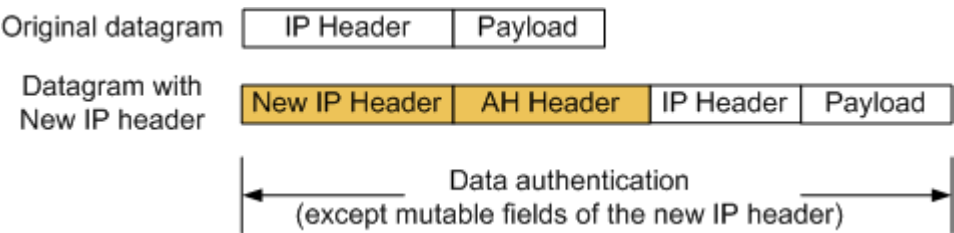


The source IP address for packets sent by a base station is the service or operation and maintenance (OM) IP address of the base station, and the destination IP address for the packets is the service or OM IP address of peer equipment.

Tunnel Mode

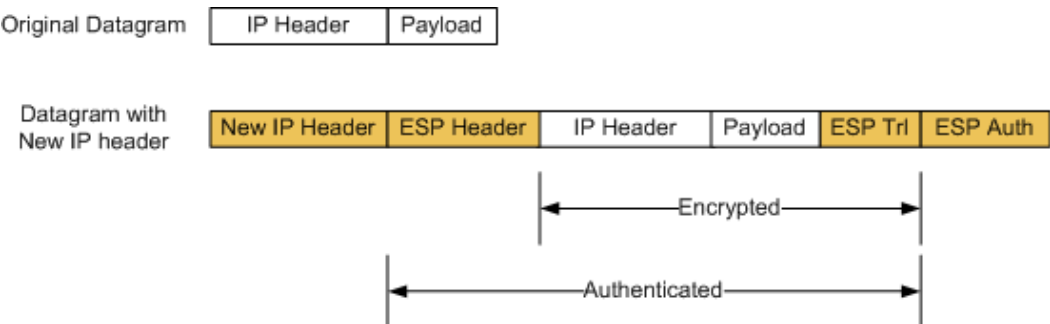
An AH header is prefixed to the IP header of the original packet, and a new IP header is prefixed to the AH header, as shown in [Figure 4-10](#).

Figure 4-10 AH packet encapsulation format used in tunnel mode



An ESP header is prefixed to the IP header of the original packet, and a new IP header is prefixed to the ESP header. An ESP trailer and an ESP authenticator are attached to the rear of the original packet, as shown in [Figure 4-11](#).

Figure 4-11 ESP packet encapsulation format used in tunnel mode



AH does not provide integrity protection for certain mutable fields in an IP packet, such as Type of Service, Time to Live, and Checksum, which may be legally modified during transmission.

In tunnel mode, IPsec encrypts the IP header of the original packet and generates a new IP header, which is used for route forwarding. The new IP header always uses the interface IP address of a base station and the IP address of the peer equipment (usually an SeGW) as the source and destination IP addresses, respectively. The IP header of the original packet contains the service or OM IP address of the base station.

NOTE

The Type of Service field in the new IP packet header is duplicated from the original IP packet header.

Summary

The transport and tunnel modes are different in security and performance. The tunnel mode provides higher security than the transport mode because the former

implements encryption and integrity protection for entire original IP packets. The transport mode provides better transmission performance than the tunnel mode because the new IP header added in tunnel mode requires more bandwidth. In addition, in tunnel mode, an SeGW must be deployed on the network to separate the trusted and untrusted domains. The SeGW must also support functions such as encapsulation, encryption, and integrity protection in tunnel mode. However, in transport mode, both communicating peers must support functions such as IKE negotiation, encryption, and integrity protection. When selecting an encapsulation mode, users need to comprehensively consider security and performance, as well as the encapsulation mode capability of the IPsec peer.

#### 4.1.2.1.3 Encryption and Authentication Algorithms

##### Encryption Algorithms

ESP encrypts IP packets to prevent unauthorized access during packet transmission. Encryption algorithms use symmetric keys to ensure that the same keys are used by IPsec peers for encryption and decryption.

The following parameters specify the ESP encryption algorithms:

- **IPSECPROPOSAL.ESPENCALG (LTE eNodeB, 5G gNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB): The parameter value can be **NULL**, **DES**, **3DES**, **AES128**, **AES192**, **AES256**, **AES\_GCM\_128**, **AES\_GCM\_256**, **NULL\_AUTH\_GMAC\_128**, or **NULL\_AUTH\_GMAC\_256**.
  - **AES\_GCM\_128** and **AES\_GCM\_256** are authenticated encryption with associated data (AEAD) encryption algorithms that provide both authentication and confidentiality protection. For details, see IETF RFC 4106.
  - **NULL\_AUTH\_GMAC\_128** and **NULL\_AUTH\_GMAC\_256** are null encryption algorithms that provide only authentication but not confidentiality protection. For details about these algorithms, see IETF RFC 4543.

##### NOTE

NULL indicates a null algorithm, and NULL\_AUTH\_GMAC\_128 and NULL\_AUTH\_GMAC\_256 indicate null encryption algorithms. The DES algorithm has been cracked, and the 3DES algorithm is considered insufficiently secure in the industry. If these algorithms of low security levels are used, encrypted data may be cracked by attackers. It is recommended that the AES\_GCM\_128 or AES\_GCM\_256 algorithm be used to ensure IPsec SA security.

The NULL, NULL\_AUTH\_GMAC\_128, and NULL\_AUTH\_GMAC\_256 encryption algorithms can be used for problem diagnosis only when the connection between the base station and SeGW is abnormal. These three encryption algorithms are not recommended in other scenarios.

You can run **SET SECPLY** to adjust the configuration security level to prohibit the parameter from being set to **DES** or **3DES**.

- **BTSIPSECPROPOSAL.ESPENCALG** (for the GBTS): The parameter value can be **NULL**, **DES**, **3DES**, **AES128**, **AES192**, or **AES256**.

 NOTE

NULL indicates a null algorithm. The DES algorithm has been cracked, and the 3DES algorithm is considered insufficiently secure in the industry. If these algorithms of low security levels are used, encrypted data may be cracked by attackers. It is recommended that the AES128 or AES256 algorithm be used to ensure IPsec SA security.

The NULL encryption algorithm can be used for problem diagnosis only when the connection between the base station and SeGW is abnormal. This encryption algorithm is not recommended in other scenarios.

## Authentication Algorithms

Both AH and ESP can check the integrity of IP packets to determine whether the IP packets were tampered with during transmission. Authentication algorithms are implemented mainly based on hash functions, which accept messages of any length and generate outputs of a fixed length. The outputs are called message digests. Upon receiving a packet from the IPsec local end, the IPsec peer calculates the digests and compares them with those carried in the packet. If the two sets of digests are identical, the packet is intact without being tampered with during transmission.

The following parameters specify the AH and ESP authentication algorithms:

- **IPSECPROPOSAL.AHAUTHALG (LTE eNodeB, 5G gNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB), which specifies the AH algorithm. The parameter value can be any of the following:
  - MD5
  - SHA1
  - SHA256
  - AES-XCBC-MAC-96
- **IPSECPROPOSAL.ESPAUTHALG (LTE eNodeB, 5G gNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB), which specifies the ESP algorithm. The parameter value can be any of the following:
  - NULL
  - MD5
  - SHA1
  - SHA256
  - AES-XCBC-MAC-96

 NOTE

The MD5 algorithm has been cracked, and the SHA1 and AES-XCBC-MAC-96 algorithms are considered insufficiently secure in the industry. If these algorithms of low security levels are used, the transmitted data may be forged by attackers. It is recommended that SHA256 be used as the AH algorithm and SHA256 be used as the ESP algorithm to ensure IPsec SA security.

You can run **SET SECPY** to adjust the configuration security level to prohibit the parameter from being set to **MD5**, **SHA1**, or **AES-XCBC-MAC-96**.

- **BTSIPSECPROPOSAL.AHAUTHALG** (for the GBTS) and **BTSIPSECPROPOSAL.ESPAUTHALG** (for the GBTS), which specify the AH and ESP algorithms, respectively. The parameter value can be any of the following:
  - NULL

- SHA256

#### NOTE

NULL indicates a null algorithm. If this algorithm of a low security level is used, the transmitted data may be forged by attackers. It is recommended that the SHA256 algorithm be used to ensure IPsec SA security.

### 4.1.2.1.4 Extended Serial Number

The base station supports the extended sequence numbers (ESNs) defined in the ESP protocol for IPsec SAs to cope with re-negotiation caused by sequence number (SN) reversion upon high-speed traffic. The SN in IPsec traffic is 32 bits. The local end can negotiate with the peer end to support 64-bit SN, reducing the number of IPsec SA re-negotiations before the life cycle expires.

The function of using ESNs defined in the ESP protocol is controlled by the **IPSECPROPOSAL.ESPESNSW (LTE eNodeB, 5G gNodeB)** parameter.

#### NOTE

- The function of using ESNs defined in the ESP protocol is applicable only to IKEv2, but not to IKEv1. That is, this function cannot be enabled when the **IKEPEER.IKEVERSION (LTE eNodeB, 5G gNodeB)** parameter is set to **IKE\_V1** or the **SECURITYTEMPLATE.IKEVERSION (LTE eNodeB, 5G gNodeB)** parameter is set to **IKE\_V1**.
- The function of using ESNs defined in the ESP protocol can be used only when the anti-replay window function is enabled. That is, this function cannot be enabled when the **IPSECPOLICY.REPLAYWND (LTE eNodeB, 5G gNodeB)** parameter is set to **WND\_DISABLE** or the **SECURITYTEMPLATE.REPLAYWND (LTE eNodeB, 5G gNodeB)** parameter is set to **WND\_DISABLE**.
- The function of using ESNs defined in the ESP protocol is applicable only to ESP scenarios, but not to AH or AH+ESP scenarios.

### 4.1.2.2 IPsec Policies

Security services offered by IPsec are based on IPsec policies defined by a security policy database (SPD). The SPD specifies which security services are available to IP packets and how to obtain these services.

The following parameters specify an IPsec policy:

- **IPSECPOLICY.SPGN** (for the eGBTS/NodeB/eNodeB/gNodeB) or **BTSIPSECPOLICY.SPGN** (for the GBTS): specifies the name of an IPsec policy group.
- **IPSECPOLICY.SPSN (5G gNodeB, LTE eNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB) or **BTSIPSECPOLICY.SPSN** (for the GBTS): specifies the sequence number of an IPsec policy.

An IPsec policy consists of access control list (ACL), perfect forward secrecy (PFS), IPsec SA lifecycle, and anti-replay window.

Base stations can negotiate one or multiple IPsec SAs based on a set of parameters related to IPsec policies. The number of negotiated IPsec SAs depends on the number of configured ACL rules. For an ACL rule whose **ACLRULE.ACTION (5G gNodeB, LTE eNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB) or **BTSACLRULE.ACTION** (for the GBTS) is set to **PERMIT**, one incoming IPsec SA and one outgoing IPsec SA can be negotiated for the ACL rule.

#### 4.1.2.2.1 ACL

An ACL consists of a series of ACL rules, which specify the data flows to be protected. Only data flows that comply with ACL rules can enter an IPsec tunnel.

#### ACL Rule Configuration Mode

IP to Any and Any to Any are two common modes for ACL rule configuration. Set parameters as follows in the **ACLRULE** and **BTSACLRULE** MOs based on the configuration mode:

- IP to Any mode  
In this mode, IPsec is used to protect data flows with the source IP address being the specified value.
  - **SIP (5G gNodeB, LTE eNodeB)**: set to the source IP address of the data flows
  - **SWC (5G gNodeB, LTE eNodeB)**: set to **0.0.0.0**
  - **DIP (5G gNodeB, LTE eNodeB)**: set to **0.0.0.0**
  - **DWC (5G gNodeB, LTE eNodeB)**: set to **255.255.255.255**
  - **ACTION (5G gNodeB, LTE eNodeB)**: set to **PERMIT**
- Any to Any mode  
In this mode, IPsec is used to protect all data flows.
  - a. Configure an ACL rule with the parameter settings as follows:
    - **SIP (5G gNodeB, LTE eNodeB)**: set to an interface IP address
    - **SWC (5G gNodeB, LTE eNodeB)**: set to **0.0.0.0**
    - **DIP (5G gNodeB, LTE eNodeB)**: set to **0.0.0.0**
    - **DWC (5G gNodeB, LTE eNodeB)**: set to **255.255.255.255**
    - **ACTION (5G gNodeB, LTE eNodeB)**: set to **PERMIT**
  - b. Configure an ACL rule for data flows that do not need to be protected by IPsec with the parameter settings as follows:
    - **SIP (5G gNodeB, LTE eNodeB)**: set to the source IP address of the data flows
    - **SWC (5G gNodeB, LTE eNodeB)**: set to **0.0.0.0**
    - **DIP (5G gNodeB, LTE eNodeB)**: set to **0.0.0.0**
    - **DWC (5G gNodeB, LTE eNodeB)**: set to **255.255.255.255**
    - **ACTION (5G gNodeB, LTE eNodeB)**: set to **DENY**
  - c. Configure an ACL rule in Any to Any mode with the parameter settings as follows:
    - **SIP (5G gNodeB, LTE eNodeB)** and **DIP (5G gNodeB, LTE eNodeB)**: set to **0.0.0.0**

- **SWC (5G gNodeB, LTE eNodeB) and DWC (5G gNodeB, LTE eNodeB):** set to 255.255.255.255
- **ACTION (5G gNodeB, LTE eNodeB):** set to **PERMIT**

The value of **RULEID (5G gNodeB, LTE eNodeB)** for an ACL rule with **ACTION (5G gNodeB, LTE eNodeB)** set to **DENY** must be smaller than that for an ACL rule with **ACTION (5G gNodeB, LTE eNodeB)** set to **PERMIT**.

The IP to Any mode is recommended. In the following example, the IP to Any mode is used. The method of configuring ACL rules depends on the network plan.

Two ACL rules cannot be applied to the same data flow.

#### NOTE

If ACL rules are configured on the base station to protect data flows and the peer IKEv2 entity carries multiple traffic selectors (TSs) in the traffic selector - initiator (TSi) and traffic selector - responder (TSr) payloads, the base station can process only the first TS in the TSi and TSr payloads on condition that the number of TSs in the two type of payloads are the same.

## Estimation of ACL Rule Reconfiguration Impact

This function estimates the impact of IPsec ACL rule reconfiguration on services based on the configuration data.

If a type of service will be added to or deleted from an IPsec tunnel after the following MML commands are executed, the system displays a message indicating the high-risk issues or terminates the operation, to avoid configuration impact on services.

- IPv4: **ADD ACLRULE/RMV ACLRULE**
- IPv6: **ADD ACLRULE6/RMV ACLRULE6**

#### NOTE

If the service data egress port corresponding to an IPsec ACL rule is not the IPsec binding port, this function takes effect after the preceding commands are executed.

This function can only estimate the impact of IPsec ACL rule reconfiguration performed by running MML commands. The **Support Forcible Execution** parameter is added to the preceding MML commands:

- If this parameter is set to **NO** in an MML command, the base station determines whether services will be added to or removed from an IPsec tunnel after the command is executed.
  - If the estimation result is that services will be affected, a failure message is returned and the failure cause is described.
  - If the estimation result is that services will not be affected, reconfiguration is performed.
- If this parameter is set to **YES** in an MML command, reconfiguration is performed without estimating the impact.



This function increases the execution time of a single MML command by up to 2s. This function does not apply to base stations using a GTMUC as the main control board.

## Source IP Address Verification for ACL Rules

This function verifies the source IP addresses in ACL rules based on the configuration data, when these rules are either being referenced by IPsec and reconfigured or referenced by IPsec.

If the source IP address in an ACL rule associated with the following MML commands does not match any IP address configured on the base station, the system either displays a high-risk warning or terminates the operation to prevent service-impacting misconfigurations.

- IPv4: **ADD ACLRULE/RMV ACLRULE/MOD ACLRULE/ADD IPSECBIND/ADD IPSECBINDITF/MOD IPSECBINDITF/ADD IPSECPOLICY/MOD IPSECPOLICY**
- IPv6: **ADD ACLRULE6/RMV ACLRULE6/MOD ACLRULE6/ADD IPSECBINDITF/MOD IPSECBINDITF/ADD IPSECPOLICY/MOD IPSECPOLICY**

Source IP address verification for IPsec ACL rules is applicable only to MML-command-based configurations. The **Support Forcible Execution** parameter is added to the preceding MML commands.

- If this parameter is set to **NO**, the source IP address in an ACL rule referenced by IPsec is verified upon MML command delivery.
  - If the source IP address does not match any IP address configured on the base station, the system returns a failure message with a description of the cause.
  - If the source IP address matches an IP address configured on the base station, the system proceeds with the configuration change.
- If this parameter is set to **YES**, the configuration change is executed directly upon MML command delivery without assessing service impact.

This function extends the execution time of a single MML command by up to 2s. This function does not apply to base stations using a GTMUC as the main control board.

### 4.1.2.2.2 ACL Narrow Down

During a single IKE negotiation between a base station and an SeGW, the base station initiates initial information exchange in most cases. The ACL rules of the base station should be included in those of the SeGW to ensure the highest security level for data flows. A single ACL rule allows negotiation results of only one incoming IPsec SA and one outgoing IPsec SA.

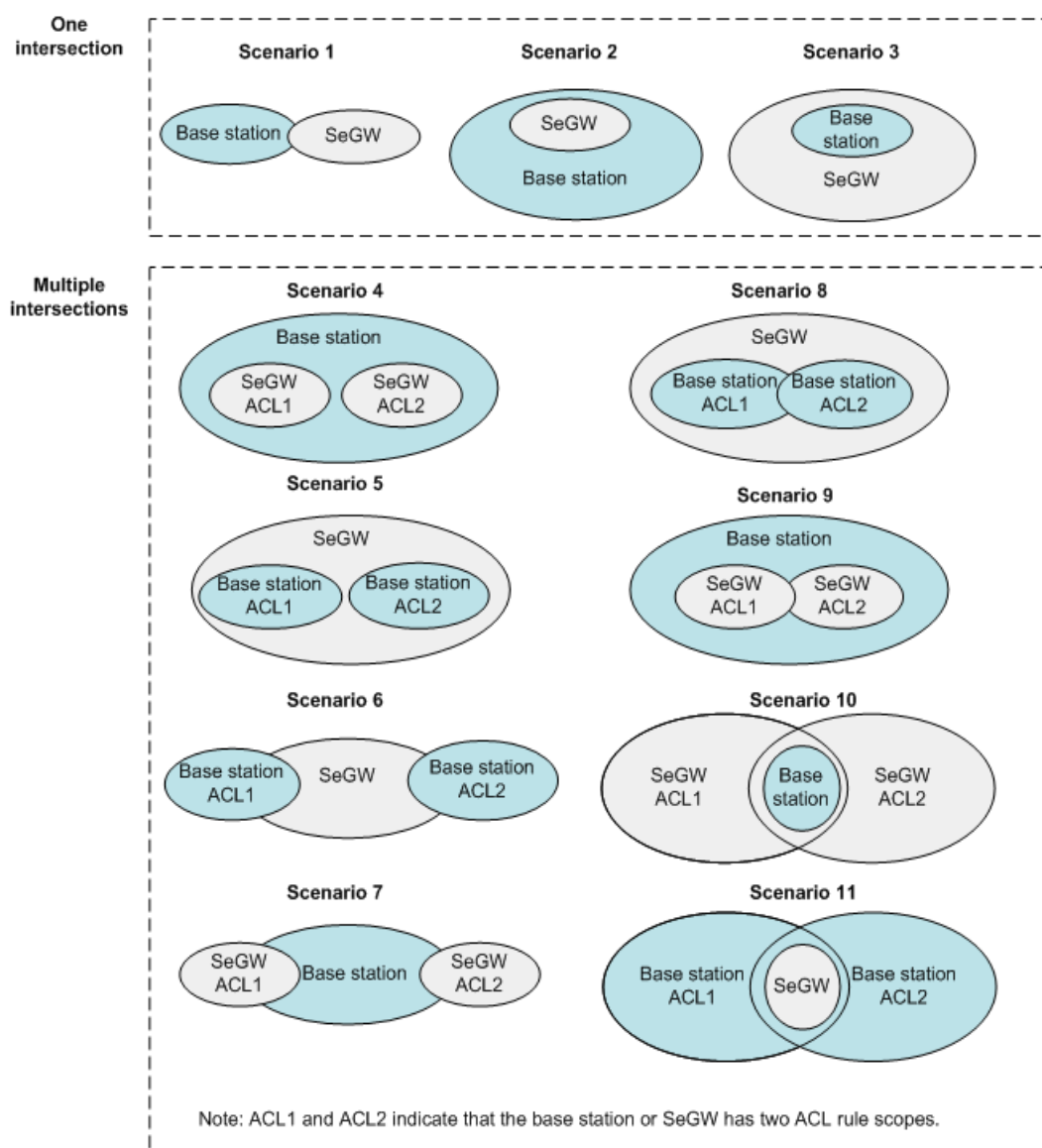
If initial information exchange is initiated by the SeGW, the SeGW serves as the initiator and the base station serves as the responder. If the ACL rules configured on the base station are inconsistent with those configured on the SeGW, IKE negotiation may fail. In this case, the base station should support IPsec ACL narrow down to obtain the ACL rules that are common to the SeGW and base station (also known as ACL rule intersections). The IPsec ACL narrow down function is controlled by the **IPSECPOLICY.NARROWDOWNSW** parameter. GBTSS do not support the IPsec ACL narrow down function.

The ACL rule configuration of the base station is different from that of an SeGW in either of the following ways:

- ACL rules overlap between the base station and the SeGW.
- The number of ACL rules configured on the base station is different from that configured on the SeGW.

In this case, there may be multiple intersections or only one intersection between the ACL rules, as shown in [Figure 4-12](#).

**Figure 4-12** ACL rule configuration of the base station different from that of the SeGW



**NOTE**

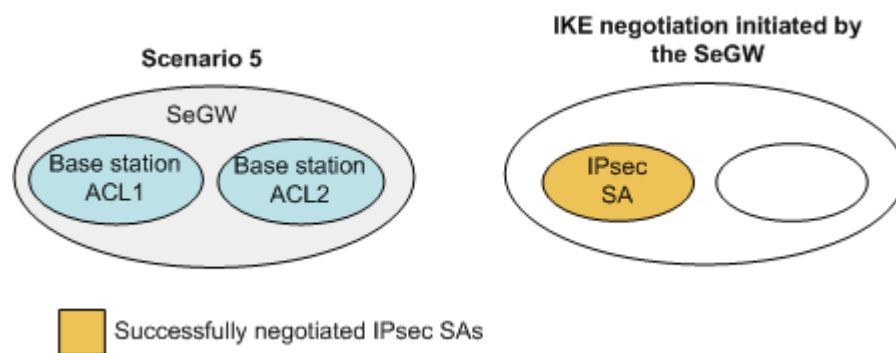
Base stations do not support the ACL narrow down function in scenarios 1, 6, and 7 among all the preceding scenarios.

If the SeGW initiates an IKE negotiation, the SA negotiation results obtained using ACL narrow down fulfill the following rules:

- If the base station is configured with multiple ACL rules, the negotiation result is the first successfully negotiated ACL rule intersection. Even if the ACL rules of the SeGW overlap other ACL rules of the base station, the negotiation will fail.
- If the SeGW is configured with multiple ACL rules, it initiates multiple IKE negotiations. The negotiation results are all successfully negotiated ACL rule intersections.

Use scenario 5 as an example. If the SeGW initiates IKE negotiation, the IPsec SA negotiation results are shown in [Figure 4-13](#).

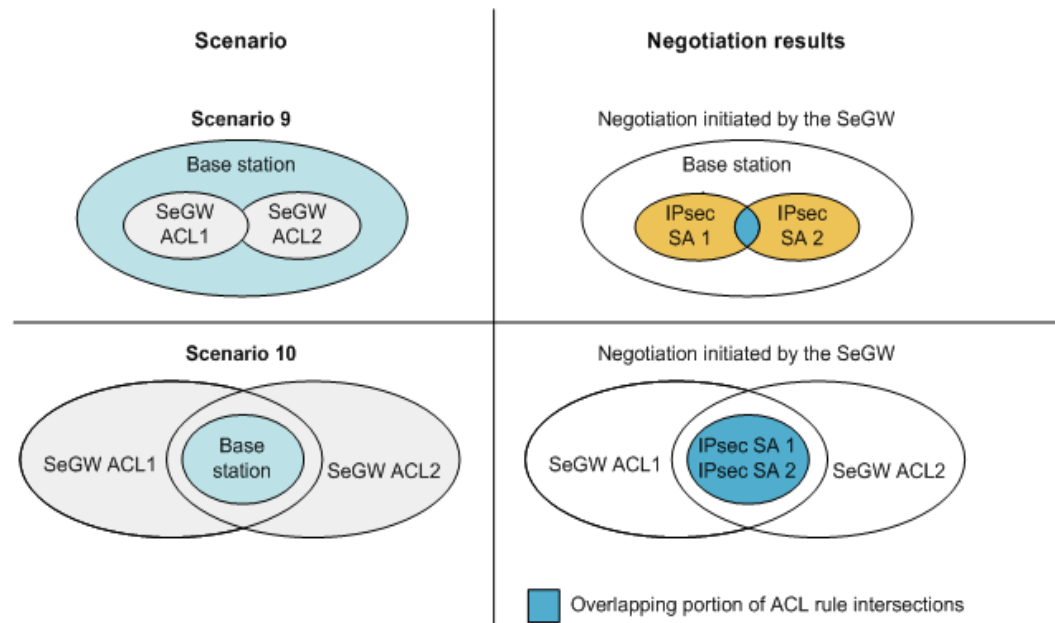
**Figure 4-13** IPsec SA negotiation results in scenario 5



In IPv4 transmission, IPsec protection is performed only on data flows matching the ACL rules in ACL rule intersections. Other data flows that do not match the ACL rules configured on the base station are transmitted in plaintext and those match the ACL rules configured on the base station are discarded. If these data flows must be protected by IPsec, services will be interrupted.

In IPv6 transmission, IPsec protection is performed only on data flows matching the ACL rules in ACL rule intersections. Other data flows are discarded.

**Figure 4-14** Multiple ACL rule intersections overlapping with each other



**NOTE**

- IPsec ACL narrow down does not consider the source and destination ports in ACL rules.
- With IPsec ACL narrow down, the relevant NEs can negotiate multiple SAs for a single ACL rule. In theory, the number of SAs that can be negotiated may exceed the maximum number of SAs that can be configured. If the number of IPsec SAs reaches the maximum, no new IPsec SA can be generated. If a base station is reset at this time, IPsec SAs may be inconsistent before and after the reset because the sequences of successful ACL rule negotiations before and after the reset may not be consistent. As a result, links may be disconnected. You can obtain the maximum SA configuration specifications by viewing the note of the **ADD IPSECPOLICY** MML command. For details, see *3900 & 5900 Series Base Station MML Command Reference* in *3900 & 5900 Series Base Station Product Documentation*.
- If the address segment mask in the traffic selector (TS) payload carried by the IKEv2 entity at the peer end is discontinuous, the base station may not completely match the TS based on the peer address segment. As a result, the negotiated address segment may be less than the peer address segment. If the peer IKEv2 entity uses the ACL mode (continuous mask mode) to configure the TS, this restriction is not involved.
- In direct IPsec scenarios, when the IPv4 IPsec configuration mode is tunnel-interface-based IPsec, it cannot share a security parameter template with IPv6 IPsec.

### 4.1.2.2.3 PFS

PFS is a security feature where the decoding of one key does not affect the security of other keys because no session key can be derived from any other key. When an IPsec SA is established through negotiation, the PFS uses the DH algorithms to generate a new key. The new key has no derivation relationship with the previously negotiated key. The PFS uses the DH algorithms to generate a new key again only when the IPsec SA lifetime ends and renegotiation is performed. PFS is guaranteed by DH algorithms and is specified by the following parameters:

- **IPSECPOLICY.PFS (LTE eNodeB, 5G gNodeB)** (eGBTS/NodeB/eNodeB/gNodeB): If this parameter is set to **PFS\_GROUP14**, **PFS\_GROUP15**, **PFS\_GROUP19**, **PFS\_GROUP20**, or **PFS\_GROUP31**, the CPU usage temporarily increases during IPsec negotiation. If a large number of calculations involving

**PFS\_GROUP19** and **PFS\_GROUP20** are performed, the base station will start flow control, which increases the IPsec negotiation duration.

 **NOTE**

It is recommended that the **PFS\_GROUP15**, **PFS\_GROUP19**, **PFS\_GROUP20**, or **PFS\_GROUP31** algorithm be used to ensure IPsec SA security.

You can run **SET SECP** to adjust the configuration security level to prohibit the parameter from being set to **PFS\_GROUP1** or **PFS\_GROUP2**.

- **BTSIPSECPOLICY.PFS** (GBTS)

#### 4.1.2.2.4 IPsec SA Lifecycle

An IPsec SA becomes invalid when its lifecycle expires. The lifecycle is determined as follows: Before an IPsec SA becomes invalid, IKE is used to negotiate a new IPsec SA.

- If **IPSECPOLICY.LTCFG (LTE eNodeB, 5G gNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB) or **BTSIPSECPOLICY.LTCFG** (for the GBTS) is set to **GLOBAL**, the lifecycle of an IPsec SA is always 3600s.
- If **IPSECPOLICY.LTCFG (LTE eNodeB, 5G gNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB) or **BTSIPSECPOLICY.LTCFG** (for the GBTS) is set to **LOCAL**, the lifecycle of an IPsec SA is specified by **IPSECPOLICY.LTS (LTE eNodeB, 5G gNodeB)** and **BTSIPSECPOLICY.LTS** (for the eGBTS/NodeB/eNodeB/gNodeB), or by **IPSECPOLICY.LTKB (LTE eNodeB, 5G gNodeB)** and **BTSIPSECPOLICY.LTKB** (for the GBTS).

#### 4.1.2.2.5 Anti-Replay Window

An anti-replay window is used to check for duplicate packets in the window. A packet is discarded if it has a duplicate packet or its sequence number is less than the lowest in the window. The window slides if the sequence number of a packet is greater than that of any packet received in the window. The following parameters specify the size of the anti-replay window:

- **IPSECPOLICY.REPLAYWND (5G gNodeB, LTE eNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB)
- **BTSIPSECPOLICY.REPLAYWND** (for the GBTS)

The configuration modification takes effect only when the next IPsec SA negotiation starts.

 **NOTE**

To ensure security, it is recommended that IPsec anti-replay be enabled. Otherwise, the IPsec tunnel of the base station may suffer replay attacks.

For details about the anti-replay window, see appendix A2 in RFC 4303.

It is recommended that the anti-replay function be disabled any time there is a severe out-of-order issue with the IPsec packets on a live network. For example, such an issue can occur when differentiated services code point (DSCP) values are attached to IPsec packets based on service types due to scheduling at network nodes. If the anti-replay function is enabled in this situation, a large number of IPsec packets may be lost, which severely affects service performance. If this function is required, it is recommended that the anti-replay window size be set to **WND\_512** to prevent IPsec packet disorder from affecting service performance.

4.1.2.3 IPsec Packet Fragmentation

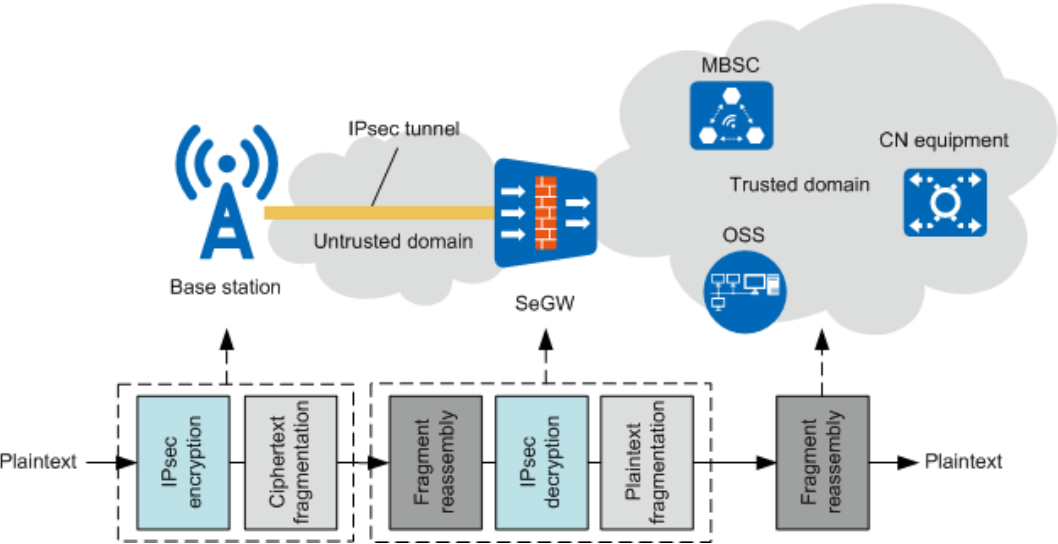
Packet fragmentation can be performed either before or after IPsec encryption. Only the tunnel mode supports packet fragmentation before IPsec encryption.

If the **IKEPEER.IPSECPREFRGSW (5G gNodeB, LTE eNodeB)** parameter is set to **OFF**, packet fragmentation after IPsec encryption will be used. If this parameter is set to **ON**, packet fragmentation before IPsec encryption will be used.

4.1.2.3.1 Packet Fragmentation After IPsec Encryption

**Figure 4-15** shows the principle of packet fragmentation after IPsec encryption. In the figure, the size of a plaintext packet after IPsec decryption is greater than the MTU of the intermediate transmission devices on the core network (CN). Therefore, the packet is fragmented again before being sent to the CN.

Figure 4-15 Packet fragmentation after IPsec encryption



**Table 4-2** lists the maximum sizes of the packet headers added for IPsec encapsulation and encryption.

**Table 4-2** Maximum sizes of the packet headers added for IPv4 IPsec encapsulation and encryption

| Security Protocol | Encryption Algorithm | Authentication Algorithm | Maximum Size of the Packet Header Added for IPsec Encapsulation and Encryption |
|-------------------|----------------------|--------------------------|--------------------------------------------------------------------------------|
| ESP               | AES                  | SHA1/AES-XCBC-MAC-96     | 73                                                                             |
|                   | AES                  | SHA256                   | 77                                                                             |
|                   | AES_GCM              | None                     | 61                                                                             |
| AH                | None                 | SHA1                     | 44                                                                             |

| Security Protocol | Encryption Algorithm | Authentication Algorithm | Maximum Size of the Packet Header Added for IPsec Encapsulation and Encryption |
|-------------------|----------------------|--------------------------|--------------------------------------------------------------------------------|
|                   | None                 | SHA256                   | 48                                                                             |

**Table 4-3** Maximum sizes of the packet headers added for IPv6 IPsec encapsulation and encryption

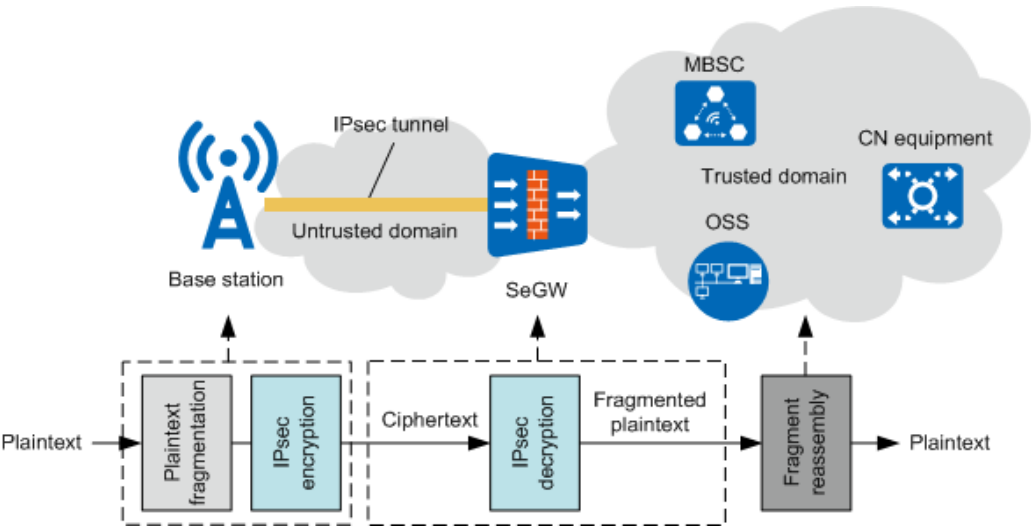
| Security Protocol | Encryption Algorithm | Authentication Algorithm | Maximum Size of the Packet Header Added for IPsec Encapsulation and Encryption |
|-------------------|----------------------|--------------------------|--------------------------------------------------------------------------------|
| ESP               | AES                  | SHA1/AES-XCBC-MAC-96     | 93                                                                             |
|                   | AES                  | SHA256                   | 97                                                                             |
|                   | AES_GCM              | None                     | 89                                                                             |
| AH                | None                 | SHA1                     | 44                                                                             |
|                   | None                 | SHA256                   | 48                                                                             |

If packet fragmentation is performed after IPsec encryption, SeGWs must reassemble packets, resulting in deterioration of their processing performance.

4.1.2.3.2 Packet Fragmentation Before IPsec Encryption

Packet fragmentation can be performed before IPsec encryption to avoid the deterioration of SeGWs' processing performance. [Figure 4-16](#) shows the principle of packet fragmentation before IPsec encryption.

**Figure 4-16** Packet fragmentation before IPsec encryption



The MTU before IPsec encryption is set using the **IKEPEER.MTU (LTE eNodeB, 5G gNodeB)** parameter. The MTU for packets to be encrypted using IPsec should meet the following requirements:

- The MTU before IPsec encryption should be small enough so that the total packet size is still smaller than the MTU of the access network after the packet header is added. This is to ensure that the access network is not required to fragment packets received from the base station and the SeGW is not required to reassemble them. The sizes of the relevant packet headers are listed in [Table 4-2](#) and [Table 4-3](#).
- The MTU before IPsec encryption should be less than the MTUs of the CN's intermediate transmission equipment. This is to ensure that packets will not be fragmented repeatedly during transmission from the SeGW to the CN, eliminating the need to reassemble fragmented packets received at the CN.

Reassembling packets at the CN impacts the CN's performance. Transmission efficiency is reduced when packets are fragmented before IPsec encryption. This is especially true when the plaintext packet whose size is greater than the MTU before IPsec encryption but less than the MTUs of the CN's intermediate transmission equipment is fragmented.

 **NOTE**

When base stations are cascaded and IPv6-over-IPv4 IPsec is used, the MTU of the outbound interface on the upper-level base station must be greater than or equal to that of the outbound interface on the lower-level base station.

## 4.1.3 IPsec Application

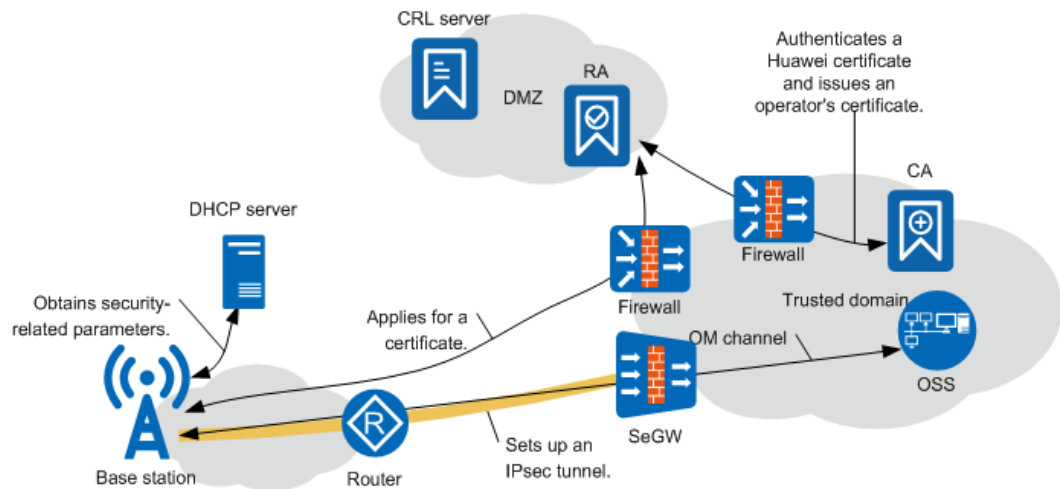
### 4.1.3.1 Typical IPsec Networking

Huawei base stations support IPsec. To secure data transmission of base stations, an SeGW must be deployed on the network.

Typically, the base station and the SeGW use digital certificates to authenticate each other. Therefore, a PKI system and a public DHCP server must be deployed on the operator's network. As stipulated in 3GPP TS 33.310, the Initialization Response message sent by the operator's CA server must contain the operator's root certificate or certificate chain. The operator's CA server must be preconfigured with the Huawei root certificate. [Figure 4-17](#) shows typical IPsec networking.



**Figure 4-17** Typical IPsec networking



In this scenario, the base station must obtain a device certificate from the CA server before establishing an IPsec tunnel to the SeGW. For details about how to apply for a device certificate, see *PKI*.

During base station deployment by PnP, the number of IPsec tunnel setups depends on whether a public DHCP server is deployed and whether the OM channel needs to be protected by IPsec:

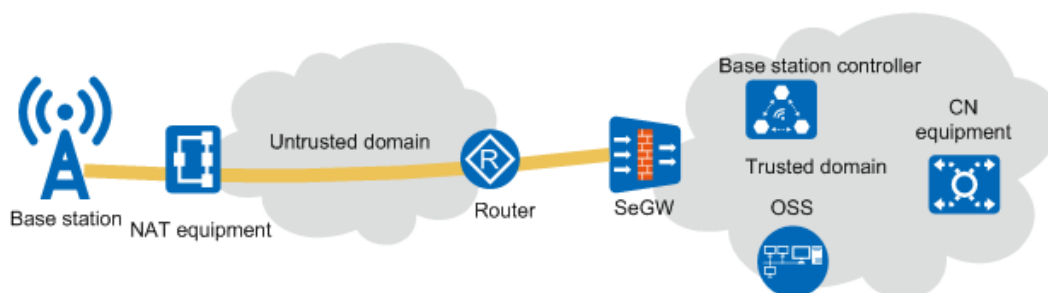
- If the OM channel needs to be protected by IPsec and a public DHCP server is deployed on the network:  
Two IPsec tunnels will be established during base station deployment by PnP. One IPsec tunnel is used to protect the messages exchanged between the base station and the MAE DHCP server. The other IPsec tunnel is used to protect the OM channel. For the automatic OM channel establishment procedure, see *Automatic OMCH Establishment*. After the base station obtains the configuration file and is reset, it negotiates with the SeGW based on the configuration data to establish a new IPsec tunnel for subsequent data transmission protection.
- If the OM channel needs to be protected by IPsec and no public DHCP server is deployed on the network:  
Only one IPsec tunnel will be established between the base station and the SeGW during base station deployment by PnP. It is used to protect the OM channel. After the base station obtains the configuration file and is reset, it negotiates with the SeGW based on the configuration data to establish a new IPsec tunnel for subsequent data transmission protection. For the automatic OM channel establishment procedure, see *Automatic OMCH Establishment*.
- If the OM channel does not need to be protected by IPsec:  
The base station directly obtains the configuration file during base station deployment by PnP. The base station then negotiates with the SeGW based on the configuration data to establish an IPsec tunnel for subsequent data transmission protection. For the automatic OM channel establishment procedure, see *Automatic OMCH Establishment*.

**NOTE**

If the UMPTb is used as the main control board, a maximum of 100 ACL rules can be configured for each **IKEPEER** MO to ensure that the feature properly takes effect.

In IPsec networks, network address translation (NAT) equipment can be deployed. For example, if a base station uses broadband to access a network, NAT equipment may be required. **Figure 4-18** shows the typical IPsec networking with NAT equipment where the IPsec tunnel of the base station is terminated on the SeGW.

**Figure 4-18** Typical IPsec networking with NAT equipment



The **IKEPEER.NATTRAV (LTE eNodeB, 5G gNodeB)** parameter specifies whether to enable the NAT traversal function for a base station. The **IKECFG.NATKLI (LTE eNodeB, 5G gNodeB)** parameter specifies the interval for sending updated NAT packets.

### 4.1.3.2 Application of IPsec on Base Stations

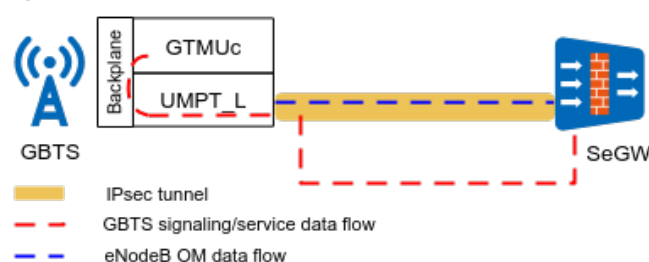
For Huawei base stations, only the Ethernet ports on the UMPT/UMDU/MDUC/UTRPe boards support IPsec. When IPsec is applied to macro base stations, boards that support IPsec must be deployed on the base stations.

### Single-Mode Base Station

For details about IPsec-capable single-mode base stations and their boards, see *BBU Hardware Description in 3900 & 5900 Series Base Station Product Documentation*.

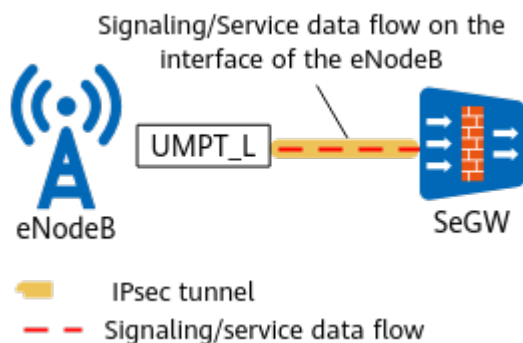
**Figure 4-19** shows an example of implementing IPsec on a GBTS configured with the GTMUc+UMPT\_L. The UMPT\_L only provides IPsec and forwards GBTS data.

**Figure 4-19** GBTS (GTMUc+UMPT\_L)



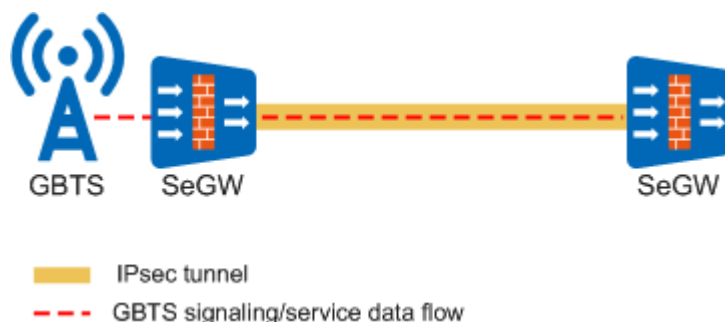
**Figure 4-20** shows an example of implementing IPsec on an eNodeB configured with the UMPT\_L. This board provides IPsec protection for data flows of the eNodeB.

**Figure 4-20** eNodeB (UMPT\_L)



An external SeGW must be deployed on the base station side to implement IPsec on a 3012 series GSM base station, BTS3900E (the 3012 series GSM base station and BTS3900E do not support the UTRPc board), 3812 series UMTS base station or other legacy base stations. An external SeGW must also be deployed on the base station side to enable such base stations to support IPsec after they are upgraded to multimode base stations. **Figure 4-21** shows the implementation.

**Figure 4-21** External SeGW deployed on the base station side

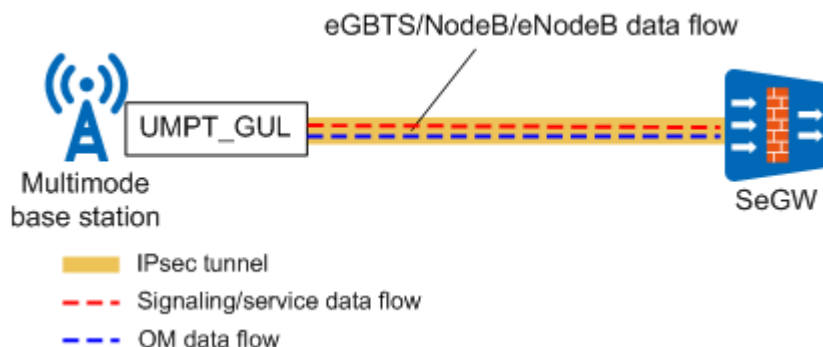


## Multimode Base Station

To reduce the number of required transmission ports, the Multi-mode BS Common IPsec feature is introduced to enable multiple RATs in a multimode base station to share the IPsec tunnel on a co-transmission port.

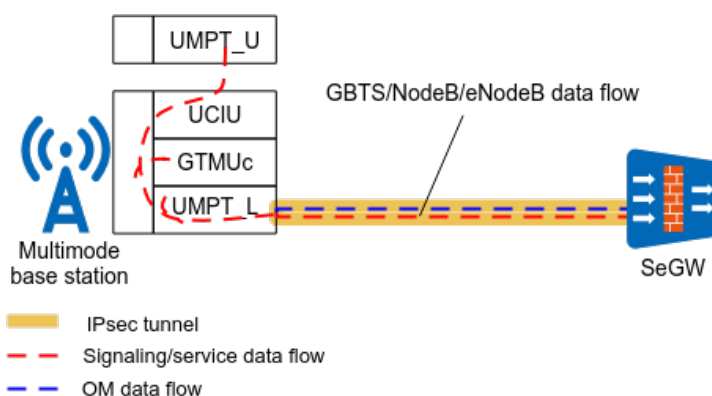
For details about which base stations and boards support IPsec when IPsec is applied to a co-MPT multimode base station, see BBU hardware description in *3900 & 5900 Series Base Station Product Documentation*. **Figure 4-22** shows how to implement co-IPsec on a co-MPT GUL multimode base station.

**Figure 4-22** Example of implementing multi-RAT co-IPsec on a co-MPT GUL multimode base station



For details about which base stations and boards support IPsec when IPsec is applied to a separate-MPT multimode base station, see BBU hardware description in *3900 & 5900 Series Base Station Product Documentation*. **Figure 4-23** shows how to implement co-IPsec on a separate-MPT GUL multimode base station that uses the UMPT\_L+GTMUc+UCIU in the root BBU and the UMPT\_U in the leaf BBU.

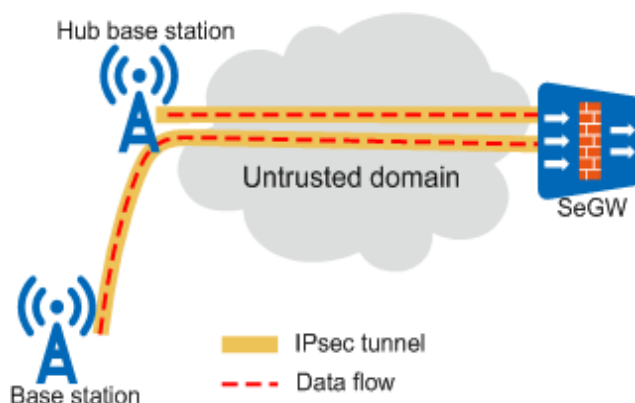
**Figure 4-23** Example of implementing multi-RAT co-IPsec on a separate-MPT GUL multimode base station



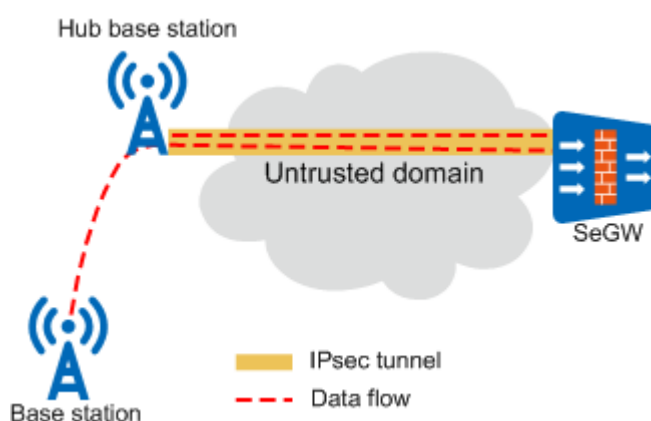
## Cascaded Base Stations

When base stations are cascaded, IPsec can be implemented in two ways, as shown in **Figure 4-24** and **Figure 4-25**.

**Figure 4-24** Separate IPsec tunnel for each base station and route forwarding by the Hub base station



**Figure 4-25** IPsec tunnel provided by the Hub base station for all cascaded base stations

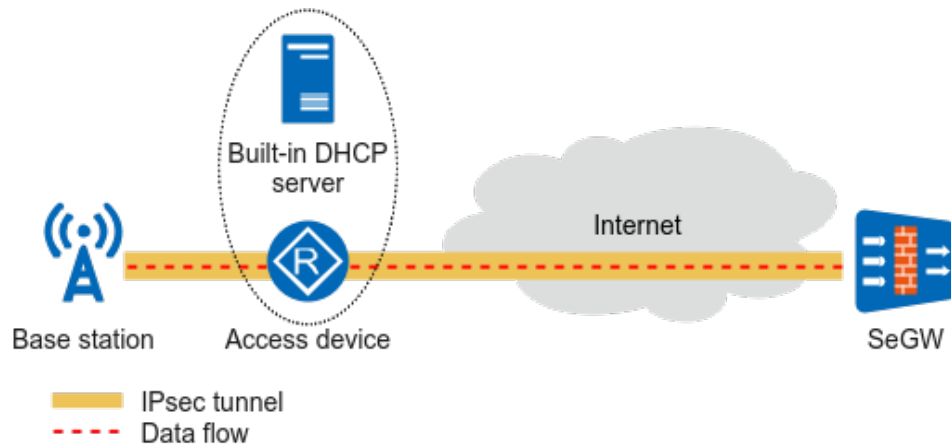


The method described in [Figure 4-24](#) is recommended to ensure the security of each cascaded base station.

## Public Network Backhaul for Base Stations

In public network backhaul scenarios, base stations access the Internet through access devices (such as cable models and APs) and dynamically obtain local IP addresses through DHCP to realize data transmission. Therefore, the local IP address of the IPsec tunnel cannot be configured in advance. Instead, the local IP address of the IPsec tunnel must be bound to the IP address dynamically assigned by DHCP to establish an IPsec connection, as shown in [Figure 4-26](#).

**Figure 4-26** The base station accesses the Internet through the access device and dynamically obtains an IP address through the built-in DHCP service of the router to establish an IPsec tunnel.



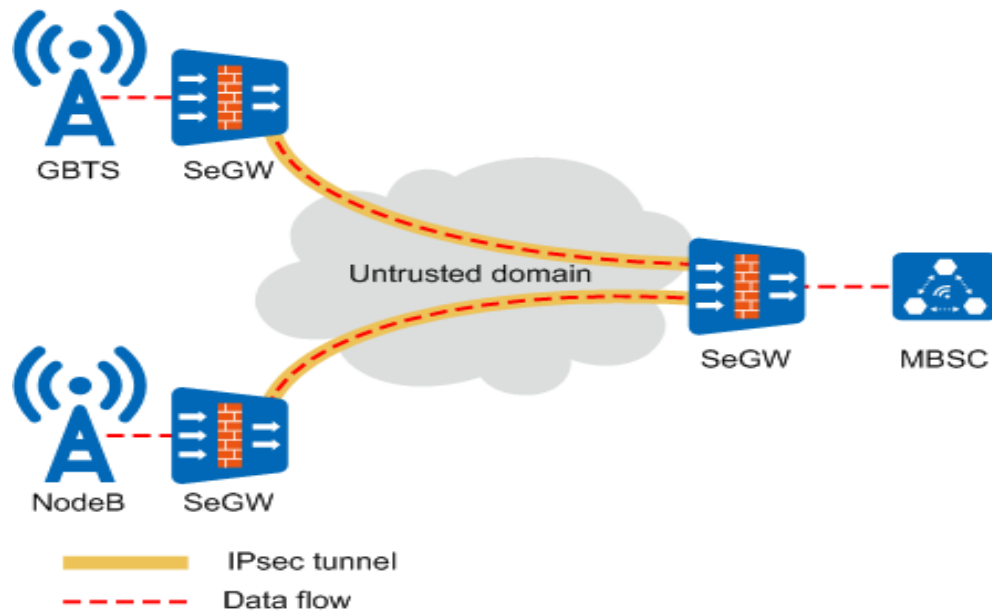
**NOTE**

- This scenario applies only to tunnel-interface-based IPsec. For details about tunnel-interface-based IPsec, see [4.1.4.2 Tunnel-Interface-based IPsec](#).
- In this scenario, only the local IPsec address can be bound to the dynamic IPv4 address obtained through DHCP. Currently, bounding to the dynamic IPv6 address is not supported.
- In this scenario, IPsec-related configuration objects do not take effect until the base station successfully obtains a dynamic IP address through DHCP.

### 4.1.3.3 Application of External IPsec on Base Station Controllers (GSM/UMTS)

If bearer networks such as leased networks and public networks encounter security threats, IPsec tunnels can be used to isolate network services for secure transmission. Currently, base station controllers do not support integrated IPsec and therefore can only use external IPsec. [Figure 4-27](#) shows an example of implementing external IPsec on a base station controller.

**Figure 4-27** Example of implementing external IPsec on a base station controller



The throughput of an external SeGW must exceed the planned total throughput on GSM and UMTS user planes.

If no SeGW is deployed on an operator's network, it is recommended that the operator use the Huawei Eudemon1000E-X or Eudemon8000E-X to implement external IPsec on the base station controller.

The recommended data configurations for the SeGW are as follows:

- It is recommended that the packet type-specific whitelist function be disabled on the SeGW. This is because interface boards on a base station controller have firewalls and provide their own whitelists.
- It is recommended that packet filtering based on the UDP port number be disabled on the SeGW. The purpose is to prevent normal packets from being filtered out.
- Configuration of IPsec on the SeGW depends on customer requirements.

#### 4.1.3.4 Network Evolution Solutions

Operators use the solutions described in [Table 4-4](#) to reconstruct existing networks.

**Table 4-4** Network evolution solutions

| <b>Solution</b>                                                                                                                                                      | <b>Device Deployment Requirements</b>                                                                                                        | <b>Impact</b>                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reconstruction from an insecure transport network to a secure network that uses digital certificate-based authentication (referred to as a PKI-based secure network) | A PKI system, public DHCP server, and SeGW must be deployed. The SeGW can use digital certificates to authenticate the identity of the peer. | Downloading and activating configuration data is required, which interrupts services.                                                                                                                                             |
| Reconstruction from an insecure transport network to a secure network that uses PSK-based authentication (referred to as a PSK-based secure network)                 | An SeGW must be deployed. The SeGW can use a PSK to authenticate the identity of the peer.                                                   | Downloading and activating configuration data is required, which interrupts services.                                                                                                                                             |
| Reconstruction from a PSK-based secure network to a PKI-based secure network                                                                                         | A PKI system, public DHCP server, and SeGW must be deployed. The SeGW can use digital certificates to authenticate the identity of the peer. | Configuration data must be modified and certificate deployment location must be configured online. To make the configured certificate deployment location take effect, the base station must be reset, which interrupts services. |



| Solution                                                                                       | Device Deployment Requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Impact                                                                                    |
|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Reconstruction from a PKI/PSK-based secure network to a PKI/PSK-based IPsec redundancy network | <ul style="list-style-type: none"><li>• Primary and secondary SeGWs must be deployed.</li><li>• DPD is enabled on the primary and secondary SeGWs.</li><li>• If the status of IPsec SAs is normal, SeGWs can advertise the base station's downlink routing information to the trusted domain. If the status of IPsec SAs is abnormal, SeGWs can advertise the base station's downlink routing revocation information to the trusted domain.</li><li>• The trusted domain can learn the base station's downlink routing information advertised by SeGWs.</li></ul> | Configuration data can be modified online and resetting the base station is not required. |

For a GBTS, run the **SET BTSCERTDEPLOY** command to set the board on which a certificate is to be deployed. For an eGBTS/NodeB/eNodeB/gNodeB, use the **SET CERTDEPLOY** command.

#### 4.1.3.5 Application of IPsec for 1588v2 Clock Packets

Data processing at the IP and Media Access Control (MAC) layers may be delayed. To eliminate the delay and provide accurate timestamps for clock packets, 1588v2 specifies that a timestamp must be attached between data processing at the MAC layer and data processing at the physical layer. When a packet is encapsulated by upper-layer protocols and MAC, an NE detects the User Datagram Protocol (UDP) port number carried in the packet before data processing at the physical layer. If the UDP port number is 319, the packet is regarded as a 1588v2 clock packet and a timestamp is added to the packet to record the leaving or arrival time of the packet.

IPsec encrypts and verifies packets at the IP layer, but timestamps are attached to 1588v2 clock packets between data processing at the MAC layer and data processing at the physical layer. As a result, two issues arise when IPsec is used to encrypt and protect data integrity for 1588v2 clock packets:

- After IPsec encryption, UDP port numbers carried in 1588v2 clock packets cannot be identified.

- After data integrity has been verified by IPsec at the sender, a 1588v2 clock packet fails the data integrity check performed by the receiver due to an attached timestamp.

The 1588v2 over IPsec solution has been introduced to address these issues. This solution enables IPsec encryption for Layer 3 (L3) unicast packets in frequency synchronization. The procedure is as follows:

1. Upon receiving an encrypted packet, the base station records the arrival time of the packet and sends the timestamp to the upper layer together with the encrypted packet because it cannot determine whether this packet is a 1588v2 clock packet.
2. The base station decrypts the packet and checks whether the packet is a 1588v2 clock packet based on the UDP port number.
3. If the packet is a 1588v2 clock packet, the base station uses the Adapter Clock Recover (ACR) algorithm to restore the clock frequency based on the timestamp and arrival time of the packet.

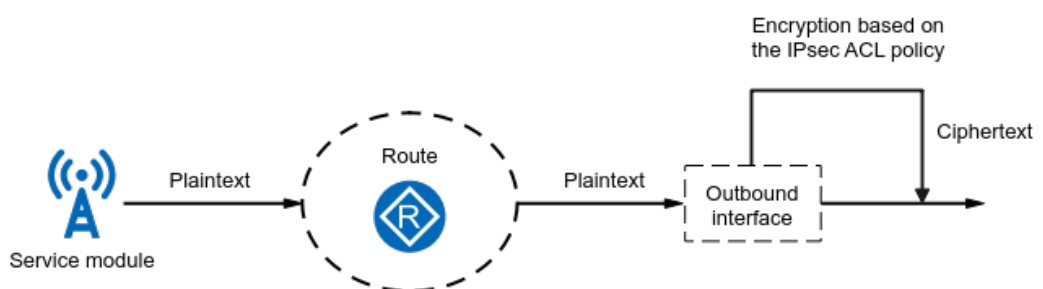
Note that this solution applies only to L3 unicast packets in frequency synchronization. This solution does not apply to time synchronization because time synchronization has the following restrictions: Timestamps are required for all L3 equipment between the base station and the SeGW. Intermediate equipment cannot identify whether encrypted packets are 1588v2 clock packets.

The eNodeB TDD does not support the 1588v2 over IPsec solution. The GBTS, eGBTS, NodeB, gNodeB, and eNodeB FDD support this solution.

## 4.1.4 IPsec Configuration Mode

The IPv4 IPsec configuration modes are ethernet-interface-based IPsec and tunnel-interface-based IPsec, which is specified by the **IKECFG.IPSECCFGMODE (LTE eNodeB, 5G gNodeB)** parameter. Tunnel-interface-based IPsec is supported only in LTE, NR, and multimode scenarios.

### 4.1.4.1 Ethernet-Interface-based IPsec



The base station's service module sends plaintext to the outbound interface. The process is as follows:

- Step 1** The base station queries the route once and sends the packet to the outbound interface.
- Step 2** The base station matches the plaintext with the IPsec ACL rule.
- If matching succeeds and the **ACTION** parameter of the ACL rule is set to **PERMIT**, the base station proceeds to the next step.

- If matching succeeds and the **ACTION** parameter of the ACL rule is set to **DENY**, the packet is sent in plaintext. The procedure ends.
- If matching fails, the packet is sent in plaintext. The procedure ends.

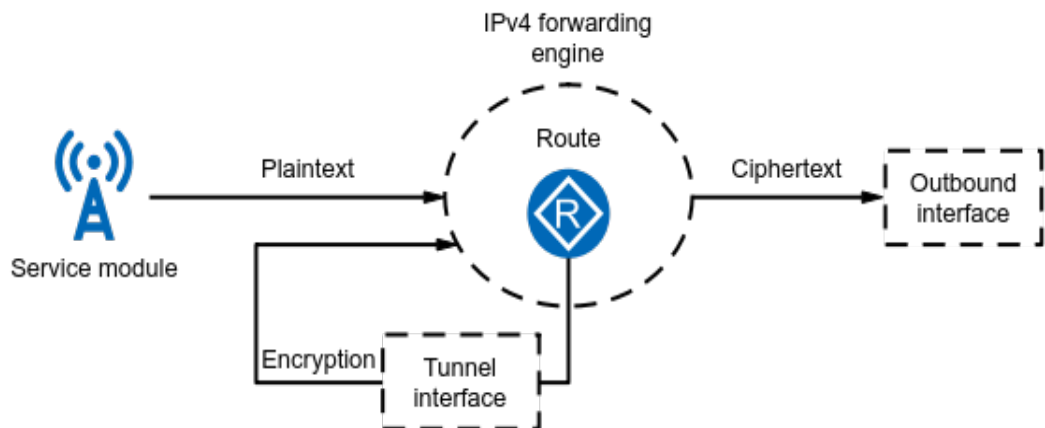
**Step 3** The base station encrypts the plaintext.

**Step 4** The base station sends the encrypted packet.

----End

## 4.1.4.2 Tunnel-Interface-based IPsec

### 4.1.4.2.1 IPv4 Tunnel Interfaces



The base station's service module sends plaintext to the IPv4 forwarding engine. The process is as follows:

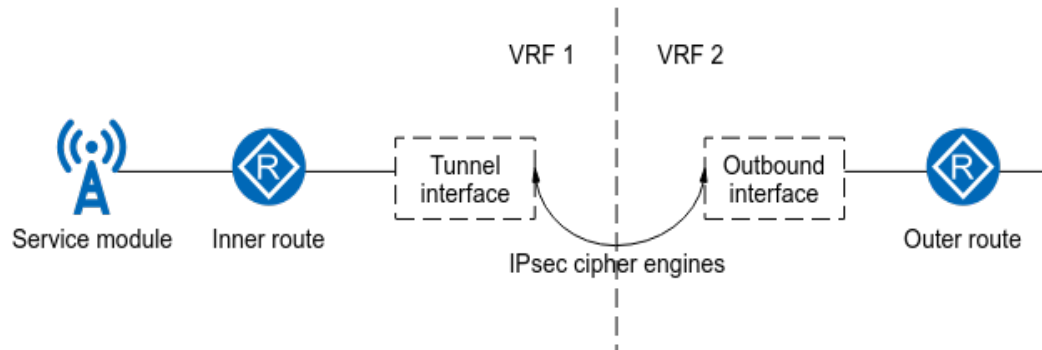
- Step 1** The base station queries the inner route first. If the outbound interface is a tunnel interface, the base station proceeds to the next step. Otherwise, the base station directly sends the packet in plain text.
- Step 2** The base station matches the plaintext with the IPsec ACL rule. If matching succeeds, the base station proceeds to the next step. If matching fails, the plaintext is discarded.
- Step 3** The base station encrypts the plaintext.
- Step 4** The base station queries the outer route and sends the encrypted packet to the outbound interface.

----End

### 4.1.4.2.2 IPv4 VRF-based Isolation

When packets are configured with IPv4 headers, VRFs can be used by IPsec to isolate inner packets from outer packets, as shown in [Figure 4-28](#).

**Figure 4-28** VRF-based isolation



Inner and outer VRFs are configured to isolate inner IPsec service packets from outer IPsec service packets, preventing the inner service modules protected by IPsec from network attacks.

**NOTE**

The **INTERFACE.VRFDIX (5G gNodeB, LTE eNodeB)** parameter can only be set to **0** for inner VRFs of GSM/UMTS services. Therefore, this parameter must be set to **0** for inner VRFs if GSM/UMTS traffic needs to be transmitted through IPsec tunnels.

## 4.2 Network Analysis

### 4.2.1 Benefits

IPsec provides transparent, end-to-end security services for IP networks, protecting them against cyber-attacks.

### 4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.1.3 Function Impacts](#), [4.3.2.2.3 Function Impacts](#), [4.3.2.3.3 Function Impacts](#), [4.3.2.4.3 Function Impacts](#), [4.3.2.5.3 Function Impacts](#), [4.3.2.6.3 Function Impacts](#), [4.3.2.7.3 Function Impacts](#), [4.3.2.8.3 Function Impacts](#), [4.3.2.9.3 Function Impacts](#), [4.3.2.10.3 Function Impacts](#), [4.3.2.11.3 Function Impacts](#), and [4.3.2.12.3 Function Impacts](#).

IPsec ensures transmission security by encapsulating and encrypting IP packets. However, this reduces the transmission efficiency of service packets on the bearer network.

Take ESP encapsulation in tunnel mode as an example. Assume that the IP payload is 500 bytes, the packet length (including the IP header and Ethernet header, with VLAN tag attached) before IPsec encapsulation is 542 bytes, the encryption algorithm AES128 is used, and the authentication algorithm SHA256 is used. The packet structure after encapsulation is as follows:

22 bytes (Ethernet header with VLAN tag attached) + 20 bytes (external IP header) + 8 bytes (ESP header) + 20 bytes (internal IP header) + 16 bytes (initialization vector, optional) + 512 bytes (payload) + 14 bytes (padding) + 2 bytes (ESP trailer) + 16 bytes (SHA256) = 630 bytes. The total length is 630 bytes.

In this scenario, the transmission efficiency would decrease from 92.25% (500/542 x 100%) to 79.37% (500/630 x 100%).

The impact of IPsec on the transmission efficiency of service data varies depending on the protocol, algorithm, and encapsulation mode as follows:

- Impact of IPsec on the transmission efficiency when SHA and SHA2 algorithms are used for data integrity check

**Table 4-5** Impact of IPsec on the transmission efficiency in transport mode

| AH              | FR     | MCS-9 | AMR 12.2 kbit/s | CS 64 kbit/s | PS 32 kbit/s | PS 128 kbit/s | PS 384 kbit/s |
|-----------------|--------|-------|-----------------|--------------|--------------|---------------|---------------|
| IPsec disabled  | 32%    | 65.5% | 29%             | 69.3%        | 51.6%        | 78.6%         | 83.5%         |
| SHA             | 24.4 % | 57.4% | 22.3%           | 60.8%        | 42.8%        | 72.9%         | 78.9%         |
| SHA2 (256 bits) | 22.4 % | 54.8% | 20.5%           | 58.2%        | 40.2%        | 71.0%         | 77.5%         |

**Table 4-6** Impact of IPsec on the transmission efficiency in tunnel mode

| AH              | FR    | MCS-9 | AMR 12.2 kbit/s | CS 64 kbit/s | PS 32 kbit/s | PS 128 kbit/s | PS 384 kbit/s |
|-----------------|-------|-------|-----------------|--------------|--------------|---------------|---------------|
| IPsec disabled  | 32%   | 65.5% | 29%             | 69.3%        | 51.6%        | 78.6%         | 83.5%         |
| SHA             | 21.3% | 53.2% | 19.4%           | 56.5%        | 38.6%        | 69.7%         | 76.6%         |
| SHA2 (256 bits) | 19.8% | 51%   | 18.4%           | 54.2%        | 36.5%        | 67.9%         | 75.1%         |

- Impact of IPsec on the transmission efficiency when ESP is used with the AES algorithm for encryption

**Table 4-7** Impact of IPsec on the transmission efficiency in transport mode

| ESP            | FR    | MCS-9 | AMR 12.2 kbit/s | CS 64 kbit/s | PS 32 kbit/s | PS 128 kbit/s | PS 384 kbit/s |
|----------------|-------|-------|-----------------|--------------|--------------|---------------|---------------|
| IPsec disabled | 32%   | 65.5% | 29%             | 69.3%        | 51.6%        | 78.6%         | 83.5%         |
| AES+SHA        | 23.2% | 54.8% | 20.3%           | 57.6%        | 40.4%        | 70.5%         | 78.2%         |

**Table 4-8** Impact of IPsec on the transmission efficiency in tunnel mode

| ESP            | FR    | MCS-9 | AMR 12.2 kbit/s | CS 64 kbit/s | PS 32 kbit/s | PS 128 kbit/s | PS 384 kbit/s |
|----------------|-------|-------|-----------------|--------------|--------------|---------------|---------------|
| IPsec disabled | 32%   | 65.5% | 29%             | 69.3%        | 51.6%        | 78.6%         | 83.5%         |
| AES+SHA        | 19.4% | 51.7% | 18.3%           | 54.4%        | 37.4%        | 68.1%         | 76.2%         |

If IPsec is enabled on an operator's network, the time required for initial base station deployment increases by less than 2 minutes. The increased time, caused by certificate requests and IPsec tunnel setups, depends on the response speed of the public DHCP server and the encryption protocol used by the SeGW.

Direct IPsec tunnels are established between neighboring base stations, and are large in number. Therefore, an intermittent network disconnection can trigger the reestablishment of a large number of IPsec tunnels, with one IPsec tunnel reestablished per second. As a result, inter-site traffic may bypass the SeGW, or inter-site services may be interrupted for a long time.

## 4.3 Requirements

### 4.3.1 Licenses

The following table lists the license items required for enabling IPsec.

| RAT     | Feature ID    | Feature Name           | Model        | Sales Unit |
|---------|---------------|------------------------|--------------|------------|
| GSM     | GBFD-113524   | BTS Integrated IPsec   | LGMIBTSIPSEC | per BTS    |
| UMTS    | WRFD-140209   | NodeB Integrated IPsec | LQW9IPSEC01  | per NodeB  |
| LTE FDD | LOFD-003009   | IPsec                  | LT1S0IPSEC00 | per eNodeB |
| NB-IoT  | MLOFD-003009  | IPsec                  | ML1S0IPSEC00 | per eNodeB |
| LTE TDD | TDLOFD-003009 | IPsec                  | LT1STIPSEC00 | per eNodeB |
| NR      | FOFD-010080   | IPsec (IPsec)          | NR0S0IPSEC00 | per gNodeB |

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

## 4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

### 4.3.2.1 GBFD-113524 BTS Integrated IPsec

#### 4.3.2.1.1 Prerequisite Functions

| RAT | Function Name      | Function Switch | Reference                | Description                                                                       |
|-----|--------------------|-----------------|--------------------------|-----------------------------------------------------------------------------------|
| GSM | Abis over IP       | None            | <i>IPv4 Transmission</i> | IPsec requires that Abis over IP be enabled.                                      |
| GSM | BTS Supporting PKI | None            | <i>PKI</i>               | This function is required if IPsec uses digital-certificate-based authentication. |

#### 4.3.2.1.2 Mutually Exclusive Functions

| RAT | Function Name    | Function Switch | Reference                                                | Description                                           |
|-----|------------------|-----------------|----------------------------------------------------------|-------------------------------------------------------|
| GSM | BTS Local Switch | None            | <i>BSS Local Switching in GBSS feature documentation</i> | IPsec is mutually exclusive with BTS local switching. |

| RAT | Function Name           | Function Switch                                                              | Reference    | Description                                                                                                                                                                                       |
|-----|-------------------------|------------------------------------------------------------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GSM | IKE version             | <b>IKEPEER.IKEV<br/>ERSION (LTE<br/>eNodeB, 5G<br/>gNodeB)</b>               | <i>IPsec</i> | If the <b>IKEPEER.IKEV<br/>ERSION (LTE<br/>eNodeB, 5G<br/>gNodeB)</b> parameter is set to <b>IKE_V1</b> , the function of using ESNs defined in the ESP protocol cannot be enabled.               |
| GSM | IKE version             | <b>SECURITYTE<br/>MPLATE.IKEV<br/>ERSION (LTE<br/>eNodeB, 5G<br/>gNodeB)</b> | None         | If the <b>SECURITYTE<br/>MPLATE.IKEV<br/>ERSION (LTE<br/>eNodeB, 5G<br/>gNodeB)</b> parameter is set to <b>IKE_V1</b> , the function of using ESNs defined in the ESP protocol cannot be enabled. |
| GSM | Anti-Replay Window Size | <b>IPSECPOLICY.<br/>REPLAYWND<br/>(LTE eNodeB,<br/>5G gNodeB)</b>            | <i>IPsec</i> | If the <b>IPSECPOLICY.<br/>REPLAYWND<br/>(LTE eNodeB,<br/>5G gNodeB)</b> parameter is set to <b>WND_DISABLE</b> , the function of using ESNs defined in the ESP protocol cannot be enabled.       |



| RAT | Function Name                | Function Switch                                            | Reference | Description                                                                                                                                                                           |
|-----|------------------------------|------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GSM | Anti-Replay Window Size      | <b>SECURITYTEMPLATE.REPLAYWIND (LTE eNodeB, 5G gNodeB)</b> | None      | If the <b>SECURITYTEMPLATE.REPLAYWIND (LTE eNodeB, 5G gNodeB)</b> parameter is set to <b>WIND_DISABLE</b> , the function of using ESNs defined in the ESP protocol cannot be enabled. |
| GSM | ESP Authentication Algorithm | <b>IPSECPROPOSAL.ESPAUTHALG (5G gNodeB, LTE eNodeB)</b>    | None      | If the <b>IPSECPROPOSAL.ESPAUTHALG (5G gNodeB, LTE eNodeB)</b> parameter is set to <b>NULL</b> , the function of using ESNs defined in the ESP protocol cannot be enabled.            |

4.3.2.1.3 Function Impacts

None

4.3.2.2 WRFD-140209 NodeB Integrated IPsec

#### 4.3.2.2.1 Prerequisite Functions

| RAT  | Function Name                                 | Function Switch | Reference                | Description                                                                       |
|------|-----------------------------------------------|-----------------|--------------------------|-----------------------------------------------------------------------------------|
| UMTS | IP Transmission Introduction on Iub Interface | None            | <i>IPv4 Transmission</i> | IPsec requires that IP Transmission Introduction on Iub Interface be enabled.     |
| UMTS | NodeB PKI Support                             | None            | <i>PKI</i>               | This function is required if IPsec uses digital-certificate-based authentication. |

#### 4.3.2.2.2 Mutually Exclusive Functions

| RAT  | Function Name | Function Switch                                                | Reference    | Description                                                                                                                                                                         |
|------|---------------|----------------------------------------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UMTS | IKE version   | <b>IKEPEER.IKEV<br/>ERSION (LTE<br/>eNodeB, 5G<br/>gNodeB)</b> | <i>IPsec</i> | If the <b>IKEPEER.IKEV<br/>ERSION (LTE<br/>eNodeB, 5G<br/>gNodeB)</b> parameter is set to <b>IKE_V1</b> , the function of using ESNs defined in the ESP protocol cannot be enabled. |

| RAT  | Function Name           | Function Switch                                            | Reference    | Description                                                                                                                                                                         |
|------|-------------------------|------------------------------------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UMTS | IKE version             | <b>SECURITYTEMPLATE.IKEVERSION (LTE eNodeB, 5G gNodeB)</b> | None         | If the <b>SECURITYTEMPLATE.IKEVERSION (LTE eNodeB, 5G gNodeB)</b> parameter is set to <b>IKE_V1</b> , the function of using ESNs defined in the ESP protocol cannot be enabled.     |
| UMTS | Anti-Replay Window Size | <b>IPSECPOLICY.REPLAYWND (LTE eNodeB, 5G gNodeB)</b>       | <i>IPsec</i> | If the <b>IPSECPOLICY.REPLAYWND (LTE eNodeB, 5G gNodeB)</b> parameter is set to <b>WND_DISABLE</b> , the function of using ESNs defined in the ESP protocol cannot be enabled.      |
| UMTS | Anti-Replay Window Size | <b>SECURITYTEMPLATE.REPLAYWND (LTE eNodeB, 5G gNodeB)</b>  | None         | If the <b>SECURITYTEMPLATE.REPLAYWND (LTE eNodeB, 5G gNodeB)</b> parameter is set to <b>WND_DISABLE</b> , the function of using ESNs defined in the ESP protocol cannot be enabled. |

| RAT  | Function Name                | Function Switch                                         | Reference | Description                                                                                                                                                                |
|------|------------------------------|---------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UMTS | ESP Authentication Algorithm | <b>IPSECPROPOSAL.ESPAUTHALG (5G gNodeB, LTE eNodeB)</b> | None      | If the <b>IPSECPROPOSAL.ESPAUTHALG (5G gNodeB, LTE eNodeB)</b> parameter is set to <b>NULL</b> , the function of using ESNs defined in the ESP protocol cannot be enabled. |

#### 4.3.2.2.3 Function Impacts

None

#### 4.3.2.3 LOFD-003009 IPsec

##### 4.3.2.3.1 Prerequisite Functions

| RAT     | Function Name                   | Function Switch | Reference  | Description                                                                       |
|---------|---------------------------------|-----------------|------------|-----------------------------------------------------------------------------------|
| LTE FDD | Public Key Infrastructure (PKI) | None            | <i>PKI</i> | This function is required if IPsec uses digital-certificate-based authentication. |

#### 4.3.2.3.2 Mutually Exclusive Functions

| RAT     | Function Name      | Function Switch                                                              | Reference    | Description                                                                                                                                                                                       |
|---------|--------------------|------------------------------------------------------------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LTE FDD | IKE version        | <b>IKEPEER.IKEV<br/>ERSION (LTE<br/>eNodeB, 5G<br/>gNodeB)</b>               | <i>IPsec</i> | If the <b>IKEPEER.IKEV<br/>ERSION (LTE<br/>eNodeB, 5G<br/>gNodeB)</b> parameter is set to <b>IKE_V1</b> , the function of using ESNs defined in the ESP protocol cannot be enabled.               |
| LTE FDD | IKE version        | <b>SECURITYTE<br/>MPLATE.IKEV<br/>ERSION (LTE<br/>eNodeB, 5G<br/>gNodeB)</b> | None         | If the <b>SECURITYTE<br/>MPLATE.IKEV<br/>ERSION (LTE<br/>eNodeB, 5G<br/>gNodeB)</b> parameter is set to <b>IKE_V1</b> , the function of using ESNs defined in the ESP protocol cannot be enabled. |
| LTE FDD | Anti-Replay Window | <b>IPSECPOLICY.<br/>REPLAYWND<br/>(LTE eNodeB,<br/>5G gNodeB)</b>            | <i>IPsec</i> | If the <b>IPSECPOLICY.<br/>REPLAYWND<br/>(LTE eNodeB,<br/>5G gNodeB)</b> parameter is set to <b>WND_DISABLE</b> , the function of using ESNs defined in the ESP protocol cannot be enabled.       |

| RAT     | Function Name                | Function Switch                                            | Reference | Description                                                                                                                                                                           |
|---------|------------------------------|------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LTE FDD | Anti-Replay Window Size      | <b>SECURITYTEMPLATE.REPLAYWIND (LTE eNodeB, 5G gNodeB)</b> | None      | If the <b>SECURITYTEMPLATE.REPLAYWIND (LTE eNodeB, 5G gNodeB)</b> parameter is set to <b>WIND_DISABLE</b> , the function of using ESNs defined in the ESP protocol cannot be enabled. |
| LTE FDD | ESP Authentication Algorithm | <b>IPSECPROPOSAL.ESPAUTHALG (5G gNodeB, LTE eNodeB)</b>    | None      | If the <b>IPSECPROPOSAL.ESPAUTHALG (5G gNodeB, LTE eNodeB)</b> parameter is set to <b>NULL</b> , the function of using ESNs defined in the ESP protocol cannot be enabled.            |

4.3.2.3.3 Function Impacts

None

4.3.2.4 MLOFD-003009 IPsec

#### 4.3.2.4.1 Prerequisite Functions

| RAT    | Function Name                   | Function Switch | Reference  | Description                                                                       |
|--------|---------------------------------|-----------------|------------|-----------------------------------------------------------------------------------|
| NB-IoT | Public Key Infrastructure (PKI) | None            | <i>PKI</i> | This function is required if IPsec uses digital-certificate-based authentication. |

#### 4.3.2.4.2 Mutually Exclusive Functions

| RAT    | Function Name | Function Switch                                            | Reference    | Description                                                                                                                                                                     |
|--------|---------------|------------------------------------------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NB-IoT | IKE version   | <b>IKEPEER.IKEVERSION (LTE eNodeB, 5G gNodeB)</b>          | <i>IPsec</i> | If the <b>IKEPEER.IKEVERSION (LTE eNodeB, 5G gNodeB)</b> parameter is set to <b>IKE_V1</b> , the function of using ESNs defined in the ESP protocol cannot be enabled.          |
| NB-IoT | IKE version   | <b>SECURITYTEMPLATE.IKEVERSION (LTE eNodeB, 5G gNodeB)</b> | None         | If the <b>SECURITYTEMPLATE.IKEVERSION (LTE eNodeB, 5G gNodeB)</b> parameter is set to <b>IKE_V1</b> , the function of using ESNs defined in the ESP protocol cannot be enabled. |

| RAT    | Function Name                | Function Switch                                           | Reference    | Description                                                                                                                                                                         |
|--------|------------------------------|-----------------------------------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NB-IoT | Anti-Replay Window Size      | <b>IPSECPOLICY.REPLAYWND (LTE eNodeB, 5G gNodeB)</b>      | <i>IPsec</i> | If the <b>IPSECPOLICY.REPLAYWND (LTE eNodeB, 5G gNodeB)</b> parameter is set to <b>WND_DISABLE</b> , the function of using ESNs defined in the ESP protocol cannot be enabled.      |
| NB-IoT | Anti-Replay Window Size      | <b>SECURITYTEMPLATE.REPLAYWND (LTE eNodeB, 5G gNodeB)</b> | None         | If the <b>SECURITYTEMPLATE.REPLAYWND (LTE eNodeB, 5G gNodeB)</b> parameter is set to <b>WND_DISABLE</b> , the function of using ESNs defined in the ESP protocol cannot be enabled. |
| NB-IoT | ESP Authentication Algorithm | <b>IPSECPROPOSAL.ESPAUTHALG (5G gNodeB, LTE eNodeB)</b>   | None         | If the <b>IPSECPROPOSAL.ESPAUTHALG (5G gNodeB, LTE eNodeB)</b> parameter is set to <b>NULL</b> , the function of using ESNs defined in the ESP protocol cannot be enabled.          |

#### 4.3.2.4.3 Function Impacts

None



### 4.3.2.5 TDLOFD-003009 IPsec

#### 4.3.2.5.1 Prerequisite Functions

| RAT     | Function Name                   | Function Switch | Reference  | Description                                                                       |
|---------|---------------------------------|-----------------|------------|-----------------------------------------------------------------------------------|
| LTE TDD | Public Key Infrastructure (PKI) | None            | <i>PKI</i> | This function is required if IPsec uses digital-certificate-based authentication. |

#### 4.3.2.5.2 Mutually Exclusive Functions

| RAT     | Function Name | Function Switch                                            | Reference    | Description                                                                                                                                                                     |
|---------|---------------|------------------------------------------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LTE TDD | IKE version   | <b>IKEPEER.IKEVERSION (LTE eNodeB, 5G gNodeB)</b>          | <i>IPsec</i> | If the <b>IKEPEER.IKEVERSION (LTE eNodeB, 5G gNodeB)</b> parameter is set to <b>IKE_V1</b> , the function of using ESNs defined in the ESP protocol cannot be enabled.          |
| LTE TDD | IKE version   | <b>SECURITYTEMPLATE.IKEVERSION (LTE eNodeB, 5G gNodeB)</b> | None         | If the <b>SECURITYTEMPLATE.IKEVERSION (LTE eNodeB, 5G gNodeB)</b> parameter is set to <b>IKE_V1</b> , the function of using ESNs defined in the ESP protocol cannot be enabled. |

| RAT     | Function Name                | Function Switch                                           | Reference    | Description                                                                                                                                                                         |
|---------|------------------------------|-----------------------------------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LTE TDD | Anti-Replay Window Size      | <b>IPSECPOLICY.REPLAYWND (LTE eNodeB, 5G gNodeB)</b>      | <i>IPsec</i> | If the <b>IPSECPOLICY.REPLAYWND (LTE eNodeB, 5G gNodeB)</b> parameter is set to <b>WND_DISABLE</b> , the function of using ESNs defined in the ESP protocol cannot be enabled.      |
| LTE TDD | Anti-Replay Window Size      | <b>SECURITYTEMPLATE.REPLAYWND (LTE eNodeB, 5G gNodeB)</b> | None         | If the <b>SECURITYTEMPLATE.REPLAYWND (LTE eNodeB, 5G gNodeB)</b> parameter is set to <b>WND_DISABLE</b> , the function of using ESNs defined in the ESP protocol cannot be enabled. |
| LTE TDD | ESP Authentication Algorithm | <b>IPSECPROPOSAL.ESPAUTHALG (5G gNodeB, LTE eNodeB)</b>   | None         | If the <b>IPSECPROPOSAL.ESPAUTHALG (5G gNodeB, LTE eNodeB)</b> parameter is set to <b>NULL</b> , the function of using ESNs defined in the ESP protocol cannot be enabled.          |

#### 4.3.2.5.3 Function Impacts

None

### 4.3.2.6 FOFD-010080 IPsec (IPsec)

#### 4.3.2.6.1 Prerequisite Functions

| RAT                                                           | Function Name            | Function Switch | Reference  | Description                                                                       |
|---------------------------------------------------------------|--------------------------|-----------------|------------|-----------------------------------------------------------------------------------|
| NR FDD<br>Low-frequency<br>NR TDD<br>High-frequency<br>NR TDD | Security Mechanism (PKI) | None            | <i>PKI</i> | This function is required if IPsec uses digital-certificate-based authentication. |

#### 4.3.2.6.2 Mutually Exclusive Functions

| RAT                                                           | Function Name | Function Switch                                            | Reference    | Description                                                                                                                                                                     |
|---------------------------------------------------------------|---------------|------------------------------------------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NR FDD<br>Low-frequency<br>NR TDD<br>High-frequency<br>NR TDD | IKE version   | <b>IKEPEER.IKEVERSION (LTE eNodeB, 5G gNodeB)</b>          | <i>IPsec</i> | If the <b>IKEPEER.IKEVERSION (LTE eNodeB, 5G gNodeB)</b> parameter is set to <b>IKE_V1</b> , the function of using ESNs defined in the ESP protocol cannot be enabled.          |
| NR FDD<br>Low-frequency<br>NR TDD<br>High-frequency<br>NR TDD | IKE version   | <b>SECURITYTEMPLATE.IKEVERSION (LTE eNodeB, 5G gNodeB)</b> | None         | If the <b>SECURITYTEMPLATE.IKEVERSION (LTE eNodeB, 5G gNodeB)</b> parameter is set to <b>IKE_V1</b> , the function of using ESNs defined in the ESP protocol cannot be enabled. |

| RAT                                                           | Function Name                | Function Switch                                           | Reference    | Description                                                                                                                                                                         |
|---------------------------------------------------------------|------------------------------|-----------------------------------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NR FDD<br>Low-frequency<br>NR TDD<br>High-frequency<br>NR TDD | Anti-Replay Window Size      | <b>IPSECPOLICY.REPLAYWND (LTE eNodeB, 5G gNodeB)</b>      | <i>IPsec</i> | If the <b>IPSECPOLICY.REPLAYWND (LTE eNodeB, 5G gNodeB)</b> parameter is set to <b>WND_DISABLE</b> , the function of using ESNs defined in the ESP protocol cannot be enabled.      |
| NR FDD<br>Low-frequency<br>NR TDD<br>High-frequency<br>NR TDD | Anti-Replay Window Size      | <b>SECURITYTEMPLATE.REPLAYWND (LTE eNodeB, 5G gNodeB)</b> | None         | If the <b>SECURITYTEMPLATE.REPLAYWND (LTE eNodeB, 5G gNodeB)</b> parameter is set to <b>WND_DISABLE</b> , the function of using ESNs defined in the ESP protocol cannot be enabled. |
| NR FDD<br>Low-frequency<br>NR TDD<br>High-frequency<br>NR TDD | ESP Authentication Algorithm | <b>IPSECPROPOSAL.ESPAUTHALG (5G gNodeB, LTE eNodeB)</b>   | None         | If the <b>IPSECPROPOSAL.ESPAUTHALG (5G gNodeB, LTE eNodeB)</b> parameter is set to <b>NULL</b> , the function of using ESNs defined in the ESP protocol cannot be enabled.          |

#### 4.3.2.6.3 Function Impacts

None

### 4.3.2.7 MRFD-121116 Multi-mode BS Common IPsec(GSM)

#### 4.3.2.7.1 Prerequisite Functions

| RAT | Function Name        | Function Switch | Reference                | Description                                                                       |
|-----|----------------------|-----------------|--------------------------|-----------------------------------------------------------------------------------|
| GSM | Abis over IP         | None            | <i>IPv4 Transmission</i> | None                                                                              |
| GSM | BTS Supporting PKI   | None            | <i>PKI</i>               | This function is required if IPsec uses digital-certificate-based authentication. |
| GSM | BTS Integrated IPsec | None            | <i>IPsec</i>             | None                                                                              |

#### 4.3.2.7.2 Mutually Exclusive Functions

| RAT | Function Name    | Function Switch | Reference (GBSS Feature Documentation) |
|-----|------------------|-----------------|----------------------------------------|
| GSM | BTS Local Switch | None            | <i>BSS Local Switching</i>             |

#### 4.3.2.7.3 Function Impacts

None

### 4.3.2.8 MRFD-121126 Multi-mode BS Common IPsec(UMTS)

#### 4.3.2.8.1 Prerequisite Functions

| RAT  | Function Name                                 | Function Switch | Reference                | Description |
|------|-----------------------------------------------|-----------------|--------------------------|-------------|
| UMTS | IP Transmission Introduction on Iub Interface | None            | <i>IPv4 Transmission</i> | None        |

| RAT  | Function Name          | Function Switch | Reference    | Description                                                                       |
|------|------------------------|-----------------|--------------|-----------------------------------------------------------------------------------|
| UMTS | NodeB PKI Support      | None            | <i>PKI</i>   | This function is required if IPsec uses digital-certificate-based authentication. |
| UMTS | NodeB Integrated IPsec | None            | <i>IPsec</i> | None                                                                              |

#### 4.3.2.8.2 Mutually Exclusive Functions

None

#### 4.3.2.8.3 Function Impacts

None

#### 4.3.2.9 MRFD-121136 Multi-mode BS Common IPsec(LTE)

##### 4.3.2.9.1 Prerequisite Functions

| RAT     | Function Name                   | Function Switch | Reference    | Description                                                                       |
|---------|---------------------------------|-----------------|--------------|-----------------------------------------------------------------------------------|
| LTE FDD | Public Key Infrastructure (PKI) | None            | <i>PKI</i>   | This function is required if IPsec uses digital-certificate-based authentication. |
| LTE FDD | IPsec                           | None            | <i>IPsec</i> | None                                                                              |

##### 4.3.2.9.2 Mutually Exclusive Functions

None

##### 4.3.2.9.3 Function Impacts

None

#### 4.3.2.10 MRFD-121146 Multi-mode BS Common IPsec(LTE TDD)

#### 4.3.2.10.1 Prerequisite Functions

| RAT     | Function Name                   | Function Switch | Reference    | Description                                                                       |
|---------|---------------------------------|-----------------|--------------|-----------------------------------------------------------------------------------|
| LTE TDD | Public Key Infrastructure (PKI) | None            | <i>PKI</i>   | This function is required if IPsec uses digital-certificate-based authentication. |
| LTE TDD | IPsec                           | None            | <i>IPsec</i> | None                                                                              |

#### 4.3.2.10.2 Mutually Exclusive Functions

None

#### 4.3.2.10.3 Function Impacts

None

#### 4.3.2.11 MRFD-121156 Multi-mode BS Common IPsec(NB-IoT)

##### 4.3.2.11.1 Prerequisite Functions

| RAT    | Function Name                   | Function Switch | Reference    | Description                                                                       |
|--------|---------------------------------|-----------------|--------------|-----------------------------------------------------------------------------------|
| NB-IoT | Public Key Infrastructure (PKI) | None            | <i>PKI</i>   | This function is required if IPsec uses digital-certificate-based authentication. |
| NB-IoT | IPsec                           | None            | <i>IPsec</i> | None                                                                              |

##### 4.3.2.11.2 Mutually Exclusive Functions

None

##### 4.3.2.11.3 Function Impacts

None

#### 4.3.2.12 MRFD-151165 Multi-mode BS Common IPsec(NR)

#### 4.3.2.12.1 Prerequisite Functions

| RAT                                                     | Function Name            | Function Switch | Reference    | Description                                                                       |
|---------------------------------------------------------|--------------------------|-----------------|--------------|-----------------------------------------------------------------------------------|
| NR FDD<br>Low-frequency NR TDD<br>High-frequency NR TDD | Security Mechanism (PKI) | None            | <i>PKI</i>   | This function is required if IPsec uses digital-certificate-based authentication. |
| NR FDD<br>Low-frequency NR TDD<br>High-frequency NR TDD | IPsec                    | None            | <i>IPsec</i> | None                                                                              |

#### 4.3.2.12.2 Mutually Exclusive Functions

None

#### 4.3.2.12.3 Function Impacts

None

### 4.3.3 Hardware

#### Base Station Models

| RAT  | Base Station Model                                                                                                                                                                                                                                                            |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GSM  | 3900 and 5900 series base stations (macro base stations)                                                                                                                                                                                                                      |
| UMTS | <ul style="list-style-type: none"><li>3900 and 5900 series base stations (macro base stations)</li><li>DBS3900 LampSite and DBS5900 LampSite</li></ul>                                                                                                                        |
| LTE  | <ul style="list-style-type: none"><li>3900 and 5900 series base stations (macro base stations)</li><li>DBS3900 LampSite and DBS5900 LampSite</li></ul>                                                                                                                        |
| NR   | <ul style="list-style-type: none"><li>3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.</li><li>DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.</li></ul> |



 NOTE

The Multi-mode BS Common IPsec(LTE TDD) feature works only on 3900 and 5900 series base stations (macro base stations).

A DH group with **IKEPROPOSAL.DHGRP (LTE eNodeB, 5G gNodeB)** set to **DH\_GROUP19**, **DH\_GROUP20**, or **DH\_GROUP31** is supported only on the BookBBU5901/BookBBU5901a/BBU5900C/BBU3910C/BTS5929R/BBU3910A.

Boards

For Huawei base stations, only the Ethernet ports on the UMPT/UMDU/MDUC/UTRPc boards support IPsec.

To support IPsec or IPsec Tunnel Backup, the base station must be configured with a board or board combination supporting IPsec to provide a co-transmission port to connect to the transport network. The board must communicate with the main control board through the backplane. The following table lists the board or board combination of each base station type to support IPsec.

**Table 4-9** Board or board combination of each base station type to support IPsec

| NE     | Board Configuration                                                                                                                                                                                                                                                 |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GBTS   | UMPT_L, which provides a co-transmission port to connect to the transport network                                                                                                                                                                                   |
| eGBTS  | <ul style="list-style-type: none"><li>UMPT_G/UMDU_G/MDUC_G, which provides a co-transmission port to connect to the transport network</li><li>UMPT_G+UTRPc, with UTRPc providing a co-transmission port to connect to the transport network</li></ul>               |
| NodeB  | <ul style="list-style-type: none"><li>UMPT_U/UMDU_U/MDUC_U, which provides a co-transmission port to connect to the transport network</li><li>UMPT_U+UTRPc, with UTRPc providing a co-transmission port to connect to the transport network</li></ul>               |
| eNodeB | <ul style="list-style-type: none"><li>UMPT_L/UMPT_T/UMDU_L/UMDU_T, which provides a co-transmission port to connect to the transport network</li><li>UMPT_L/UMPT_T+UTRPc, with UTRPc providing a co-transmission port to connect to the transport network</li></ul> |
| MBTS   | UMPT/UTRPc/UMDU/MDUC, which provides a co-transmission port to connect to the transport network                                                                                                                                                                     |
| gNodeB | UMPT_N, which provides a co-transmission port to connect to the transport network                                                                                                                                                                                   |

#### NOTICE

This function requires compliance with hardware requirements defined in the meanings of the parameters used for function activation (listed in [4.4.2.3.1 Data Preparation](#)).

The function of using ESNs defined in the ESP protocol is supported only on UMPTb series boards.

When the IPsec configuration mode is tunnel-interface-based:

- USU NEs are not supported.
- The UTRPc and MDUC boards are not supported.
- UMPTa/UMPTb+UTRPc, with UTRPc providing a co-transmission port to connect to the transport network, is not supported.

A DH group with **IKEPROPOSAL.DHGRP (LTE eNodeB, 5G gNodeB)** set to **DH\_GROUP19**, **DH\_GROUP20**, or **DH\_GROUP31** is supported only on the UMPTb/UMPTe/UMPTg/UMPTga/UMPTba/UMPTb/UBBPg (excluding the UBBPg1) series boards.

## RF Modules

This function does not depend on RF modules.

## 4.3.4 Networking

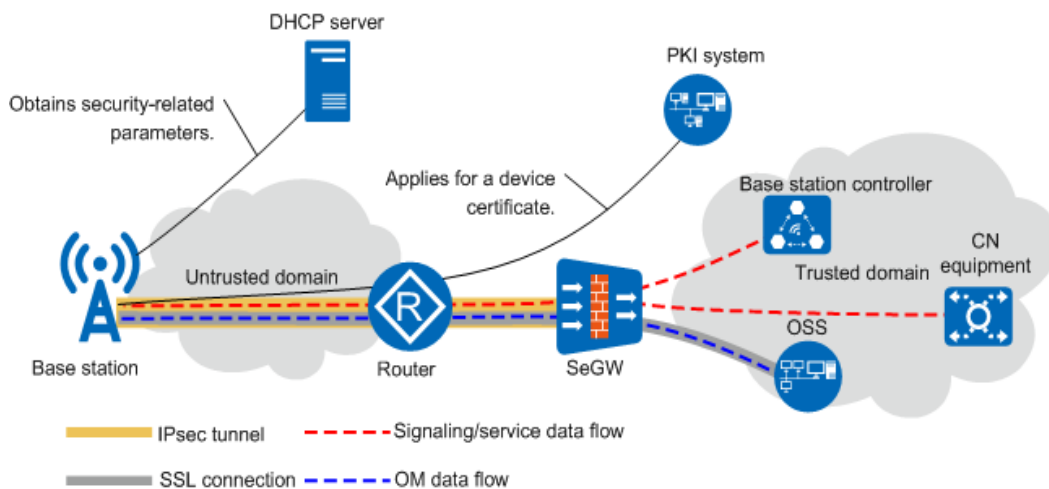
The following requirements must be met to deploy IPsec:

- An SeGW is deployed on the operator's network.
- The SeGW complies with the encryption protocols defined in RFC 2409 or RFC 7296 and supports PKI- or PSK-based authentication.

IPsec networking is confronted with major challenges in the secure domain, authentication method, and data flow protection, which are described as follows:

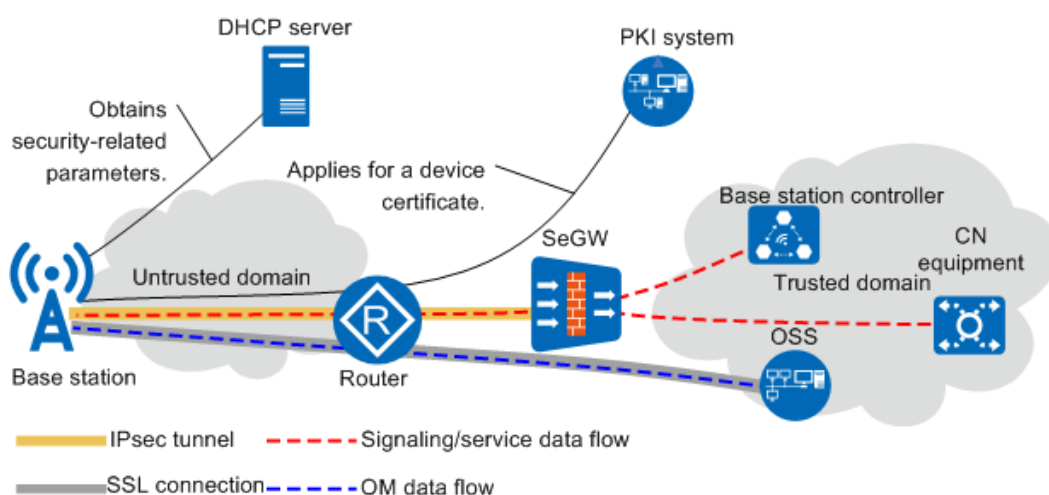
- An operator's network is divided into trusted and untrusted domains. IPsec protects only the untrusted domain. Generally, the core network (CN) is considered secure and the access network is considered insecure. SeGWs are deployed between the two domains to provide IPsec protection for data flows transmitted between the base station and SeGW.
- Two authentication methods can be used between the base station and SeGW: PKI- and PSK-based authentication. Accordingly, IPsec networks are classified into PKI- and PSK-based secure networks.
- Data flows on the base station include signaling, service, and OM data flows. In network planning, users must identify data flows to be protected and specify protection policies. Huawei base stations provide IPsec and Secure Sockets Layer (SSL) protection for OM data flows.
  - [Figure 4-29](#) shows an example of the network in which OM data flows are protected by IPsec and SSL.

**Figure 4-29** Example of the PKI-based secure network in which OM data flows are protected by IPsec



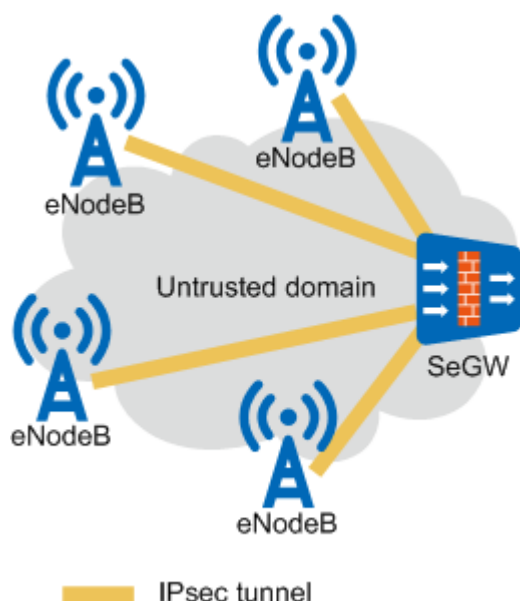
- **Figure 4-30** shows an example of the network in which OM data flows are protected by SSL instead of IPsec.

**Figure 4-30** Example of the PKI-based secure network in which OM data flows are not protected by IPsec



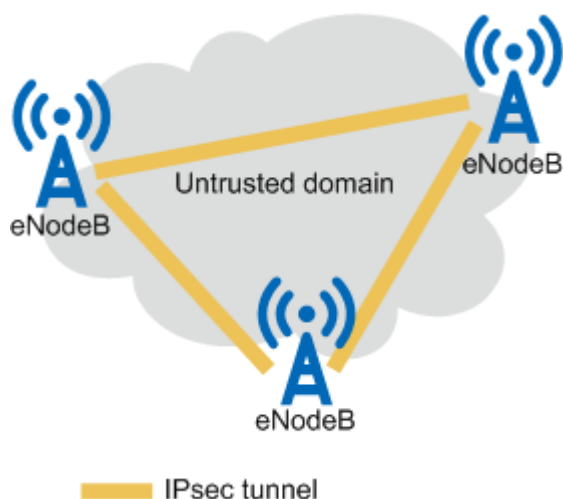
In centralized secure networking, an IPsec tunnel is established between each eNodeB and the SeGW. Data flows transmitted over X2 interfaces between eNodeBs are protected by IPsec tunnels, as shown in [Figure 4-31](#).

**Figure 4-31** Example of secure networking over the X2 interface between eNodeBs



If direct IPsec tunnels can be set up over the X2 interface between eNodeBs, distributed secure networking can be used to reduce the transmission delay over the X2 interface, as shown in [Figure 4-32](#). For details, see *S1 and X2 Self-Management* in eRAN feature documentation.

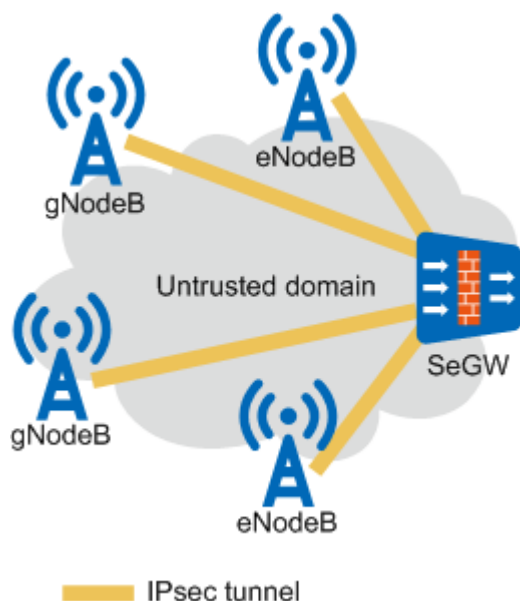
**Figure 4-32** Example of the direct IPsec tunnel over the X2 interface between eNodeBs



When the eNodeB uses a UMPT/UTRPe to provide an eX2 interface, the data protection policy is the same as that for the X2 interface, as shown in [Figure 4-31](#). In this scenario, IPsec applies to the signaling and service data flows over the eX2 interface.

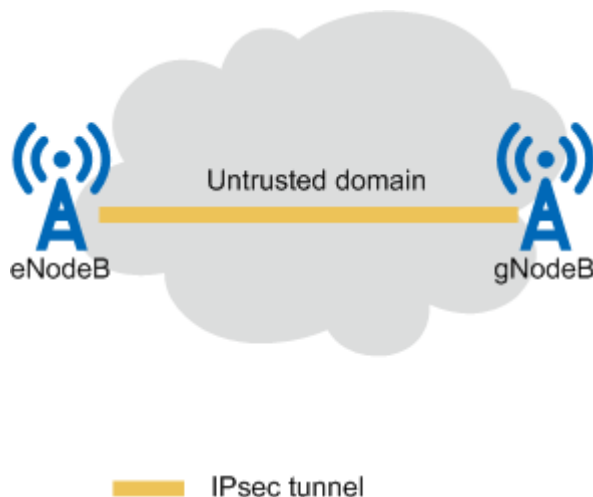
In centralized secure NSA DC networking, the eNodeB and gNodeB separately establish IPsec tunnels with the SeGW to protect the data flows over the X2 interface between the eNodeB and the gNodeB, as shown in [Figure 4-33](#).

**Figure 4-33** Example of secure networking over the X2 interface between the eNodeB and gNodeB



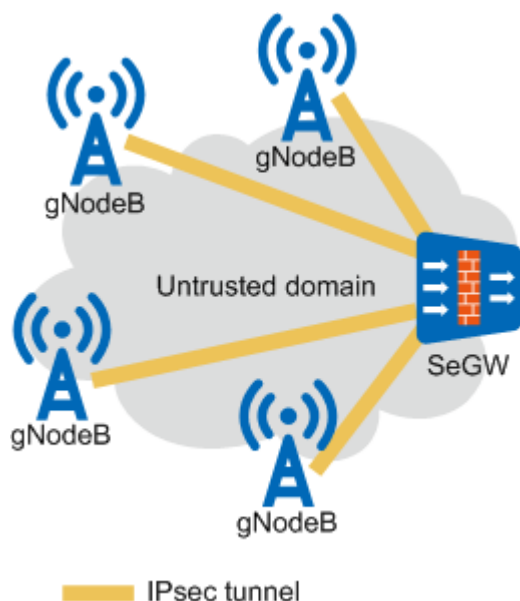
In NSA DC networking, if direct IPsec tunnels can be set up over the X2 interface between the eNodeB and gNodeB, distributed secure networking can be used to reduce the transmission delay over the X2 interface, as shown in [Figure 4-34](#). For details, see *X2 and S1 Self-Management in NSA Networking* in eRAN feature documentation or 5G RAN feature documentation.

**Figure 4-34** Example of the direct IPsec tunnel over the X2 interface between the eNodeB and gNodeB



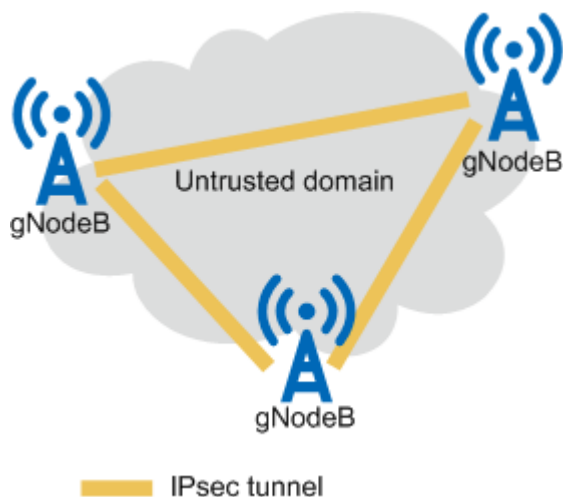
In centralized secure SA networking, an IPsec tunnel is established between each gNodeB and the SeGW. Data flows transmitted over Xn interfaces between gNodeBs are protected by IPsec tunnels, as shown in [Figure 4-35](#).

**Figure 4-35** Example of secure networking over the Xn interface between gNodeBs



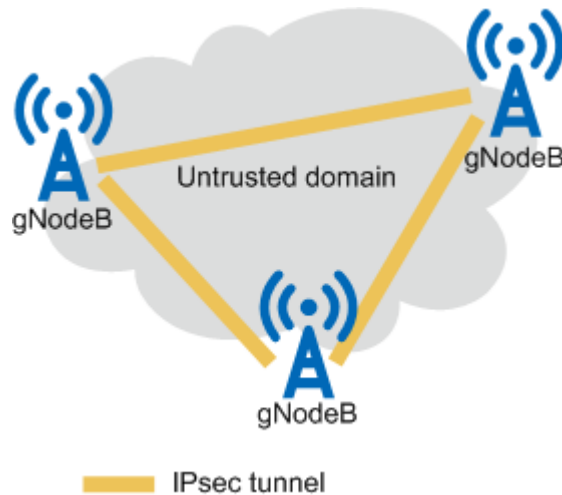
In SA networking, if direct IPsec tunnels can be set up over the Xn interface between gNodeBs, distributed secure networking can be used to reduce the transmission delay over the Xn interface, as shown in [Figure 4-36](#). For details, see *NG and Xn Self-Management* in 5G RAN feature documentation.

**Figure 4-36** Example of the direct IPsec tunnel over the Xn interface between gNodeBs



[Figure 4-37](#) shows an example of data flows over the eXn interface between gNodeBs in SA networking. For details, see *eXn Self-Management* in 5G RAN feature documentation.

**Figure 4-37** Example of the direct IPsec tunnel over the eXn interface between gNodeBs



### 4.3.5 Others

None

## 4.4 Operation and Maintenance (New Deployment)

### 4.4.1 When to Use

Unlike time division multiplexing (TDM) networks, IP networks cannot ensure transmission security. If an operator uses a public IP network, it is recommended that the IPsec feature be activated to provide integrity and confidentiality protection for wireless services. If the operator requires that IPsec negotiation use digital certificate-based authentication, activate the PKI feature. For details about how to activate PKI, see *PKI*. Currently, the IPsec feature is disabled by default. After IPsec is enabled, the associated policies, such as those configured on neighboring base stations and SeGWs, must be consistent across the entire network. If the IPsec enabling setting and associated policy configurations are inconsistent, IPsec link setup may fail and service loss may occur.

### 4.4.2 Data Configuration

#### 4.4.2.1 Required Information

Before activating the IPsec feature, confirm the peer SeGW configuration information listed in [Table 4-10](#) with the operator to ensure configuration data consistency and successful IPsec negotiation between the communicating peers.

**Table 4-10** Information for IPsec negotiation

| Information to Be Collected |                                                                                                                                                                                                                                                                                                                                   | Parameter on the Base Station Side                                                                                                                                                    |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IKE information             | IKE version                                                                                                                                                                                                                                                                                                                       | Version                                                                                                                                                                               |
|                             | IKEv1 exchange mode                                                                                                                                                                                                                                                                                                               | Exchange Mode                                                                                                                                                                         |
|                             | Type of the local ID                                                                                                                                                                                                                                                                                                              | Local ID Type                                                                                                                                                                         |
|                             | IP address of the SeGW                                                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"><li>Ethernet-interface-based configuration mode: Remote IP Address</li><li>Tunnel-interface-based configuration mode: Remote IPv4 Address</li></ul> |
|                             | IKE name of the SeGW <ul style="list-style-type: none"><li>If a PSK is used for identity authentication, obtain the local name of the SeGW.</li><li>If digital certificates are used for identity authentication, obtain information about the <b>subjectaltnname</b> field in the device certificate used by the SeGW.</li></ul> | Remote Name                                                                                                                                                                           |
|                             | IKE encryption algorithm                                                                                                                                                                                                                                                                                                          | Encryption Algorithm                                                                                                                                                                  |
|                             | IKE authentication algorithm                                                                                                                                                                                                                                                                                                      | Authentication Algorithm                                                                                                                                                              |
|                             | IKE PRF algorithm                                                                                                                                                                                                                                                                                                                 | PRF Algorithm                                                                                                                                                                         |
|                             | IKE DH group                                                                                                                                                                                                                                                                                                                      | Diffie-Hellman Group                                                                                                                                                                  |
|                             | DPD mode                                                                                                                                                                                                                                                                                                                          | DPD Mode                                                                                                                                                                              |
|                             | DPD idle time                                                                                                                                                                                                                                                                                                                     | DPD Idle Time                                                                                                                                                                         |
|                             | DPD packet retransmission interval                                                                                                                                                                                                                                                                                                | DPD Retransmission Interval                                                                                                                                                           |
|                             | Number of DPD packet retransmissions                                                                                                                                                                                                                                                                                              | DPD Retransmission Count                                                                                                                                                              |
|                             | Authentication method                                                                                                                                                                                                                                                                                                             | Authentication Method                                                                                                                                                                 |
|                             | Whether to use OCSP to check the peer certificate revocation status                                                                                                                                                                                                                                                               | OCSP Switch                                                                                                                                                                           |
|                             | CRL using policy                                                                                                                                                                                                                                                                                                                  | CRL Using Policy                                                                                                                                                                      |
| IPsec information           | IPsec encapsulation mode                                                                                                                                                                                                                                                                                                          | Encapsulation Mode                                                                                                                                                                    |



| Information to Be Collected |                                    | Parameter on the Base Station Side  |
|-----------------------------|------------------------------------|-------------------------------------|
|                             | IPsec protocol type                | Transform                           |
|                             | ESP encryption algorithm           | ESP Encryption Algorithm            |
|                             | ESP integrity protection algorithm | ESP Authentication Algorithm        |
|                             | AH integrity protection algorithm  | AH Authentication Algorithm         |
|                             | Perfect forward secrecy (PFS) flag | Perfect Forward Secrecy             |
|                             | Extended Serial Number             | ESP Extended Sequence Number Switch |

In addition to the preceding information, obtain the configurations of the ACLRULE on the SeGW.

## 4.4.2.2 Data Preparation

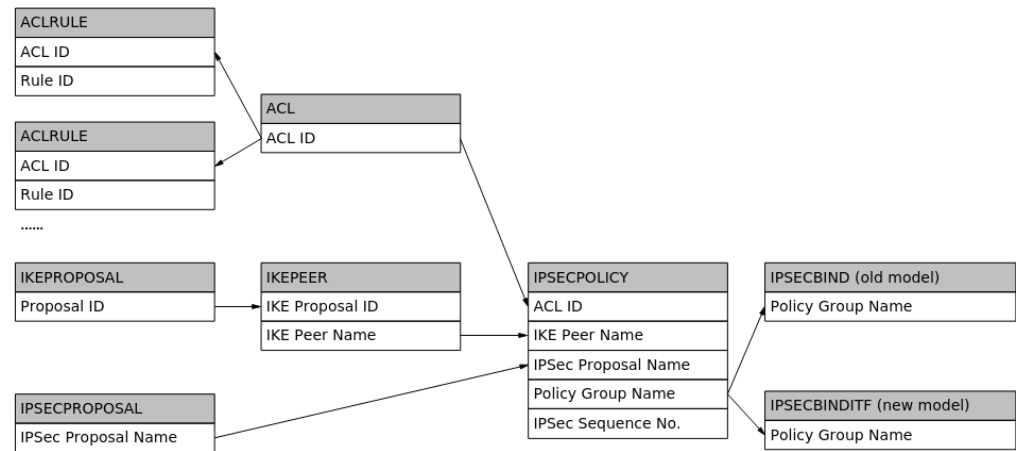
### 4.4.2.2.1 Ethernet-Interface-based Configuration Mode

One or multiple IPsec policies can be configured according to actual network conditions. Each IPsec policy is bound to one ACL. If multiple IPsec tunnels need to be bound to service interfaces, multiple ACLs need to be added. In an ACL, one or multiple ACL rules can be configured for data flows that need to be protected by IPsec. The base station provides IPsec for the data flows that comply with the ACL rules. An IPsec policy takes effect only after it is bound to a transmission port.

- An ACL rule is added to an ACL by using the **ACLID (LTE eNodeB, 5G gNodeB)** parameter.
- An IKE proposal is bound to an IKE peer by using the **PROPID (LTE eNodeB, 5G gNodeB)** parameter.
- An ACL is bound to an IPsec policy by using the **ACLID (LTE eNodeB, 5G gNodeB)** parameter.
- An IKE peer is bound to an IPsec policy by using the **PEERNAME (LTE eNodeB, 5G gNodeB)** parameter.
- An IPsec proposal is bound to an IPsec policy by using the **PROPNAME (LTE eNodeB, 5G gNodeB)** parameter.
- An IPsec policy is bound to a transmission port by using the **SPGN (LTE eNodeB, 5G gNodeB)** parameter.

**Figure 4-38** shows IPsec configuration principles. This figure uses MOs on the base station side as an example. The configuration principles on the base station controller side are similar.

**Figure 4-38** IPv4 IPsec configuration principles (Ethernet-interface-based configuration mode)



**NOTE**

If multiple IPsec tunnels are required to connect to multiple SeGWs, multiple IPsec policies must be configured, each of which is bound to an IKE peer and ACL. If multiple IPsec policies need to be bound to one port, these policies must have the same **SPGN (LTE eNodeB, 5G gNodeB)** value (through which they are bound to the port) but different **SPSN (LTE eNodeB, 5G gNodeB)** values.

**Table 4-11** Parameters that must be set to configure an IKE proposal

| Parameter Name       | Parameter ID                                      | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proposal ID          | <b>IKEPROPOSAL.PROPID (LTE eNodeB, 5G gNodeB)</b> | Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Encryption Algorithm | <b>IKEPROPOSAL.ENCALG (LTE eNodeB, 5G gNodeB)</b> | <p>The parameter settings on the base station and SeGW must be the same.</p> <p>The DES algorithm has been cracked, and the 3DES algorithm is considered insufficiently secure in the industry. If these algorithms of low security levels are used, encrypted data may be cracked by attackers. It is recommended that the AES_GCM_128 or AES_GCM_256 encryption algorithm be used to ensure IKE SA security.</p> <p>You can run <b>SET SECPLY</b> to adjust the configuration security level to prohibit the parameter from being set to <b>MD5</b>, <b>SHA1</b>, or <b>AES-XCBC-MAC-96</b>.</p> |

| Parameter Name           | Parameter ID                                                 | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Authentication Algorithm | <b>IKEPROPOSAL.AUTH<br/>ALG (LTE eNodeB, 5G<br/>gNodeB)</b>  | <p>The parameter settings on the base station and SeGW must be the same.</p> <p>The MD5, SHA1, and AES-XCBC-96 algorithms are considered insufficiently secure in the industry. If these algorithms of low security levels are used, the transmitted data may be forged by attackers. It is recommended that the SHA256 or SHA384 algorithm for authentication be used to ensure IKE SA security.</p> <p>You can run <b>SET SECPLY</b> to adjust the configuration security level to prohibit the parameter from being set to <b>MD5</b>, <b>SHA1</b>, or <b>AES-XCBC-MAC-96</b>.</p> |
| Authentication Method    | <b>IKEPROPOSAL.AUTH<br/>METH (LTE eNodeB,<br/>5G gNodeB)</b> | <p>This parameter must be set to <b>IKE_CERT_SIG</b> if digital-certificate-based authentication is required. The parameter settings on the base station and SeGW must be the same.</p>                                                                                                                                                                                                                                                                                                                                                                                               |
| Diffie-Hellman Group     | <b>IKEPROPOSAL.DHGR<br/>P (LTE eNodeB, 5G<br/>gNodeB)</b>    | <p>The parameter settings on the base station and SeGW must be the same.</p> <p>The DH_GROUP1 and DH_GROUP2 algorithms have been cracked in the industry. If these algorithms of low security levels are used, the negotiated keys may be cracked by attackers. It is recommended that the DH_GROUP14, DH_GROUP15, DH_GROUP19, DH_GROUP20, or DH_GROUP31 algorithm be used.</p> <p>You can run <b>SET SECPLY</b> to adjust the configuration security level to prohibit the parameter from being set to <b>DH_GROUP1</b> or <b>DH_GROUP2</b>.</p>                                     |

| Parameter Name                                                                                                                                                                                                                                                                                                                                                              | Parameter ID                                         | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PRF Algorithm                                                                                                                                                                                                                                                                                                                                                               | <b>IKEPROPOSAL.PRFAQ (LTE eNodeB, 5G gNodeB)</b>     | <p>This parameter must be set when the <b>IKEPEER.IKEVERSION (LTE eNodeB, 5G gNodeB)</b> parameter is set to <b>IKE_V2</b>. The parameter settings on the base station and SeGW must be the same.</p> <p>The HMAC_MD5 algorithm has been cracked. As this algorithm is weak in security, the pseudorandom numbers generated using this algorithm may be guessed by attackers. It is recommended that the HMAC_SHA256 or HMAC_SHA384 algorithm be used.</p> <p>You can run <b>SET SECPLY</b> to adjust the configuration security level to prohibit the parameter from being set to <b>HMAC_MD5</b>, <b>HMAC_SHA1</b>, or <b>AES128_XCBC</b>.</p> |
| ISAKMP SA Duration                                                                                                                                                                                                                                                                                                                                                          | <b>IKEPROPOSAL.DURATION (LTE eNodeB, 5G gNodeB)</b>  | <p>Retain the default value.</p> <p>It is recommended that this parameter be set to a value greater than <b>3600</b>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| IKE Reauthentication Switch                                                                                                                                                                                                                                                                                                                                                 | <b>IKEPROPOSAL.REAUTHSW (5G gNodeB, LTE eNodeB)</b>  | <p>When this switch is turned on, the base station supports both active and passive IKE reauthentication. When this switch is turned off, the base station supports only passive IKE reauthentication. Set this parameter based on the network plan.</p>                                                                                                                                                                                                                                                                                                                                                                                         |
| IKE Reauthentication Lifetime                                                                                                                                                                                                                                                                                                                                               | <b>IKEPROPOSAL.REAUTHHLT (5G gNodeB, LTE eNodeB)</b> | <p>It is recommended that this parameter be set to a value greater than the value of the <b>IKEPROPOSAL.DURATION (5G gNodeB, LTE eNodeB)</b> parameter.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p>Note: In base station deployment by PnP, the SeGWs from some vendors have special requirements for the combinations of the following IKEv2 negotiation algorithms: <b>Encryption Algorithm</b>, <b>Authentication Algorithm</b>, <b>Diffie-Hellman Group</b>, and <b>PRF Algorithm</b>. For details about the requirements, see <i>Automatic OMCH Establishment</i>.</p> |                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

**Table 4-12** Parameters that must be set to configure an IKE peer

| Parameter Name    | Parameter ID                                             | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IKE Peer Name     | <b><i>IKEPEER.PEERNAME (LTE eNodeB, 5G gNodeB)</i></b>   | Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| IKE Proposal ID   | <b><i>IKEPEER.PROPID (LTE eNodeB, 5G gNodeB)</i></b>     | The value of this parameter must be the same as that of the <b><i>IKEPROPOSAL.PROPID (LTE eNodeB, 5G gNodeB)</i></b> parameter.                                                                                                                                                                                                                                                                                                                                                                                |
| Version           | <b><i>IKEPEER.IKEVERSION (LTE eNodeB, 5G gNodeB)</i></b> | The parameter settings on the base station and SeGW must be the same. You can run <b>SET SECPLY</b> to adjust the configuration security level to prohibit the parameter from being set to <b>IKE_V1</b> .                                                                                                                                                                                                                                                                                                     |
| Exchange Mode     | <b><i>IKEPEER.EXCHMODE (5G gNodeB, LTE eNodeB)</i></b>   | This parameter must be set when the <b><i>IKEPEER.IKEVERSION (5G gNodeB, LTE eNodeB)</i></b> parameter is set to <b>IKE_V1</b> . The parameter settings on the base station and SeGW must be the same. The value <b>MAIN</b> is recommended.                                                                                                                                                                                                                                                                   |
| IP Version        | <b><i>IKEPEER.IPVERSION (5G gNodeB, LTE eNodeB)</i></b>  | Set this parameter to <b>IPV4</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Local ID Type     | <b><i>IKEPEER.IDTYPE (LTE eNodeB, 5G gNodeB)</i></b>     | Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Remote IP Address | <b><i>IKEPEER.REMOTEIP (LTE eNodeB, 5G gNodeB)</i></b>   | The value of this parameter must be the same as that of the <b><i>ACLRULE.DIP (LTE eNodeB, 5G gNodeB)</i></b> parameter if the <b><i>IPSECPROPOSAL.ENCAPMODE (LTE eNodeB, 5G gNodeB)</i></b> parameter is set to <b>TRANSPORT</b> . If they are different, encrypted packets cannot be decrypted by the peer.<br><br>If the <b><i>IPSECPROPOSAL.ENCAPMODE (LTE eNodeB, 5G gNodeB)</i></b> parameter is set to <b>TUNNEL</b> , the value of this parameter must be the same as the IP address of the peer SeGW. |

| Parameter Name              | Parameter ID                                              | Setting Notes                                                                                                                                                                                                                                                                                                |
|-----------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Remote Name                 | <b><i>IKEPEER.REMOTENAME (LTE eNodeB, 5G gNodeB)</i></b>  | If this parameter is specified and the <b><i>IKEPROPOSAL.AUTHMETH (5G gNodeB, LTE eNodeB)</i></b> parameter is set to <b>IKE_CERT_SIG</b> , the base station will verify the peer ID during IKE negotiation with the SeGW. If this parameter is not specified, the base station will not verify the peer ID. |
| Pre-shared Key              | <b><i>IKEPEER.PKEY (LTE eNodeB, 5G gNodeB)</i></b>        | This parameter is valid only when the <b><i>IKEPROPOSAL.AUTHMETH (5G gNodeB, LTE eNodeB)</i></b> parameter is set to <b>PRE_SHARED_KEY</b> .                                                                                                                                                                 |
| DPD Mode                    | <b><i>IKEPEER.DPD (LTE eNodeB, 5G gNodeB)</i></b>         | DPD is enabled by default. Set the DPD-related parameters to the same values at both IKE ends. If the values are different, the end with a shorter timer length will detect that the peer is offline prior to the peer, and an IKE renegotiation will be triggered.                                          |
| DPD Idle Time               | <b><i>IKEPEER.DPDIDLETIME (LTE eNodeB, 5G gNodeB)</i></b> |                                                                                                                                                                                                                                                                                                              |
| DPD Retransmission Interval | <b><i>IKEPEER.DPDRETRI (5G gNodeB, LTE eNodeB)</i></b>    |                                                                                                                                                                                                                                                                                                              |
| DPD Retransmission Count    | <b><i>IKEPEER.DPDRETRN (LTE eNodeB, 5G gNodeB)</i></b>    |                                                                                                                                                                                                                                                                                                              |
| NAT traversal               | <b><i>IKEPEER.NATTRAV (5G gNodeB, LTE eNodeB)</i></b>     | It is recommended that this parameter be set to <b>ENABLE</b> in networks deployed with NAT equipment. For details, see <a href="#">4.1.3.1 Typical IPsec Networking</a> .                                                                                                                                   |

| Parameter Name            | Parameter ID                                                 | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Local IP Address          | <b>IKEPEER.LOCALIP</b><br><i>(LTE eNodeB, 5G gNodeB)</i>     | <p>This parameter specifies the local IP address for IKE negotiation. The device IP address is used by default when this parameter is set to <b>0.0.0.0</b>. If multiple IPv4 addresses on the same network segment are configured on the same interface, the local IPv4 address in the <b>IKEPEER</b> MO must be configured. Otherwise, IKE negotiation may fail.</p> <p>The value of this parameter must be the same as that of the <b>ACLRULE.SIP</b> <i>(LTE eNodeB, 5G gNodeB)</i> parameter if the <b>IPSECPROPOSAL.ENCAPMODE</b> <i>(LTE eNodeB, 5G gNodeB)</i> parameter is set to <b>TRANSPORT</b>. If they are different, encrypted packets cannot be decrypted by the peer.</p> |
| Redundancy Flag           | <b>IKEPEER.REDUNDANCYFLAG</b> <i>(5G gNodeB, LTE eNodeB)</i> | Set this parameter to <b>NONE</b> when the IPsec redundancy among multiple SeGWs feature is not enabled.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Certificate Source        | <b>IKEPEER.CERTSOURCE</b> <i>(5G gNodeB, LTE eNodeB)</i>     | Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Certificate File Name     | <b>IKEPEER.CERTNAME</b> <i>(5G gNodeB, LTE eNodeB)</i>       | This parameter must be set when the <b>IKEPEER.CERTSOURCE</b> <i>(5G gNodeB, LTE eNodeB)</i> parameter is set to <b>CERTMK</b> . Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| IPSec PreFragment Switch  | <b>IKEPEER.IPSECPREFRGSW</b> <i>(5G gNodeB, LTE eNodeB)</i>  | If the network plan requires that SeGW performance be preferentially guaranteed, it is recommended that this parameter be set to <b>ON</b> . If the network plan requires that the CN performance and transmission efficiency be preferentially guaranteed, the default value <b>OFF</b> is recommended.                                                                                                                                                                                                                                                                                                                                                                                   |
| Maximum Transmission Unit | <b>IKEPEER.MTU</b> <i>(5G gNodeB, LTE eNodeB)</i>            | This parameter is valid only when the <b>IKEPEER.IPSECPREFRGSW</b> <i>(5G gNodeB, LTE eNodeB)</i> parameter is set to <b>ON</b> . For details about the setting notes, see <a href="#">4.1.2.3.2 Packet Fragmentation Before IPsec Encryption</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                        |

| Parameter Name             | Parameter ID                                         | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Trust Group ID             | <b>IKEPEER.TRUSTGROUP ID (LTE eNodeB, 5G gNodeB)</b> | This parameter specifies the ID of a trust group. When certificate-based authentication is used for link negotiation or establishment, the trust certificates in the trust group configured in the <b>TRUSTGROUP</b> MO are used to authenticate the peer end. If this parameter is set to <b>65535</b> , no trust group is associated, and all loaded trust certificates in the system are used to authenticate the device certificates of the peer end.                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| IKEv2 Fragmentation Switch | <b>IKEPEER.IKEV2FRGSW (5G gNodeB, LTE eNodeB)</b>    | This parameter specifies whether to enable IKEv2 fragmentation negotiation. For details about this function, see RFC 7383.<br><br>Whether IKEv2 fragmentation takes effect depends on the negotiation result, which can be viewed in the <b>DSP IKESA</b> command output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| IKEv2 Fragmentation MTU    | <b>IKEPEER.IKEV2FRGMTU (LTE eNodeB, 5G gNodeB)</b>   | This parameter specifies the maximum transmission unit of IKEv2 fragments for the IKE peer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| OCSP Switch                | <b>IKEPEER.OCSPSW (LTE eNodeB, 5G gNodeB)</b>        | This parameter specifies whether the local end uses the IKE OCSP (for details, see RFC 4806) to check the certificate revocation status of the peer end. When this switch is turned on, IKE messages sent by the local end contain OCSP requests, requiring the IKE messages sent by the peer end to contain OCSP responses. This switch applies only to IKEv2 and takes effect only when the <b>IKEPROPOSAL.AUTHMETH (LTE eNodeB, 5G gNodeB)</b> parameter is set to <b>IKE_CERT_SIG</b> . The effective status of this switch can be queried by running the <b>DSP IKESA</b> command.<br><br>You are advised to check whether the peer SeGW supports IKE OCSP, and set this parameter to <b>ON</b> if the peer SeGW supports IKE OCSP and retain the default value <b>OFF</b> if it does not. Setting this parameter to <b>ON</b> while the peer SeGW does not support IKE OCSP brings IKE disconnection risks. |



| Parameter Name   | Parameter ID                                            | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CRL using policy | <b><i>IKEPEER.CRLPOLICY (LTE eNodeB, 5G gNodeB)</i></b> | <p>This parameter specifies the CRL using policy for the base station in IKE scenarios.</p> <ul style="list-style-type: none"><li>• When this parameter is set to <b>GLOBAL</b>, the CRL using policy for the base station in IKE scenarios is the same as that specified in the <b>SET CRLPOLICY</b> command.</li><li>• When this parameter is set to <b>ALARM</b>, the base station in IKE scenarios reports ALM-26834 Peer Certificate Invalid if the CRL- or OCSP-based check result indicates that the peer certificate has been revoked.</li><li>• When this parameter is set to <b>DISCONNECT</b>, the base station in IKE scenarios terminates link negotiation and reports ALM-26834 Peer Certificate Invalid if the CRL- or OCSP-based check result indicates that the peer certificate has been revoked; the base station in IKE scenarios only terminates link negotiation if the check result indicates that the CRL has not taken effect or has expired.</li></ul> |

**Table 4-13** Parameters that must be set to configure an ACL

| Parameter Name | Parameter ID                                      | Setting Notes                                                                                                                                                  |
|----------------|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ACL ID         | <b><i>ACL.ACLID (5G gNodeB, LTE eNodeB)</i></b>   | <p>At least one ACL rule must be defined for each ACL.</p> <p>If an ACL is bound to an IPsec policy, the value of this parameter ranges from 3000 to 3999.</p> |
| Description    | <b><i>ACL.ACLDESC (LTE eNodeB, 5G gNodeB)</i></b> | <p>Set this parameter based on the network plan.</p>                                                                                                           |

**Table 4-14** Parameters that must be set to configure an ACL rule

| Parameter Name    | Parameter ID                                  | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ACL ID            | <b>ACLRULE.ACLID (LTE eNodeB, 5G gNodeB)</b>  | The value of this parameter must be the same as that of the <b>ACL.ACLID (5G gNodeB, LTE eNodeB)</b> parameter.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Rule ID           | <b>ACLRULE.RULEID (LTE eNodeB, 5G gNodeB)</b> | Each ACL rule in an ACL must have a unique ID.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Action            | <b>ACLRULE.ACTION (LTE eNodeB, 5G gNodeB)</b> | <p>Set this parameter based on the network plan. It can be set to <b>DENY</b> or <b>PERMIT</b>. If this parameter is set to <b>PERMIT</b>, the base station encrypts or decrypts the data that matches the ACL rule. If this parameter is set to <b>DENY</b>, the base station does not encrypt or decrypt the data that matches the ACL rule.</p> <p>The IDs of the ACL rules with this parameter set to <b>DENY</b> must be smaller than the IDs of the ACL rules with this parameter set to <b>PERMIT</b>.</p> <ul style="list-style-type: none"><li>• If a packet matches an ACL rule in an ACL, the base station determines whether to encrypt the packet by IPsec based on the value of this parameter. If the packet does not match an ACL rule, the base station tries the next ACL rule until all ACL rules in the ACL are tried.</li><li>• A packet is not encrypted if it does not match any ACL rule.</li></ul> |
| Protocol Type     | <b>ACLRULE.PT (LTE eNodeB, 5G gNodeB)</b>     | Set this parameter to <b>IP</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Source IP Address | <b>ACLRULE.SIP (LTE eNodeB, 5G gNodeB)</b>    | The value of this parameter must be a configured device IP address if the <b>IPSECPROPOSAL.ENCAPMODE (LTE eNodeB, 5G gNodeB)</b> parameter is set to <b>TRANSPORT</b> . Otherwise, encrypted packets cannot be decrypted by the peer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Parameter Name           | Parameter ID                                 | Setting Notes                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Destination IP Address   | <b>ACLRULE.DIP (LTE eNodeB, 5G gNodeB)</b>   | If the <b>IPSECPROPOSAL.ENCAPMODE (LTE eNodeB, 5G gNodeB)</b> parameter is set to <b>TRANSPORT</b> , the value of this parameter must be a host IP address instead of a network segment address. The value of this parameter must be the same as that of the <b>IKEPEER.REMOTEIP (LTE eNodeB, 5G gNodeB)</b> parameter. Otherwise, encrypted packets cannot be decrypted by the peer. |
| Source Wildcard          | <b>ACLRULE.SWC (LTE eNodeB, 5G gNodeB)</b>   | Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                                                                         |
| Destination Wildcard     | <b>ACLRULE.DWC (LTE eNodeB, 5G gNodeB)</b>   | Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                                                                         |
| Match Source Port        | <b>ACLRULE.SMPT (LTE eNodeB, 5G gNodeB)</b>  | Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                                                                         |
| Source Port Operate      | <b>ACLRULE.SOP (LTE eNodeB, 5G gNodeB)</b>   | These parameters must be set when the <b>ACLRULE.SMPT (5G gNodeB, LTE eNodeB)</b> parameter is set to <b>YES</b> .                                                                                                                                                                                                                                                                    |
| Source Port 1            | <b>ACLRULE.SPT1 (LTE eNodeB, 5G gNodeB)</b>  |                                                                                                                                                                                                                                                                                                                                                                                       |
| Source Port 2            | <b>ACLRULE.SPT2 (LTE eNodeB, 5G gNodeB)</b>  |                                                                                                                                                                                                                                                                                                                                                                                       |
| Match Destination Port   | <b>ACLRULE.DMPT (LTE eNodeB, 5G gNodeB)</b>  | Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                                                                         |
| Destination Port Operate | <b>ACLRULE.DOP (5G gNodeB, LTE eNodeB)</b>   | These parameters must be set when the <b>ACLRULE.DMPT (5G gNodeB, LTE eNodeB)</b> parameter is set to <b>YES</b> .                                                                                                                                                                                                                                                                    |
| Destination Port 1       | <b>ACLRULE.DPT1 (LTE eNodeB, 5G gNodeB)</b>  |                                                                                                                                                                                                                                                                                                                                                                                       |
| Destination Port 2       | <b>ACLRULE.DPT2 (LTE eNodeB, 5G gNodeB)</b>  |                                                                                                                                                                                                                                                                                                                                                                                       |
| Match DSCP               | <b>ACLRULE.MDSCP (5G gNodeB, LTE eNodeB)</b> | Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                                                                         |
| DSCP                     | <b>ACLRULE.DSCP (LTE eNodeB, 5G gNodeB)</b>  | This parameter must be set when the <b>ACLRULE.MDSCP (5G gNodeB, LTE eNodeB)</b> parameter is set to <b>YES</b> .                                                                                                                                                                                                                                                                     |

**Table 4-15** Parameters that must be set to configure an IPsec proposal

| Parameter Name               | Parameter ID                                                     | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IPSec Proposal Name          | <b>IPSECPROPOSAL.PRO<br/>PNAME (5G gNodeB,<br/>LTE eNodeB)</b>   | Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Encapsulation Mode           | <b>IPSECPROPOSAL.ENC<br/>APMODE (LTE<br/>eNodeB, 5G gNodeB)</b>  | The parameter settings on the base station and SeGW must be the same.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Transform                    | <b>IPSECPROPOSAL.TRA<br/>NMODE (LTE<br/>eNodeB, 5G gNodeB)</b>   | The parameter settings on the base station and SeGW must be the same.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| AH Authentication Algorithm  | <b>IPSECPROPOSAL.AHA<br/>UTHALG (LTE<br/>eNodeB, 5G gNodeB)</b>  | <p>The parameter settings on the base station and SeGW must be the same.</p> <p>The MD5 algorithm has been cracked, and the SHA1 and AES-XCBC-MAC-96 algorithms are considered insufficiently secure in the industry. If these algorithms of low security levels are used, data may be forged by attackers. It is recommended that the SHA256 algorithm be used.</p> <p>You can run <b>SET SECLY</b> to adjust the configuration security level to prohibit the parameter from being set to <b>MD5</b>, <b>SHA1</b>, or <b>AES-XCBC-MAC-96</b>.</p>                                  |
| ESP Authentication Algorithm | <b>IPSECPROPOSAL.ESP<br/>AUTHALG (LTE<br/>eNodeB, 5G gNodeB)</b> | <p>The parameter settings on the base station and SeGW must be the same.</p> <p>NULL indicates a null algorithm. The MD5 algorithm has been cracked, and the SHA1 and AES-XCBC-MAC-96 algorithms are considered insufficiently secure in the industry. If these algorithms of low security levels are used, data may be forged by attackers. It is recommended that the SHA256 algorithm be used.</p> <p>You can run <b>SET SECLY</b> to adjust the configuration security level to prohibit the parameter from being set to <b>MD5</b>, <b>SHA1</b>, or <b>AES-XCBC-MAC-96</b>.</p> |

| Parameter Name                      | Parameter ID                                                    | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ESP Encryption Algorithm            | <b>IPSECPROPOSAL.ESP<br/>ENCALG (LTE<br/>eNodeB, 5G gNodeB)</b> | The parameter settings on the base station and SeGW must be the same.<br>NULL indicates a null algorithm, and NULL_AUTH_GMAC_128 and NULL_AUTH_GMAC_256 indicate null encryption algorithms. The DES algorithm has been cracked, and the 3DES algorithm is considered insufficiently secure in the industry. If these algorithms of low security levels are used, encrypted data may be cracked by attackers. It is recommended that the AES_GCM_128 or AES_GCM_256 algorithm be used. |
| ESP Extended Sequence Number Switch | <b>IPSECPROPOSAL.ESP<br/>ESNSW (LTE eNodeB,<br/>5G gNodeB)</b>  | The parameter settings on the base station and SeGW must be the same.                                                                                                                                                                                                                                                                                                                                                                                                                  |

**Table 4-16** Parameters that must be set to configure an IPsec policy

| Parameter Name      | Parameter ID                                                 | Setting Notes                                                                                                                              |
|---------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Policy Group Name   | <b>IPSECPOLICY.SPGN<br/>(LTE eNodeB, 5G<br/>gNodeB)</b>      | Set this parameter based on the network plan.                                                                                              |
| IPSec Sequence No.  | <b>IPSECPOLICY.SPSN<br/>(LTE eNodeB, 5G<br/>gNodeB)</b>      | Set this parameter based on the network plan.                                                                                              |
| ACL Type            | <b>IPSECPOLICY.ACLTYP<br/>E (5G gNodeB, LTE<br/>eNodeB)</b>  | Set this parameter based on the network plan.                                                                                              |
| ACL ID              | <b>IPSECPOLICY.ACLID<br/>(LTE eNodeB, 5G<br/>gNodeB)</b>     | This parameter binds an IPsec policy to an ACL. Only data flows that comply with rules in the ACL are processed based on the IPsec policy. |
| IPSec Proposal Name | <b>IPSECPOLICY.PROPN<br/>AME (5G gNodeB,<br/>LTE eNodeB)</b> | The value of this parameter must be the same as that of the <b>IPSECPROPOSAL.PROPNAME (5G gNodeB, LTE eNodeB)</b> parameter.               |

| Parameter Name            | Parameter ID                                            | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IKE Peer Name             | <b>IPSECPOLICY.PEERNAME (LTE eNodeB, 5G gNodeB)</b>     | The value of this parameter must be the same as that of the <b>IKEPEER.PEERNAME (LTE eNodeB, 5G gNodeB)</b> parameter.                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Perfect Forward Secrecy   | <b>IPSECPOLICY.PFS (LTE eNodeB, 5G gNodeB)</b>          | <p>The parameter settings on the base station and SeGW must be the same.</p> <p>The PFS_GROUP1 and PFS_GROUP2 algorithms have been cracked in the industry. If these algorithms of low security levels are used, the negotiated keys may be cracked by attackers. It is recommended that the PFS_GROUP15, PFS_GROUP19, PFS_GROUP20, or PFS_GROUP31 algorithm be used.</p> <p>You can run <b>SET SECPLY</b> to adjust the configuration security level to prohibit the parameter from being set to <b>PFS_GROUP1</b> or <b>PFS_GROUP2</b>.</p> |
| SA Duration Mode          | <b>IPSECPOLICY.LTCFG (LTE eNodeB, 5G gNodeB)</b>        | The default value is recommended.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Lifetime Based On Time    | <b>IPSECPOLICY.LTS (5G gNodeB, LTE eNodeB)</b>          | Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Lifetime Based On Traffic | <b>IPSECPOLICY.LTKB (5G gNodeB, LTE eNodeB)</b>         | The default value is recommended.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Anti-Replay Window Size   | <b>IPSECPOLICY.REPLAY WND (LTE eNodeB, 5G gNodeB)</b>   | <p>If serious packet disorder occurs on the network, it is recommended that the anti-replay function be disabled by setting this parameter to <b>WND_DISABLE</b>.</p> <p>If the anti-replay function is required, it is recommended that this parameter be set to <b>WND_512</b>.</p> <p>If the anti-replay function is disabled, the base station may encounter replay attacks.</p>                                                                                                                                                          |
| Narrow Down Switch        | <b>IPSECPOLICY.NARROWDOWNSW (5G gNodeB, LTE eNodeB)</b> | It is recommended that this parameter be set to <b>ON</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

An IPsec policy takes effect only after it is bound to a port. [Table 4-17](#) describes the parameters that must be set when the **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** parameter is set to **OLD**.

**Table 4-17** Parameters that must be set to configure the binding between an IPsec policy and a port

| Parameter Name    | Parameter ID                                  | Setting Notes                                                                                                                                                                                                |
|-------------------|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subboard Type     | <b>IPSECBIND.SBT (5G gNodeB, LTE eNodeB)</b>  | Set this parameter based on the network plan.                                                                                                                                                                |
| Port Type         | <b>IPSECBIND.PT (5G gNodeB, LTE eNodeB)</b>   | If the <b>IPSECBIND.PT (5G gNodeB, LTE eNodeB)</b> parameter is set to <b>ETH</b> , the port specified by the <b>IPSECBIND.PN (LTE eNodeB, 5G gNodeB)</b> parameter cannot be a member of an Ethernet trunk. |
| Port No.          | <b>IPSECBIND.PN (LTE eNodeB, 5G gNodeB)</b>   |                                                                                                                                                                                                              |
| Policy Group Name | <b>IPSECBIND.SPGN (LTE eNodeB, 5G gNodeB)</b> | The value of this parameter must be the same as that of the <b>IPSECPOLICY.SPGN (5G gNodeB, LTE eNodeB)</b> parameter.                                                                                       |

[Table 4-18](#) describes the parameters that must be set when the **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** parameter is set to **NEW**.

**Table 4-18** Parameters that must be set to configure the binding between an IPsec policy and an interface

| Parameter Name          | Parameter ID                                               | Setting Notes                                                                                                          |
|-------------------------|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| IPSec Bind Interface ID | <b>IPSECBINDITF.IPSECBINDITFID (5G gNodeB, LTE eNodeB)</b> | Set this parameter based on the network plan.                                                                          |
| Policy Group Name       | <b>IPSECBINDITF.SPGN (5G gNodeB, LTE eNodeB)</b>           | The value of this parameter must be the same as that of the <b>IPSECPOLICY.SPGN (5G gNodeB, LTE eNodeB)</b> parameter. |
| Interface ID            | <b>IPSECBINDITF.ITFID (5G gNodeB, LTE eNodeB)</b>          | Set this parameter based on the network plan.                                                                          |

(Optional) [Table 4-19](#) describes the parameters that must be set for basic IKE configurations.

**Table 4-19** Parameters that must be set for basic IKE configurations

| Parameter Name                             | Parameter ID                                                 | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Local Name                                 | <b>IKECFG.IKELNM</b> ( <i>LTE eNodeB, 5G gNodeB</i> )        | <b>Local Name</b> needs to be configured in the following scenarios: <ul style="list-style-type: none"><li>The <b>IKEPEER.IDTYPE</b> (<i>5G gNodeB, LTE eNodeB</i>) and <b>IKEPROPOSAL.AUTHMETH</b> (<i>5G gNodeB, LTE eNodeB</i>) are set to <b>FQDN</b> and <b>PRE_SHARED_KEY</b>, respectively.</li><li>The <b>IKEPROPOSAL.AUTHMETH</b> (<i>5G gNodeB, LTE eNodeB</i>) parameter is set to <b>IKE_CERT_SIG</b>, and the <b>Local Name</b> of the certificate is <b>NULL</b>.</li></ul> |
| Interval for Sending Keepalive Packets     | <b>IKECFG.IKEKLI</b> ( <i>LTE eNodeB, 5G gNodeB</i> )        | Set these parameters if IKEv1 entities require the keepalive function. If both parameters are set to <b>0</b> , the keepalive function is disabled.<br><br>IKEv2 does not support the keepalive function. Therefore, the two parameters do not need to be set for IKEv2.                                                                                                                                                                                                                  |
| Keepalive Timeout                          | <b>IKECFG.IKEKLT</b> ( <i>LTE eNodeB, 5G gNodeB</i> )        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Interval for Sending NAT Keepalive Packets | <b>IKECFG.NATKLI</b> ( <i>5G gNodeB, LTE eNodeB</i> )        | This parameter is valid when the <b>IKEPEER.NATTRAV</b> ( <i>5G gNodeB, LTE eNodeB</i> ) parameter is set to <b>ENABLE</b> . The value <b>20</b> is recommended.                                                                                                                                                                                                                                                                                                                          |
| DSCP                                       | <b>IKECFG.DSCP</b> ( <i>5G gNodeB, LTE eNodeB</i> )          | This parameter specifies the differentiated services code point (DSCP) for IKE negotiation packets. The recommended value of this parameter is <b>48</b> .                                                                                                                                                                                                                                                                                                                                |
| IPSec Bind Mode                            | <b>IKECFG.IPSECBINDMODE</b> ( <i>5G gNodeB, LTE eNodeB</i> ) | This parameter is valid only when the new model is used and VLAN sub-interfaces are supported.                                                                                                                                                                                                                                                                                                                                                                                            |
| IPSec Config Mode                          | <b>IKECFG.IPSECCFGMODE</b> ( <i>LTE eNodeB, 5G gNodeB</i> )  | Set this parameter to <b>ETHINTERFACE_BASED</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                         |



 NOTE

When the **IKEPROPOSAL.AUTHMETH (5G gNodeB, LTE eNodeB)** parameter is set to **IKE\_CERT\_SIG**, the following are true: (1) If IKEv1 is used, the **IKECFG.IKELNM (5G gNodeB, LTE eNodeB)** parameter does not need to be set because its setting does not take effect. (2) If IKEv2 is used, the **IKECFG.IKELNM (5G gNodeB, LTE eNodeB)** parameter takes effect. In this case, if the **IKEPEER.IDTYPE (5G gNodeB, LTE eNodeB)** parameter is set to **FQDN**, the base station uses the value of the SubjectAltName field in its digital certificate as the local IKE name. If the **IKEPEER.IDTYPE (5G gNodeB, LTE eNodeB)** parameter is set to **DN**, the base station uses the value of the DomainName field in its digital certificate as the local IKE name.

(Optional) [Table 4-20](#) describes the parameters that must be set to configure IPsec replay alarm detection.

**Table 4-20** Parameters that must be set to configure IPsec replay alarm detection

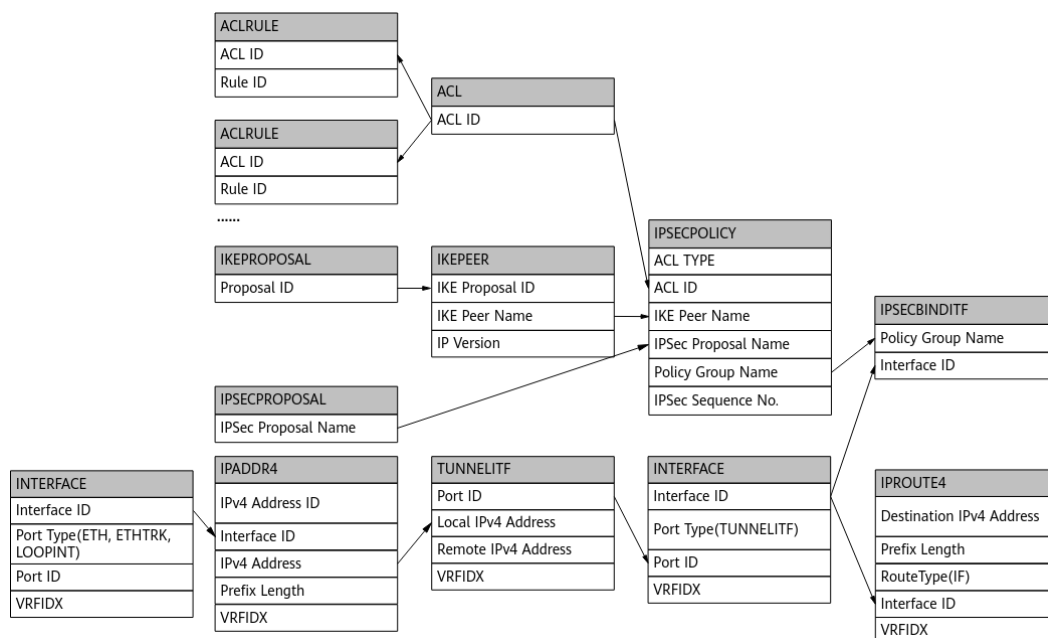
| Parameter Name               | Parameter ID                                             | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IPsec Replay Alarm Switch    | <b>IPGUARD.IPSECREPLAYCHKSW (5G gNodeB, LTE eNodeB)</b>  | It is recommended that this parameter be set to <b>ENABLE</b> to enable IPsec replay alarm detection. In this case, the anti-replay function must also be enabled, that is, the <b>IPSECPOLICY.REPLAYWND (LTE eNodeB, 5G gNodeB)</b> parameter cannot be set to <b>WND_DISABLE</b> .<br><br>If the IPsec replay alarm detection function is disabled, the administrator cannot receive an alarm indicating that the base station is under an IPsec replay attack. |
| IPsec Replay Alarm Threshold | <b>IPGUARD.IPSECREPLAYALMTHD (LTE eNodeB, 5G gNodeB)</b> | Set this parameter based on the network plan. If the number of IPsec replay attack packets reaches this threshold, ALM-25950 Base Station Being Attacked is reported with <b>Specific Problem</b> being <b>IPsec Replay</b> .                                                                                                                                                                                                                                     |

**4.4.2.2.2 Tunnel-Interface-based Configuration Mode**

In tunnel-interface-based IPv4 IPsec configuration mode, a local IP address can be specified or dynamically assigned through DHCP.

## Specified IP Address

**Figure 4-39** Configuration principles of tunnel-interface-based IPv4 IPsec (with a specified IP address)



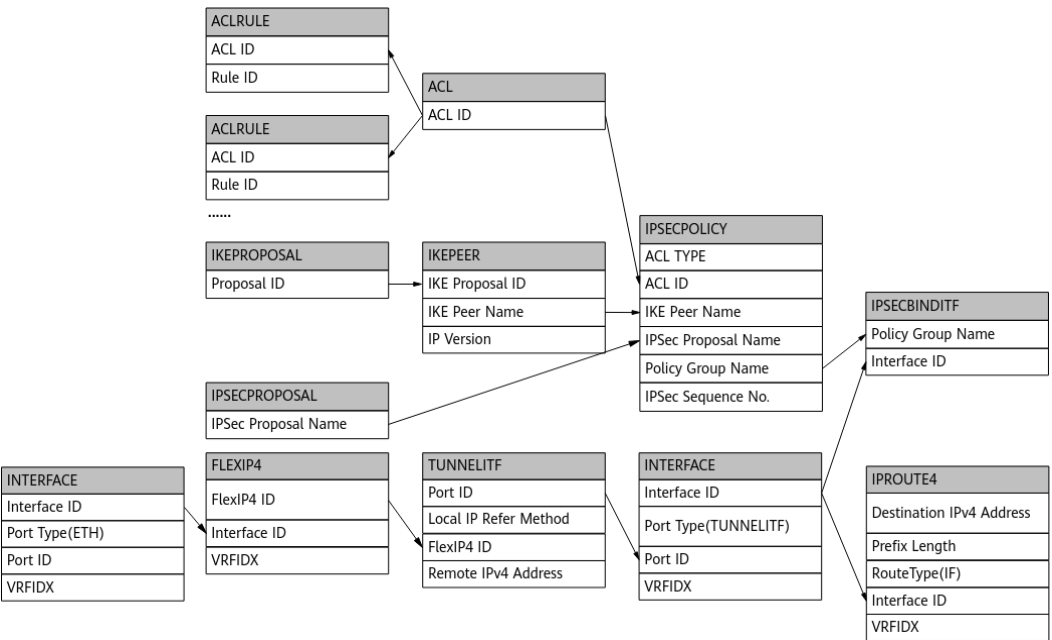
As shown in [Figure 4-39](#), the binding between the **TUNNELITF** and **INTERFACE** MOs and the binding between the **IPROUTE4** and **INTERFACE** MOs are added for in tunnel-interface-based IPv4 IPsec configuration mode with a specified IP address. Other binding relationships are the same as those for Ethernet-interface-based IPv4 IPsec configuration mode.

The functions of the configuration models are as follows:

- The **IKEPROPOSAL**, **IKEPEER**, **IPSECPROPOSAL**, and **IPSECPOLICY** MOs define information necessary for security negotiation between the base station and peer security equipment.
- The **ACL** and **ACLRULE** MOs define ACL rules.
- The **IPROUTE4** MO defines a static IPv4 route.
- The **TUNNELITF** and **INTERFACE** MOs define IPsec tunnel interfaces.

## IP Address Dynamically Assigned Through DHCP

**Figure 4-40** Configuration principles of tunnel-interface-based IPv4 IPsec (with an IP address dynamically assigned through DHCP)



As shown in [Figure 4-40](#), when a local IP address need to be dynamically assigned through DHCP in tunnel-interface-based IPv4 IPsec configuration mode, binding between the **FLEXIP4** and **INTERFACE** MOs and binding between the **TUNNELITF** and **FLEXIP4** MOs are added. Other binding relationships are the same as those for tunnel-interface-based IPv4 IPsec configuration mode with a specified IP address.

The functions of the configuration models are as follows:

- If the **FLEXIP4** MO references the **INTERFACE** MO, the DHCP function is enabled on the **INTERFACE** MO to obtain dynamic IPv4 addresses.
- If the **TUNNELITF** MO references the **FLEXIP4** MO, the local IP address of the IPsec tunnel is set to the dynamic IPv4 address obtained from the **INTERFACE** MO referenced by the **FLEXIP4** MO.
- The functions of other MOs are the same as those in the "Specified IP Address" section.

Configure the **IPGUARD**, **IKEPROPOSAL**, and **IPSECPROPOSAL** MOs. For details, see [4.4.2.2.1 Ethernet-Interface-based Configuration Mode](#).

[Table 4-21](#) lists the data to be prepared for basic IKE configurations. Specifically, the **IPSec Config Mode** parameter should be set to **TUNNELITF\_BASED**.

**Table 4-21** Parameters that must be set for basic IKE configurations

| Parameter Name    | Parameter ID                                              | Setting Notes                                  |
|-------------------|-----------------------------------------------------------|------------------------------------------------|
| IPSec Config Mode | <b><i>IKECFG.IPSECCFGMODE (LTE eNodeB, 5G gNodeB)</i></b> | Set this parameter to <b>TUNNELITF_BASED</b> . |

Configure an **IKEPEER** MO with **IP Version** set to **IPV4\_TUNNELITF**. [Table 4-22](#) describes the key parameters in this MO.

**Table 4-22** Data to be prepared for an IKE peer

| Parameter Name  | Parameter ID                                             | Setting Notes                                                                                                                                                                                                                                                                                                                                      |
|-----------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IKE Peer Name   | <b><i>IKEPEER.PEERNAME (5G gNodeB, LTE eNodeB)</i></b>   | Set this parameter based on the network plan. This parameter specifies the name that is used to represent this MO in the base station. The value of this parameter is case-insensitive.                                                                                                                                                            |
| IKE Proposal ID | <b><i>IKEPEER.PROPID (5G gNodeB, LTE eNodeB)</i></b>     | Set this parameter based on the network plan. This parameter specifies the ID of the IKE proposal used by the base station. This parameter is used to associate the <b>IKEPEER</b> MO with the <b>IKEPROPOSAL.PROPID</b> parameter. Before configuring an <b>IKEPEER</b> MO, ensure that the associated <b>IKEPROPOSAL</b> MO has been configured. |
| Version         | <b><i>IKEPEER.IKEVERSION (5G gNodeB, LTE eNodeB)</i></b> | Set this parameter based on the network plan.<br><br>You can run <b>SET SECPLY</b> to adjust the configuration security level to prohibit the parameter from being set to <b>IKE_V1</b> .                                                                                                                                                          |
| Exchange Mode   | <b><i>IKEPEER.EXCHMODE (LTE eNodeB, 5G gNodeB)</i></b>   | This parameter needs to be set only when <b>IKEPEER.IKEVERSION (5G gNodeB, LTE eNodeB)</b> is set to <b>IKE_V1</b> . Set this parameter based on the network plan.                                                                                                                                                                                 |
| IP Version      | <b><i>IKEPEER.IPVERSION (5G gNodeB, LTE eNodeB)</i></b>  | Set this parameter to <b>IPV4_TUNNELITF</b> .                                                                                                                                                                                                                                                                                                      |

| Parameter Name              | Parameter ID                                          | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Local ID Type               | <b>IKEPEER.IDTYPE (5G gNodeB, LTE eNodeB)</b>         | Set this parameter based on the network plan. This parameter specifies the authentication mode used when the base station performs IKE negotiation with the peer.                                                                                                                                                                                                                                                                                                                     |
| Remote Name                 | <b>IKEPEER.REMOTENAME (5G gNodeB, LTE eNodeB)</b>     | If this parameter is specified and the <b>IKEPROPOSAL.AUTHMETH (5G gNodeB, LTE eNodeB)</b> parameter is set to <b>IKE_CERT_SIG</b> , the base station will verify the peer ID during IKE negotiation with the SeGW. If this parameter is not specified, the base station will not verify the peer ID.                                                                                                                                                                                 |
| Pre-shared Key              | <b>IKEPEER.PKEY (5G gNodeB, LTE eNodeB)</b>           | If the <b>IKEPROPOSAL.AUTHMETH (5G gNodeB, LTE eNodeB)</b> parameter is set to <b>PRE_SHARED_KEY</b> , set this parameter based on the network plan.                                                                                                                                                                                                                                                                                                                                  |
| DPD Mode                    | <b>IKEPEER.DPD (5G gNodeB, LTE eNodeB)</b>            | If dead peer detection (DPD) is used to check the status of the IKE SA that consists of the local base station and the peer, these parameters must be set based on the network plan.<br><br>DPD is recommended because it occupies a relatively small bandwidth. It is also recommended that the values of these parameters be the same at two ends of the IKE SA. If the values are different, the DPD configuration with a shorter timer length takes effect before the other does. |
| DPD Idle Time               | <b>IKEPEER.DPDIDLETIME (5G gNodeB, LTE eNodeB)</b>    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| DPD Retransmission Interval | <b>IKEPEER.DPDRETRI (5G gNodeB, LTE eNodeB)</b>       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| DPD Retransmission Count    | <b>IKEPEER.DPDRETRN (5G gNodeB, LTE eNodeB)</b>       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Redundancy Flag             | <b>IKEPEER.REDUNDANCYFLAG (5G gNodeB, LTE eNodeB)</b> | Set this parameter based on the network plan.<br><br>If this parameter is set to <b>NONE</b> , IPsec redundancy among multiple SeGWs is disabled.<br><br>If the <b>TUNNELITF</b> MO associated with the <b>INTERFACE</b> MO bound to an <b>IPSECBINDITF</b> MO references an <b>FLEXIP4</b> MO, <b>IKEPEER.REDUNDANCYFLAG (LTE eNodeB, 5G gNodeB)</b> in the <b>IPSECPOLICY</b> MO referenced by the <b>IPSECBINDITF</b> MO cannot be set to <b>MASTER</b> .                          |

| Parameter Name             | Parameter ID                                                  | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Certificate Source         | <b>IKEPEER.CERTSOUR<br/>CE (5G gNodeB, LTE<br/>eNodeB)</b>    | Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                                                                                                                                             |
| Certificate File Name      | <b>IKEPEER.CERTNAME<br/>(5G gNodeB, LTE<br/>eNodeB)</b>       | This parameter must be set when the <b>IKEPEER.CERTSOURCE (5G gNodeB, LTE eNodeB)</b> parameter is set to <b>CERTMK</b> . Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                   |
| IPSec PreFragment Switch   | <b>IKEPEER.IPSECPREFR<br/>GSW (5G gNodeB,<br/>LTE eNodeB)</b> | If the network plan requires that SeGW performance be preferentially guaranteed, it is recommended that this parameter be set to <b>ON</b> . If the network plan requires that the CN performance and transmission efficiency be preferentially guaranteed, the default value <b>OFF</b> is recommended.                                                                                                                                                  |
| Trust Group ID             | <b>IKEPEER.TRUSTGRO<br/>UPID (LTE eNodeB,<br/>5G gNodeB)</b>  | This parameter specifies the ID of a trust group. When certificate-based authentication is used for link negotiation or establishment, the trust certificates in the trust group configured in the <b>TRUSTGROUP MO</b> are used to authenticate the peer end. If this parameter is set to <b>65535</b> , no trust group is associated, and all loaded trust certificates in the system are used to authenticate the device certificates of the peer end. |
| IKEv2 Fragmentation Switch | <b>IKEPEER.IKEV2FRGS<br/>W (5G gNodeB, LTE<br/>eNodeB)</b>    | This parameter specifies whether to enable IKEv2 fragmentation negotiation. For details about this function, see RFC 7383.<br><br>Whether IKEv2 fragmentation takes effect depends on the negotiation result, which can be viewed in the <b>DSP IKESA</b> command output.                                                                                                                                                                                 |
| IKEv2 Fragmentation MTU    | <b>IKEPEER.IKEV2FRGM<br/>TU (LTE eNodeB, 5G<br/>gNodeB)</b>   | This parameter specifies the maximum transmission unit of IKEv2 fragments for the IKE peer.                                                                                                                                                                                                                                                                                                                                                               |

| Parameter Name | Parameter ID                                         | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OCSP Switch    | <b><i>IKEPEER.OCSPSW (LTE eNodeB, 5G gNodeB)</i></b> | <p>This parameter specifies whether the local end uses the IKE OCSP (for details, see RFC 4806) to check the certificate revocation status of the peer end. When this switch is turned on, IKE messages sent by the local end contain OCSP requests, requiring the IKE messages sent by the peer end to contain OCSP responses. This switch applies only to IKEv2 and takes effect only when the <b><i>IKEPROPOSAL.AUTHMETH (LTE eNodeB, 5G gNodeB)</i></b> parameter is set to <b>IKE_CERT_SIG</b>. The effective status of this switch can be queried by running the <b>DSP IKESA</b> command.</p> <p>You are advised to check whether the peer SeGW supports IKE OCSP, and set this parameter to <b>ON</b> if the peer SeGW supports IKE OCSP and retain the default value <b>OFF</b> if it does not. Setting this parameter to <b>ON</b> while the peer SeGW does not support IKE OCSP brings IKE disconnection risks.</p> |

| Parameter Name   | Parameter ID                                        | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CRL using policy | <b>IKEPEER.CRLPOLICY</b><br>(LTE eNodeB, 5G gNodeB) | <p>This parameter specifies the CRL using policy for the base station in IKE scenarios.</p> <ul style="list-style-type: none"><li>• When this parameter is set to <b>GLOBAL</b>, the CRL using policy for the base station in IKE scenarios is the same as that specified in the <b>SET CRLPOLICY</b> command.</li><li>• When this parameter is set to <b>ALARM</b>, the base station in IKE scenarios reports ALM-26834 Peer Certificate Invalid if the CRL- or OCSP-based check result indicates that the peer certificate has been revoked.</li><li>• When this parameter is set to <b>DISCONNECT</b>, the base station in IKE scenarios terminates link negotiation and reports ALM-26834 Peer Certificate Invalid if the CRL- or OCSP-based check result indicates that the peer certificate has been revoked; the base station in IKE scenarios only terminates link negotiation if the check result indicates that the CRL has not taken effect or has expired.</li></ul> |

Configure **ACL** MOs to manage **ACLRULE** MOs. [Table 4-23](#) describes the key parameters.

**Table 4-23** Data preparation for ACL

| Parameter Name | Parameter ID                             | Setting Notes                                                                                                                                                                           |
|----------------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ACL ID         | <b>ACL.ACLID</b> (5G gNodeB, LTE eNodeB) | Set this parameter based on the network plan. If an ACL is referenced by an IPsec policy or packet filtering, the <b>ACL</b> MO must be associated with at least one <b>ACLRULE</b> MO. |

Configure IPv4 ACL rules. Add an **ACLRULE** MO to define data flows to be protected. The configurations of the **ACLRULE** MOs under the same **ACL** MO must be different. [Table 4-24](#) describes the key parameters.



**Table 4-24** Data preparation for ACLRULE

| Parameter Name             | Parameter ID                                              | Setting Notes                                                                                                                 |
|----------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| ACL ID                     | <b>ACLRULE.ACLID</b> ( <i>LTE eNodeB, 5G gNodeB</i> )     | This parameter specifies the ID of the <b>ACL</b> to which the <b>ACLRULE</b> MO belongs.                                     |
| Rule ID                    | <b>ACLRULE.RULEID</b> ( <i>LTE eNodeB, 5G gNodeB</i> )    | Set this parameter based on the network plan. In one ACL, each ACL rule must have a unique ID.                                |
| Action                     | <b>ACLRULE.ACTION</b> ( <i>LTE eNodeB, 5G gNodeB</i> )    | Set this parameter to <b>PERMIT</b> . Do not set it to <b>DENY</b> .                                                          |
| Protocol Type              | <b>ACLRULE.PT</b> ( <i>LTE eNodeB, 5G gNodeB</i> )        | Set this parameter based on the network plan. This parameter specifies the type of the protocol corresponding to an ACL rule. |
| Source IP Address          | <b>ACLRULE.SIP</b> ( <i>LTE eNodeB, 5G gNodeB</i> )       | Set this parameter based on the network plan.                                                                                 |
| Destination IP Address     | <b>ACLRULE.DIP</b> ( <i>LTE eNodeB, 5G gNodeB</i> )       | Set this parameter based on the network plan.                                                                                 |
| Source Wildcard            | <b>ACLRULE.SWC</b> ( <i>LTE eNodeB, 5G gNodeB</i> )       | Set this parameter based on the network plan.                                                                                 |
| Destination Wildcard       | <b>ACLRULE.DWC</b> ( <i>LTE eNodeB, 5G gNodeB</i> )       |                                                                                                                               |
| Support Forcible Execution | <b>ACLRULE.FORCEEXEC</b> ( <i>5G gNodeB, LTE eNodeB</i> ) | Set this parameter based on the network plan.                                                                                 |

Configure IPv4 IPsec policies. Add an **IPSECPOLICY** MO with **ACL Type** set to **IPV4** to configure the range of data to be protected and the protection policy. [Table 4-25](#) describes the key parameters.

**Table 4-25** Data preparation for IPSECPOLICY

| Parameter Name          | Parameter ID                                                   | Setting Notes                                                                                                                                                                                                                                                                                                                 |
|-------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Policy Group Name       | <b>IPSECPOLICY.SPGN</b><br><i>(5G gNodeB, LTE eNodeB)</i>      | Set this parameter based on the network plan. This parameter specifies the name of an IPsec policy group. The value of this parameter is case-insensitive. An IPsec policy group consists of multiple IPsec policies. Each IPsec policy is configured with an IPsec proposal.                                                 |
| IPSec Sequence No.      | <b>IPSECPOLICY.SPSN</b><br><i>(5G gNodeB, LTE eNodeB)</i>      | Set this parameter based on the network plan. Only one IPsec policy can be configured in an IPv4 security policy group.                                                                                                                                                                                                       |
| ACL Type                | <b>IPSECPOLICY.ACLTYP</b><br><i>E (5G gNodeB, LTE eNodeB)</i>  | Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                 |
| ACL ID                  | <b>IPSECPOLICY.ACLID</b><br><i>(5G gNodeB, LTE eNodeB)</i>     | Set this parameter based on the network plan. This parameter can be set to an IPv4 or IPv6 ACL ID. The ACL specified by this parameter must have been configured.                                                                                                                                                             |
| IPSec Proposal Name     | <b>IPSECPOLICY.PROPN</b><br><i>AME (5G gNodeB, LTE eNodeB)</i> | Set this parameter based on the network plan. Ensure that the name is correct when this parameter is referenced. An incorrect setting of this parameter will result in implementation failures.                                                                                                                               |
| IKE Peer Name           | <b>IPSECPOLICY.PEERNAME</b><br><i>(5G gNodeB, LTE eNodeB)</i>  | Set this parameter based on the network plan. The IKE peer specified by this parameter must have been configured.                                                                                                                                                                                                             |
| Perfect Forward Secrecy | <b>IPSECPOLICY.PFS</b> <i>(5G gNodeB, LTE eNodeB)</i>          | Set this parameter based on the network plan.<br><br>It is recommended that the PFS_GROUP15, PFS_GROUP19, PFS_GROUP20, or PFS_GROUP31 algorithm be used.<br><br>You can run <b>SET SECPLY</b> to adjust the configuration security level to prohibit the parameter from being set to <b>PFS_GROUP1</b> or <b>PFS_GROUP2</b> . |
| SA Duration Mode        | <b>IPSECPOLICY.LTCFG</b><br><i>(5G gNodeB, LTE eNodeB)</i>     | Set this parameter based on the network plan. The value <b>GLOBAL</b> is recommended.                                                                                                                                                                                                                                         |

| Parameter Name         | Parameter ID                                   | Setting Notes                                                                                                                                                           |
|------------------------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lifetime Based On Time | <b>IPSECPOLICY.LTS (5G gNodeB, LTE eNodeB)</b> | Set this parameter based on the network plan. This parameter is valid only when the <b>IPSECPOLICY.LTCFG (5G gNodeB, LTE eNodeB)</b> parameter is set to <b>LOCAL</b> . |

Configure a **FLEXIP4** MO. [Table 4-26](#) describes the key parameters.

**Table 4-26** Data preparation for FLEXIP4

| Parameter Name           | Parameter ID                                     | Setting Notes                                                                                                                                                                      |
|--------------------------|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Flexible IPv4 Address ID | <b>FLEXIP4.FLEXIP4ID (LTE eNodeB, 5G gNodeB)</b> | Set this parameter to the flexible IPv4 address ID of the interface.                                                                                                               |
| Interface ID             | <b>FLEXIP4.ITFID (LTE eNodeB, 5G gNodeB)</b>     | Set this parameter to the planned number of the interface whose local IP address needs to be dynamically assigned by DHCP. This interface must be a normal interface on the panel. |
| VRF Index                | <b>FLEXIP4.VRFIDX (LTE eNodeB, 5G gNodeB)</b>    | Set this parameter to the planned VRFID. The VRFID cannot be set to <b>0</b> .                                                                                                     |

Configure a **TUNNELITF** MO to create an IPsec tunnel interface. [Table 4-27](#) describes the key parameters.

**Table 4-27** Data preparation for TUNNELITF

| Parameter Name            | Parameter ID                                               | Setting Notes                                                                                                                                                                                      |
|---------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mode                      | <b>TUNNELITF.MODE (5G gNodeB, LTE eNodeB)</b>              | Set this parameter to <b>IPSEC_IPV4</b> .                                                                                                                                                          |
| Local IP Reference Method | <b>TUNNELITF.LOCALIP REFMETHOD (LTE eNodeB, 5G gNodeB)</b> | Set this parameter based on the network plan. Set this parameter to <b>IPID</b> if the local IP address needs to be dynamically assigned by DHCP. Otherwise, set this parameter to <b>IPADDR</b> . |

| Parameter Name           | Parameter ID                                       | Setting Notes                                                                                                                                                                                                                               |
|--------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Flexible IPv4 Address ID | <b>TUNNELITF.FLEXIP4ID (LTE eNodeB, 5G gNodeB)</b> | This parameter is valid when the <b>TUNNELITF.LOCALIPREFMETHOD (LTE eNodeB, 5G gNodeB)</b> parameter is set to <b>IPID</b> . Set this parameter to the value of <b>FLEXIP4.FLEXIP4ID (LTE eNodeB, 5G gNodeB)</b> .                          |
| Local IPv4 Address       | <b>TUNNELITF.LOCALIP4 (5G gNodeB, LTE eNodeB)</b>  | Set this parameter based on the network plan. This parameter specifies the IPv4 address of a base station.                                                                                                                                  |
| Remote IPv4 Address      | <b>TUNNELITF.REMOTEIP4 (5G gNodeB, LTE eNodeB)</b> | Set this parameter based on the network plan.                                                                                                                                                                                               |
| VRF Index                | <b>TUNNELITF.VRFIDX (5G gNodeB, LTE eNodeB)</b>    | The value of this parameter must be different from that of <b>INTERFACE.VRFIDX (5G gNodeB, LTE eNodeB)</b> . This parameter is valid when the <b>TUNNELITF.LOCALIPREFMETHOD (LTE eNodeB, 5G gNodeB)</b> parameter is set to <b>IPADDR</b> . |

Configure an **INTERFACE** MO with **Port Type** set to **TUNNELITF** to create an IPv4 interface of the tunnel type. [Table 4-28](#) describes the key parameters.

**Table 4-28** Data preparation for INTERFACE

| Parameter Name | Parameter ID                                    | Setting Notes                                                                                                             |
|----------------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Port Type      | <b>INTERFACE.PT (5G gNodeB, LTE eNodeB)</b>     | Set this parameter to <b>TUNNELITF</b> .                                                                                  |
| VRF Index      | <b>INTERFACE.VRFIDX (5G gNodeB, LTE eNodeB)</b> | The value of this parameter must be different from that of the <b>TUNNELITF.VRFIDX (5G gNodeB, LTE eNodeB)</b> parameter. |

Configure an **IPROUTE4** or a **SRCIPROUTE4** MO. Set the next hop of plaintext data flows to the TUNNELITF interface generated in the previous step so that the plaintext data flows can enter the IPsec tunnel. [Table 4-29](#) and [Table 4-30](#) describe the key parameters.

**Table 4-29** Data preparation for IPRROUTE4

| Parameter Name | Parameter ID                                             | Setting Notes                                                                                            |
|----------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Route Type     | <b>IPROUTE4.RTTYPE</b><br><i>(LTE eNodeB, 5G gNodeB)</i> | Set this parameter to <b>IF</b> .                                                                        |
| Port Type      | <b>IPROUTE4.PT</b> <i>(LTE eNodeB, 5G gNodeB)</i>        | Set this parameter to <b>INTERFACE</b> .                                                                 |
| Port ID        | <b>IPROUTE4.PORTID</b><br><i>(LTE eNodeB, 5G gNodeB)</i> | Set this parameter to the number of the TUNNELITF interface generated in the previous step.              |
| VRF Index      | <b>IPROUTE4.VRFIDX</b><br><i>(LTE eNodeB, 5G gNodeB)</i> | Set this parameter to the value of the <b>INTERFACE.VRFIDX</b> <i>(5G gNodeB, LTE eNodeB)</i> parameter. |

**Table 4-30** Data preparation for SRCIPROUTE4

| Parameter Name    | Parameter ID                                                | Setting Notes                                                                                                                  |
|-------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| VRF Index         | <b>SRCIPROUTE4.VRFIDX</b> <i>(LTE eNodeB, 5G gNodeB)</i>    | Set this parameter to the value of the <b>INTERFACE.VRFIDX</b> <i>(5G gNodeB, LTE eNodeB)</i> parameter.                       |
| Source IP Address | <b>SRCIPROUTE4.SRCIP</b><br><i>(LTE eNodeB, 5G gNodeB)</i>  | This parameter specifies the source IPv4 address, which is the IP address of the service that needs to enter the IPsec tunnel. |
| Route Type        | <b>SRCIPROUTE4.RTTYPE</b><br><i>(LTE eNodeB, 5G gNodeB)</i> | Set this parameter to <b>IF</b> .                                                                                              |
| Port Type         | <b>SRCIPROUTE4.PT</b> <i>(LTE eNodeB, 5G gNodeB)</i>        | Set this parameter to <b>INTERFACE</b> .                                                                                       |
| Port ID           | <b>SRCIPROUTE4.PORTID</b> <i>(LTE eNodeB, 5G gNodeB)</i>    | Set this parameter to the number of the TUNNELITF IPv4 interface generated in the previous step.                               |

 NOTE

To enable the IPsec redundancy among multi SeGWs in this configuration mode,

- For IPv4 IPsec, a route with the corresponding **IKEPEER.REDUNDANCYFLAG (5G gNodeB, LTE eNodeB)** parameter set to **MASTER** needs to be configured as the active route by running the **ADD IPROUTE4** or **ADD SRCIPROUTE4** command, and a standby route with the corresponding **IKEPEER.REDUNDANCYFLAG (5G gNodeB, LTE eNodeB)** parameter set to **SLAVE** can be automatically generated. The standby route is automatically generated only when the standby IKE peer takes effect. The generated routes can be queried by running the **DSP IPROUTE4** or **DSP SRCIPROUTE4** command.
- For IPv6-over-IPv4 IPsec, the active route needs to be configured by running the **ADD IPROUTE6** or **ADD SRCIPROUTE6** command, and a standby route is automatically generated. All these routes can be queried by running the **DSP IPROUTE6** or **DSP SRCIPROUTE6** command.

Configure IPv4 IPsec policy group binding relationships. Add an **IPSECBINDITF** MO to bind an IPsec policy group to an IPv4 tunnel interface. [Table 4-31](#) describes the key parameters.

**Table 4-31** Data preparation for the binding between an IPsec policy and a port

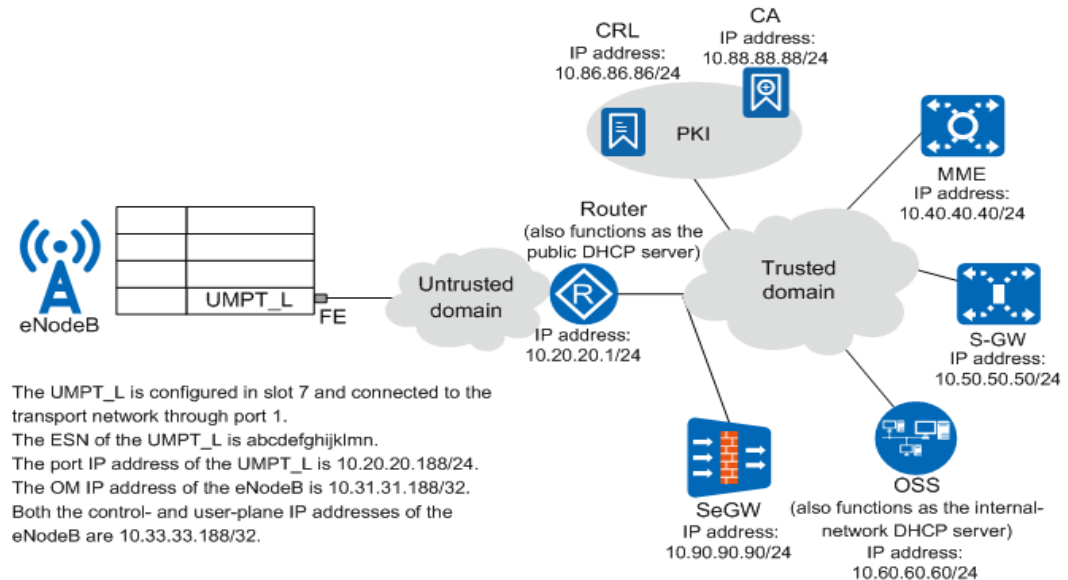
| Parameter Name          | Parameter ID                                               | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IPSec Bind Interface ID | <b>IPSECBINDITF.IPSECBINDITFID (5G gNodeB, LTE eNodeB)</b> | Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                                                                                                           |
| Policy Group Name       | <b>IPSECBINDITF.SPGN (5G gNodeB, LTE eNodeB)</b>           | Set this parameter based on the network plan. When an IPsec policy group is bound to a physical port, all policies in the group are applied to the port. This enables different SAs to protect different types of packets.<br><br>After an IPsec policy group is bound to an Ethernet port, no new IPsec policy can be added to the group and no modifications can be made to the existing IPsec policies in the group. |
| Interface ID            | <b>IPSECBINDITF.ITFID (5G gNodeB, LTE eNodeB)</b>          | Set this parameter based on the network plan. This parameter specifies the index of the interface whose port type is <b>TUNNELITF</b> .                                                                                                                                                                                                                                                                                 |

### 4.4.2.3 Deploying IPsec for an eGBTS/NodeB/eNodeB/gNodeB on a PKI-based Secure Network

On a PKI-based secure network, IPsec configurations are the same for eGBTSs using the UMPT\_G/UMDU\_G/MDUC\_G to provide a transmission port for connecting to the transport network, NodeBs, gNodeBs, and eNodeBs.

**Figure 4-41** shows an example of IPsec deployment for an eNodeB on a PKI-based secure network.

**Figure 4-41** Example of IPsec deployment for an eNodeB on a PKI-based secure network



In this networking scenario, the UMPT\_L provides IPsec protection for the following data flows:

- eNodeB signaling and service data flows
- eNodeB OM data flows
- Certificate management-related data flows between the eNodeB and CA server
- Data flows generated when the eNodeB obtains CRLs or certificate files from the CRL server

#### 4.4.2.3.1 Data Preparation

For details, see [4.4.2.2 Data Preparation](#).

#### 4.4.2.3.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

### Ethernet-Interface-based Configuration Mode

The MML configuration is as follows:

```
//Setting the IPsec configuration mode
SET IKECFG: IPSEC CFGMODE=ETHINTERFACE_BASED;
//Adding an IKE proposal
ADD IKEPROPOSAL: PROPID=10, ENCALG=AES128, AUTHALG=SHA256, AUTHMETH=IKE_CERT_SIG,
DHGRP=DH_GROUP14, PRFALG=HMAC_SHA256, DURATION=86400, REAUTHSW=ON, REAUTHLT=604800;
//Adding an IKE peer
ADD IKEPEER: PEERNAME="ike", PROPID=10, IKEVERSION=IKE_V2, IPVERSION=IPV4, IDTYPE=IPV4,
REMOTEIP="10.90.90.90", REMOTENAME="secgw", DPD=PERIODIC, DPDIDLETIME=10, DPDRETRI=5,
DPDRETRN=5, LOCALIP="10.20.20.188", REDUNDANCYFLAG=NONE, CERTSOURCE=APPCERT,
IPSECPREFRGSW=OFF, TRUSTGROUPID=65535, OCSPSW=OFF, CRLPOLICY=GLOBAL;
//Adding an ACL
ADD ACL: ACLID=3000, ACLDESC="for IPsec";
//Adding ACL rules to the ACL so that the UMPD encrypts or decrypts the following data flows that pass
through the transmission ports on the UMPD
//Setting ACL rule 1 (RULEID = 1) for eNodeB signaling and service data flows
ADD ACLRULE: ACLID=3000, RULEID=1, ACTION=PERMIT, PT=IP, SIP="10.33.33.188", SWC="0.0.0.0",
DIP="0.0.0.0", DWC="255.255.255.255", MDSCP=NO;
//Setting ACL rule 2 (RULEID = 2) for OM data flows and certificate management-related data flows
(including those generated during the interaction with the digital certificate and CRL database) of the
eNodeB when OM channels are protected by IPsec. If the OM channel does not need to be protected by
IPsec, the source IP address for certificate management-related data flows must be different from the OM
IP address. You are advised to configure a new internal-network IP address (for example, 10.45.45.45) as
the source IP address for certificate management-related data flows. Therefore, the SIP in the following ACL
rule needs to be changed to 10.45.45.45.
ADD ACLRULE: ACLID=3000, RULEID=2, ACTION=PERMIT, PT=IP, SIP="10.31.31.188", SWC="0.0.0.0",
DIP="0.0.0.0", DWC="255.255.255.255", MDSCP=NO;
//Adding an IPsec proposal
ADD IPSECPROPOSAL: PROPNAME="prop0", ENCAPMODE=TUNNEL, TRANMODE=ESP,
ESPENCALG=AES128, ESPAUTHALG=SHA256;
//Adding an IPsec policy
ADD IPSECPOLICY: SPGN="Policy0", SPSN=1, ACLTYPE=IPV4, ACLID=3000, PROPNAME="prop0",
PEERNAME="ike", LTCFG=LOCAL, LTS=3600, LTKB=0, REPLAYWND=WND_DISABLE, NARROWDOWNSW=ON;

• Scenario 1:
//When GTRANSPARA.TRANSCFGMODE is set to OLD
//Binding the IPsec policy to an outbound port
ADD IPSECBIND: SN=7, CN=0, SRN=0, SBT=BASE_BOARD, PT=ETH, PN=1, SPGN="Policy0";

• Scenario 2:
//When GTRANSPARA.TRANSCFGMODE is set to NEW
//Binding the IPsec policy to an outbound port
ADD IPSECBINDITF: IPSECBINDITFID=0, ITFID=0, SPGN="Policy0";

//(Optional) Setting basic IKE information
SET IKECFG: IKELNM="IKECFG1", DSCP=48, IPSEC SWITCHBACK=OFF;
//(Optional) Setting the IPsec replay alarm switch and threshold
SET IPGUARD: IPSECREPLAYCHKSW=ENABLE, IPSECREPLAYALMTHD=10;
```

## Tunnel-Interface-based Configuration Mode

The MML-based configuration is as follows: (The **INTERFACE.VRFID** (**5G gNodeB**, **LTE eNodeB**) parameter can only be set to **0** for inner VRFs of GSM/UMTS services. Therefore, this parameter must be set to **0** for inner VRFs if GSM/UMTS traffic needs to be transmitted through IPsec tunnels.)

```
//Setting the transmission configuration mode
SET GTRANSPARA: TRANSCFGMODE=NEW;
//Setting the IPsec configuration mode
SET IKECFG: IPSEC CFGMODE=TUNNELITF_BASED;
//Adding an IKE proposal
ADD IKEPROPOSAL: PROPID=1, ENCALG=AES128, AUTHALG=SHA256, AUTHMETH=IKE_CERT_SIG,
DHGRP=DH_GROUP14, PRFALG=HMAC_SHA256, REAUTHSW=OFF;
//Adding an IKE peer
ADD IKEPEER: PEERNAME="4in4_IPSEC11111", PROPID=1, IKEVERSION=IKE_V2,
IPVERSION=IPV4_TUNNELITF, IDTYPE=IPV4, REMOTENAME="secgw", DPD=PERIODIC,
REDUNDANCYFLAG=NONE, CERTSOURCE=APPCERT, IPSECPREFRGSW=OFF;
//Adding an ACL
ADD ACL: ACLID=3000;
//Setting ACL rule 1 (RULEID = 1) for eNodeB signaling and service data flows
```



```
ADD ACLRULE: ACLID=3000, RULEID=1, ACTION=PERMIT, PT=IP, SIP="10.33.33.188", SWC="0.0.0.0",
DIP="0.0.0.0", DWC="255.255.255.255", MDSCP=NO;
//Setting ACL rule 2 (RULEID = 2) for OM data flows and certificate management-related data flows
(including those generated during the interaction with the digital certificate and CRL database) of the
eNodeB when OM channels are protected by IPsec. If the OM channel does not need to be protected by
IPsec, the source IP address for certificate management-related data flows must be different from the OM
IP address. You are advised to configure a new internal-network IP address (for example, 10.45.45.45) as
the source IP address for certificate management-related data flows. Therefore, the SIP in the following ACL
rule needs to be changed to 10.45.45.45.
ADD ACLRULE: ACLID=3000, RULEID=2, ACTION=PERMIT, PT=IP, SIP="10.31.31.188", SWC="0.0.0.0",
DIP="0.0.0.0", DWC="255.255.255.255", MDSCP=NO;
//Adding an IPsec proposal
ADD IPSECPROPOSAL: PROPNAME="4in4_IPSEC11111", TRANMODE=ESP, ESPENCALG=AES128,
ESPAUTHALG=SHA256;
//Adding an IPsec policy
ADD IPSECPOLICY: SPGN="4in4_IPSEC11111", SPSN=1, ACLTYPE=IPV4, ACLID=3000,
PROPNAME="4in4_IPSEC11111", PEERNAME="4in4_IPSEC11111", PFS=DISABLE, LTCFG=LOCAL;
```

- Scenario 1: Dynamic IP addresses being used (**VRFIDX** set to **0** for the inner VRF and **2** for the outer VRF)

```
//Adding FLEXIP4 (The interface ID referenced by FLEXIP4 must be the ID of a normal interface on the
panel.)
ADD INTERFACE: ITFID=10, ITFTYPE=NORMAL, PT=ETH, PORTID=1, VRFIDX=2, IPV6SW=DISABLE;
ADD FLEXIP4: FLEXIP4ID=6, ITFID=10, VRFIDX=2;
//Adding a tunnel interface (with an IP address dynamically assigned by DHCP)
ADD TUNNELITF: PORTID=1, MODE=IPSEC_IPV4, LOCALIPREFMETHOD=IPID, FLEXIP4ID=6,
REMOTEIP4="10.90.90.90";
ADD INTERFACE: ITFID=11, ITFTYPE=NORMAL, PT=TUNNELITF, PORTID=1, VRFIDX=0,
IPV6SW=DISABLE;
//Binding the IPsec policy to an outbound interface
ADD IPSECBINDITF: IPSECBINDITFID=0, SPGN="4in4_IPSEC11111", ITFID=11, FORCEEXECUTE=YES;
//Configuring routes for the control-plane and user-plane IP addresses of the internal network. Routes
enter the IPsec tunnel through the following routes. The source-based route is used as an example.
ADD SRCIPROUTE4: SRCRTIDX=0, VRFIDX=0, SRCIP="10.33.33.188", RTTYPE=IF, PT=INTERFACE,
PORTID=11;
//If OM data is protected by IPsec, configure a route for the OM IP address to enter the IPsec tunnel.
If OM data is not protected by IPsec, no configuration is required. The source-based route is used as
an example.
ADD SRCIPROUTE4: SRCRTIDX=1, VRFIDX=0, SRCIP="10.31.31.188", RTTYPE=IF, PT=INTERFACE,
PORTID=11;
```

- Scenario 2: Specific IP addresses being used (**VRFIDX** set to **0** for the inner VRF and **1** for the outer VRF)

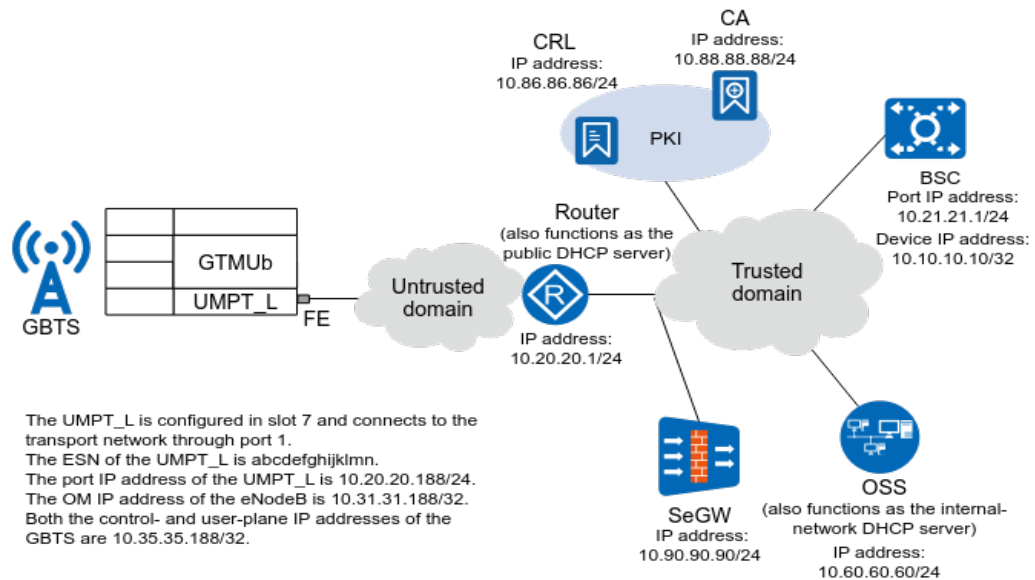
```
//The local end uses a fixed IP address. There is no need to run ADD FLEXIP4.
ADD INTERFACE: ITFID=10, ITFTYPE=NORMAL, PT=ETH, PORTID=1, VRFIDX=1, IPV6SW=DISABLE;
ADD IPADDR4: ITFID=10, IP="10.20.20.188", MASK="255.255.255.248", VRFIDX=1;
//Adding a tunnel interface (with a specified IP address)
ADD TUNNELITF: PORTID=2, MODE=IPSEC_IPV4, LOCALIPREFMETHOD=IPADDR, VRFIDX=1,
LOCALIP4="10.20.20.188", REMOTEIP4="10.90.90.90";
ADD INTERFACE: ITFID=12, ITFTYPE=NORMAL, PT=TUNNELITF, PORTID=2, VRFIDX=0,
IPV6SW=DISABLE;
//Binding the IPsec policy to an outbound interface
ADD IPSECBINDITF: IPSECBINDITFID=0, SPGN="4in4_IPSEC11111", ITFID=12, FORCEEXECUTE=YES;
//Configuring routes for the control-plane and user-plane IP addresses of the internal network. Routes
enter the IPsec tunnel through the following routes. The source-based route is used as an example.
ADD SRCIPROUTE4: SRCRTIDX=0, VRFIDX=0, SRCIP="10.33.33.188", RTTYPE=IF, PT=INTERFACE,
PORTID=12;
//If OM data is protected by IPsec, configure a route for the OM IP address to enter the IPsec tunnel.
If OM data is not protected by IPsec, no configuration is required. The source-based route is used as
an example.
ADD SRCIPROUTE4: SRCRTIDX=1, VRFIDX=0, SRCIP="10.31.31.188", RTTYPE=IF, PT=INTERFACE,
PORTID=12;
```

#### 4.4.2.4 Deploying IPsec for a GBTS/eGBTS on a PKI-based Secure Network

This section describes the IPsec configurations of a GBTS/eGBTS on a PKI-based secure network with the UMPT\_L providing a co-transmission port to connect to

the transport network. **Figure 4-42** shows an example of implementing IPsec on a GBTS configured with the GTMUc+UMPT\_L.

**Figure 4-42** Example of deployment for a GBTS using the GTMUc+UMPT\_L on a PKI-based secure network



In this networking scenario:

- The GTMUc and UMPT\_L communicate with each other through the backplane.
- The UMPT\_L only provides IPsec and forwards GBTS data.
- All IPsec data on the UMPT\_L is configured on the eNodeB.
- The UMPT\_L provides IPsec protection for the following data flows:
  - GBTS signaling and service data flows
  - eNodeB OM data flows (only if the OM channel needs to be protected by IPsec)
  - Certificate management-related data flows between the eNodeB and CA server
  - CRL- or certificate file-related data flows obtained by the eNodeB from the CRL server

#### 4.4.2.4.1 Data Preparation

For details, see [4.4.2.2 Data Preparation](#).

#### NOTE

The GBTS does not support IPsec ACL narrow down. Therefore, the **Narrow Down Switch** parameter in [4.4.2.2 Data Preparation](#) is invalid for the GBTS.

#### 4.4.2.4.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After

Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Most MML-based configurations are the same as those in [4.4.2.3.2 Using MML Commands](#). The differences are as follows:

- In Ethernet-interface-based configuration mode, only the ACL rule configuration is different.
- In tunnel-interface-based configuration mode, the ACL rule configuration is different, and routes from the GBTS control-plane IP address or user-plane IP address to the IPsec tunnel also need to be added.

The following only describes the differences.

The network shown in [Figure 4-42](#) is used as an example. The ACL rules to be added are as follows:

- Rule 1 (RULEID = 1) applies to GBTS signaling and service data flows.
- Rule 2 (RULEID = 2) applies to eNodeB OM data flows and certificate management-related data flows, including those generated during the interaction with the digital certificate and CRL database.

In this scenario, the MML command examples of adding ACL rules are as follows:

```
ADD ACLRULE: ACLID=3000, RULEID=1, PT=IP, SIP="10.35.35.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=2, PT=IP, SIP="10.31.31.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;
```

In tunnel-interface-based configuration mode, a route to the IPsec tunnel also needs to be added.

```
ADD SRCIPROUTE4: SRCRTIDX=0, VRFIDX=1, SRCIP="10.35.35.188", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=1, VRFIDX=1, SRCIP="10.31.31.188", RTTYPE=IF, PT=INTERFACE, PORTID=11;
```

If IPsec is not required for OM data flows, the MML command examples of adding ACL rules are as follows where the IP address 10.45.45.45 in Rule 2 (RULEID = 2) is the source IP address for certificate management-related data flows:

```
ADD ACLRULE: ACLID=3000, RULEID=1, PT=IP, SIP="10.35.35.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=2, PT=IP, SIP="10.45.45.45", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;
```

In tunnel-interface-based configuration mode, a route to the IPsec tunnel also needs to be added.

```
ADD SRCIPROUTE4: SRCRTIDX=0, VRFIDX=1, SRCIP="10.35.35.188", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=1, VRFIDX=1, SRCIP="10.45.45.45", RTTYPE=IF, PT=INTERFACE, PORTID=11;
```

#### 4.4.2.5 Deploying Co-IPsec for a Dual-Mode Base Station on a PKI-based Secure Network

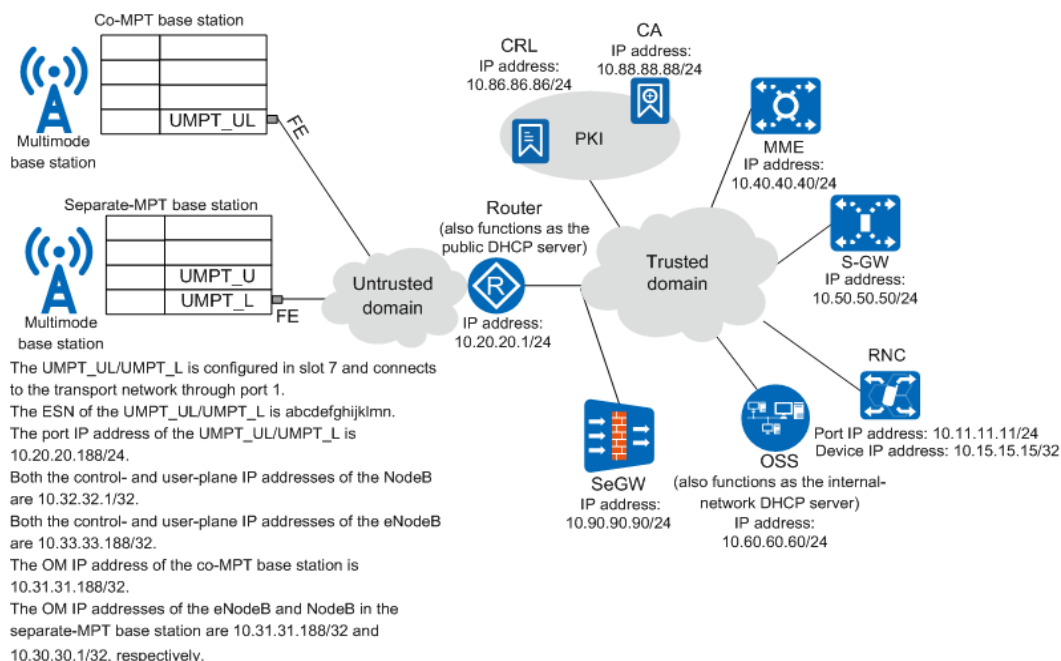
This section describes how to deploy co-IPsec for a dual-mode base station on a PKI-based secure network. The configuration procedures are similar for dual-mode

base stations. The only differences are the board configurations. [Table 4-32](#) lists typical board configurations for co-MPT and separate-MPT dual-mode base stations.

**Table 4-32** Typical board configurations for dual-mode base stations

| Base Station Type                                                                | Co-MPT  | Separate-MPT                                                                                                  |
|----------------------------------------------------------------------------------|---------|---------------------------------------------------------------------------------------------------------------|
| GL dual-mode base station                                                        | UMPT_GL | GSM: GTMUc/UMPT_G<br>LTE: UMPT_L (LTE provides a co-transmission port to connect to the transport network.)   |
| GU dual-mode base station                                                        | UMPT_GU | GSM: GTMUc/UMPT_G<br>UMTS: UMPT_U (UMTS provides a co-transmission port to connect to the transport network.) |
| UL dual-mode base station                                                        | UMPT_UL | UMTS: UMPT_U<br>LTE: UMPT_L (LTE provides a co-transmission port to connect to the transport network.)        |
| LN dual-mode base station<br>In this case, the BBU must be a BBU5900 or BBU3910. | UMPT_LN | LTE: UMPT_L<br>NR: UMPT_N (NR provides a co-transmission port to connect to the transport network.)           |

[Figure 4-43](#) shows an example of a UL dual-mode base station on a PKI-based secure network.

**Figure 4-43** Example of deploying co-IPsec on a UL dual-mode base station on a PKI-based secure network

On the co-MPT base station, the UMPT provides IPsec protection for the following data flows:

- Signaling and service data flows of the two single-mode base stations (NodeB/eNodeB signaling and service data flows in this example)
- OM data flows of the dual-mode base station (only if the OM channel needs to be protected by IPsec)
- Certificate management-related data flows between the dual-mode base station and CA server
- Data flows generated when the dual-mode base station obtains CRLs from the CRL server

#### NOTE

In a co-MPT base station, the method of deploying a UMDU/MDUC is the same as that of deploying a UMPT. MDUCs apply only to GU dual-mode.

In separate-MPT mode, a GU dual-mode base station does not support tunnel-interface-based configuration mode.

On the separate-MPT dual-mode base station:

- Single-mode main control boards communicate with each other through the backplane. In this example, the UMPT\_U and UMPT\_L communicate with each other through the backplane.
- A more recent RAT provides IPsec and forwards data for the other mode.
- A more recent RAT provides a transmission board to connect to the transport network and provides IPsec protection for the following data flows:
  - Signaling and service data flows of the two single-mode base stations (NodeB/eNodeB signaling and service data flows in this example)

- OM data flows of the two single-mode base stations (only if the OM channel needs to be protected by IPsec)
- Certificate management-related data flows between a more recent mode and the CA server
- Data flows generated when a more recent mode obtains CRLs from the CRL server

Pay attention to the following when an IPsec network with a single-mode base station is reconstructed into a co-IPsec network with a dual-mode base station. The following uses an IPsec network with an eNodeB reconstructed into a co-IPsec network with a UL dual-mode base station as an example:

- If an IPsec network with an eNodeB using the UMPT\_L is reconstructed into a co-IPsec network with a UL co-MPT base station using the UMPT\_UL:
  - Security reconfiguration is not required if an existing ACL rule applies to NodeB signaling and service data flows, indicating that an ACL rule in Any to Any mode with the **ACLRULE.ACTION (5G gNodeB, LTE eNodeB)** parameter set to **PERMIT** has been configured.
  - Adding an ACL rule with the **ACLRULE.ACTION (5G gNodeB, LTE eNodeB)** parameter set to **PERMIT** for NodeB signaling and service data flows is required if existing ACL rules do not apply to these data flows.
  - In tunnel-interface-based configuration mode, additional routes configured with the signaling and service IP addresses of the NodeB to the IPsec tunnel need to be added.
- If an IPsec network with an eNodeB using the UMPT\_L is reconstructed into a co-IPsec network with a UL separate-MPT base station using the UMPT\_U +UMPT\_L:
  - If an existing ACL rule applies to NodeB signaling, service, and OM data flows, indicating that an ACL rule in Any to Any mode with the **ACLRULE.ACTION (5G gNodeB, LTE eNodeB)** parameter set to **PERMIT** has been configured, the following rules apply:
    - In Ethernet-interface-based configuration mode:
      - Security reconfiguration is not required when the OM channel needs to be protected by IPsec.
      - Adding an ACL rule with the **ACLRULE.ACTION (5G gNodeB, LTE eNodeB)** parameter set to **DENY** for NodeB OM data flows is required when the OM channel does not need to be protected by IPsec. The ACL ID for this ACL rule must be smaller than that for any ACL rule in Any to Any mode.
    - In tunnel-interface-based configuration mode:
      - If OM data is protected by IPsec, routes configured with the signaling, service, and OM IP addresses of the NodeB to the IPsec tunnel need to be added.
      - If OM data is not protected by IPsec, routes configured with the signaling and service IP addresses of the NodeB to the IPsec tunnel need to be added.
  - If existing ACL rules do not apply to NodeB signaling, service, and OM data flows, adding an ACL rule with the **ACLRULE.ACTION (5G gNodeB, LTE eNodeB)** parameter set to **PERMIT** for these data flows is required.

In tunnel-interface-based configuration mode, routes configured with the signaling, service, and OM IP addresses of the NodeB to the IPsec tunnel need to be added.

#### 4.4.2.5.1 Data Preparation

For details, see [4.4.2.2 Data Preparation](#).

#### 4.4.2.5.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Most MML-based configurations are the same as those described in [4.4.2.3.2 Using MML Commands](#). The following only describes the differences.

### Co-MPT Base Station

For a co-MPT dual-mode base station, configure ACL rules as follows:

- Configure an ACL rule for each mode to protect signaling and service data flows.
- Configure an ACL rule for the dual-mode base station to protect OM data flows and certificate management-related data flows, including those generated during the interaction with the digital certificate and CRL database. If the OM channel does not need to be protected by IPsec, the source IP address of this ACL rule must be different from the OM IP address. It is required that a new internal-network IP address be configured as the source IP address for certificate management-related data flows.

The network shown in [4.4.2.5 Deploying Co-IPsec for a Dual-Mode Base Station on a PKI-based Secure Network](#) is used as an example. The ACL rules to be added are as follows:

- Rule 1 (RULEID = 1) applies to NodeB signaling and service data flows.
- Rule 2 (RULEID = 2) applies to eNodeB signaling and service data flows.
- Rule 3 (RULEID = 3) applies to MBTS OM data flows and certificate management-related data flows, including those generated during the interaction with the digital certificate and CRL database.

In this scenario, the MML command examples of adding ACL rules are as follows:

```
ADD ACLRULE: ACLID=3000, RULEID=1, PT=IP, SIP="10.32.32.1", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=2, PT=IP, SIP="10.33.33.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=3, PT=IP, SIP="10.31.31.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;
```

In tunnel-interface-based configuration mode, a route to the IPsec tunnel also needs to be added.

```
ADD SRCIPROUTE4: SRCRTIDX=0,VRFIDX=1,SRCIP="10.32.32.1",RTTYPE=IF,PT=INTERFACE,PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=1,VRFIDX=1,SRCIP="10.33.33.188",RTTYPE=IF,PT=INTERFACE,PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=2,VRFIDX=1,SRCIP="10.31.31.188",RTTYPE=IF,PT=INTERFACE,PORTID=11;
```

If IPsec is not required for OM data flows, the MML command examples of adding ACL rules are as follows where the IP address 10.45.45.45 in Rule 3 (RULEID = 3) is the source IP address for certificate management-related data flows:

```
ADD ACLRULE: ACLID=3000, RULEID=1, PT=IP, SIP="10.32.32.1", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=2, PT=IP, SIP="10.33.33.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=3, PT=IP, SIP="10.45.45.45", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;
```

In tunnel-interface-based configuration mode, a route to the IPsec tunnel also needs to be added. The following configurations are needed:

```
ADD SRCIPROUTE4: SRCRTIDX=0,VRFIDX=1,SRCIP="10.32.32.1",RTTYPE=IF,PT=INTERFACE,PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=1,VRFIDX=1,SRCIP="10.33.33.188",RTTYPE=IF,PT=INTERFACE,PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=2,VRFIDX=1,SRCIP="10.45.45.45",RTTYPE=IF,PT=INTERFACE,PORTID=11;
```

## Separate-MPT Base Station

For a separate-MPT dual-mode base station, configure ACL rules as follows:

- Configure an ACL rule for each mode to protect signaling and service data flows.
- Configure an ACL rule for the less recent RAT to protect the OM channel. Do not configure this ACL rule if the OM channel does not need to be protected by IPsec.
- Configure an ACL rule for the more recent RAT to protect OM data flows and certificate management-related data flows, including those generated during the interaction with the digital certificate and CRL database. If the OM channel does not need to be protected by IPsec, the source IP address of this ACL rule must be different from the OM IP address. It is required that a new internal-network IP address be configured as the source IP address for certificate management-related data flows.

The network shown in [4.4.2.5 Deploying Co-IPsec for a Dual-Mode Base Station on a PKI-based Secure Network](#) is used as an example. The ACL rules to be added are as follows:

- Rule 1 (RULEID = 1) applies to NodeB signaling and service data flows.
- Rule 2 (RULEID = 2) applies to eNodeB signaling and service data flows.
- Rule 3 (RULEID = 3) applies to NodeB OM data flows.
- Rule 4 (RULEID = 4) applies to eNodeB OM data flows and certificate management-related data flows, including those generated during the interaction with the digital certificate and CRL database.

In this scenario, the MML command examples of adding ACL rules are as follows:

```
ADD ACLRULE: ACLID=3000, RULEID=1, PT=IP, SIP="10.32.32.1", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=2, PT=IP, SIP="10.33.33.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;
```



```
ADD ACLRULE: ACLID=3000, RULEID=3, PT=IP, SIP="10.30.30.1", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=4, PT=IP, SIP="10.31.31.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;
```

In tunnel-interface-based configuration mode, a route to the IPsec tunnel also needs to be added.

```
ADD SRCIPROUTE4: SRCRTIDX=0, VRFID=1, SRCIP="10.32.32.1", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=1, VRFID=1, SRCIP="10.33.33.188", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=2, VRFID=1, SRCIP="10.30.30.1", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=3, VRFID=1, SRCIP="10.31.31.188", RTTYPE=IF, PT=INTERFACE, PORTID=11;
```

If IPsec is not required for OM data flows, the MML command examples of adding ACL rules are as follows where the IP address 10.45.45.45 in Rule 3 (RULEID = 3) is the source IP address for certificate management-related data flows:

```
ADD ACLRULE: ACLID=3000, RULEID=1, PT=IP, SIP="10.32.32.1", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=2, PT=IP, SIP="10.33.33.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=3, PT=IP, SIP="10.45.45.45", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;
```

In tunnel-interface-based configuration mode, a route to the IPsec tunnel also needs to be added. The following configurations are needed:

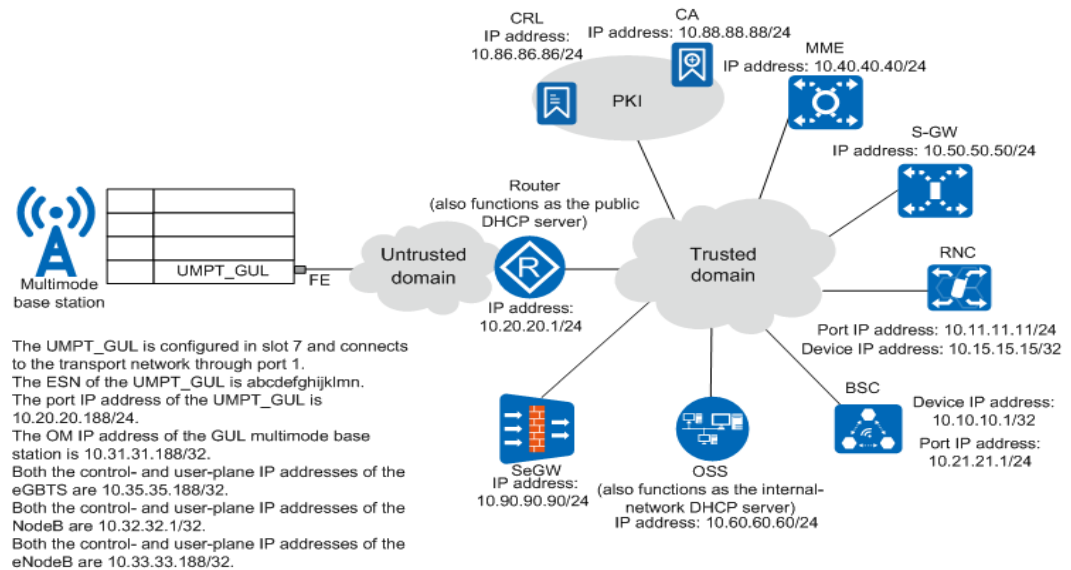
```
ADD SRCIPROUTE4: SRCRTIDX=0, VRFID=1, SRCIP="10.32.32.1", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=1, VRFID=1, SRCIP="10.33.33.188", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=2, VRFID=1, SRCIP="10.45.45.45", RTTYPE=IF, PT=INTERFACE, PORTID=11;
```

#### 4.4.2.6 Deploying Co-IPsec for a Triple-Mode Base Station on a PKI-based Secure Network

This section describes how to deploy co-IPsec for a triple-mode base station on a PKI-based secure network. For typical board configurations of co-MPT and separate-MPT base stations, see *BBU Hardware Description in 3900 & 5900 Series Base Station Product Documentation*. The following takes a GUL triple-mode base station as an example.

##### Co-MPT Base Station

**Figure 4-44** shows an example of a co-MPT GUL triple-mode base station on a PKI-based secure network.

**Figure 4-44** Example of deployment on a co-MPT GUL triple-mode base station on a PKI-based secure network

In this networking scenario, the UMPT provides IPsec protection for the following data flows:

- eGBTS/NodeB/eNodeB signaling and service data flows
- OM data flows of the GUL multimode base station (only if the OM channel needs to be protected by IPsec)
- Certificate management-related data flows between the GUL multimode base station and CA server
- Data flows generated when the GUL multimode base station obtains CRLs from the CRL server

If an IPsec network with an eNodeB using the UMPT\_L is reconstructed into a co-IPsec network with a GUL triple-mode base station using the UMPT\_GUL:

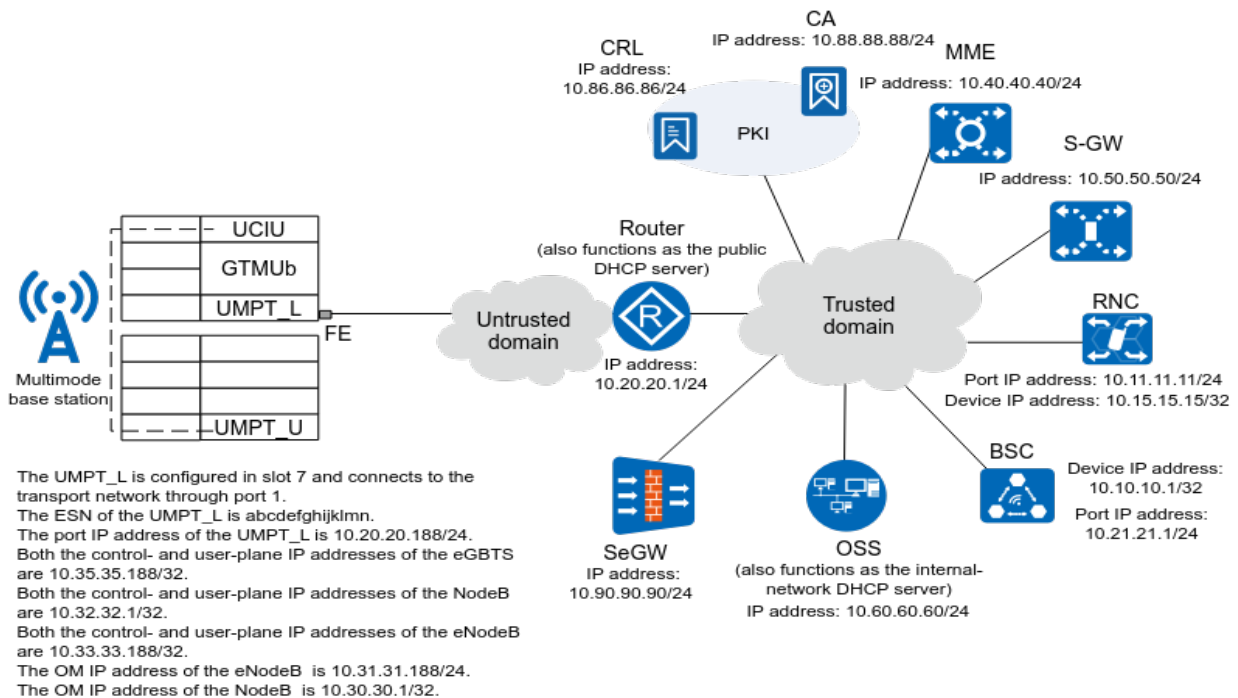
- IPsec reconfiguration is not required if an existing ACL rule applies to eGBTS/NodeB signaling and service data flows, that is, an ACL rule in Any to Any mode with **ACLRULE.ACTION (LTE eNodeB, 5G gNodeB)** set to **PERMIT** has been configured.
- Add an ACL rule with **ACLRULE.ACTION (LTE eNodeB, 5G gNodeB)** set to **PERMIT** for eGBTS/NodeB signaling and service data flows if existing ACL rules do not apply to these data flows.

#### NOTE

In a co-MPT base station, the method of deploying a UMDU\_GUL is the same as that of deploying a UMPT\_GUL.

## Separate-MPT Base Station

**Figure 4-45** shows an example of a separate-MPT GUL triple-mode base station on a PKI-based secure network.

**Figure 4-45** Example of deployment on a separate-MPT GUL triple-mode base station on a PKI-based secure network

In this networking scenario:

- In the root BBU, the GTMUc and UCIU communicate with the UMPT\_L through the BBU backplane.
- The UMPT\_U is configured in the leaf BBU.
- The root and leaf BBUs are interconnected by connecting the UCIU and UMPT\_U.
- The UMPT\_L provides IPsec protection for the following data flows:
  - eGBTS/NodeB/eNodeB signaling and service data flows
  - NodeB/eNodeB OM data flows (only if the OM channel needs to be protected by IPsec)
  - Certificate management-related data flows between the eNodeB and CA server
  - Data flows generated when the eNodeB obtains CRLs from the CRL server

If an IPsec network with an eNodeB using the UMPT\_L is reconstructed into a co-IPsec network with a GUL triple-mode base station (UMPT\_L+GTMUc+UCIU in the root BBU and UMPT\_U in the leaf BBU):

- If an existing ACL rule applies to eGBTS/NodeB signaling and service data flows and NodeB OM data flows (that is, an ACL rule in Any to Any mode with **ACLRULE.ACTION (LTE eNodeB, 5G gNodeB)** set to **PERMIT** has been configured): IPsec reconfiguration is not required when the OM channel is protected by IPsec. Add an ACL rule with **ACLRULE.ACTION (LTE eNodeB, 5G gNodeB)** set to **DENY** for NodeB OM data flows when the OM channel does not need to be protected by IPsec. The ACL ID for this ACL rule must be smaller than that for any ACL rule in Any to Any mode.

- If existing ACL rules do not apply to eGBTS/NodeB signaling and service data flows and NodeB OM data flows, add ACL rules with **ACLRULE.ACTION (LTE eNodeB, 5G gNodeB)** set to **PERMIT** for eGBTS/NodeB signaling and service data flows and NodeB OM data flows.
- In tunnel-interface-based configuration mode, the configurations of ACL rules and routes to the IPsec tunnel need to be added.

#### 4.4.2.6.1 Data Preparation

For details, see [4.4.2.2 Data Preparation](#).

#### 4.4.2.6.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Most MML-based configurations are the same as those described in [4.4.2.3.2 Using MML Commands](#). The following only describes the differences.

### Co-MPT Base Station

For a co-MPT triple-mode base station, configure ACL rules as follows:

- Configure an ACL rule for each mode to protect signaling and service data flows.
- Configure an ACL rule for the triple-mode base station to protect OM data flows and certificate management-related data flows, including those generated during the interaction with the digital certificate and CRL database. If the OM channel does not need to be protected by IPsec, the source IP address of this ACL rule must be different from the OM IP address. It is required that a new internal-network IP address be configured as the source IP address for certificate management-related data flows.

The network shown in [Figure 4-44](#) is used as an example. The ACL rules to be added are as follows:

- Rule 1 (RULEID = 1) applies to eGBTS signaling and service data flows.
- Rule 2 (RULEID = 2) applies to NodeB signaling and service data flows.
- Rule 3 (RULEID = 3) applies to eNodeB signaling and service data flows.
- Rule 4 (RULEID = 4) applies to OM data flows and certificate management-related data flows, including those generated during the interaction with the digital certificate and CRL database, of the triple-mode base station.

In this scenario, the MML command examples of adding ACL rules are as follows:

```
ADD ACLRULE: ACLID=3000, RULEID=1, PT=IP, SIP="10.35.35.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;
```

```
ADD ACLRULE: ACLID=3000, RULEID=2, PT=IP, SIP="10.32.32.1", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=3, PT=IP, SIP="10.33.33.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=4, PT=IP, SIP="10.31.31.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;
```

In tunnel-interface-based configuration mode, a route to the IPsec tunnel also needs to be added.

```
ADD SRCIPROUTE4: SRCRTIDX=0, VRFID=1, SRCIP="10.35.35.188", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=1, VRFID=1, SRCIP="10.32.32.1", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=2, VRFID=1, SRCIP="10.33.33.188", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=3, VRFID=1, SRCIP="10.31.31.188", RTTYPE=IF, PT=INTERFACE, PORTID=11;
```

If IPsec is not required for OM data flows, the MML command examples of adding ACL rules are as follows where the IP address 10.45.45.45 in Rule 4 (RULEID = 4) is the source IP address for certificate management-related data flows:

```
ADD ACLRULE: ACLID=3000, RULEID=1, PT=IP, SIP="10.35.35.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=2, PT=IP, SIP="10.32.32.1", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=3, PT=IP, SIP="10.33.33.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=4, PT=IP, SIP="10.45.45.45", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;
```

In tunnel-interface-based configuration mode, a route to the IPsec tunnel also needs to be added.

```
ADD SRCIPROUTE4: SRCRTIDX=0, VRFID=1, SRCIP="10.35.35.188", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=1, VRFID=1, SRCIP="10.32.32.1", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=2, VRFID=1, SRCIP="10.33.33.188", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=3, VRFID=1, SRCIP="10.45.45.45", RTTYPE=IF, PT=INTERFACE, PORTID=11;
```

## Separate-MPT Base Station

For a separate-MPT triple-mode base station, configure ACL rules as follows:

- Configure an ACL rule for each mode to protect signaling and service data flows.
- Configure an ACL rule for the mode in the leaf BBU to protect OM data flows. Do not configure this ACL rule if the OM channel does not need to be protected by IPsec.
- Configure an ACL rule for a more recent RAT in the root BBU to protect OM data flows and certificate management-related data flows, including those generated during the interaction with the digital certificate and CRL database. If the OM channel does not need to be protected by IPsec, the source IP address of this ACL rule must be different from the OM IP address. It is required that a new internal-network IP address be configured as the source IP address for certificate management-related data flows.

The network shown in [Figure 4-45](#) is used as an example. The ACL rules to be added are as follows:

- Rule 1 (RULEID = 1) applies to eGBTS signaling and service data flows.
- Rule 2 (RULEID = 2) applies to NodeB signaling and service data flows.
- Rule 3 (RULEID = 3) applies to eNodeB signaling and service data flows.
- Rule 4 (RULEID = 4) applies to NodeB OM data flows.

- Rule 5 (RULEID = 5) applies to eNodeB OM data flows and certificate management-related data flows, including those generated during the interaction with the digital certificate and CRL database.

In this scenario, the MML command examples of adding ACL rules are as follows:

```
ADD ACLRULE: ACLID=3000, RULEID=1, PT=IP, SIP="10.35.35.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=2, PT=IP, SIP="10.32.32.1", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=3, PT=IP, SIP="10.33.33.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=4, PT=IP, SIP="10.30.30.1", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=5, PT=IP, SIP="10.31.31.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;
```

In tunnel-interface-based configuration mode, a route to the IPsec tunnel also needs to be added.

```
ADD SRCIPROUTE4: SRCRTIDX=0, VRFIDX=1, SRCIP="10.35.35.188", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=1, VRFIDX=1, SRCIP="10.32.32.1", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=2, VRFIDX=1, SRCIP="10.33.33.188", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=3, VRFIDX=1, SRCIP="10.30.30.1", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=4, VRFIDX=1, SRCIP="10.31.31.188", RTTYPE=IF, PT=INTERFACE, PORTID=11;
```

If IPsec is not required for OM data flows, the MML command examples of adding ACL rules are as follows where the IP address 10.45.45.45 in Rule 4 (RULEID = 4) is the source IP address for certificate management-related data flows:

```
ADD ACLRULE: ACLID=3000, RULEID=1, PT=IP, SIP="10.35.35.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=2, PT=IP, SIP="10.32.32.1", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=3, PT=IP, SIP="10.33.33.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=4, PT=IP, SIP="10.45.45.45", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;
```

In tunnel-interface-based configuration mode, a route to the IPsec tunnel also needs to be added.

```
ADD SRCIPROUTE4: SRCRTIDX=0, VRFIDX=1, SRCIP="10.35.35.188", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=1, VRFIDX=1, SRCIP="10.32.32.1", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=2, VRFIDX=1, SRCIP="10.33.33.188", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=3, VRFIDX=1, SRCIP="10.45.45.45", RTTYPE=IF, PT=INTERFACE, PORTID=11;
```

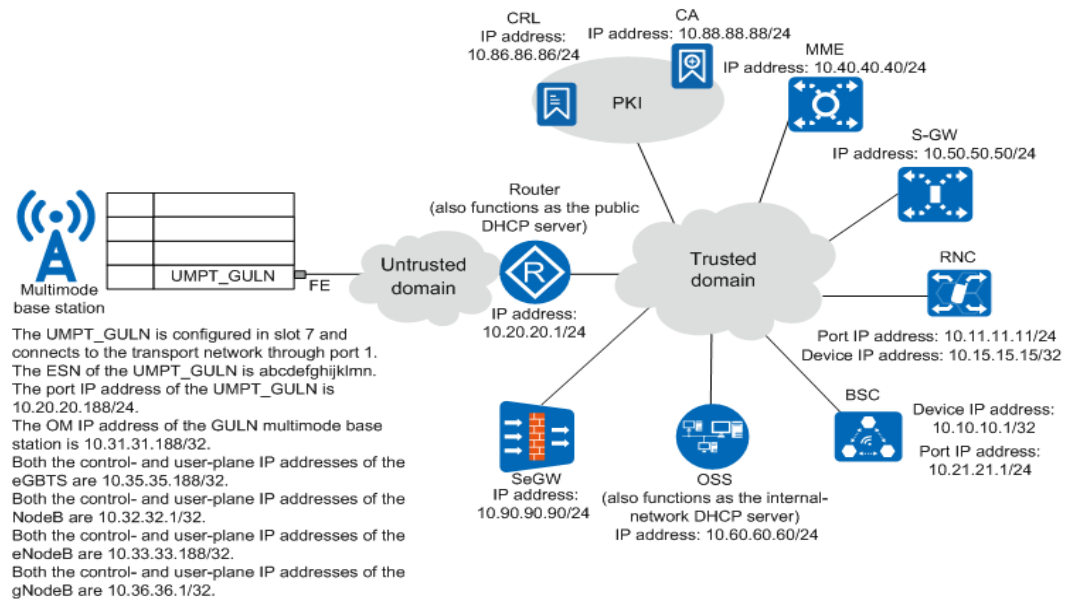
#### 4.4.2.7 Deploying Co-IPsec for a Quadruple-Mode Base Station on a PKI-based Secure Network

This section describes how to deploy co-IPsec for a quadruple-mode base station on a PKI-based secure network. For typical board configurations of co-MPT and separate-MPT base stations, see *BBU Hardware Description in 3900 & 5900 Series Base Station Product Documentation*. The following takes a GULN quadruple-mode base station as an example.

### Co-MPT Base Station

**Figure 4-46** shows an example of a co-MPT GULN quadruple-mode base station on a PKI-based secure network.

**Figure 4-46** Example of deployment on a co-MPT GULN quadruple-mode base station on a PKI-based secure network



In this networking scenario, the UMPT provides IPsec protection for the following data flows:

- eGBTS/NodeB/eNodeB/gNodeB signaling and service data flows
- OM data flows of the GULN multimode base station (only if the OM channel needs to be protected by IPsec)
- Certificate management-related data flows between the GULN multimode base station and CA server
- Data flows generated when the GULN multimode base station obtains CRLs from the CRL server

If a co-IPsec network with an LN dual-mode base station using the UMPT\_LN is reconstructed into a co-IPsec network with a GULN quadruple-mode base station using the UMPT\_GULN:

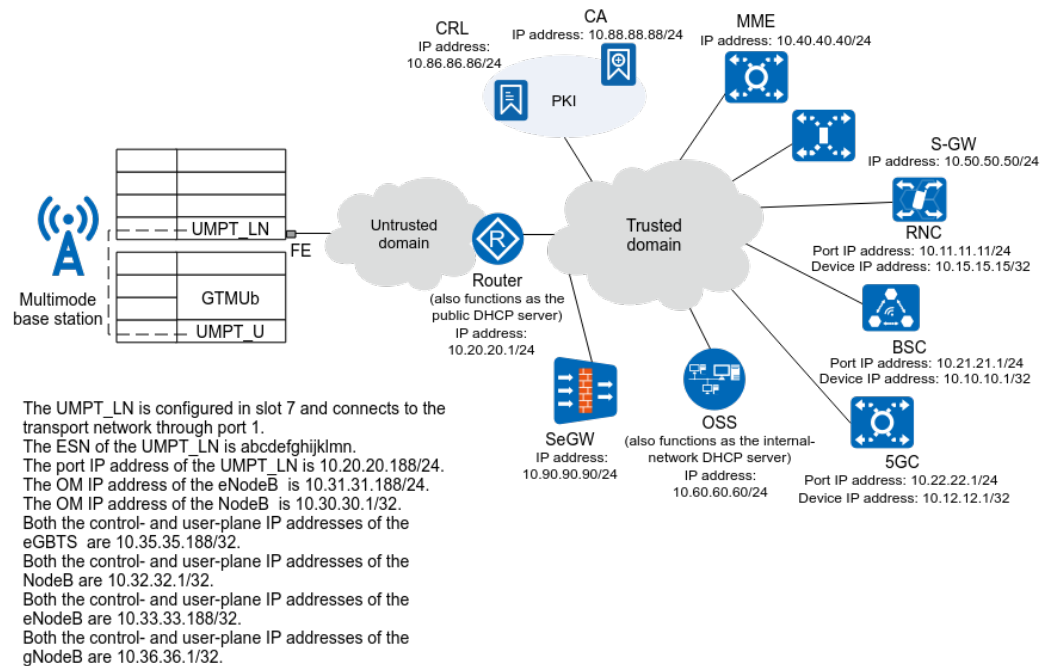
- IPsec reconfiguration is not required if an existing ACL rule applies to eGBTS/NodeB signaling and service data flows, that is, an ACL rule in Any to Any mode with **ACLRULE.ACTION (LTE eNodeB, 5G gNodeB)** set to **PERMIT** has been configured.
- Add an ACL rule with **ACLRULE.ACTION (LTE eNodeB, 5G gNodeB)** set to **PERMIT** for eGBTS/NodeB signaling and service data flows if existing ACL rules do not apply to these data flows.
- In tunnel-interface-based configuration mode, the configurations of ACL rules and routes to the IPsec tunnel need to be added.

## Separate-MPT Base Station

**Figure 4-47** shows an example of a separate-MPT GULN quadruple-mode base station on a PKI-based secure network.



**Figure 4-47** Example of deployment on a separate-MPT GULN quadruple-mode base station on a PKI-based secure network



In this networking scenario:

- The UMPT\_LN is configured in the root BBU.
- In the leaf BBU, the UMPT\_U communicates with the GTMUc through the BBU backplane.
- The root and leaf BBUs are interconnected by connecting the UMPT\_LN and UMPT\_U.
- The UMPT\_LN provides IPsec protection for the following data flows:
  - eGBTS/NodeB/eNodeB/gNodeB signaling and service data flows
  - NodeB/eNodeB/gNodeB OM data flows (only if the OM channel needs to be protected by IPsec)
  - Certificate management-related data flows between the eNodeB/gNodeB and CA server
  - Data flows generated when the eNodeB/gNodeB obtains CRLs from the CRL server

If a co-IPsec network with an LN dual-mode base station using the UMPT\_LN is reconstructed into a co-IPsec network with a GULN quadruple-mode base station (UMPT\_LN in the root BBU and UMPT\_U+GTMUc in the leaf BBU):

- If an existing ACL rule applies to eGBTS/NodeB signaling and service data flows and NodeB OM data flows (that is, an ACL rule in Any to Any mode with **ACLRULE.ACTION (LTE eNodeB, 5G gNodeB)** set to **PERMIT** has been configured): IPsec reconfiguration is not required when the OM channel is protected by IPsec. In Ethernet-interface-based configuration mode, an ACL rule with **ACLRULE.ACTION (LTE eNodeB, 5G gNodeB)** set to **DENY** for NodeB OM data flows needs to be added when the OM channel is not protected by IPsec. The ACL ID for this ACL rule must be smaller than that for any ACL rule in Any to Any mode.



- If existing ACL rules do not apply to eGBTS/NodeB signaling and service data flows and NodeB OM data flows, add ACL rules with **ACLRULE.ACTION (LTE eNodeB, 5G gNodeB)** set to **PERMIT** for eGBTS/NodeB signaling and service data flows and NodeB OM data flows.
- In tunnel-interface-based configuration mode, the configurations of ACL rules and routes to the IPsec tunnel need to be added.

#### 4.4.2.7.1 Data Preparation

For details, see [4.4.2.2 Data Preparation](#).

#### 4.4.2.7.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Most MML-based configurations are the same as those described in [4.4.2.3.2 Using MML Commands](#). The following only describes the differences.

### Co-MPT Base Station

For a co-MPT quadruple-mode base station, configure ACL rules as follows:

- Configure an ACL rule for each mode to protect signaling and service data flows.
- Configure an ACL rule for the quadruple-mode base station to protect OM data flows and certificate management-related data flows, including those generated during the interaction with the digital certificate and CRL database. If the OM channel does not need to be protected by IPsec, the source IP address of this ACL rule must be different from the OM IP address. It is required that a new internal-network IP address be configured as the source IP address for certificate management-related data flows.

The network shown in [Figure 4-46](#) is used as an example. The ACL rules to be added are as follows:

- Rule 1 (RULEID = 1) applies to eGBTS signaling and service data flows.
- Rule 2 (RULEID = 2) applies to NodeB signaling and service data flows.
- Rule 3 (RULEID = 3) applies to eNodeB signaling and service data flows.
- Rule 4 (RULEID = 4) applies to gNodeB signaling and service data flows.
- Rule 5 (RULEID = 5) applies to MBTS OM data flows and certificate management-related data flows, including those generated during the interaction with the digital certificate and CRL database.

In this scenario, the MML command examples of adding ACL rules are as follows:

```
ADD ACLRULE: ACLID=3000, RULEID=1, PT=IP, SIP="10.35.35.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;
```

```
ADD ACLRULE: ACLID=3000, RULEID=2, PT=IP, SIP="10.32.32.1", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=3, PT=IP, SIP="10.33.33.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=4, PT=IP, SIP="10.36.36.1", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=5, PT=IP, SIP="10.31.31.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;
```

In tunnel-interface-based configuration mode, a route to the IPsec tunnel also needs to be added.

```
ADD SRCIPROUTE4: SRCRTIDX=0, VRFID=1, SRCIP="10.35.35.188", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=1, VRFID=1, SRCIP="10.32.32.1", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=2, VRFID=1, SRCIP="10.33.33.188", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=3, VRFID=1, SRCIP="10.36.36.1", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=4, VRFID=1, SRCIP="10.31.31.188", RTTYPE=IF, PT=INTERFACE, PORTID=11;
```

If IPsec is not required for OM data flows, the MML command examples of adding ACL rules are as follows where the IP address 10.45.45.45 in Rule 5 (RULEID = 5) is the source IP address for certificate management-related data flows:

```
ADD ACLRULE: ACLID=3000, RULEID=1, PT=IP, SIP="10.35.35.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=2, PT=IP, SIP="10.32.32.1", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=3, PT=IP, SIP="10.33.33.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=4, PT=IP, SIP="10.36.36.1", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=5, PT=IP, SIP="10.45.45.45", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;
```

In tunnel-interface-based configuration mode, a route to the IPsec tunnel also needs to be added.

```
ADD SRCIPROUTE4: SRCRTIDX=0, VRFID=1, SRCIP="10.35.35.188", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=1, VRFID=1, SRCIP="10.32.32.1", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=2, VRFID=1, SRCIP="10.33.33.188", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=3, VRFID=1, SRCIP="10.36.36.1", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=4, VRFID=1, SRCIP="10.45.45.45", RTTYPE=IF, PT=INTERFACE, PORTID=11;
```

## Separate-MPT Base Station

For a separate-MPT quadruple-mode base station, configure ACL rules as follows:

- Configure an ACL rule for each mode to protect signaling and service data flows.
- Configure an ACL rule for the mode in the leaf BBU to protect OM data flows. Do not configure this ACL rule if the OM channel does not need to be protected by IPsec.
- Configure an ACL rule for the mode in the root BBU to protect OM data flows and certificate management-related data flows, including those generated during the interaction with the digital certificate and CRL database. If the OM channel does not need to be protected by IPsec, the source IP address of this ACL rule must be different from the OM IP address. It is required that a new internal-network IP address be configured as the source IP address for certificate management-related data flows.

The network shown in [Figure 4-47](#) is used as an example. The ACL rules to be added are as follows:

- Rule 1 (RULEID = 1) applies to eGBTS signaling and service data flows.
- Rule 2 (RULEID = 2) applies to NodeB signaling and service data flows.
- Rule 3 (RULEID = 3) applies to eNodeB signaling and service data flows.
- Rule 4 (RULEID = 4) applies to gNodeB signaling and service data flows.
- Rule 5 (RULEID = 5) applies to NodeB OM data flows.
- Rule 6 (RULEID = 6) applies to the LN dual-mode base station's OM data flows and certificate management-related data flows, including those generated during the interaction with the digital certificate and CRL database.

In this scenario, the MML command examples of adding ACL rules are as follows:

```
ADD ACLRULE: ACLID=3000, RULEID=1, PT=IP, SIP="10.35.35.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=2, PT=IP, SIP="10.32.32.1", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=3, PT=IP, SIP="10.33.33.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=4, PT=IP, SIP="10.36.36.1", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=5, PT=IP, SIP="10.30.30.1", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=6, PT=IP, SIP="10.31.31.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;
```

In tunnel-interface-based configuration mode, a route to the IPsec tunnel also needs to be added.

```
ADD SRCIPROUTE4: SRCRTIDX=0, VRFIDX=1, SRCIP="10.35.35.188", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=1, VRFIDX=1, SRCIP="10.32.32.1", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=2, VRFIDX=1, SRCIP="10.33.33.188", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=3, VRFIDX=1, SRCIP="10.36.36.1", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=4, VRFIDX=1, SRCIP="10.30.30.1", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=5, VRFIDX=1, SRCIP="10.31.31.188", RTTYPE=IF, PT=INTERFACE, PORTID=11;
```

If IPsec is not required for OM data flows, the MML command examples of adding ACL rules are as follows where the IP address 10.45.45.45 in Rule 5 (RULEID = 5) is the source IP address for certificate management-related data flows:

```
ADD ACLRULE: ACLID=3000, RULEID=1, PT=IP, SIP="10.35.35.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=2, PT=IP, SIP="10.32.32.1", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=3, PT=IP, SIP="10.33.33.188", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=4, PT=IP, SIP="10.36.36.1", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;  
ADD ACLRULE: ACLID=3000, RULEID=5, PT=IP, SIP="10.45.45.45", SWC="0.0.0.0", DIP="0.0.0.0",  
DWC="255.255.255.255", MDSCP=NO;
```

In tunnel-interface-based configuration mode, a route to the IPsec tunnel also needs to be added.

```
ADD SRCIPROUTE4: SRCRTIDX=0, VRFIDX=1, SRCIP="10.35.35.188", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=1, VRFIDX=1, SRCIP="10.32.32.1", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=2, VRFIDX=1, SRCIP="10.33.33.188", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=3, VRFIDX=1, SRCIP="10.36.36.1", RTTYPE=IF, PT=INTERFACE, PORTID=11;  
ADD SRCIPROUTE4: SRCRTIDX=4, VRFIDX=1, SRCIP="10.45.45.45", RTTYPE=IF, PT=INTERFACE, PORTID=11;
```

#### 4.4.2.8 Deploying IPsec on a PSK-based Secure Network

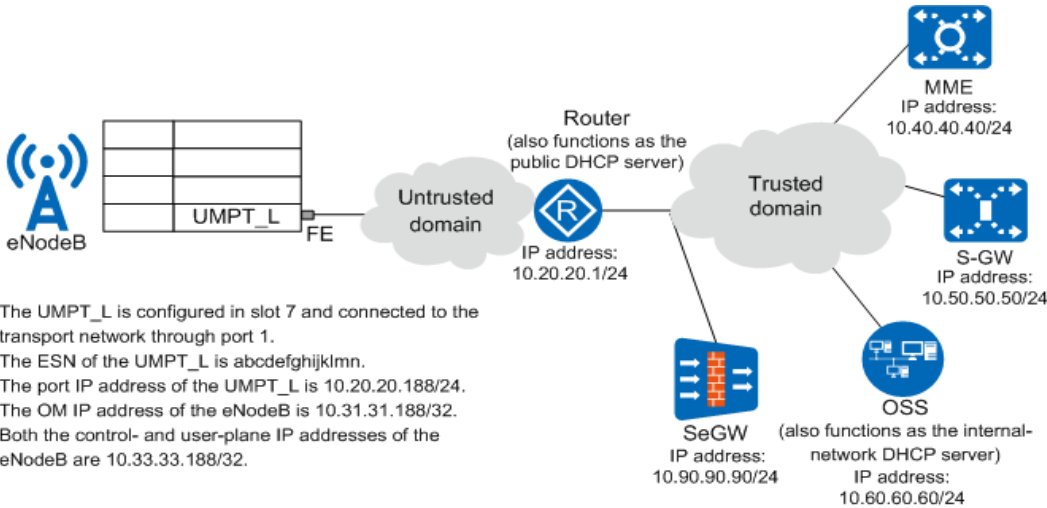
On PSK-based secure networks, a PKI system does not need to be deployed.

IPsec configurations on a PSK-based secure network differ from those on a PKI-based secure network in the following ways when the eNodeB is used as an example:

- The parameter settings in the **IKEPROPOSAL**, **IKEPEER**, and **IKECFG** MOs are different. For details, see [4.4.2.8.1 Data Preparation](#).
- ACL rules do not need to be configured for certificate management-related data flows and data flows generated during the interaction with the digital certificate and CRL database.

This section uses the network shown in [Figure 4-48](#) as an example to describe how to deploy IPsec for an eNodeB on a PSK-based secure network. IPsec configurations in other scenarios are similar and are not described in this document.

**Figure 4-48** Example of deployment for an eNodeB on a PSK-based secure network



4.4.2.8.1 Data Preparation

For details about data preparation, see [4.4.2.2 Data Preparation](#). The following describes only the differences in data preparation.

**Table 4-33** Parameters that must be set to configure an IKE proposal

| Parameter Name        | Parameter ID                                         | Setting Notes                                                                        |
|-----------------------|------------------------------------------------------|--------------------------------------------------------------------------------------|
| Authentication Method | <b>IKEPROPOSAL.AUTH METH (5G gNodeB, LTE eNodeB)</b> | Set this parameter to <b>PRE_SHARED_KEY</b> if PSK-based authentication is required. |

**Table 4-34** Parameters that must be set to configure an IKE peer

| Parameter Name | Parameter ID                                      | Setting Notes                                                                                                             |
|----------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Local ID Type  | <b>IKEPEER.IDTYPE (5G gNodeB, LTE eNodeB)</b>     | Set this parameter based on the network plan.                                                                             |
| Remote Name    | <b>IKEPEER.REMOTENAME (5G gNodeB, LTE eNodeB)</b> | If PSK-based authentication is required, set this parameter to a value same as the IKE local name configured at the SeGW. |
| Pre-shared Key | <b>IKEPEER.PKEY (5G gNodeB, LTE eNodeB)</b>       | If PSK-based authentication is required, set this parameter to the same value as that of the IKE peer.                    |

(Optional) If PSK-based authentication is required and the **IDTYPE (5G gNodeB, LTE eNodeB)** parameter is set to **FQDN**, engineering personnel also need to configure the name of the IKE local end, as listed in [Table 4-35](#).

**Table 4-35** Parameters for basic IKE configurations

| Parameter Name | Parameter ID                                 | Setting Notes                                                                                                                                                                                                                                                                    |
|----------------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Local Name     | <b>IKECFG.IKELNM (5G gNodeB, LTE eNodeB)</b> | Set this parameter when the <b>AUTHMETH (5G gNodeB, LTE eNodeB)</b> and <b>IDTYPE (5G gNodeB, LTE eNodeB)</b> parameters are set to <b>PRE_SHARED_KEY</b> and <b>FQDN</b> , respectively.<br><br>Set this parameter to a value same as the IKE peer name configured at the SeGW. |

 **NOTE**

When the **IKEPROPOSAL.AUTHMETH (5G gNodeB, LTE eNodeB)** parameter is set to **IKE\_CERT\_SIG**, the following are true: (1) If IKEv1 is used, the **IKECFG.IKELNM (5G gNodeB, LTE eNodeB)** parameter does not need to be set because its setting does not take effect. (2) If IKEv2 is used, the **IKECFG.IKELNM (5G gNodeB, LTE eNodeB)** parameter takes effect. If the **IKEPEER.IDTYPE (5G gNodeB, LTE eNodeB)** parameter is set to **FQDN**, the base station uses the **SubjectAltName** field in the digital certificate as the local IKE name. Especially, if the **SubjectAltName** field in the certificate file is empty in IKEv2, the base station uses the value of the **IKECFG.IKELNM (LTE eNodeB, 5G gNodeB)** field as the FQDN. If the **IKEPEER.IDTYPE (5G gNodeB, LTE eNodeB)** parameter is set to **DN**, the base station uses the value of the **DomainName** field in its digital certificate as the name of the IKE local end.

#### 4.4.2.8.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

The following describes only the differences between the PKI- and PSK-based secure networks in initial configuration.

```
//Before running the ADD IKEPROPOSAL command, running the SET IKECFG command to set local IKE configurations
SET IKECFG: IKELNM="IKECFG1", IKECLI=0, IKEKLT=0, DSCP=48;
ADD IKEPROPOSAL:PROPID=10, ENCALG=AES256, AUTHALG=SHA256, AUTHMETH=PRE_SHARED_KEY,
DHGRP=DH_GROUP14, PRFALG=HMAC_SHA256, DURATION=86400, REAUTHSW=OFF;
ADD IKEPEER: PEERNAME="ike", PROPID=10, IKEVERSION=IKE_V2, IPVERSION=IPV4, IDTYPE=FQDN,
REMOTEIP="10.90.90.90", REMOTENAME="secgw", PKEY="*****", DPD=PERIODIC, DPDIDLETIME=10,
DPDRETRI=5, DPDRETRN=5, LOCALIP="10.20.20.188", REDUNDANCYFLAG=NONE, CERTSOURCE=APPCERT,
IPSECPREFRGSW=OFF;
ADD ACLRULE: ACLID=3000, RULEID=1, ACTION=PERMIT, PT=IP, SIP="10.33.33.188", SWC="0.0.0.0",
DIP="0.0.0.0", DWC="255.255.255.255", MDSCP=NO;
ADD ACLRULE: ACLID=3000, RULEID=2, ACTION=PERMIT, PT=IP, SIP="10.31.31.188", SWC="0.0.0.0",
DIP="0.0.0.0", DWC="255.255.255.255", MDSCP=NO;
```

#### NOTE

- Rule 1 (RULEID = 1) applies to eNodeB signaling and service data flows.
- Rule 2 (RULEID = 2) applies to eNodeB OM data flows.

When the preceding data flows pass through a transmission port on the UMPT, the UMPT encrypts or decrypts them.

These ACL rules are required when the OM channel needs to be protected by IPsec.

If the OM channel does not need to be protected by IPsec, rule 2 does not need to be configured.

#### 4.4.2.9 Deactivation

Run the **RMV IPSECBIND** (old model)/**RMV IPSECBINDITF** (new model) command to remove the binding of an IPsec policy group from a port. In this way, the IPsec function is disabled on the port.

- Scenario 1:  
//When GTRANSPARA.TRANSFCGMODE is set to OLD  
//Removing the binding of an IPsec policy group named Policy0 from an Ethernet port numbered 0 connected to a base sub-board on the board in slot 6  
RMV IPSECBIND: SN=6, CN=0, SRN=0, SBT=BASE\_BOARD, PT=ETH,PN=0, SPGN="Policy0";
- Scenario 2:  
//When GTRANSPARA.TRANSFCGMODE is set to NEW  
RMV IPSECBINDITF: IPSECBINDITFID=0;

#### 4.4.2.10 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

## 4.4.3 Activation Verification

### Observing the IPsec Tunnel

**Step 1** Run the **DSP IKESA** command to check the IKE SA status.

All SAs are successfully established if the command output indicates both of the following:

- The status of SAs in phases 1 and 2 is **Ready|StayAlive** or **Ready**.

 NOTE

The IKE SAs negotiated in phase 1 do not contain **ACL ID** and **ACL Rule ID**. Therefore, the values of both **ACL ID** and **ACL Rule ID** in the **DSP IKESA** command output are **NULL** in phase 1.

- The number of SAs in phase 2 is the same as the number of ACL rules whose **ACTION** is **PERMIT** in the **LST ACLRULE** command.

**Step 2** Run the **DSP IPSECSA** command to check the IPsec SA status.

**Step 3** Check whether services protected by the IPsec tunnel are normal.

- Initiate voice and data services and then check whether services are running properly.
- Check whether the corresponding base station is online on the MAE topology view.

**Step 4** Check the IPsec replay status.

- Check whether ALM-25950 Base Station Being Attacked is reported with **Specific Problem** being **IPsec Replay**. If this alarm exists, the base station is under IPsec replay attacks.
- If IPsec replay attacks exist, run the **DSP INVALIDPKTINFO** command to query IPsec replay packets.

**Step 5** Check the IKE- and IPsec-related performance counters. For details, see [4.4.4 Network Monitoring](#).

-----End

### Observing the Active and Standby IPsec Tunnels

**Step 1** Check the status of the active and standby IPsec SAs.

Run the **DSP IPSECSA** command to separately check the status of the active and standby IPsec SAs. If data about the active and standby IPsec SAs is displayed in the command output, the active and standby IPsec tunnels have been established successfully.

**Step 2** Disable the active IPsec tunnel and then check whether services are running properly.

1. Remove the network cable to disable the active IPsec tunnel.
2. Run the **DSP IPSECDTNL** command to check whether the IPsec policy in use is the secondary IPsec policy.
3. Initiate voice and data services and then check whether services are running properly.
4. Check whether the corresponding base station is online on the MAE topology view.

----End

## Observing IKE Reauthentication

Run the **DSP IKESA** command to check the IKE SA status. Check whether the values of **IKE Reauthentication Lifetime(s)** and **IKE Reauthentication Remaining Time(s)** are **NULL**. If both values are not **NULL**, IKE reauthentication has been enabled.

## Observing IPsec ACL Narrow Down

Run the **DSP IPSECSA** or **DSP IKESA** command to check the values of the following parameters: **Source IP**, **Protected Flow Source Wildcard**, **Destination IP**, and **Protected Flow Destination Wildcard**. If the values are not **NULL**, the IPsec ACL narrow down function has been enabled.

## 4.4.4 Network Monitoring

You can use the MAE signaling tracing function or performance counters to monitor IPsec performance.

### MAE Signaling Tracing-based Monitoring

On the **Signaling Trace Management** tab page of the MAE-Access, select **IPSec Performance Monitoring**. Then, check the values of the following fields in the query results: **IPSec Send Rate(pps)**, **IPSec Receive Rate(pps)**, and **IPSec Receive DropRate(0.001%)**. For details, see the "RAN Trace and Monitoring" section in *MAE Product Documentation*.

### Counter-based Monitoring

[Table 4-36](#) lists the counters used to monitor IKE status. [Table 4-37](#) lists the counters used to monitor IPsec status.

#### NOTE

- The eGBTS/NodeB/eNodeB/gNodeB supports IKE and IPsec performance monitoring, whereas the GBTS does not.
- On the base station, only the UMPT/UMDU/MDUC/UTRPc board supports the **IKE Measurement** subset (the counter names are prefixed with **VS.IKE.**).
- After IKE- and IPsec-related data is configured on the base station, you can select all base station boards supporting IPsec and IKE in the MAE-Access performance management window. If a board that does not use IKE or IPsec is selected, the value of the corresponding performance counter is 0.



**Table 4-36** Counters indicating IKE status

| Performance Counter                                                   | Description                                                                      |
|-----------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>VS.IKE.RxPackets (5G gNodeB, LTE eNodeB)</b>                       | Number of received IKE packets                                                   |
| <b>VS.IKE.TxPackets (5G gNodeB, LTE eNodeB)</b>                       | Number of transmitted IKE packets                                                |
| <b>VS.IKE.SubSARekey.Times (5G gNodeB, LTE eNodeB)</b>                | Number of Re-keys Performed by IKE Child SA                                      |
| <b>VS.IKE.DPDSessionFail.Times (5G gNodeB, LTE eNodeB)</b>            | Number of Failed DPD Sessions                                                    |
| <b>VS.IKE.Init.ChildSARekeyFail.Times (5G gNodeB, LTE eNodeB)</b>     | Number of Failed Initiations of Re-key Negotiations for IKE Child SA             |
| <b>VS.IKE.Response.ChildSARekeyFail.Times (5G gNodeB, LTE eNodeB)</b> | Number of Failed Responses to Re-key Negotiations for IKE Child SA               |
| <b>VS.IKE.Init.ChildSARekeySucc.Times (5G gNodeB, LTE eNodeB)</b>     | Number of Successful Initiations of Re-key Negotiations for IKE Child SA         |
| <b>VS.IKE.Response.ChildSARekeySucc.Times (5G gNodeB, LTE eNodeB)</b> | Number of Successful Responses to Re-key Negotiations for IKE Child SA           |
| <b>VS.IKE.Init.SARekeySucc.Times (5G gNodeB, LTE eNodeB)</b>          | Number of Successful Initiations of Re-key Negotiations for IKE SA               |
| <b>VS.IKE.Response.SARekeySucc.Times (5G gNodeB, LTE eNodeB)</b>      | Number of Successful Responses to Re-key Negotiations for IKE SA                 |
| <b>VS.IKE.Init.SARekeyFail.Times (5G gNodeB, LTE eNodeB)</b>          | Number of Failed Initiations of Re-key Negotiations for IKE SA                   |
| <b>VS.IKE.Response.SARekeyFail.Times (5G gNodeB, LTE eNodeB)</b>      | Number of Failed Responses to Re-key Negotiations for IKE SA                     |
| <b>VS.IKE.Init.V2ReauthSucc.Times (5G gNodeB, LTE eNodeB)</b>         | Number of Successful Initiations of IKEv2 Re-authentications                     |
| <b>VS.IKE.Response.V2ReauthSucc.Times (5G gNodeB, LTE eNodeB)</b>     | Number of Successful Responses to IKEv2 Re-authentications                       |
| <b>VS.IKE.Init.V2ReauthFail.Times (5G gNodeB, LTE eNodeB)</b>         | Number of Failed Initiations of IKEv2 Re-authentications                         |
| <b>VS.IKE.Response.V2ReauthFail.Times (5G gNodeB, LTE eNodeB)</b>     | Number of Failed Responses to IKEv2 Re-authentications                           |
| <b>VS.IKE.Receive.InvalidSPIPackets (5G gNodeB, LTE eNodeB)</b>       | Number of Received IP or ESP Packets with SPI That Does Not Match an Existing SA |
| <b>VS.IKEPEER.RxPackets (5G gNodeB, LTE eNodeB)</b>                   | Number of IKE Packets Received by an IKE Peer                                    |

| Performance Counter                                                    | Description                                                                       |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>VS.IKEPEER.TxPackets (5G gNodeB, LTE eNodeB)</b>                    | Number of IKE Packets Transmitted by an IKE Peer                                  |
| <b>VS.IKEPEER.SubSARkey.Times (5G gNodeB, LTE eNodeB)</b>              | Number of Re-key Negotiations Initiated for IKE Peer Child SA                     |
| <b>VS.IKEPEER.DPDSessionFail.Times (5G gNodeB, LTE eNodeB)</b>         | Number of IKE Disconnections Caused by DPD Sessions of an IKE Peer                |
| <b>VS.IKEPEER.Resp.Recv.AuthReq.Times (5G gNodeB, LTE eNodeB)</b>      | Number of IKE Authentication Requests Received by an IKE Peer                     |
| <b>VS.IKEPEER.Init.Recv.AuthResp.Times (5G gNodeB, LTE eNodeB)</b>     | Number of Responses Received by an IKE Peer to IKE Authentication Requests        |
| <b>VS.IKEPEER.Resp.Recv.CrChildSAReq.Times (5G gNodeB, LTE eNodeB)</b> | Number of IKE Create Child SA Requests Received by an IKE Peer                    |
| <b>VS.IKEPEER.Init.Recv.CrChildSAResponse (5G gNodeB, LTE eNodeB)</b>  | Number of Responses to IKE Create Child SA Requests Received by an IKE Peer       |
| <b>VS.IKEPEER.RespRecvErrorNotify (5G gNodeB, LTE eNodeB)</b>          | Number of IKE Error Type Notify Messages Received by an IKE Peer                  |
| <b>VS.IKEPEER.Resp.Recv.StateNotify.Times (5G gNodeB, LTE eNodeB)</b>  | Number of IKE State Type Notify Messages Received by an IKE Peer                  |
| <b>VS.IKEPEER.Resp.Recv.DelSAReq.Times (5G gNodeB, LTE eNodeB)</b>     | Number of IKE Delete SA Requests Received by an IKE Peer                          |
| <b>VS.IKEPEER.Init.Recv.DelSAResponses (5G gNodeB, LTE eNodeB)</b>     | Number of Responses Received by an IKE Peer to IKE Delete SA Requests             |
| <b>VS.IKEPEER.Init.Recv.ConfigReply.Times (5G gNodeB, LTE eNodeB)</b>  | Number of Responses Received by an IKE Peer to IKE Configuration Payload Requests |
| <b>VS.IKEPEER.Resp.Recv.IKEInitReq.Times (5G gNodeB, LTE eNodeB)</b>   | Number of IKE Initialization Requests Received by an IKE Peer                     |
| <b>VS.IKEPEER.Init.Recv.IKEInitResp.Times (5G gNodeB, LTE eNodeB)</b>  | Number of Responses Received by an IKE Peer to IKE Initialization Requests        |
| <b>VS.IKEPEER.Recv.InvalidPackets (5G gNodeB, LTE eNodeB)</b>          | Number of Invalid IKE Packets Received by an IKE Peer                             |
| <b>VS.IKEPEER.Init.Send.AuthReq.Times (5G gNodeB, LTE eNodeB)</b>      | Number of IKE Authentication Requests Sent by an IKE Peer                         |
| <b>VS.IKEPEER.Resp.Send.AuthResp.Times (5G gNodeB, LTE eNodeB)</b>     | Number of Responses Sent by an IKE Peer to IKE Authentication Requests            |
| <b>VS.IKEPEER.Init.Send.CrChildSAReq.Times (5G gNodeB, LTE eNodeB)</b> | Number of IKE Create Child SA Requests Sent by an IKE Peer                        |

| Performance Counter                                                     | Description                                                             |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------|
| <b>VS.IKEPEER.Resp.Send.CrChildSAResp.Times (5G gNodeB, LTE eNodeB)</b> | Number of Responses Sent by an IKE Peer to IKE Create Child SA Requests |
| <b>VS.IKEPEER.Init.Send.ErrorNotify.Times (5G gNodeB, LTE eNodeB)</b>   | Number of IKE Error Type Notify Messages Sent by an IKE Peer            |
| <b>VS.IKEPEER.Init.Send.StateNotify.Times (5G gNodeB, LTE eNodeB)</b>   | Number of IKE State Type Notify Messages Sent by an IKE Peer            |
| <b>VS.IKEPEER.Init.Send.DelSAReq.Times (5G gNodeB, LTE eNodeB)</b>      | Number of Responses Sent by an IKE Peer to IKE Delete SA Requests       |
| <b>VS.IKEPEER.Resp.Send.DelSAResp.Times (5G gNodeB, LTE eNodeB)</b>     | Number of IKE Delete SA Requests Sent by an IKE Peer                    |
| <b>VS.IKEPEER.Init.Send.ConfigReq.Times (5G gNodeB, LTE eNodeB)</b>     | Number of IKE Configuration Payload Requests Sent by an IKE Peer        |
| <b>VS.IKEPEER.Init.Send.IKEInitReq.Times (5G gNodeB, LTE eNodeB)</b>    | Number of IKE Initialization Requests Sent by an IKE Peer               |
| <b>VS.IKEPEER.Resp.Send.IKEInitResp.Times (5G gNodeB, LTE eNodeB)</b>   | Number of Responses Sent by an IKE Peer to IKE Initialization Requests  |

**Table 4-37** Counters indicating IPsec status

| Performance Counter                                                    | Description                                                                 |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| <b>VS.IPsec.RxCheckReplayFailDropPkts (5G gNodeB, LTE eNodeB)</b>      | Number of discarded packets received due to replay checking fail            |
| <b>VS.IPsec.RxAHCheckFailDropPkts (5G gNodeB, LTE eNodeB)</b>          | Number of discarded packets received due to AH checking fail                |
| <b>VS.IPsec.RxESPAuthFailDropPkts (5G gNodeB, LTE eNodeB)</b>          | Number of received but discarded packets due to ESP authentication failures |
| <b>VS.IPsec.RxESPDecryptionFail-DropPkts (5G gNodeB, LTE eNodeB)</b>   | Number of received but discarded packets due to ESP decryption failures     |
| <b>VS.IPsec.RxDecryptACLFailDropPkts (5G gNodeB, LTE eNodeB)</b>       | Number of discarded packets received due to decrypt ACL fail                |
| <b>VS.IPsec.RxDecryptSuccessPkts (5G gNodeB, LTE eNodeB)</b>           | Number of decrypt success packets received                                  |
| <b>VS.IPsec.TxOutboundSAMiss-DropPkts (5G gNodeB, LTE eNodeB)</b>      | Number of discarded packets transmitted due to outbound SA miss             |
| <b>VS.IPsec.TxAntiReplaySnWrapped-DropPkts (5G gNodeB, LTE eNodeB)</b> | Number of discarded packets transmitted due to sequence number overflow     |

| Performance Counter                                   | Description                                   |
|-------------------------------------------------------|-----------------------------------------------|
| VS.IPsec.TxEncryptSuccessPkts (5G gNodeB, LTE eNodeB) | Number of encrypt success packets transmitted |
| VS.IPsec.SABuilt.Times (5G gNodeB, LTE eNodeB)        | Number of IPsec SA Establishments             |
| VS.IPsec.SADel.Times (5G gNodeB, LTE eNodeB)          | Number of IPsec SA Deletions                  |

## Alarm Monitoring

After the IPsec feature is activated, the base station may report the following alarms:

- ALM-25891 IKE Negotiation Failure
- ALM-25950 Base Station Being Attacked with "Specific Problem" displayed as IPsec Replay

If alarms or services cannot be recovered after the alarm handling or services are abnormal although no alarms are reported, the local or peer IKE SA or IPsec SA may be faulty. In this case, you are advised to run the **RST IKESA** command to manually trigger an IKE SA or IPsec SA renegotiation.

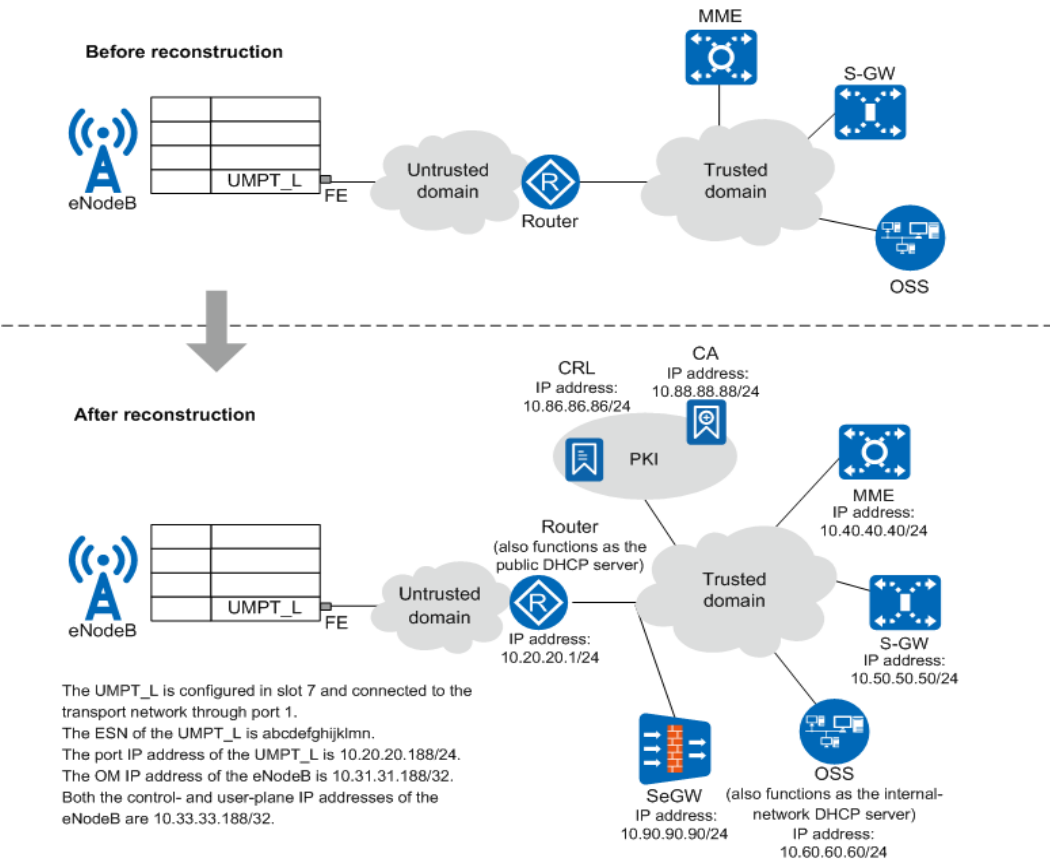
## 4.5 Operation and Maintenance (Reconstruction)

### 4.5.1 Reconstruction from an Insecure Network to a PKI-based Secure Network

In this scenario, the reconstruction requirements and IPsec reconfiguration procedure are the same for the eGBTS/NodeB/eNodeB/gNodeB/multimode base station. This section uses an eNodeB as an example to describe the reconstruction requirements and IPsec reconfiguration procedure when an insecure network is reconstructed into a PKI-based secure network.

**Figure 4-49** shows the network topologies before and after the reconstruction.

**Figure 4-49** Example of reconstructing an insecure network into a PKI-based secure network for an eNodeB



Before the reconstruction, the base station must meet the hardware requirements described in [4.3.3 Hardware](#).

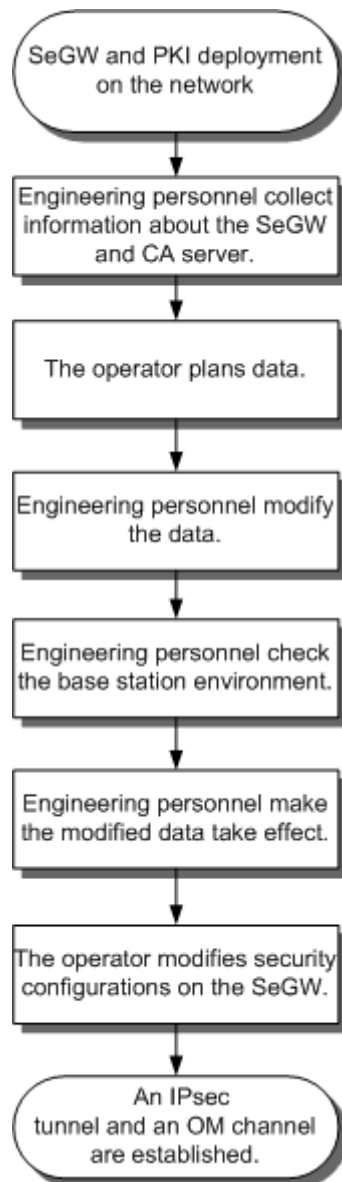
After the reconstruction, all data flows in the PKI-based secure network shares one IPsec tunnel. [Table 4-38](#) describes the IP address planning for reconstructing an insecure network into a PKI-based secure network.

**Table 4-38** IP address planning for reconstructing an insecure network into a PKI-based secure network

| Item                                  | Example         | Remarks                                                                                                                                                                                                                      |
|---------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IP address of FE port 1 on the UMPT_L | 10.20.20.188/24 | This IP address is used as the source IP address at the outer layer of the IPsec tunnel on the base station side.<br><br>It is recommended that multiple port IP addresses be planned if multiple IPsec tunnels are planned. |

| Item                                                    | Example                                              | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Control-plane and user-plane IP addresses of the eNodeB | 10.33.33.188/32                                      | This IP address must be a logical IP address.<br>If signaling data and service data need to be transmitted separately, a signaling IP address and a service IP address should be planned separately.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| OM IP address of the eNodeB                             | 10.31.31.188/32                                      | This IP address must be a logical IP address if OM data is transmitted in an IPsec tunnel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Source IP address for certificate updates               | 10.31.31.188/32<br>10.45.45.45/32<br>10.20.20.188/24 | If the OM channel needs to be protected by IPsec and the CA server can be accessed through either the internal or public network, the source IP address for certificate updates must be set to the OM IP address, for example, 10.31.31.188.<br>If the OM channel does not need to be protected by IPsec and the CA server can be accessed through either the internal or public network, it is recommended that the source IP address for certificate updates be set to a new internal-network logical IP address, for example, 10.45.45.45.<br>If the CA server can be accessed through only the public network, the source IP address for certificate updates must be set to a port IP address, for example, 10.20.20.188. |

## General Procedure



## Network Deployment and Information Collection

- An SeGW has been deployed on the network and has been configured with an operator-issued device certificate, an operator's root certificate, and security-related parameters.
- A PKI system has been deployed on the network. The CA server in the PKI system is preconfigured with the Huawei root certificate.
- Engineering personnel have collected information about the SeGW. For details, see [4.4.2.1 Required Information](#).
- Engineering personnel have collected information about the CA server. For details, see the "Precautions" section in *PKI*.

## Security Data Planning

IPsec data planning is similar to that described in [4.4.2.2 Data Preparation](#).

PKI data planning is the same as that for newly deployed base stations. For details, see the "Data Preparation" section in *PKI*.

## Preparing the Incremental Script

An incremental script is generated based on data of existing base stations and includes configuration modifications.

For details about how to modify IPsec configurations, see [4.4.2.10 Using the MAE-Deployment](#). ACL rules are configured according to the network plan.

For details about how to modify PKI configurations, see "Using the MAE-Deployment" in *PKI*.

## Checking the Base Station Environment

- The base station meets the hardware requirements described in [4.3.3 Hardware](#).
- The license for the IPsec feature has been activated on the base station.
- The base station has been preconfigured with a Huawei initial certificate and the Huawei root certificate.
  - Check for Huawei initial certificate  
Run the **DSP BRDCERTMK** command with **CERTFILETYPE** set to **DEVCERT**. If **VendorCertFlag = YES** is displayed in the command output, the board has a Huawei initial certificate.
  - Check for Huawei root certificate  
Run the **DSP BRDCERTMK** command with **CERTFILETYPE** set to **TRUSTCERT**. If **VendorCertFlag = YES** is displayed in the command output, the board has a Huawei root certificate.

## Downloading the Incremental Script

1. Log in the MAE.
2. On the MAE-Deployment, choose **Area Management > Planned Area**, and export the incremental script.
3. In the **Export Incremental Scripts** dialog box, choose a specific base station from which the script is exported, specify **Output Path** and **Script Executor Operation**, and click **OK**.
4. On the displayed **Script Executor** page, observe the export progress.
5. After the export is complete, restart the base station to make the script take effect.

## Modifying Routing Information

The operator modifies routing information to enable data flows that need to be protected by IPsec to pass through the SeGW before reaching their final destination.



## Establishing IPsec Tunnels

The base station and the SeGW perform IKE negotiation to establish an IPsec tunnel. When the OM channel between the base station and the MAE is successfully established, network reconstruction succeeds.

### Activation Verification

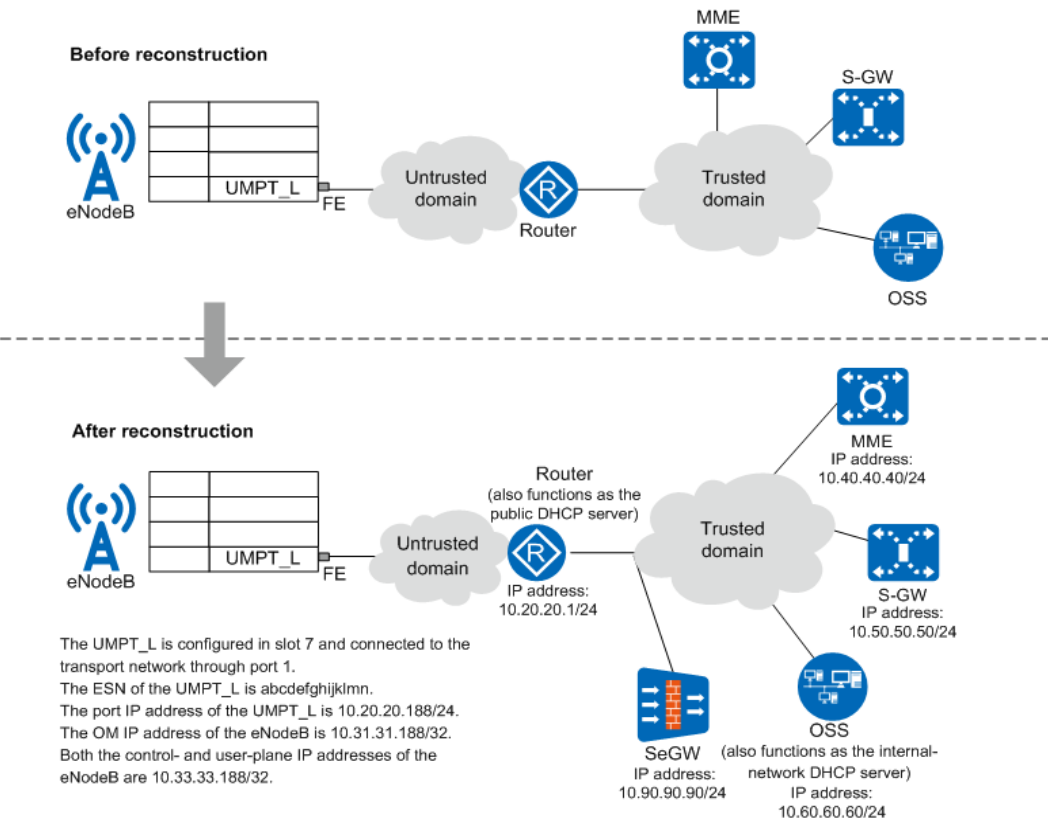
- Run the **DSP IKESA** command to check the SA status. All SAs are successfully established when the command output indicates both of the following: The value of **SA Flag** for all SAs in phases 1 and 2 is **Ready|StayAlive** or **Ready**. The number of SAs in phase 2 is the same as the number of ACL rules with **ACTION (LTE eNodeB, 5G gNodeB)** set to **PERMIT**.
- Run the **DSP IPSECSA** command to check the status of IPsec SAs.
- Observe the base station and test services as follows:
  - a. Check whether the corresponding base station is online on the MAE topology view.
  - b. Initiate voice and data services and then check whether services are running properly.

## 4.5.2 Reconstruction from an Insecure Network to a PSK-based Secure Network

In this scenario, the reconstruction requirements and IPsec reconfiguration procedure are the same for eGBTs, NodeBs, eNodeBs, gNodeBs, and multimode base stations. This section uses an eNodeB as an example to describe the reconstruction requirements and IPsec reconfiguration procedure when an insecure network is reconstructed into a PSK-based secure network.

**Figure 4-50** shows the network topologies before and after the reconstruction.

**Figure 4-50** Example of reconstructing an insecure network into a PSK-based secure network for an eNodeB



Before the reconstruction, the base station must meet the hardware requirements described in [4.3.3 Hardware](#).

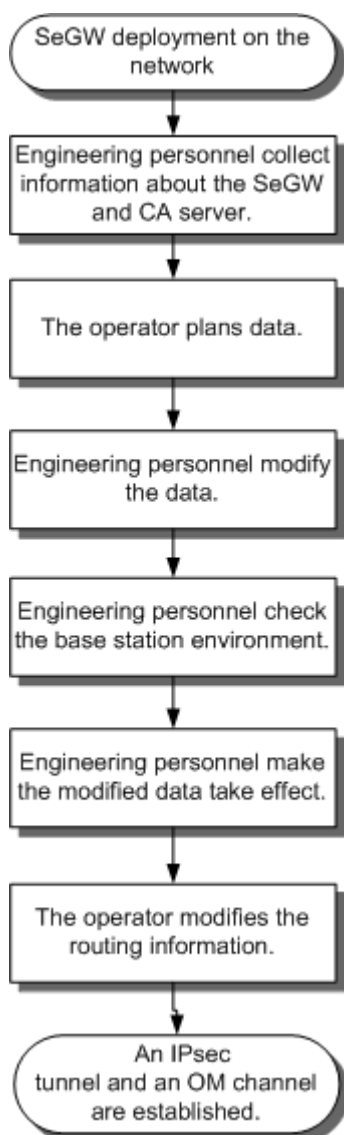
After the reconstruction, all data flows in the PSK-based secure network shares one IPsec tunnel. [Table 4-39](#) describes the IP address planning for reconstructing an insecure network into a PSK-based secure network.

**Table 4-39** IP address planning for reconstructing an insecure network into a PSK-based secure network

| Item                                        | Example         | Remarks                                                                                                                                                                                                                  |
|---------------------------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IP address of FE port 1 on the UMPT_L       | 10.20.20.188/24 | This IP address is used as the source IP address at the outer layer of the IPsec tunnel on the base station side.<br>It is recommended that multiple port IP addresses be planned if multiple IPsec tunnels are planned. |
| Signaling/ service IP address of the eNodeB | 10.33.33.188/32 | This IP address must be a logical IP address.<br>If signaling data and service data need to be transmitted separately, a signaling IP address and a service IP address should be planned separately.                     |

| Item                        | Example         | Remarks                                                                                    |
|-----------------------------|-----------------|--------------------------------------------------------------------------------------------|
| OM IP address of the eNodeB | 10.31.31.188/32 | This IP address must be a logical IP address if OM data is transmitted in an IPsec tunnel. |

## General Procedure



## Network Deployment and Information Collection

Engineering personnel have collected information about the SeGW. For details, see [4.4.2.1 Required Information](#).

## Security Data Planning

IPsec data planning is similar to that described in [4.4.2.3.1 Data Preparation](#). The following tables describe only the differences.

Data to be prepared for an IKE proposal:

| Parameter Name        | Parameter ID                                               | Setting Notes                                                                        |
|-----------------------|------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Authentication Method | <b><i>IKEPROPOSAL.AUTHMETH (LTE eNodeB, 5G gNodeB)</i></b> | Set this parameter to <b>PRE_SHARED_KEY</b> if PSK-based authentication is required. |

Data to be prepared for an IKE peer:

| Parameter Name | Parameter ID                                             | Setting Notes                                                                                                             |
|----------------|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Local ID Type  | <b><i>IKEPEER.IDTYPE (LTE eNodeB, 5G gNodeB)</i></b>     | Set this parameter based on the network plan.                                                                             |
| Remote Name    | <b><i>IKEPEER.REMOTENAME (LTE eNodeB, 5G gNodeB)</i></b> | If PSK-based authentication is required, set this parameter to a value same as the IKE local name configured at the SeGW. |
| Pre-shared Key | <b><i>IKEPEER.PKEY (LTE eNodeB, 5G gNodeB)</i></b>       | If PSK-based authentication is required, set this parameter to the same value as that of the IKE peer.                    |

(Optional) If PSK-based authentication is required and the ***IDTYPE (LTE eNodeB, 5G gNodeB)*** parameter is set to **FQDN**, engineering personnel also need to configure the IKE local name.

| Parameter Name | Parameter ID                                        | Setting Notes                                                                                                                                                                                                                                                                                  |
|----------------|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Local Name     | <b><i>IKECFG.IKELNM (LTE eNodeB, 5G gNodeB)</i></b> | Set this parameter when the <b><i>AUTHMETH (LTE eNodeB, 5G gNodeB)</i></b> and <b><i>IDTYPE (LTE eNodeB, 5G gNodeB)</i></b> parameters are set to <b>PRE_SHARED_KEY</b> and <b>FQDN</b> , respectively.<br><br>Set this parameter to a value same as the IKE peer name configured at the SeGW. |

#### NOTE

When the **IKEPROPOSAL.AUTHMETH (5G gNodeB, LTE eNodeB)** parameter is set to **IKE\_CERT\_SIG**, if IKEv1 is used, the **IKECFG.IKELNM (5G gNodeB, LTE eNodeB)** parameter does not need to be set because its setting does not take effect. If IKEv2 is used, the **IKECFG.IKELNM (5G gNodeB, LTE eNodeB)** parameter takes effect. If the **IKEPEER.IDTYPE (5G gNodeB, LTE eNodeB)** parameter is set to **FQDN**, the base station uses the **SubjectAltName** field in the digital certificate as the IKE local name. Especially, if the **SubjectAltName** field in the certificate file is empty in IKEv2, the base station uses the value of the **IKECFG.IKELNM (LTE eNodeB, 5G gNodeB)** field as the FQDN. If the **IKEPEER.IDTYPE (5G gNodeB, LTE eNodeB)** parameter is set to **DN**, the base station uses the value of the **DomainName** field in its digital certificate as the IKE local name.

## Preparing the Incremental Script

An incremental script is generated based on data of existing base stations and includes configuration modifications.

For details about how to modify IPsec configurations, see [4.4.2.10 Using the MAE-Deployment](#). ACL rules are configured according to the network plan.

## Checking the Base Station Environment

- The base station meets the hardware requirements described in [4.3.3 Hardware](#).
- The license for the IPsec feature has been activated on the base station.

## Downloading the Incremental Script

For details, see [Downloading the Incremental Script](#) in [4.5.1 Reconstruction from an Insecure Network to a PKI-based Secure Network](#).

## Modifying Routing Information

The operator modifies routing information to enable data flows that need to be protected by IPsec to pass through the SeGW before reaching their final destination.

## Establishing IPsec Tunnels

The base station and the SeGW perform IKE negotiation to establish an IPsec tunnel. When the OM channel between the base station and the MAE is successfully established, network reconstruction succeeds.

## Activation Verification

For details, see [Activation Verification](#) in [4.5.1 Reconstruction from an Insecure Network to a PKI-based Secure Network](#).

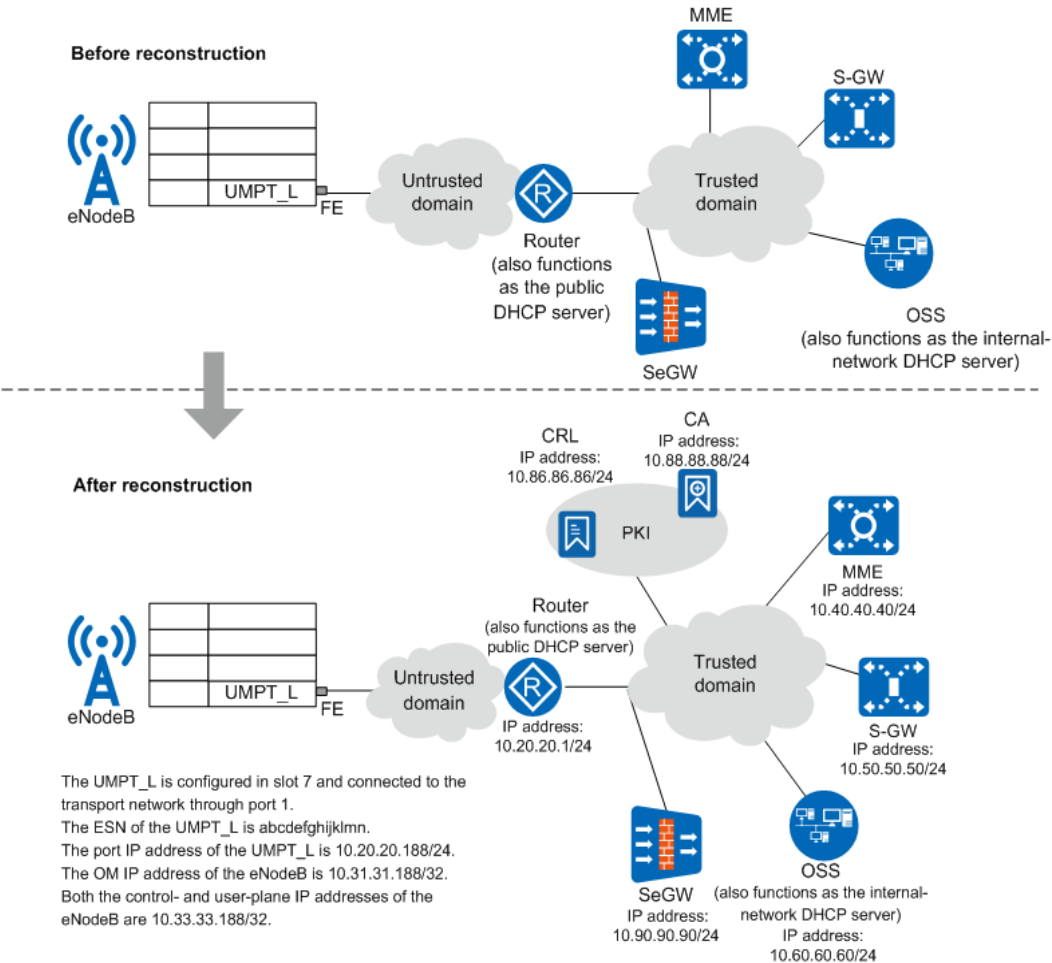
## 4.5.3 Reconstruction from a PSK-based Secure Network to a PKI-based Secure Network

In this scenario, the reconstruction requirements and IPsec reconfiguration procedure are the same for the eGBTS/NodeB/eNodeB/gNodeB/multimode base

station. This section uses an eNodeB as an example to describe the reconstruction requirements and IPsec reconfiguration procedure when a PSK-based secure network is reconstructed into a PKI-based secure network.

Figure 4-51 shows the network topologies before and after the reconstruction.

Figure 4-51 Example of reconstructing a PSK-based secure network into a PKI-based secure network for an eNodeB

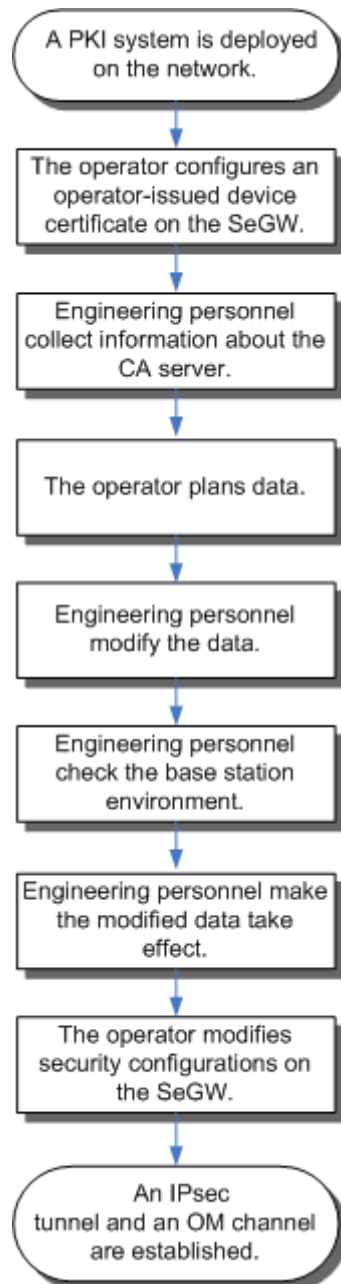


After the reconstruction, the data flows for certificate management and other data flows share the same IPsec tunnel in the PKI-based secure network.

| Item                                      | Example                                              | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source IP address for certificate updates | 10.31.31.188/32<br>10.45.45.45/32<br>10.20.20.188/24 | <p>If the OM channel needs to be protected by IPsec and the CA server can be accessed through either the internal or public network, the source IP address for certificate updates must be set to the OM IP address, for example, 10.31.31.188.</p> <p>If the OM channel does not need to be protected by IPsec and the CA server can be accessed through either the internal or public network, it is recommended that the source IP address for certificate updates be set to a new internal-network logical IP address, for example, 10.45.45.45.</p> <p>If the CA server can be accessed through only the public network, the source IP address for certificate updates must be set to a port IP address, for example, 10.20.20.188.</p> |

The IP addresses for other data flows remain unchanged before and after the reconstruction.

## General Procedure



## Network Deployment and Information Collection

- The operator deploys a PKI system on the network and preconfigures the Huawei root certificate on a CA server in the system.
- The operator configures an operator-issued device certificate and the operator's root certificate on the SeGW.
- Engineering personnel have collected information about the CA server. For details, see the "Precautions" section in *PKI*.

## Security Data Planning

The requirements for IPsec data planning are as follows:



- **AUTHMETH (LTE eNodeB, 5G gNodeB)** in the **IKEPROPOSAL** MO must be set to **IKE\_CERT\_SIG**.
- It is recommended that **IDTYPE (LTE eNodeB, 5G gNodeB)** in the **IKEPEER** MO be set to **FQDN**. **REMOTENAME (LTE eNodeB, 5G gNodeB)** in the **IKEPEER** MO must be the same as **Local Name** of the SeGW.
- Other IPsec parameter settings on the eNodeB remain unchanged.

PKI data planning is the same as that for newly deployed base stations. For details, see the "Data Preparation" section in *PKI*.

## Preparing the Incremental Script

For details, see [Preparing the Incremental Script](#) in [Preparing the Incremental Script](#).

## Checking the Base Station Environment

- The base station meets the hardware requirements described in [4.3.3 Hardware](#).
- The license for the IPsec feature has been activated on the base station.
- The base station has been preconfigured with a Huawei initial certificate and the Huawei root certificate.
  - Check for Huawei initial certificates  
Run the **DSP BRDCERTMK** command with **CERTFILETYPE** set to **DEVCERT**. If **VendorCertFlag = YES** is displayed in the command output, the board has a Huawei initial certificate.
  - Check for Huawei root certificate  
Run the **DSP BRDCERTMK** command with **CERTFILETYPE** set to **TRUSTCERT**. If **VendorCertFlag = YES** is displayed in the command output, the board has a Huawei root certificate.

## Downloading the Incremental Script

For details, see [Downloading the Incremental Script](#) in [4.5.1 Reconstruction from an Insecure Network to a PKI-based Secure Network](#).

## Modifying SeGW Configurations

Security-related parameters on the SeGW, such as the authentication method, need to be modified by the SeGW engineering personnel, in order to be consistent with the base station security configuration.

## Establishing IPsec Tunnels

The base station and the SeGW perform IKE negotiation to establish an IPsec tunnel. When the OM channel between the base station and the MAE is successfully established, network reconstruction succeeds.

## Activation Verification

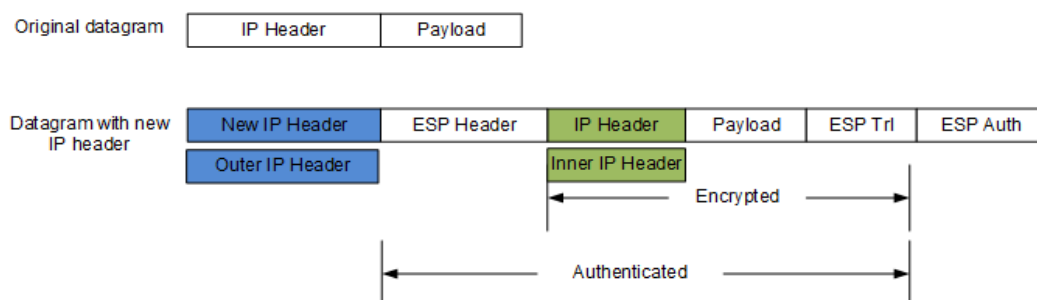
For details, see [Activation Verification](#) in [4.5.1 Reconstruction from an Insecure Network to a PKI-based Secure Network](#).

# 5 IPv6 IPsec

## 5.1 Principles

### 5.1.1 Introduction

IPsec applies to both IPv4 and IPv6 packets.



Based on the IP versions of the inner and outer IP headers, the following functions in tunnel mode are available:

- IPv4 IPsec: IPv4 for both the inner and outer IP headers
- IPv6-over-IPv4 IPsec: IPv6 for the inner IP header and IPv4 for the outer IP header
- IPv4-over-IPv6 IPsec: IPv4 for the inner IP header and IPv6 for the outer IP header
- IPv6 IPsec: IPv6 for both the inner and outer IP headers

For details about IPv4 IPsec, see [4 IPv4 IPsec](#). IPv6-over-IPv4 IPsec and IPv4-over-IPv6 IPsec are mainly used for evolution from IPv4 networking to IPv6 networking. They are not recommended for deployment over a long period. IPv6-over-IPv4 IPsec, IPv4-over-IPv6 IPsec, and IPv6 IPsec support only the tunnel mode.

 NOTE

- The IPv4-over-IPv6 IPsec configuration is similar to the IPv6 IPsec configuration. Compared with the IPv6 IPsec configuration, the IPv4-over-IPv6 IPsec configuration additionally requires the configuration of the **IPROUTE4** or **SRCIPROUTE4** MO, which enables import of service data flows to a TUNNELITF interface to which an IPsec policy is bound and is used for association with the **ACL** MO.
- The IPv6-over-IPv4 IPsec configuration is similar to the IPv4 IPsec configuration. Compared with the IPv4 IPsec configuration, the IPv6-over-IPv4 IPsec configuration additionally requires the configuration of the **IPROUTE6** or **SRCIPROUTE6** MO, which enables import of service data flows to an IPv4 interface (in Ethernet-interface-based configuration mode) or a TUNNELITF interface (in tunnel-interface-based configuration mode) to which an IPsec policy is bound and is used for association with the **ACL6** MO.

For details about common functions of IPv4 IPsec, IPv6-over-IPv4 IPsec, IPv4-over-IPv6 IPsec, and IPv6 IPsec, see [4.1 Principles](#). [Table 5-1](#) describes the function differences.

**Table 5-1** Function differences for IPv4 IPsec, IPv6-over-IPv4 IPsec, IPv4-over-IPv6 IPsec, and IPv6 IPsec

| Function                              | IPv4 IPsec (in Ethernet-Interface-based Configuration Mode) | IPv4 IPsec (in Tunnel-Interface-based Configuration Mode) | IPv6-over-IPv4 IPsec (in Ethernet-Interface-based Configuration Mode) | IPv6-over-IPv4 IPsec (in Tunnel-Interface-based Configuration Mode) | IPv4-over-IPv6 IPsec | IPv6 IPsec    |
|---------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------|----------------------|---------------|
| IPsec redundancy among multiple SeGWs | Supported                                                   | Supported                                                 | Supported                                                             | Supported                                                           | Supported            | Supported     |
| IPsec tunnel backup                   | Supported                                                   | Not supported                                             | Not supported                                                         | Not supported                                                       | Not supported        | Not supported |
| Tunnel interface                      | Not supported                                               | Supported                                                 | Not supported                                                         | Supported                                                           | Supported            | Supported     |
| IPsec redirection                     | Supported                                                   | Not supported                                             | Supported                                                             | Not supported                                                       | Not supported        | Not supported |
| NAT traversal                         | Supported                                                   | Supported                                                 | Supported                                                             | Supported                                                           | Not supported        | Not supported |

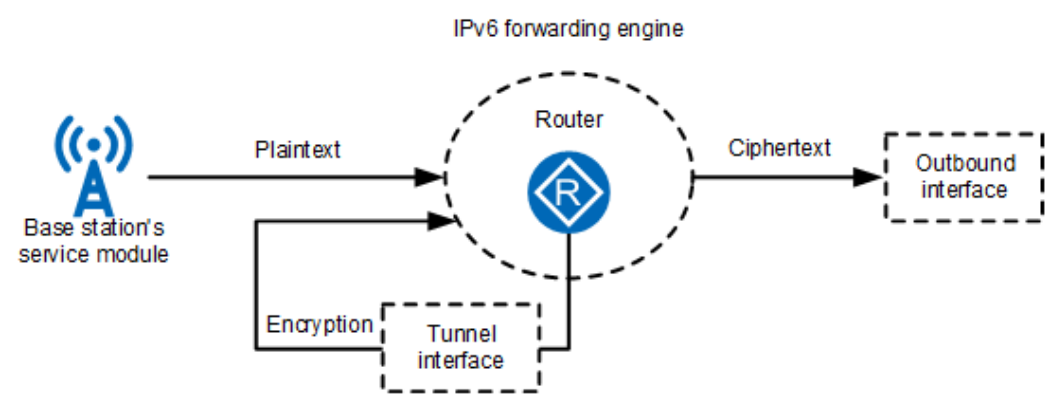
| Function                                                                                                                                                                                                                                               | IPv4 IPsec (in Ethernet-Interface-based Configuration Mode) | IPv4 IPsec (in Tunnel-Interface-based Configuration Mode) | IPv6-over-IPv4 IPsec (in Ethernet-Interface-based Configuration Mode) | IPv6-over-IPv4 IPsec (in Tunnel-Interface-based Configuration Mode) | IPv4-over-IPv6 IPsec | IPv6 IPsec    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------|----------------------|---------------|
| Old transmission model                                                                                                                                                                                                                                 | Supported                                                   | Not supported                                             | Not supported                                                         | Not supported                                                       | Not supported        | Not supported |
| New transmission model                                                                                                                                                                                                                                 | Supported                                                   | Supported                                                 | Supported                                                             | Supported                                                           | Supported            | Supported     |
| Note: The old and new transmission models are specified by the <b>GTRANSPARA.TRANS CFGMODE (LTE eNodeB, 5G gNodeB)</b> parameter. The IPv4 IPsec model configuration is specified by the <b>IKECFG.IPSECCFGMODE (LTE eNodeB, 5G gNodeB)</b> parameter. |                                                             |                                                           |                                                                       |                                                                     |                      |               |

5.1.2 ACL Rules

For details about how to configure IPv6 ACL rules, see [ACL Rule Configuration Mode](#).

If an IPv6 ACL rule used by an IPsec policy has **Protocol Type** set to a non-IP type, **Match Source Port** set to **YES**, or **Match Destination Port** set to **YES**, ensure that the IPv6 ACL rule contains path maximum transmission unit (PMTU) data flows. Otherwise, PMTU data flows will be discarded and PMTU discovery will fail.

5.1.3 IPv6 Tunnel Interfaces



The base station's service module sends plaintext to the IPv6 forwarding engine. The process is as follows:

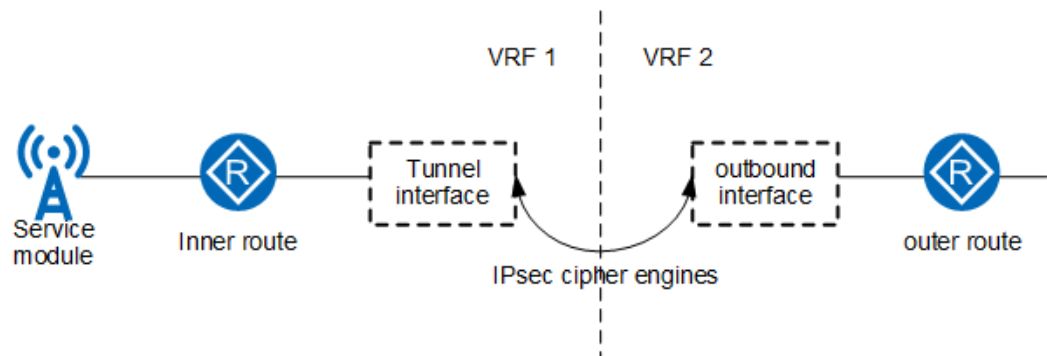
- Step 1** The base station queries the route first. If the next hop of the route is a tunnel interface, the base station proceeds to the next step. Otherwise, the plaintext is sent directly.
- Step 2** The base station matches the plaintext with the IPsec ACL rule. If matching succeeds, the base station proceeds to the next step. If matching fails, the plaintext is discarded.
- Step 3** The base station encrypts the plaintext.
- Step 4** The base station queries the route again and sends the encrypted packet to the outbound interface.

----End

## 5.1.4 IPv6 VRF-based isolation

When outer packets are configured with IPv6 headers, VRFs can be used by IPsec to isolate inner packets from outer packets, as shown in [Figure 5-1](#).

**Figure 5-1** VRF-based Isolation



Inner and outer VRFs are configured to isolate inner IPsec service packets from outer IPsec service packets, preventing the inner service modules protected by IPsec from network attacks.

### NOTE

To ensure intra-site communications in co-MPT scenarios, it is recommended that the IPv4 service addresses of all RATs in a co-MPT base station be configured in one VRF and the IPv6 service addresses of all RATs be configured in another VRF.

## 5.2 Network Analysis

### 5.2.1 Benefits

This feature improves the security of base station data transmission in IPv6 networking.

## 5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.1.3 Function Impacts](#), [5.3.2.2.3 Function Impacts](#), [5.3.2.3.3 Function Impacts](#), and [5.3.2.4.3 Function Impacts](#).

For details about the impact on the network, see [4.2.2 Impacts](#).

## 5.3 Requirements

### 5.3.1 Licenses

| RAT     | Feature ID    | Feature Name           | Model        | Sales Unit |
|---------|---------------|------------------------|--------------|------------|
| LTE FDD | LOFD-003024   | IPsec for IPv6         | LT1SIIPV600  | Per eNodeB |
| LTE FDD | MLOFD-003024  | IPsec for IPv6         | ML1SIIPV600  | Per eNodeB |
| LTE TDD | TDLOFD-003024 | IPsec for IPv6         | LT1STIPSEC06 | Per eNodeB |
| NR      | FOFD-010080   | IPsec (IPsec for IPv6) | NR0S0IPSEC00 | Per gNodeB |

#### NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

### 5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

#### 5.3.2.1 LOFD-003024 IPsec for IPv6

### 5.3.2.1.1 Prerequisite Functions

| RAT     | Function Name                   | Function Switch | Reference  | Description                                                                       |
|---------|---------------------------------|-----------------|------------|-----------------------------------------------------------------------------------|
| LTE FDD | Public Key Infrastructure (PKI) | None            | <i>PKI</i> | This function is required if IPsec uses digital-certificate-based authentication. |

### 5.3.2.1.2 Mutually Exclusive Functions

None

### 5.3.2.1.3 Function Impacts

None

## 5.3.2.2 TDLOFD-003024 IPsec for IPv6

### 5.3.2.2.1 Prerequisite Functions

| RAT     | Function Name                   | Function Switch | Reference  | Description                                                                       |
|---------|---------------------------------|-----------------|------------|-----------------------------------------------------------------------------------|
| LTE TDD | Public Key Infrastructure (PKI) | None            | <i>PKI</i> | This function is required if IPsec uses digital-certificate-based authentication. |

### 5.3.2.2.2 Mutually Exclusive Functions

None

### 5.3.2.2.3 Function Impacts

None

## 5.3.2.3 MLOFD-003024 IPsec for IPv6



### 5.3.2.3.1 Prerequisite Functions

| RAT    | Function Name                   | Function Switch | Reference  | Description                                                                       |
|--------|---------------------------------|-----------------|------------|-----------------------------------------------------------------------------------|
| NB-IoT | Public Key Infrastructure (PKI) | None            | <i>PKI</i> | This function is required if IPsec uses digital-certificate-based authentication. |

### 5.3.2.3.2 Mutually Exclusive Functions

None

### 5.3.2.3.3 Function Impacts

None

### 5.3.2.4 FOFD-010080 IPsec (IPsec for IPv6)

#### 5.3.2.4.1 Prerequisite Functions

| RAT | Function Name            | Function Switch | Reference  | Description                                                                       |
|-----|--------------------------|-----------------|------------|-----------------------------------------------------------------------------------|
| NR  | Security Mechanism (PKI) | None            | <i>PKI</i> | This function is required if IPsec uses digital-certificate-based authentication. |

#### 5.3.2.4.2 Mutually Exclusive Functions

None

#### 5.3.2.4.3 Function Impacts

None

## 5.3.3 Hardware

### Base Station Models

| RAT | Base Station Model                                                                                                                                                                                                                                                            |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LTE | <ul style="list-style-type: none"><li>3900 and 5900 series base stations (macro base stations)</li><li>DBS3900 LampSite and DBS5900 LampSite</li></ul>                                                                                                                        |
| NR  | <ul style="list-style-type: none"><li>3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.</li><li>DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.</li></ul> |

### Boards

Only the UMPT/UMDU boards support this function.

### RF Modules

This function does not depend on RF modules.

## 5.3.4 Others

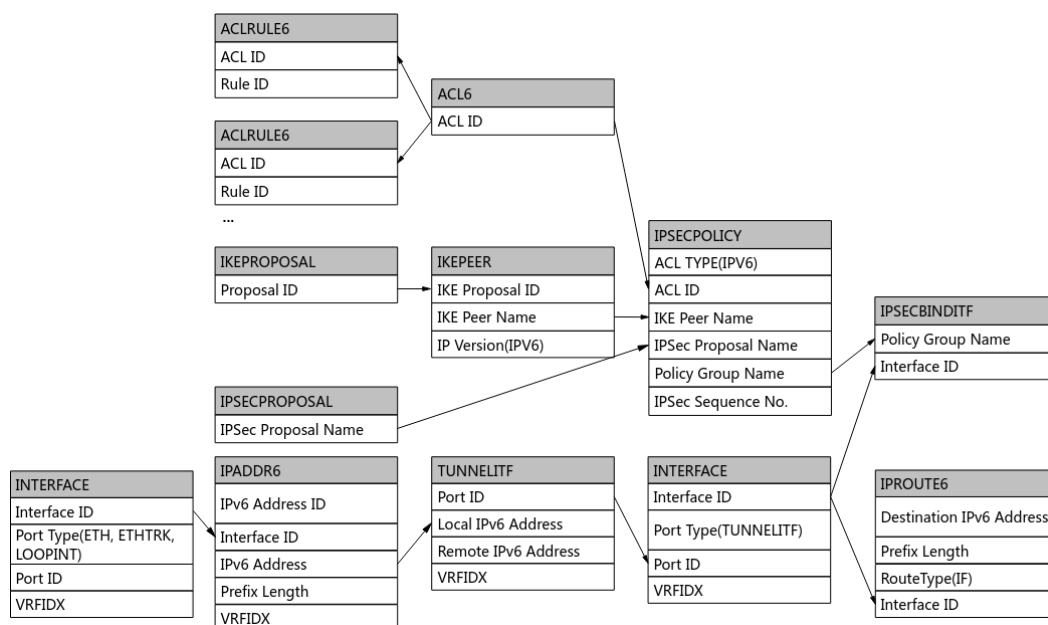
Peer NEs, such as the core network, MAE, SeGW, and clock server, must support the IPv6 protocol. If the peer NEs do not support the IPv6 protocol, IPv6-related MOs cannot be configured. Only IPv4-related MOs can be configured.

## 5.4 Operation and Maintenance

### 5.4.1 Data Configuration

#### 5.4.1.1 Data Preparation

IPsec tunnels can be configured to secure transmission of base station data. The following figure shows the relationship between configuration models.

**Figure 5-2** IPv6 IPsec configuration principles

As shown in [Figure 5-2](#), binding between the **TUNNELITF** and **INTERFACE** MOs and binding between the **IPROUTE6** and **INTERFACE** MOs are added for IPv6 IPsec. Other binding relationships are the same as those for IPv4 IPsec. For details, see [4.4.2 Data Configuration](#). Secure networking scenario examples are similar in IPv6 and IPv4 IPsec networking. The only difference lies in IP addresses. For details, see secure networking scenarios in [4.4.2 Data Configuration](#).

The functions of the configuration models are as follows:

- The **IKEPROPOSAL**, **IKEPEER**, **IPSECPROPOSAL**, and **IPSECPOLICY** MOs define information necessary for security negotiation between the base station and peer security equipment.
- The **ACL6** and **ACLRULE6** MOs define ACL rules.
- The **IPROUTE6** MO defines static IPv6 routes.
- The **TUNNELITF** and **INTERFACE** MOs define IPsec tunnel interfaces.

Configure the **IPGUARD**, **IKEPROPOSAL**, and **IPSECPROPOSAL** MOs. For details, see [4.4.2 Data Configuration](#).

Configure an **IKEPEER** MO with **IP Version** set to **IPV6**. [Table 5-2](#) describes the key parameters in this MO.

**Table 5-2** Data to be prepared for an IKE peer

| Parameter Name | Parameter ID                                       | Setting Notes                                                                                                                                                                           |
|----------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IKE Peer Name  | <b>IKEPEER.PEERNAME</b><br>(5G gNodeB, LTE eNodeB) | Set this parameter based on the network plan. This parameter specifies the name that is used to represent this MO in the base station. The value of this parameter is case-insensitive. |

| Parameter Name  | Parameter ID                                             | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IKE Proposal ID | <b><i>IKEPEER.PROPID (5G gNodeB, LTE eNodeB)</i></b>     | Set this parameter based on the network plan. This parameter specifies the ID of the IKE proposal used by the base station. This parameter is used to associate the <b><i>IKEPEER</i></b> MO with the <b><i>IKEPROPOSAL.PROPID (5G gNodeB, LTE eNodeB)</i></b> parameter. Before configuring an <b><i>IKEPEER</i></b> MO, ensure that the associated <b><i>IKEPROPOSAL</i></b> MO has been configured. |
| Version         | <b><i>IKEPEER.IKEVERSION (5G gNodeB, LTE eNodeB)</i></b> | Set this parameter based on the network plan.<br>You can run <b>SET SECPLY</b> to adjust the configuration security level to prohibit the parameter from being set to <b>IKE_V1</b> .                                                                                                                                                                                                                  |
| Exchange Mode   | <b><i>IKEPEER.EXCHMODE (LTE eNodeB, 5G gNodeB)</i></b>   | This parameter needs to be set only when <b><i>IKEPEER.IKEVERSION (5G gNodeB, LTE eNodeB)</i></b> is set to <b>IKE_V1</b> . Set this parameter based on the network plan.                                                                                                                                                                                                                              |
| IP Version      | <b><i>IKEPEER.IPVERSION (5G gNodeB, LTE eNodeB)</i></b>  | Set this parameter to <b>IPV6</b> .                                                                                                                                                                                                                                                                                                                                                                    |
| Local ID Type   | <b><i>IKEPEER.IDTYPE (5G gNodeB, LTE eNodeB)</i></b>     | Set this parameter based on the network plan. This parameter specifies the authentication mode used when the base station performs IKE negotiation with the peer. If the <b><i>IKEPEER.EXCHMODE (5G gNodeB, LTE eNodeB)</i></b> parameter is set to <b>MAIN</b> , this parameter must be set to <b>IPV6</b> .                                                                                          |
| Remote Name     | <b><i>IKEPEER.REMOTENAME (5G gNodeB, LTE eNodeB)</i></b> | If this parameter is specified and the <b><i>IKEPROPOSAL.AUTHMETH (5G gNodeB, LTE eNodeB)</i></b> parameter is set to <b>IKE_CERT_SIG</b> , the base station will verify the peer ID during IKE negotiation with the SeGW. If this parameter is not specified, the base station will not verify the peer ID.                                                                                           |
| Pre-shared Key  | <b><i>IKEPEER.PKEY (5G gNodeB, LTE eNodeB)</i></b>       | If the <b><i>IKEPROPOSAL.AUTHMETH (5G gNodeB, LTE eNodeB)</i></b> parameter is set to <b>PRE_SHARED_KEY</b> , set this parameter based on the network plan.                                                                                                                                                                                                                                            |

| Parameter Name              | Parameter ID                                          | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DPD Mode                    | <b>IKEPEER.DPD (5G gNodeB, LTE eNodeB)</b>            | If dead peer detection (DPD) is used to check the status of the IKE SA that consists of the local base station and the peer, these parameters must be set based on the network plan.<br><br>DPD is recommended because it occupies a relatively small bandwidth. It is also recommended that the values of these parameters be the same at two ends of the IKE SA. If the values are different, the DPD configuration with a shorter timer length takes effect before the other does.                         |
| DPD Idle Time               | <b>IKEPEER.DPDIDLETIME (5G gNodeB, LTE eNodeB)</b>    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| DPD Retransmission Interval | <b>IKEPEER.DPDRETRI (5G gNodeB, LTE eNodeB)</b>       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| DPD Retransmission Count    | <b>IKEPEER.DPDRETRN (5G gNodeB, LTE eNodeB)</b>       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Redundancy Flag             | <b>IKEPEER.REDUNDANCYFLAG (5G gNodeB, LTE eNodeB)</b> | If this parameter is set to <b>NONE</b> , IPsec redundancy among multiple SeGWs is disabled.                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Certificate Source          | <b>IKEPEER.CERTSOURCE (5G gNodeB, LTE eNodeB)</b>     | Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Certificate File Name       | <b>IKEPEER.CERTNAME (5G gNodeB, LTE eNodeB)</b>       | This parameter must be set when the <b>IKEPEER.CERTSOURCE (5G gNodeB, LTE eNodeB)</b> parameter is set to <b>CERTMK</b> . Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                                                                       |
| IPSec PreFragment Switch    | <b>IKEPEER.IPSECPREFRGSW (5G gNodeB, LTE eNodeB)</b>  | In PMTU passive mode (that is, <b>PMTUCFG.MODE (5G gNodeB, LTE eNodeB)</b> is set to <b>PASSIVE</b> ), if <b>IKEPEER.IPSECPREFRGSW (5G gNodeB, LTE eNodeB)</b> is set to <b>OFF</b> , inner IP packets will still be fragmented and the MTU size is always set to 1380. If the MTU 1380 does not meet customer requirements, set <b>IKEPEER.IPSECPREFRGSW (5G gNodeB, LTE eNodeB)</b> to <b>ON</b> and specify the MTU size of the tunnel interface.<br><br>In other cases, the default value is recommended. |

| Parameter Name             | Parameter ID                                                  | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Trust Group ID             | <b>IKEPEER.TRUSTGROUPID (LTE eNodeB, 5G gNodeB)</b>           | This parameter specifies the ID of a trust group. When certificate-based authentication is used for link negotiation or establishment, the trust certificates in the trust group configured in the <b>TRUSTGROUP</b> MO are used to authenticate the peer end. If this parameter is set to <b>65535</b> , no trust group is associated, and all loaded trust certificates in the system are used to authenticate the device certificates of the peer end.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| IKEv2 Fragmentation Switch | <b>IKEPEER.IKEV2FRAGMENTATION (5G gNodeB, LTE eNodeB)</b>     | This parameter specifies whether to enable IKEv2 fragmentation negotiation. For details about this function, see RFC 7383.<br><br>Whether IKEv2 fragmentation takes effect depends on the negotiation result, which can be viewed in the <b>DSP IKESA</b> command output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| IKEv2 Fragmentation MTU    | <b>IKEPEER.IKEV2FRAGMENTATION_MTU (LTE eNodeB, 5G gNodeB)</b> | This parameter specifies the maximum transmission unit of IKEv2 fragments for the IKE peer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| OCSP Switch                | <b>IKEPEER.OCSPSW (LTE eNodeB, 5G gNodeB)</b>                 | This parameter specifies whether the local end uses the IKE OCSP (for details, see RFC 4806) to check the certificate revocation status of the peer end. When this switch is turned on, IKE messages sent by the local end contain OCSP requests, requiring the IKE messages sent by the peer end to contain OCSP responses. This switch applies only to IKEv2 and takes effect only when the <b>IKEPROPOSAL.AUTHMETHOD (LTE eNodeB, 5G gNodeB)</b> parameter is set to <b>IKE_CERT_SIG</b> . The effective status of this switch can be queried by running the <b>DSP IKESA</b> command.<br><br>You are advised to check whether the peer SeGW supports IKE OCSP, and set this parameter to <b>ON</b> if the peer SeGW supports IKE OCSP and retain the default value <b>OFF</b> if it does not. Setting this parameter to <b>ON</b> while the peer SeGW does not support IKE OCSP brings IKE disconnection risks. |

| Parameter Name   | Parameter ID                                        | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CRL using policy | <b>IKEPEER.CRLPOLICY</b><br>(LTE eNodeB, 5G gNodeB) | <p>This parameter specifies the CRL using policy for the base station in IKE scenarios.</p> <ul style="list-style-type: none"><li>When this parameter is set to <b>GLOBAL</b>, the CRL using policy for the base station in IKE scenarios is the same as that specified in the <b>SET CRLPOLICY</b> command.</li><li>When this parameter is set to <b>ALARM</b>, the base station in IKE scenarios reports ALM-26834 Peer Certificate Invalid if the CRL- or OCSP-based check result indicates that the peer certificate has been revoked.</li><li>When this parameter is set to <b>DISCONNECT</b>, the base station in IKE scenarios terminates link negotiation and reports ALM-26834 Peer Certificate Invalid if the CRL- or OCSP-based check result indicates that the peer certificate has been revoked; the base station in IKE scenarios only terminates link negotiation if the check result indicates that the CRL has not taken effect or has expired.</li></ul> |

Configure **ACL6** MOs to manage **ACLRULE6** MOs. [Table 5-3](#) describes the key parameters in the **ACL6** MO.

**Table 5-3** Data preparation for ACL6

| Parameter Name | Parameter ID                              | Setting Notes                                                                                                                                                                             |
|----------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IPv6 ACL ID    | <b>ACL6.ACLID</b> (5G gNodeB, LTE eNodeB) | Set this parameter based on the network plan. If an ACL is referenced by an IPsec policy or packet filtering, the <b>ACL6</b> MO must be associated with at least one <b>ACLRULE6</b> MO. |

Configure IPv6 ACL rules. Add an **ACLRULE6** MO to define data flows to be protected. The configurations of the **ACLRULE6** MOs under the same **ACL6** MO must be different. [Table 5-4](#) describes the key parameters.

**Table 5-4** Data preparation for ACLRULE6

| Parameter Name                    | Parameter ID                                         | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IPv6 ACL ID                       | <b>ACLRULE6.ACLID</b> (5G gNodeB, LTE eNodeB)        | This parameter specifies the ID of the IPv6 ACL to which the <b>ACLRULE6</b> MO belongs.                                                                                                                                                                                                                                                                                                     |
| IPv6 Rule ID                      | <b>ACLRULE6.RULEID</b> (5G gNodeB, LTE eNodeB)       | Set this parameter based on the network plan. In one ACL, each ACL rule must have a unique ID.                                                                                                                                                                                                                                                                                               |
| IPv6 ACL Action                   | <b>ACLRULE6.ACTION</b> (5G gNodeB, LTE eNodeB)       | Set this parameter to <b>PERMIT</b> . Do not set it to <b>DENY</b> .                                                                                                                                                                                                                                                                                                                         |
| Protocol Type                     | <b>ACLRULE6.PT</b> (5G gNodeB, LTE eNodeB)           | Set this parameter based on the network plan. This parameter specifies the type of the protocol corresponding to an ACL rule.                                                                                                                                                                                                                                                                |
| Source IPv6 Address               | <b>ACLRULE6.SIP</b> (5G gNodeB, LTE eNodeB)          | Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                                                                                |
| Destination IPv6 Address          | <b>ACLRULE6.DIP</b> (5G gNodeB, LTE eNodeB)          | Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                                                                                |
| Support Forcible Execution        | <b>ACLRULE6.FORCEEXECUTE</b> (5G gNodeB, LTE eNodeB) | Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                                                                                |
| Source Address Prefix Length      | <b>ACLRULE6.SPFLEN</b> (5G gNodeB, LTE eNodeB)       | Set this parameter based on the network plan. The <b>ACLRULE6.SPFLEN</b> (5G gNodeB, LTE eNodeB) and <b>ACLRULE6.DPFLEN</b> (5G gNodeB, LTE eNodeB) parameters are specified in each ACL rule. The IPv6 prefix length is not a wildcard mask. When the prefix length of the source address or destination address is set to 0, the source or destination IPv6 address can match any address. |
| Destination Address Prefix Length | <b>ACLRULE6.DPFLEN</b> (5G gNodeB, LTE eNodeB)       |                                                                                                                                                                                                                                                                                                                                                                                              |

Configure IPv6 IPsec policies. Add an **IPSECPOLICY** MO with **ACL Type** set to **IPv6** to configure the range of data to be protected and the protection policy. [Table 5-5](#) describes the key parameters.



**Table 5-5** Data preparation for IPSECPOLICY

| Parameter Name          | Parameter ID                                                   | Setting Notes                                                                                                                                                                                                                                                                                                                 |
|-------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Policy Group Name       | <b>IPSECPOLICY.SPGN</b><br><i>(5G gNodeB, LTE eNodeB)</i>      | Set this parameter based on the network plan. This parameter specifies the name of an IPsec policy group. The value of this parameter is case-insensitive. An IPsec policy group consists of multiple IPsec policies. Each IPsec policy is configured with an IPsec proposal.                                                 |
| IPSec Sequence No.      | <b>IPSECPOLICY.SPSN</b><br><i>(5G gNodeB, LTE eNodeB)</i>      | Set this parameter based on the network plan. Only one IPsec policy can be configured in an IPv6 security policy group.                                                                                                                                                                                                       |
| ACL Type                | <b>IPSECPOLICY.ACLTYP</b><br><i>E (5G gNodeB, LTE eNodeB)</i>  | Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                 |
| ACL ID                  | <b>IPSECPOLICY.ACLID</b><br><i>(5G gNodeB, LTE eNodeB)</i>     | Set this parameter based on the network plan. This parameter can be set to an IPv4 or IPv6 ACL ID. The ACL specified by this parameter must have been configured.                                                                                                                                                             |
| IPSec Proposal Name     | <b>IPSECPOLICY.PROPN</b><br><i>AME (5G gNodeB, LTE eNodeB)</i> | Set this parameter based on the network plan. Ensure that the name is correct when this parameter is referenced. An incorrect setting of this parameter will result in implementation failures.                                                                                                                               |
| IKE Peer Name           | <b>IPSECPOLICY.PEERNAME</b><br><i>(5G gNodeB, LTE eNodeB)</i>  | Set this parameter based on the network plan. The IKE peer specified by this parameter must have been configured.                                                                                                                                                                                                             |
| Perfect Forward Secrecy | <b>IPSECPOLICY.PFS</b> <i>(5G gNodeB, LTE eNodeB)</i>          | Set this parameter based on the network plan.<br><br>It is recommended that the PFS_GROUP15, PFS_GROUP19, PFS_GROUP20, or PFS_GROUP31 algorithm be used.<br><br>You can run <b>SET SECPLY</b> to adjust the configuration security level to prohibit the parameter from being set to <b>PFS_GROUP1</b> or <b>PFS_GROUP2</b> . |
| SA Duration Mode        | <b>IPSECPOLICY.LTCFG</b><br><i>(5G gNodeB, LTE eNodeB)</i>     | Set this parameter based on the network plan. The value <b>GLOBAL</b> is recommended.                                                                                                                                                                                                                                         |

| Parameter Name         | Parameter ID                                   | Setting Notes                                                                                                                                                           |
|------------------------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lifetime Based On Time | <b>IPSECPOLICY.LTS</b> (5G gNodeB, LTE eNodeB) | Set this parameter based on the network plan. This parameter is valid only when the <b>IPSECPOLICY.LTCFG</b> (5G gNodeB, LTE eNodeB) parameter is set to <b>LOCAL</b> . |

Configure a **TUNNELITF** MO to create an IPsec tunnel interface. [Table 5-6](#) describes the key parameters.

**Table 5-6** Data preparation for TUNNELITF

| Parameter Name      | Parameter ID                                       | Setting Notes                                                                                                       |
|---------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Mode                | <b>TUNNELITF.MODE</b> (5G gNodeB, LTE eNodeB)      | Set this parameter to <b>IPSEC_IPV6</b> .                                                                           |
| Local IPv6 Address  | <b>TUNNELITF.LOCALIP6</b> (5G gNodeB, LTE eNodeB)  | Set this parameter based on the network plan. This parameter specifies the IPv6 address of a base station.          |
| Remote IPv6 Address | <b>TUNNELITF.REMOTEIP6</b> (5G gNodeB, LTE eNodeB) | Set this parameter based on the network plan.                                                                       |
| VRF Index           | <b>TUNNELITF.VRFIDX</b> (5G gNodeB, LTE eNodeB)    | Set this parameter to a value different from that of the <b>INTERFACE.VRFIDX</b> (5G gNodeB, LTE eNodeB) parameter. |

Configure an **INTERFACE** MO with **Port Type** set to **TUNNELITF**. [Table 5-7](#) describes the key parameters.

**Table 5-7** Data preparation for INTERFACE

| Parameter Name | Parameter ID                                    | Setting Notes                                                                                                             |
|----------------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Port Type      | <b>INTERFACE.PT</b> (5G gNodeB, LTE eNodeB)     | Set this parameter to <b>TUNNELITF</b> .                                                                                  |
| VRF Index      | <b>INTERFACE.VRFIDX</b> (5G gNodeB, LTE eNodeB) | The value of this parameter must be different from that of the <b>TUNNELITF.VRFIDX</b> (5G gNodeB, LTE eNodeB) parameter. |

Configure an **IPROUTE6** MO. Set the next hop of plaintext data flows to the TUNNELITF interface generated in the previous step so that the plaintext data flows can enter the IPsec tunnel. [Table 5-8](#) describes the key parameters.

**Table 5-8** Data preparation for IPROUTE6

| Parameter Name | Parameter ID                                      | Setting Notes                                                                                     |
|----------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Route Type     | <b>IPROUTE6.RTTYPE</b><br>(5G gNodeB, LTE eNodeB) | Set this parameter to <b>IF</b> .                                                                 |
| Interface ID   | <b>IPROUTE6.ITFID</b> (5G gNodeB, LTE eNodeB)     | Set this parameter to the number of the TUNNELITF interface generated in the previous step.       |
| VRF Index      | <b>IPROUTE6.VRFIDX</b><br>(5G gNodeB, LTE eNodeB) | Set this parameter to the value of the <b>INTERFACE.VRFIDX</b> (5G gNodeB, LTE eNodeB) parameter. |

 **NOTE**

When IPsec redundancy among multiple SeGWs is enabled in IPv6 scenarios:

- For IPv6 IPsec, a route with the corresponding **IKEPEER.REDUNDANCYFLAG** (5G gNodeB, LTE eNodeB) parameter set to **MASTER** needs to be configured as the active route by running the **ADD IPROUTE6** command, and a standby route with the corresponding **IKEPEER.REDUNDANCYFLAG** (5G gNodeB, LTE eNodeB) parameter set to **SLAVE** can be automatically generated. The standby route is automatically generated only when the standby IKE peer takes effect. The generated routes can be queried by running the **DSP IPROUTE6** command.
- For IPv4-over-IPv6 IPsec, the active route needs to be configured by running the **ADD IPROUTE4** command, and a standby route is automatically generated. All these routes can be queried by running the **DSP IPROUTE4** command.

Configure IPv6 IPsec policy group binding relationships. Add an **IPSECBINDITF** MO to bind an IPsec policy group to an IPv6 tunnel interface. [Table 5-9](#) describes the key parameters.

**Table 5-9** Data preparation for the binding between an IPsec policy and a port

| Parameter Name          | Parameter ID                                               | Setting Notes                                                                                                                                                                                                |
|-------------------------|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IPSec Bind Interface ID | <b>IPSECBINDITF.IPSECBINDITFID</b> (5G gNodeB, LTE eNodeB) | Set this parameter based on the network plan. This parameter specifies the index of the IPv6 interface to which the IPsec policy group is applied. The port type of the IPv6 interface is <b>TUNNELITF</b> . |

| Parameter Name    | Parameter ID                                         | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Policy Group Name | <b>IPSECBINDITF.SPGN<br/>(5G gNodeB, LTE eNodeB)</b> | Set this parameter based on the network plan. When an IPsec policy group is bound to a physical port, all policies in the group are applied to the port. This enables different SAs to protect different types of packets.<br><br>After an IPsec policy group is bound to an Ethernet port, no new IPsec policy can be added to the group and no modifications can be made to the existing IPsec policies in the group. |

### 5.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

### Activation Command Examples

```
//Adding an IKE proposal
ADD IKEPROPOSAL: PROPID=10, ENCALG=AES128, AUTHALG=SHA256, AUTHMETH=IKE_CERT_SIG,
DHGRP=DH_GROUP14, DURATION=86400, REAUTHSW=ON, REAUTHLT=604800;
//Adding an IKE peer
ADD IKEPEER: PEERNAME="ike", PROPID=10, IKEVERSION=IKE_V2, IPVERSION=IPV6, IDTYPE=IPV6,
REMOTENAME="secgw", DPD=PERIODIC, DPDIDLETIME=10, DPDRETRI=5, DPDRETRN=5,
CERTSOURCE=APPCERT, IPSECPREFRGSW=OFF, TRUSTGROUPID=65535, OCSPSW=OFF, CRLPOLICY=GLOBAL;
//Adding an ACL
ADD ACL6: ACLID=3000, ACLDESC="for IPsec";
//Adding ACL rules to the ACL so that the UMPT encrypts or decrypts the following data flows that pass
through the transmission ports on the UMPT
//Setting ACL rule 1 (RULEID = 1) for eNodeB signaling and service data flows
ADD ACLRULE6: ACLID=3000, RULEID=1, ACTION=PERMIT, PT=IP6, SIP="2002::10:0:1", SPFXLEN=96,
DIP="0::0", DPFXLEN=0, MDSCP=NO;
//Setting ACL rule 2 (RULEID = 2) for OM data flows and certificate management-related data flows
(including those generated during the interaction with the digital certificate and CRL database) of the
eNodeB when OM channels are protected by IPsec. If the OM channel does not need to be protected by
IPsec, the source IP address for certificate management-related data flows must be different from the OM
IP address. You are advised to configure a new internal-network IP address (for example, 2002::11:45:45) as
the source IP address for certificate management-related data flows. Therefore, the SIP in the following ACL
rule needs to be changed to 2002::11:45:45.
ADD ACLRULE6: ACLID=3000, RULEID=2, ACTION=PERMIT, PT=IP6, SIP="2002::11:0:1", SPFXLEN=96,
DIP="0::0", DPFXLEN=0, MDSCP=NO;
//Adding an IPsec proposal
ADD IPSECPROPOSAL: PROPNAME="prop0", TRANMODE=ESP;
//Adding an IPsec policy
ADD IPSECPOLICY: SPGN="Policy0", SPSN=1, ACLTYPE=IPV6, ACLID=3000, PROPNAME="prop0",
PEERNAME="ike", LTCFG=LOCAL, LTS=86400, NARROWDOWNSW=ON;
```

```
//Adding a tunnel interface
ADD TUNNELITF: PORTID=8, MODE=IPSEC_IPV6, LOCALIPREFMETHOD=IPADDR, VRFIDX=1,
LOCALIP6="2002::12:0:1", REMOTEIP6="2002::12:0:2";
ADD INTERFACE: ITFID=8, ITFTYPE=NORMAL, PT=TUNNELITF, PORTID=8, VRFIDX=2;
//Binding the IPsec policy to an outbound interface
ADD IPSECBINDITF: IPSECBINDITFID=0, ITFID=8, SPGN="Policy0";
//Configuring routes to the internal-network IP addresses (data flows matching the routes can enter the
IPsec tunnel)
ADD IPROUTE6: RTIDX=1, DSTIP="2183::", PFXLEN=16, RTTYPE=IF, ITFID=8, PREF=1, VRFIDX=2;
//(Optional) Setting basic IKE information
SET IKECFG: IKELNM="IKECFG1", NATKLI=20, DSCP=48, IPSECSWITCHBACK=OFF;
//(Optional) Setting the IPsec replay alarm switch and threshold
SET IPGUARD: IPSECREPLAYCHKSW=ENABLE, IPSECREPLAYALMTHD=10;
```

### 5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

#### NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

## 5.4.2 Activation Verification

### Observing the IPsec Tunnel

**Step 1** Run the **DSP IKESA** command to check the IKE SA status.

All SAs are successfully established if the command output indicates both of the following:

- The status of SAs in phases 1 and 2 is **Ready|StayAlive** or **Ready**.

#### NOTE

The IKE SAs negotiated in phase 1 do not contain **IPv6 ACL ID** and **IPv6 Rule ID**. Therefore, the values of both **IPv6 ACL ID** and **IPv6 Rule ID** in the **DSP IKESA** command output are **NULL** in phase 1.

- The number of SAs in phase 2 is the same as the number of ACL rules whose **IPv6 ACL Action** is **Permit** in the **LST ACLRULE6** command.

**Step 2** Run the **DSP IPSECSA** command to check the IPsec SA status.

**Step 3** Check whether services protected by the IPsec tunnel are normal.

- Initiate voice and data services and then check whether services are running properly.
- Check whether the corresponding base station is online on the MAE topology view.

**Step 4** Check the IPsec replay status.

- Check whether ALM-25950 Base Station Being Attacked is reported with **Specific Problem** being **IPsec Replay**. If this alarm exists, the base station is under IPsec replay attacks.
- If IPsec replay attacks exist, run the **DSP INVALIDPKTINFO** command to query IPsec replay packets.

**Step 5** Check the IKE- and IPsec-related performance counters. For details, see [4.4.4 Network Monitoring](#).

----End

### 5.4.3 Network Monitoring

For details, see [4.4.4 Network Monitoring](#).

# 6 IPsec Tunnel Backup

## 6.1 Principles

### Working Principles

The IPsec Tunnel Backup feature enables a base station to establish active and standby IPsec tunnels with two SeGWs, respectively. If the active IPsec tunnel is faulty, services are automatically switched over to the standby IPsec tunnel, which improves reliability.

If IPsec is activated, the base station can communicate with the SeGWs through the active and standby IPsec tunnels, which operate in hot backup mode. The base station negotiates with the primary and secondary SeGWs about individual IKE tunnels and IPsec tunnels. This ensures the reliability of the IPsec tunnels.

This function is not supported when the **GTRANSPARA.TRANS CFGMODE (LTE eNodeB, 5G gNodeB)** parameter is set to **NEW** and the **IKECFG.IPSECBINDMODE (LTE eNodeB, 5G gNodeB)** parameter is set to **ALL**.

**Figure 6-1** illustrates a switchover between active and standby IPsec tunnels. Normally, IPsec traffic is transmitted through the active IPsec tunnel. If BFD or DPD detects that the active IPsec tunnel is faulty (the SeGW is unreachable or the IPsec SAs of the active IPsec tunnel are abnormal) and the standby IPsec tunnel works properly:

- In the uplink, the base station automatically switches services to the standby IPsec tunnel.
- In the downlink, the SeGW must work with the base station to switch services to the standby IPsec tunnel. If the SeGW detects that the active IPsec tunnel is faulty, the SeGW automatically switches services to the standby IPsec tunnel.

Whether the services are switched back to the active IPsec tunnel after it is recovered is controlled as follows:

- The switchback of uplink services from the standby IPsec tunnel to the active IPsec tunnel is controlled by the **IPSECDTNL.IPSEC SWITCHBACK (LTE eNodeB, 5G gNodeB)** parameter. The **IPSECDTNL.IPSEC SWITCHBACK TM (LTE eNodeB, 5G gNodeB)** parameter specifies the wait time before services

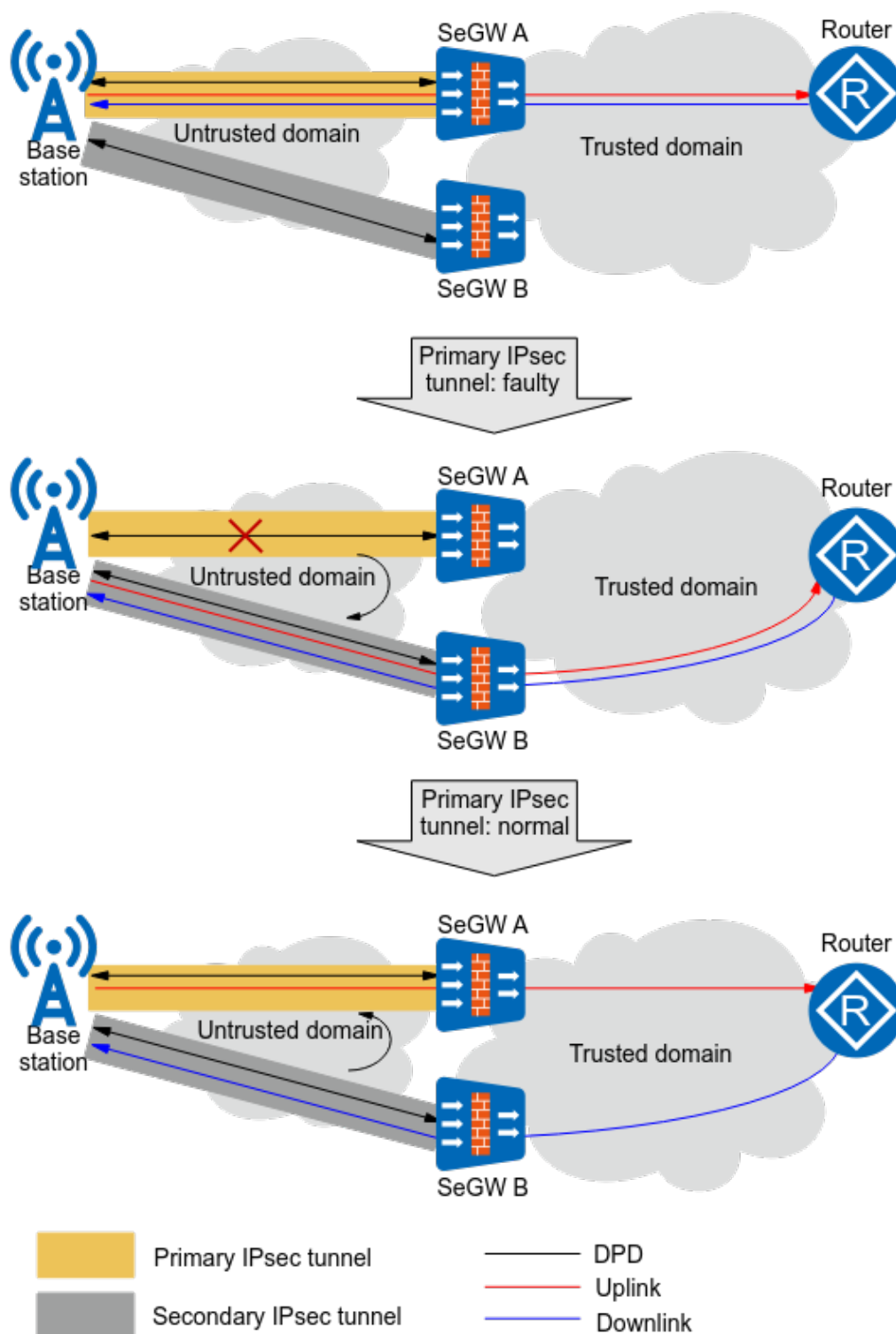
are switched over from the standby IPsec tunnel to the active IPsec tunnel. To avoid frequent service switchovers between the active and standby IPsec tunnels due to intermittent disconnection of the active IPsec tunnel, it is recommended that the service switchover wait time be longer than 10s.

- The router determines whether to switch back downlink services from the standby IPsec tunnel to the active IPsec tunnel.

If both the active and standby IPsec tunnels become faulty, services are carried on whichever IPsec tunnel recovers first. In this case, service switchover is not controlled by the **IPSECDTNL.IPSECSWITCHBACK (LTE eNodeB, 5G gNodeB)** parameter.



**Figure 6-1** Switchover between active and standby IPsec tunnels



In the uplink, the base station usually uses the active IPsec tunnel to send packets according to user configurations. The following parameters specify the active IPsec tunnel on the base station side:

- **IPSECDTNL.MSPGN (LTE eNodeB, 5G gNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB) or **BTSIPSECDTNL.MSPGN** (for the GBTS): specifies the name of the primary IPsec policy group.
- **IPSECDTNL.MSPSN (LTE eNodeB, 5G gNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB) or **BTSIPSECDTNL.MSPSN** (for the GBTS): specifies the ID of the primary IPsec policy.

The following parameters specify the standby IPsec tunnel:

- **IPSECDTNL.SSPGN (LTE eNodeB, 5G gNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB) or **BTSIPSECDTNL.SSPGN** (for the GBTS): specifies the name of the secondary IPsec policy group.
- **IPSECDTNL.SSPSN (LTE eNodeB, 5G gNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB) or **BTSIPSECDTNL.SSPSN** (for the GBTS): specifies the ID of the secondary IPsec policy.

In the downlink, the router must support dynamic routing protocols to select routes.

## IPsec Tunnel Detection

When IPsec Tunnel Backup is activated, the base station uses BFD and DPD to detect the status of connection between the base station and the SeGW.

If the **IPSECDTNL.BFDDTCTSW (LTE eNodeB, 5G gNodeB)** parameter is set to **ON**, there must be a BFD session that has been set up. In this case:

- If DPD is enabled, both BFD and DPD are used to detect the connection status.
- If DPD is disabled, only BFD is used to detect the connection status.

To use BFD to detect the connection status, an IPsec tunnel must be bound to a BFD session ID. On the base station side, the source and destination IP addresses of the BFD session must be the same as the local and peer IP addresses of the associated active or standby IPsec tunnel, respectively. When BFD is used to detect the connectivity of an IPsec tunnel, ensure that BFD packets do not enter the IPsec tunnel. For details about BFD, see *IPv4 Transmission*.

If the **IPSECDTNL.BFDDTCTSW (LTE eNodeB, 5G gNodeB)** parameter is set to **OFF**, DPD must be enabled. For details about DPD, see [4.1.1.3 IKE DPD](#).

## Application Constraints

- If BFD is used to detect the status of IPsec tunnels, the SeGWs must not work in hot backup mode. This is because that SeGWs working in hot backup mode use logical IP addresses for communication, and cannot provide physical IP addresses required when BFD is used to detect the connection status. When DPD is used to detect the status of IPsec tunnels, this restriction does not apply.
- IPsec Tunnel Backup is not applicable when the base station provides two transmission ports, one with VLAN configurations and the other without VLAN configurations.
- During base station deployment by PnP, only DPD can be used on IPsec tunnels. BFD cannot be used.

- The SeGW should support dynamic route advertisement and withdrawal so that the dynamic routes can be quickly withdrawn when the corresponding IKE or SeGW becomes abnormal.
- The routes advertised by the primary SeGW have higher priorities than those advertised by the secondary SeGW.

## 6.2 Network Analysis

### 6.2.1 Benefits

IPsec Tunnel Backup improves the reliability of IPsec data transmission.

### 6.2.2 Impacts

The impacts between this function and other functions are described in [6.3.2.1.3 Function Impacts](#), [6.3.2.2.3 Function Impacts](#), [6.3.2.3.3 Function Impacts](#), and [6.3.2.4.3 Function Impacts](#).

For details about the impact on the network, see [4.2.2 Impacts](#).

## 6.3 Requirements

### 6.3.1 Licenses

To enable the IPsec Tunnel Backup feature, the following licenses need to be activated for the gNodeBs and FDD, TDD, and NB-IoT eNodeBs, while no licenses need to be activated for the GBTs, eGBTs, and NodeBs.

| RAT     | Feature ID    | Feature Name                | Model        | Sales Unit |
|---------|---------------|-----------------------------|--------------|------------|
| LTE FDD | LOFD-003019   | IPsec Tunnel Backup         | LT1S00IPTB00 | Per eNodeB |
| NB-IoT  | MLOFD-003019  | IPsec Tunnel Backup         | ML1S00IPTB00 | Per eNodeB |
| LTE TDD | TDLOFD-131212 | IPsec Tunnel Backup         | LT1SIPTBCK01 | Per eNodeB |
| NR      | FOFD-010080   | IPsec (IPsec Tunnel Backup) | NR0S0IPSEC00 | Per gNodeB |

#### NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

## 6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

### 6.3.2.1 LOFD-003019 IPsec Tunnel Backup

#### 6.3.2.1.1 Prerequisite Functions

| RAT     | Function Name                      | Function Switch | Reference                | Description                                                             |
|---------|------------------------------------|-----------------|--------------------------|-------------------------------------------------------------------------|
| LTE FDD | Bidirectional Forwarding Detection | None            | <i>IPv4 Transmission</i> | This function must be enabled when BFD is used for IPsec Tunnel Backup. |
| LTE FDD | IPsec                              | None            | <i>IPsec</i>             | None                                                                    |

#### 6.3.2.1.2 Mutually Exclusive Functions

| RAT     | Function Name                         | Function Switch                                            | Reference    |
|---------|---------------------------------------|------------------------------------------------------------|--------------|
| LTE FDD | IPsec redundancy among multiple SeGWs | None                                                       | <i>IPsec</i> |
| LTE FDD | eNodeB Supporting IPsec Redirection   | <b>IKEPEER.REDIREC<br/>TSW (5G gNodeB,<br/>LTE eNodeB)</b> | <i>IPsec</i> |

#### 6.3.2.1.3 Function Impacts

None

### 6.3.2.2 MLOFD-003019 IPsec Tunnel Backup

### 6.3.2.2.1 Prerequisite Functions

| RAT    | Function Name                      | Function Switch | Reference                | Description                                                             |
|--------|------------------------------------|-----------------|--------------------------|-------------------------------------------------------------------------|
| NB-IoT | Bidirectional Forwarding Detection | None            | <i>IPv4 Transmission</i> | This function must be enabled when BFD is used for IPsec Tunnel Backup. |
| NB-IoT | IPsec                              | None            | <i>IPsec</i>             | None                                                                    |

### 6.3.2.2.2 Mutually Exclusive Functions

| RAT    | Function Name                         | Function Switch                                   | Reference    |
|--------|---------------------------------------|---------------------------------------------------|--------------|
| NB-IoT | IPsec redundancy among multiple SeGWs | None                                              | <i>IPsec</i> |
| NB-IoT | eNodeB Supporting IPsec Redirection   | <b>IKEPEER.REDIRECTSW (5G gNodeB, LTE eNodeB)</b> | <i>IPsec</i> |

### 6.3.2.2.3 Function Impacts

None

## 6.3.2.3 TDLOFD-131212 IPsec Tunnel Backup

### 6.3.2.3.1 Prerequisite Functions

| RAT     | Function Name                      | Function Switch | Reference                | Description                                                             |
|---------|------------------------------------|-----------------|--------------------------|-------------------------------------------------------------------------|
| LTE TDD | Bidirectional Forwarding Detection | None            | <i>IPv4 Transmission</i> | This function must be enabled when BFD is used for IPsec Tunnel Backup. |
| LTE TDD | IPsec                              | None            | <i>IPsec</i>             | None                                                                    |

### 6.3.2.3.2 Mutually Exclusive Functions

| RAT     | Function Name                         | Function Switch                                   | Reference    |
|---------|---------------------------------------|---------------------------------------------------|--------------|
| LTE TDD | IPsec redundancy among multiple SeGWs | None                                              | <i>IPsec</i> |
| LTE TDD | eNodeB Supporting IPsec Redirection   | <b>IKEPEER.REDIRECTSW (5G gNodeB, LTE eNodeB)</b> | <i>IPsec</i> |

### 6.3.2.3.3 Function Impacts

None

### 6.3.2.4 FOFD-010080 IPsec (IPsec Tunnel Backup)

#### 6.3.2.4.1 Prerequisite Functions

| RAT                                                     | Function Name                                              | Function Switch | Reference                | Description                                                             |
|---------------------------------------------------------|------------------------------------------------------------|-----------------|--------------------------|-------------------------------------------------------------------------|
| NR FDD<br>Low-frequency NR TDD<br>High-frequency NR TDD | Transmission Network Detection and Reliability Improvement | None            | <i>IPv4 Transmission</i> | This function must be enabled when BFD is used for IPsec Tunnel Backup. |

#### 6.3.2.4.2 Mutually Exclusive Functions

| RAT                                                     | Function Name                                 | Function Switch                                   | Reference    |
|---------------------------------------------------------|-----------------------------------------------|---------------------------------------------------|--------------|
| NR FDD<br>Low-frequency NR TDD<br>High-frequency NR TDD | IPsec (IPsec Redundancy Among Multiple SeGWs) | None                                              | <i>IPsec</i> |
| NR FDD<br>Low-frequency NR TDD<br>High-frequency NR TDD | IPsec (IPsec Redirection)                     | <b>IKEPEER.REDIRECTSW (5G gNodeB, LTE eNodeB)</b> | <i>IPsec</i> |

### 6.3.2.4.3 Function Impacts

None

## 6.3.3 Hardware

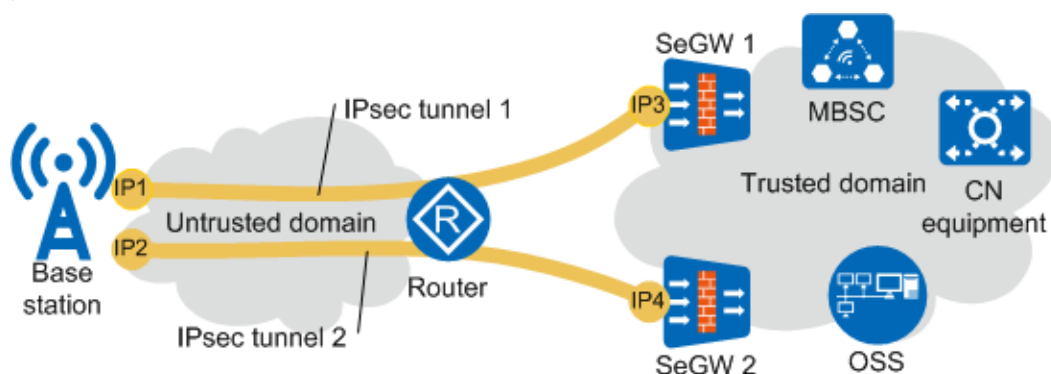
For details, see [4.3.3 Hardware](#).

## 6.3.4 Networking

There are two typical networking schemes for IPsec Tunnel Backup, distinguished by whether the base station uses one or two physical ports.

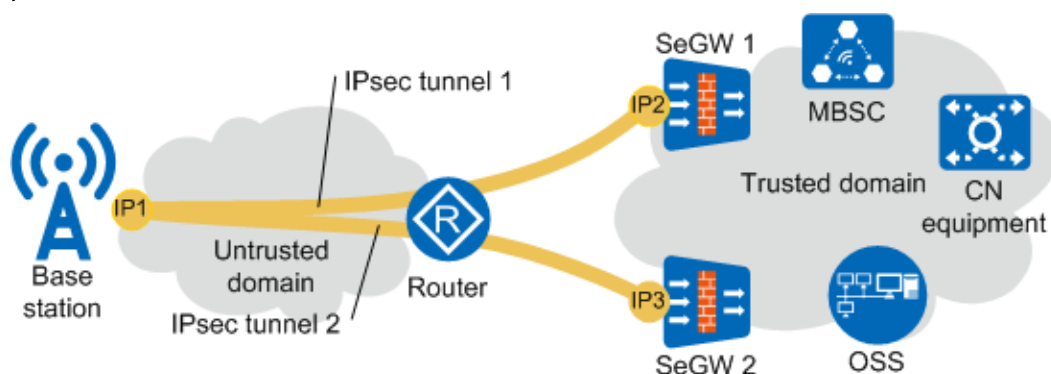
**Figure 6-2** shows the networking where the base station uses two ports. In this scenario, two parallel IPsec tunnels are set up to two SeGWs, and security policies are bound to the two ports separately, with BFD enabled.

**Figure 6-2** Example of networking in which the base station provides two physical ports



**Figure 6-3** shows the networking where the base station uses one physical port. In this scenario, two IPsec tunnels are set up from the single port to two SeGWs, and security policies are bound to the port, with BFD enabled.

**Figure 6-3** Example of networking in which the base station provides one physical port



## 6.3.5 Others

None

## 6.4 Operation and Maintenance

### 6.4.1 Data Configuration

#### 6.4.1.1 Data Preparation

If active and standby IPsec tunnels are planned, after the two IPsec tunnels are configured as instructed in [4.4.2.3 Deploying IPsec for an eGBTS/NodeB/eNodeB/gNodeB on a PKI-based Secure Network](#), BFD must be enabled to detect the connectivity of the active and standby IPsec tunnels. [Table 6-1](#) describes the parameters that must be set to configure a BFD session when **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**.

**Table 6-1** Parameters that must be set to configure a BFD session

| Parameter Name | Parameter ID                                    | Setting Notes                                                                                                                                                                                                                                                                                   |
|----------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Session ID     | <b>BFDSESSION.BFDSN (5G gNodeB, LTE eNodeB)</b> | Set this parameter based on the network plan.                                                                                                                                                                                                                                                   |
| Source IP      | <b>BFDSESSION.SRCIP (5G gNodeB, LTE eNodeB)</b> | If active and standby IPsec tunnels are used, BFD needs to be enabled to detect their connectivity. In this case, <b>BFDSESSION.SRCIP (5G gNodeB, LTE eNodeB)</b> and <b>BFDSESSION.DSTIP (5G gNodeB, LTE eNodeB)</b> specify the IP addresses of the IKE local end and IKE peer, respectively. |
| Destination IP | <b>BFDSESSION.DSTIP (5G gNodeB, LTE eNodeB)</b> |                                                                                                                                                                                                                                                                                                 |



| Parameter Name                   | Parameter ID                                             | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| My Discriminator Generation Mode | <b>BFDSESSION.MYDSCRMGENMODE (LTE eNodeB, 5G gNodeB)</b> | <p>This parameter specifies the generation mode of my discriminator. Set this parameter based on the network plan.</p> <p>The following generation modes are supported:</p> <ul style="list-style-type: none"><li>• <b>STATIC_CONFIG</b>. In this mode, my discriminator equals the session ID (<b>BFDSESSION.BFDSN (5G gNodeB, LTE eNodeB)</b>) plus one. It is recommended that your discriminator not be set on the router. If your discriminator is set on the router, your discriminator needs to be the same as my discriminator of the base station.</li><li>• <b>DYNAMIC_GEN</b>. In this mode, my discriminator is generated randomly when negotiation of a session starts.</li><li>• <b>DYNAMIC_GEN_ENHANCE</b>. In this mode, my discriminator is generated randomly when negotiation of a session starts and, if my discriminator is the same as your discriminator, the base station discards this session and generates another my discriminator randomly to start negotiation.</li></ul> <p>If your discriminator is set on the router, it can be only set to <b>STATIC_CONFIG</b>.</p> |
| Hop Type                         | <b>BFDSESSION.HT (5G gNodeB, LTE eNodeB)</b>             | Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Min TX Interval                  | <b>BFDSESSION.MINTI (5G gNodeB, LTE eNodeB)</b>          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Min RX Interval                  | <b>BFDSESSION.MINRI (5G gNodeB, LTE eNodeB)</b>          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Detection Multiplier             | <b>BFDSESSION.DM (5G gNodeB, LTE eNodeB)</b>             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

| Parameter Name   | Parameter ID                                         | Setting Notes |
|------------------|------------------------------------------------------|---------------|
| Session Catalog  | <b>BFDSESSION.CATALOG</b><br>(5G gNodeB, LTE eNodeB) |               |
| DSCP             | <b>BFDSESSION.DSCP</b><br>(5G gNodeB, LTE eNodeB)    |               |
| Protocol Version | <b>BFDSESSION.VER</b> (5G gNodeB, LTE eNodeB)        |               |

**Table 6-2** describes the parameters that must be set to configure a BFD session when **GTRANSPARA.TRANS CFGMODE** (5G gNodeB, LTE eNodeB) is set to **NEW**.

**Table 6-2** Parameters that must be set to configure a BFD session

| Parameter Name | Parameter ID                              | Setting Notes                                                                                                                                                                                                                                                                     |
|----------------|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Session ID     | <b>BFD.BFDSN</b> (5G gNodeB, LTE eNodeB)  | Set this parameter based on the network plan.                                                                                                                                                                                                                                     |
| VRF Index      | <b>BFD.VRFIDX</b> (5G gNodeB, LTE eNodeB) |                                                                                                                                                                                                                                                                                   |
| Source IP      | <b>BFD.SRCIP</b> (5G gNodeB, LTE eNodeB)  | If active and standby IPsec tunnels are used, BFD needs to be enabled to detect their connectivity. In this case, <b>BFD.SRCIP</b> (5G gNodeB, LTE eNodeB) and <b>BFD.DSTIP</b> (5G gNodeB, LTE eNodeB) specify the IP addresses of the IKE local end and IKE peer, respectively. |
| Destination IP | <b>BFD.DSTIP</b> (5G gNodeB, LTE eNodeB)  |                                                                                                                                                                                                                                                                                   |

| Parameter Name                   | Parameter ID                                        | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| My Discriminator Generation Mode | <b>BFD.MYDISCRMGENMODE (LTE eNodeB, 5G gNodeB)</b>  | <p>This parameter specifies the generation mode of my discriminator. Set this parameter based on the network plan.</p> <p>The following generation modes are supported:</p> <ul style="list-style-type: none"><li>• <b>STATIC_CONFIG</b>. In this mode, my discriminator is generated based on the configured value of <b>BFD.MYDISCREAMINATOR (5G gNodeB, LTE eNodeB)</b>. It is recommended that your discriminator not be set on the router. If your discriminator is set on the router, your discriminator needs to be the same as my discriminator of the base station.</li><li>• <b>DYNAMIC_GEN</b>. In this mode, my discriminator is generated randomly when negotiation of a session starts.</li><li>• <b>DYNAMIC_GEN_ENHANCE</b>. In this mode, my discriminator is generated randomly when negotiation of a session starts and, if my discriminator is the same as your discriminator, the base station discards this session and generates another my discriminator randomly to start negotiation.</li></ul> <p>If your discriminator is set on the router, it can be only set to <b>STATIC_CONFIG</b>.</p> |
| My Discriminator                 | <b>BFD.MYDISCREAMINATOR (5G gNodeB, LTE eNodeB)</b> | Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Hop Type                         | <b>BFD.HT (5G gNodeB, LTE eNodeB)</b>               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Min TX Interval                  | <b>BFD.MINTI (5G gNodeB, LTE eNodeB)</b>            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Min RX Interval                  | <b>BFD.MINRI (5G gNodeB, LTE eNodeB)</b>            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| Parameter Name                  | Parameter ID                                   | Setting Notes                                                                                      |
|---------------------------------|------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Detection Multiplier            | <b>BFD.DM (5G gNodeB, LTE eNodeB)</b>          |                                                                                                    |
| Session Catalog                 | <b>BFD.CATALOG (5G gNodeB, LTE eNodeB)</b>     |                                                                                                    |
| DSCP                            | <b>BFD.DSCP (5G gNodeB, LTE eNodeB)</b>        |                                                                                                    |
| Protocol Version                | <b>BFD.VER (5G gNodeB, LTE eNodeB)</b>         |                                                                                                    |
| BFD Authentication Switch       | <b>BFD.BFDAUTHSW (5G gNodeB, LTE eNodeB)</b>   | This parameter is valid when <b>BFD.HT (5G gNodeB, LTE eNodeB)</b> is set to <b>SINGLE_HOP</b> .   |
| BFD Authentication Algorithm    | <b>BFD.BFDAUTHTYPE (5G gNodeB, LTE eNodeB)</b> | These parameters are valid when <b>BFD.BFDAUTHSW (5G gNodeB, LTE eNodeB)</b> is set to <b>ON</b> . |
| BFD Authentication Key Chain ID | <b>BFD.KEYCHAINID (5G gNodeB, LTE eNodeB)</b>  |                                                                                                    |

**Table 6-3** describes the parameters that must be set to configure a pair of active and standby IPsec tunnels as planned.

**Table 6-3** Parameters that must be set to configure a pair of active and standby IPsec tunnels

| Parameter Name            | Parameter ID                                    | Setting Notes                                 |
|---------------------------|-------------------------------------------------|-----------------------------------------------|
| IPSec Dual Tunnel ID      | <b>IPSECDTNL.DUALID (5G gNodeB, LTE eNodeB)</b> | Set this parameter based on the network plan. |
| Master Policy Group Name  | <b>IPSECDTNL.MSPGN (5G gNodeB, LTE eNodeB)</b>  | Set this parameter based on the network plan. |
| Master IPSec Sequence No. | <b>IPSECDTNL.MSPSN (5G gNodeB, LTE eNodeB)</b>  | Set this parameter based on the network plan. |

| Parameter Name                 | Parameter ID                                                | Setting Notes                                                             |
|--------------------------------|-------------------------------------------------------------|---------------------------------------------------------------------------|
| Slave Policy Group Name        | <b>IPSECDTNL.SSPGN</b><br>(5G gNodeB, LTE eNodeB)           | Set this parameter based on the network plan.                             |
| Slave IPsec Sequence No.       | <b>IPSECDTNL.SSPSN</b><br>(5G gNodeB, LTE eNodeB)           | Set this parameter based on the network plan.                             |
| BFD Detect Switch              | <b>IPSECDTNL.BFDDTCT SW</b> (5G gNodeB, LTE eNodeB)         | For details about the setting notes, see <a href="#">6.1 Principles</a> . |
| Master Tunnel's BFD Session ID | <b>IPSECDTNL.MBFDSN</b><br>(5G gNodeB, LTE eNodeB)          | Set this parameter based on the network plan.                             |
| Slave Tunnel's BFD Session ID  | <b>IPSECDTNL.SBFDSN</b><br>(5G gNodeB, LTE eNodeB)          | Set this parameter based on the network plan.                             |
| IPsec Dual Tunnel Switch Back  | <b>IPSECDTNL.IPSECSWI TCHBACK</b> (5G gNodeB, LTE eNodeB)   | For details about the setting notes, see <a href="#">6.1 Principles</a> . |
| IPsec Switch Back Wait Time    | <b>IPSECDTNL.IPSECSWI TCHBACKTM</b> (5G gNodeB, LTE eNodeB) | For details about the setting notes, see <a href="#">6.1 Principles</a> . |

#### 6.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

```
//Configuring one IPsec tunnel by referring to the preceding MML command example and then configuring data as follows
//Adding the second IKE peer
ADD IKEPEER: PEERNAME="Ike2", PROPID=10, IKEVERSION=IKE_V2, IPVERSION=IPV4, IDTYPE=IPV4, REMOTEIP="10.80.80.80", REMOTENAME="Secgw2", DPD=PERIODIC, DPDIDLETIME=10, DPDRETRI=5, DPDRETRN=5, LOCALIP="10.21.21.188", REDUNDANCYFLAG=NONE, CERTSOURCE=APPCERT, IPSECPREFRGSW=OFF;
//Adding the second IPsec policy
//If one port is used, the two IPsec policies have the same group name but different serial numbers, and they need to be bound to this port at a time.
```

```
//If two ports are used, the two IPsec policies have different group names but may have the same serial number. The two IPsec policies need to be separately bound to the two ports.  
//The following is an example for two ports.  
ADD IPSECPOLICY: SPGN="Policy1", SPSN=2, ACLTYPE=IPV4, ACLID=3000, PROPNAME="prop0",  
PEERNAME="Ike2", LTCFG=LOCAL, LTS=3600, LTKB=0, REPLAYWND=WND_DISABLE,  
NARROWDOWNSW=OFF;
```

- Scenario 1:

```
//When GTRANSPARA.TRANSFCFGMODE is set to OLD  
//Binding the second IPsec policy to an outbound interface  
ADD IPSECBIND: SN=7, CN=0, SRN=0, SBT=BASE_BOARD, PT=ETH, PN=0, SPGN="Policy1";  
//Adding BFD sessions for the two IPsec tunnels  
ADD BFDSESSION: SN=7, CN=0, SRN=0, BFDSN=1, SRCIP="10.20.20.188", DSTIP="10.90.90.90",  
MYDSCRMGENMODE=STATIC_CONFIG, HT=MULTI_HOP, DSCP=0;  
ADD BFDSESSION: SN=7, CN=0, SRN=0, BFDSN=2, SRCIP="10.21.21.188", DSTIP="10.80.80.80",  
MYDSCRMGENMODE=STATIC_CONFIG, HT=MULTI_HOP, DSCP=0;  
//Configuring the two IPsec tunnels as active and standby IPsec tunnels  
ADD IPSECDTNL: DUALID=0, MSPGN="Policy0", MSPSN=1, SSPGN="Policy1", SSPSN=2,  
BFDDTCTSW=ON, MBFDSN=1, SBFDSN=2, IPSECSWITCHBACK=ON, IPSECSWITCHBACKTM=180;
```

- Scenario 2:

```
//When GTRANSPARA.TRANSFCFGMODE is set to NEW  
//Binding the second IPsec policy to an outbound interface  
ADD IPSECBINDITF: IPSECBINDITFID=1, ITFID=0, SPGN="Policy1";  
//Adding BFD sessions for the two IPsec tunnels  
ADD BFD: BFDSN=100, VRFIDX=0, SRCIP="10.20.20.188", DSTIP="10.90.90.90",  
MYDSCRMGENMODE=STATIC_CONFIG, MYDISCREAMINATOR=1, HT=MULTI_HOP, DSCP=0;  
ADD BFD: BFDSN=101, VRFIDX=0, SRCIP="10.21.21.188", DSTIP="10.80.80.80",  
MYDSCRMGENMODE=STATIC_CONFIG, MYDISCREAMINATOR=2, HT=MULTI_HOP, DSCP=0;  
//Configuring the two IPsec tunnels as active and standby IPsec tunnels  
ADD IPSECDTNL: DUALID=0, MSPGN="Policy0", MSPSN=1, SSPGN="Policy1", SSPSN=2,  
BFDDTCTSW=ON, MBFDSN=100, SBFDSN=101, IPSECSWITCHBACK=ON, IPSECSWITCHBACKTM=180;
```

### 6.4.1.3 Using the MAE-Deployment

For details, see [4.4.2.10 Using the MAE-Deployment](#).

## 6.4.2 Activation Verification

For details, see [4.4.3 Activation Verification](#).

## 6.4.3 Network Monitoring

For details, see [4.4.4 Network Monitoring](#).

# 7 IPsec Redundancy Among Multiple SeGWs

---

## 7.1 Principles

The IPsec redundancy among multiple SeGWs feature allows the base station to set up a standby IPsec tunnel to a secondary SeGW once any fault is detected on the IPsec tunnel to the primary SeGW, realizing IPsec redundancy among multiple SeGWs.

The differences between IPsec Tunnel Backup and IPsec redundancy among multiple SeGWs are as follows:

- IPsec Tunnel Backup features a shorter service recovery time than IPsec redundancy among multiple SeGWs.  
IPsec Tunnel Backup is a hot backup mode which requires that the standby IPsec tunnel keep connected at all times and both the active and standby SAs be maintained. IPsec redundancy among multiple SeGWs is a cold backup mode which requires that only the tunnel to the primary SeGW keep connected and only the active SA be maintained.
- IPsec Tunnel Backup supports both BFD and DPD, and supports only one standby IPsec tunnel. IPsec redundancy among multiple SeGWs supports only DPD, but supports a maximum of seven standby IPsec tunnels (priorities of standby IPsec tunnels are configurable, and only one standby tunnel is supported when the IPv4 IPsec configuration mode is tunnel-interface-based IPsec).

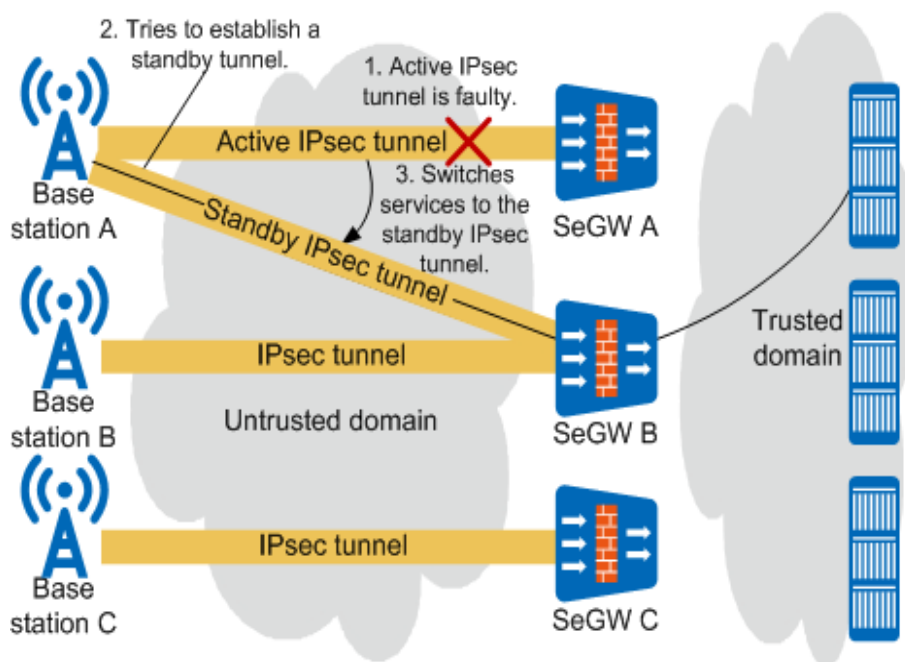
The IPsec redundancy among multiple SeGWs feature uses IKE DPD to detect the status of the IPsec tunnels between the base station and the SeGWs. If the active IPsec tunnel between the base station and the primary SeGW is faulty, the base station attempts to establish a standby IPsec tunnel with a secondary SeGW, realizing IPsec redundancy among multiple SeGWs, as shown in [Figure 7-1](#).

The following base stations or board combinations support IPsec redundancy among multiple SeGWs:

- NodeB

- eNodeB
- gNodeB
- GBTS configured with GTMUc+UMPT\_L
- eGBTS configured with a UMPT

**Figure 7-1** IPsec redundancy among multiple SeGWs



IPsec redundancy among multiple SeGWs involves IPsec tunnel switchover and IPsec tunnel switchback.

## IPsec Tunnel Switchover

If the active IPsec tunnel between the base station and the primary SeGW becomes faulty, the base station attempts to perform an IKE renegotiation with the primary SeGW. If the IKE renegotiation fails, the base station reports an IKE negotiation failure alarm and starts an IPsec tunnel switchover procedure. An IPsec tunnel switchover will not be triggered if an IKE SA negotiation between the base station and a SeGW succeeds but the IPsec SA negotiation between them fails.

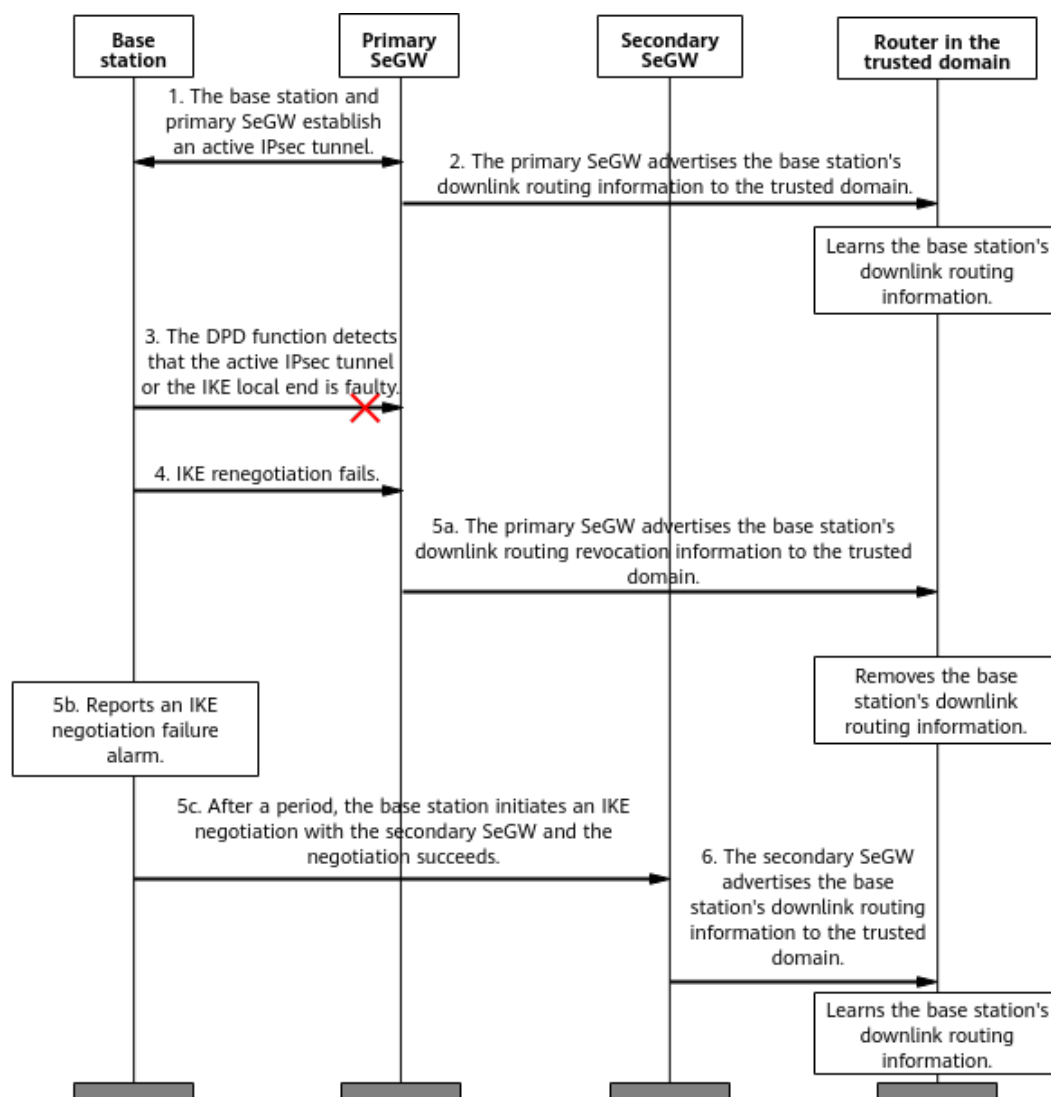
The ***IKEPEER.REDUNDANCYFLAG (5G gNodeB, LTE eNodeB)*** and ***IKEPEER.MASTERPEERNAME (5G gNodeB, LTE eNodeB)*** parameters specify the binding relationship between active and standby IPsec tunnels. If multiple secondary SeGWs are configured, each SeGW should be configured with a unique priority (specified by the ***IKEPEER.PRIORITY (5G gNodeB, LTE eNodeB)*** parameter). One primary SeGW corresponds to at most seven secondary SeGWs.

If the base station fails IKE negotiations with all secondary SeGWs, it reports an IKE negotiation failure alarm. The alarm cause indicates that IKE redundancy fails. Then, the base station starts a new round of IKE negotiations with each secondary SeGW in descending order of priority. If the IKE negotiation between the base station and a secondary SeGW succeeds or the active IPsec tunnel recovers, the base station clears the IKE negotiation failure alarm.



Figure 7-2 shows the IPsec tunnel switchover procedure.

Figure 7-2 IPsec tunnel switchover triggered by an active IPsec tunnel failure



The base station initiates IKE negotiations with the secondary SeGWs in sequence and attempts to establish a standby IPsec tunnel. During the switchover from the primary SeGW to the secondary SeGW, if **IKECFG.IPSEC SWITCHBACK (5G gNodeB, LTE eNodeB)** is set to **OFF**, the negotiation with the primary SeGW is stopped. Otherwise, the negotiation with the primary SeGW is resumed.

The base station initiates IKE negotiations for standby IPsec tunnel setup with the secondary SeGWs in sequence by descending order of SeGW priority after waiting for a random delay within the period specified by the sum of the **IKECFG.IPSEC SOWAITTIME (LTE eNodeB, 5G gNodeB)** and **IKECFG.IPSEC SORANDTIME (5G gNodeB, LTE eNodeB)** parameter values when the active IPsec tunnel is faulty. This prevents base stations from simultaneously initiating multiple IKE negotiations with a single secondary SeGW. The waiting period for each secondary SeGW is a random delay within the period specified by the sum of the **IKECFG.IPSEC SOWAITTIME (LTE eNodeB, 5G gNodeB)** and **IKECFG.IPSEC SORANDTIME (5G gNodeB, LTE eNodeB)** parameter values.

If the negotiation with the secondary SeGW with the lowest priority fails, the subsequent negotiation policy is still controlled by the **IKECFG.IPSECSWITCHBACK (5G gNodeB, LTE eNodeB)** parameter.

- When this parameter is set to **ON**, the negotiation starts again from the primary SeGW in turn.
- When this parameter is set to **OFF**, the negotiation starts from the secondary SeGW with the highest priority in turn.
- When this parameter is set to **IPV4\_ON+IPV6\_OFF** or **IPV4\_OFF+IPV6\_ON**, IPv4 switchback is disabled but IPv6 switchback is enabled. The negotiation policy is the same as that when the preceding parameters are set to **On** or **OFF**.

If the standby IPsec tunnel is successfully established, services are switched over to the standby IPsec tunnel.

When modifying or deleting the secondary SeGW in use, the preceding switchover policy must be complied.

#### NOTE

- The **IKECFG.IPSECSORANDTIME (5G gNodeB, LTE eNodeB)** parameter specifies the maximum random delay window. Each base station randomly selects a period of time within the window for delay.
- If the **RST IKESA** command fails to be executed, the IPsec redundancy switchover is not triggered.
- The IPv4 link dynamically assigned by DHCP cannot be used as the IPsec tunnel between the base station and the primary SeGW.

## IPsec Tunnel Switchback

The IPsec tunnel switchback policy is controlled by the **IKECFG.IPSECSWITCHBACK (5G gNodeB, LTE eNodeB)** parameter.

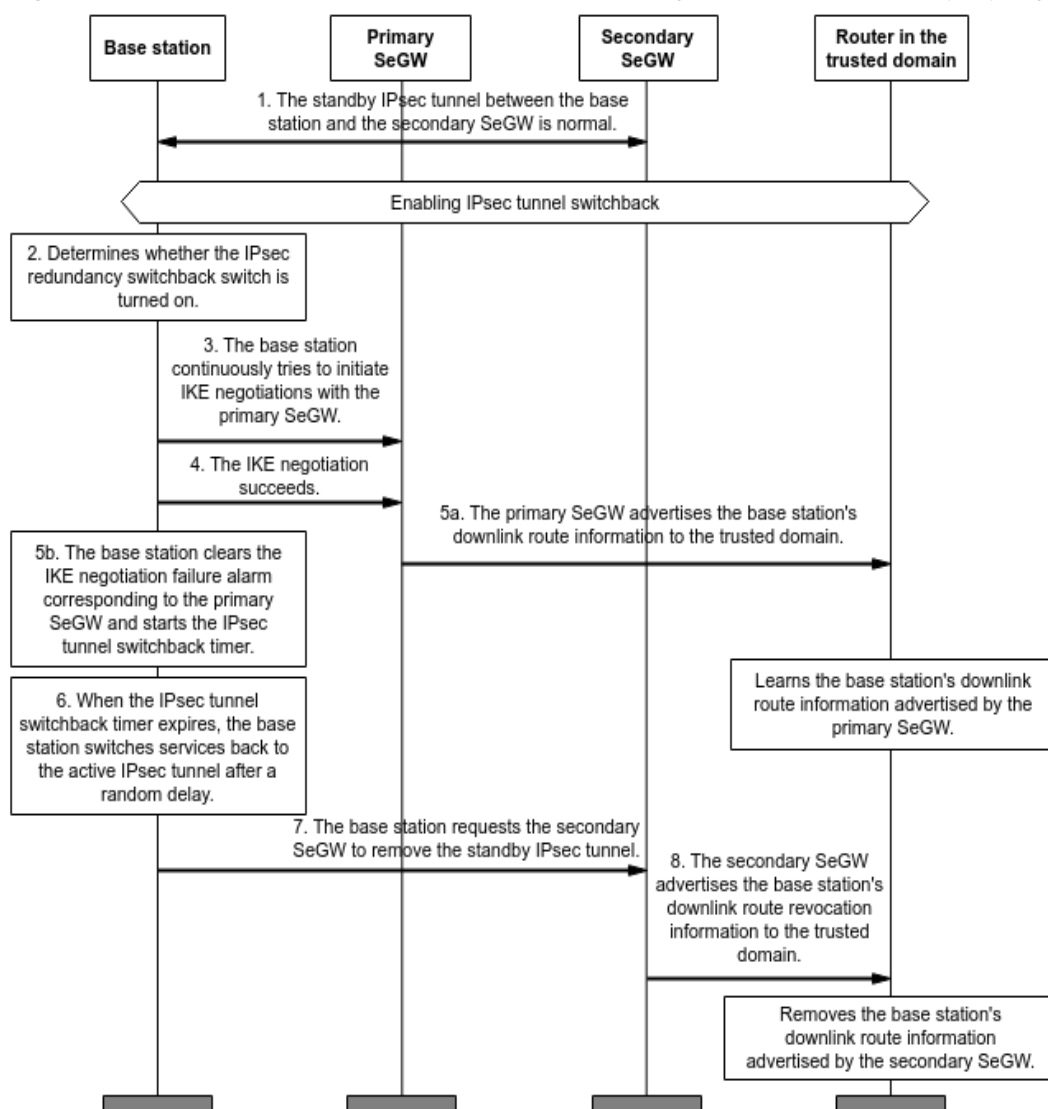
- If this parameter is set to **ON**, the base station continuously attempts to initiate IKE negotiations with the primary SeGW to recover the active IPsec tunnel.
  - If the standby IPsec tunnel in use is normal and the IKE negotiation between the base station and the primary SeGW succeeds, the base station clears the IKE negotiation failure alarm corresponding to the primary SeGW and starts a timer, the length of which is specified by the **IKECFG.IPSECSBWAITTIME (5G gNodeB, LTE eNodeB)** parameter. When the IPsec tunnel switchback timer expires, the base station waits for a random period within the range specified by the **IKECFG.IPSECSORANDTIME (5G gNodeB, LTE eNodeB)** parameter, switches upper-layer services back to the active IPsec tunnel, and removes the standby IPsec tunnel. This prevents simultaneous switchback to the primary SeGW in a short time.
  - If the standby IPsec tunnel in use is faulty, services are switched back to the primary SeGW without waiting.
- If this parameter is set to **OFF**, the base station stops IKE negotiation with the primary SeGW. If a normal standby IPsec tunnel is available, the base station does not attempt to establish an IPsec tunnel with the primary SeGW. In this case, IPv4 and IPv6 switchback are both disabled.

- When this parameter is set to **IPV4\_ON+IPV6\_OFF** or **IPV4\_OFF+IPV6\_ON**, either IPv4 or IPv6 switchback is enabled. The switchback policy is the same as that when the preceding parameters are set to **ON** or **OFF**.

If the base station fails a round of IKE negotiations with all secondary SeGWs, the base station attempts to initiate IKE negotiations with the primary SeGW to recover the active IPsec tunnel, regardless of whether the **IKECFG.IPSEC SWITCHBACK (5G gNodeB, LTE eNodeB)** parameter is set to **ON**. If an IKE negotiation for the active IPsec tunnel with the primary SeGW succeeds before any IKE negotiation with the secondary SeGWs, the base station clears the IKE negotiation failure alarm corresponding to the primary SeGW and switches services back to the active IPsec tunnel.

If the active IPsec tunnel is unstable, setting the **IKECFG.IPSEC SWITCHBACK (5G gNodeB, LTE eNodeB)** parameter to **OFF** will stop the negotiation with the primary SeGW, avoiding ping-pong switchovers between active and standby IPsec tunnels.

**Figure 7-3** IPsec tunnel switchback when the standby IPsec tunnel works properly



 NOTE

During IPsec tunnel switchback, services may be temporarily interrupted and then recovered.

Specifically, the OM channel of the base station will be interrupted for five minutes before recovery during IPsec tunnel switchback if the stateful firewall is enabled on the primary SeGW, and for at least five minutes if the stateful firewall is enabled on the secondary SeGW.

In Ethernet-interface-based configuration mode, when the active IPsec tunnel is faulty but the standby IPsec tunnel is working properly on an IPv4 network, modifying the parameters in the **IPSECPOLICY** and **IKEPEER** MOs for the active IPsec tunnel will result in renegotiation of the active and standby IPsec tunnels, leading to service interruption.

The restrictions for using the IPsec redundancy among multiple SeGWs feature are as follows:

- The IPsec redundancy among multiple SeGWs feature is not supported during PnP base station deployment.
- The IPsec redundancy among multiple SeGWs feature is not compatible with the IPsec tunnel backup function.
- On the base station side, the ACL rules for IPsec tunnels must be set to the "IP to IP" or "IP to Any" mode. If ACL rules for IPsec tunnels are set to the "Any to Any" mode, the SeGWs cannot dynamically advertise the base station's downlink routing information to the trusted domain based on the status of IPsec SAs, and downlink routes are unreachable.
- On the base station side, the ACL IDs referenced by IPsec policies that all active and standby IKE peers belong to must be the same.
- On the base station side, DPD must be enabled for all active and standby IKE peers.
- Redundancy between IPv4 and IPv6 tunnels is not supported.

## 7.2 Network Analysis

### 7.2.1 Benefits

This feature improves IPsec reliability.

If the negotiation with the standby SeGW in use fails, a switchover from the standby SeGW to another standby SeGW in the redundancy group will be performed. The waiting time for the switchover is affected by the sum of **IKECFG.IPSEC SOWAITTIME (LTE eNodeB, 5G gNodeB)** and **IKECFG.IPSEC SORANDTIME (LTE eNodeB, 5G gNodeB)**. During the switchover waiting time, if the fault of the standby SeGW in use is rectified, services are not switched to another standby SeGW. This reduces frequent switchovers caused by intermittent disconnection of the standby SeGW in use.

### 7.2.2 Impacts

The impacts between this function and other functions are described in [7.3.2.1.3 Function Impacts](#), [7.3.2.2.3 Function Impacts](#), [7.3.2.3.3 Function Impacts](#), [7.3.2.4.3 Function Impacts](#), [7.3.2.5.3 Function Impacts](#), and [7.3.2.6.3 Function Impacts](#).

If the negotiation with the standby SeGW in use fails, a switchover from the standby SeGW to another standby SeGW in the redundancy group will be performed after the switchover waiting time specified by the sum of **IKECFG.IPSEC SOWAITTIME (LTE eNodeB, 5G gNodeB)** and **IKECFG.IPSEC SORANDTIME (LTE eNodeB, 5G gNodeB)**. If the standby SeGW in use remains faulty throughout the switchover waiting time, the switchover will take a longer time and consequently services will be interrupted for a longer time.

## 7.3 Requirements

### 7.3.1 Licenses

The following table lists the licenses to be activated.

| RAT     | Feature ID    | Feature Name                                  | Model        | Sales Unit     |
|---------|---------------|-----------------------------------------------|--------------|----------------|
| GSM     | GBFD-160209   | IPSec Redundancy Among Multiple SeGWs         | LGB3IPRMSGW  | Per eGBTS/GBTS |
| UMTS    | WRFD-160274   | IPSec Redundancy Among Multiple SeGWs         | LQW9MSIPS01  | Per NodeB      |
| LTE FDD | LOFD-070211   | IPSec Redundancy Among Multiple SeGWs         | LT1SIPRMSG00 | Per eNodeB     |
| NB-IoT  | MLOFD-070211  | IPSec Redundancy Among Multiple SeGWs         | ML1SIPSERA00 | Per eNodeB     |
| LTE TDD | TDLOFD-070211 | IPSec Redundancy Among Multiple SeGWs         | LT1SIPSRAM00 | Per eNodeB     |
| NR      | FOFD-010080   | IPsec (IPsec Redundancy Among Multiple SeGWs) | NR0S0IPSEC00 | Per gNodeB     |

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

## 7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

### 7.3.2.1 GBFD-160209 IPsec Redundancy Among Multiple SeGWs

#### 7.3.2.1.1 Prerequisite Functions

| RAT | Function Name        | Function Switch | Reference    |
|-----|----------------------|-----------------|--------------|
| GSM | BTS Integrated IPsec | None            | <i>IPsec</i> |

#### 7.3.2.1.2 Mutually Exclusive Functions

| RAT | Function Name       | Function Switch | Reference    |
|-----|---------------------|-----------------|--------------|
| GSM | IPsec Tunnel Backup | None            | <i>IPsec</i> |

#### 7.3.2.1.3 Function Impacts

Active and standby routes cannot be configured for pre-encryption data in an IPv6 redundancy group.

### 7.3.2.2 WRFD-160274 IPsec Redundancy Among Multiple SeGWs

#### 7.3.2.2.1 Prerequisite Functions

| RAT  | Function Name          | Function Switch | Reference    |
|------|------------------------|-----------------|--------------|
| UMTS | NodeB Integrated IPsec | None            | <i>IPsec</i> |

### 7.3.2.2.2 Mutually Exclusive Functions

| RAT  | Function Name       | Function Switch | Reference    |
|------|---------------------|-----------------|--------------|
| UMTS | IPsec Tunnel Backup | None            | <i>IPsec</i> |

### 7.3.2.2.3 Function Impacts

Active and standby routes cannot be configured for pre-encryption data in an IPv6 redundancy group.

## 7.3.2.3 LOFD-070211 IPsec Redundancy Among Multiple SeGWs

### 7.3.2.3.1 Prerequisite Functions

| RAT     | Function Name  | Function Switch | Reference    |
|---------|----------------|-----------------|--------------|
| LTE FDD | IPsec          | None            | <i>IPsec</i> |
| LTE FDD | IPsec for IPv6 | None            | <i>IPsec</i> |

### 7.3.2.3.2 Mutually Exclusive Functions

| RAT     | Function Name       | Function Switch | Reference    |
|---------|---------------------|-----------------|--------------|
| LTE FDD | IPsec Tunnel Backup | None            | <i>IPsec</i> |

### 7.3.2.3.3 Function Impacts

Active and standby routes cannot be configured for pre-encryption data in an IPv6 redundancy group.

## 7.3.2.4 MLOFD-070211 IPsec Redundancy Among Multiple SeGWs

### 7.3.2.4.1 Prerequisite Functions

| RAT    | Function Name  | Function Switch | Reference    |
|--------|----------------|-----------------|--------------|
| NB-IoT | IPsec          | None            | <i>IPsec</i> |
| NB-IoT | IPsec for IPv6 | None            | <i>IPsec</i> |

#### 7.3.2.4.2 Mutually Exclusive Functions

| RAT    | Function Name       | Function Switch | Reference    |
|--------|---------------------|-----------------|--------------|
| NB-IoT | IPsec Tunnel Backup | None            | <i>IPsec</i> |

#### 7.3.2.4.3 Function Impacts

Active and standby routes cannot be configured for pre-encryption data in an IPv6 redundancy group.

### 7.3.2.5 TDLOFD-070211 IPsec Redundancy Among Multiple SeGWs

#### 7.3.2.5.1 Prerequisite Functions

| RAT     | Function Name  | Function Switch | Reference    |
|---------|----------------|-----------------|--------------|
| LTE TDD | IPsec          | None            | <i>IPsec</i> |
| LTE TDD | IPsec for IPv6 | None            | <i>IPsec</i> |

#### 7.3.2.5.2 Mutually Exclusive Functions

| RAT     | Function Name       | Function Switch | Reference    |
|---------|---------------------|-----------------|--------------|
| LTE TDD | IPsec Tunnel Backup | None            | <i>IPsec</i> |

#### 7.3.2.5.3 Function Impacts

Active and standby routes cannot be configured for pre-encryption data in an IPv6 redundancy group.

### 7.3.2.6 FOFD-010080 IPsec (IPsec Redundancy Among Multiple SeGWs)

#### 7.3.2.6.1 Prerequisite Functions

| RAT                                                           | Function Name | Function Switch | Reference    |
|---------------------------------------------------------------|---------------|-----------------|--------------|
| NR FDD<br>Low-frequency<br>NR TDD<br>High-frequency<br>NR TDD | IPsec         | None            | <i>IPsec</i> |



### 7.3.2.6.2 Mutually Exclusive Functions

| RAT                                                           | Function Name               | Function Switch | Reference    |
|---------------------------------------------------------------|-----------------------------|-----------------|--------------|
| NR FDD<br>Low-frequency<br>NR TDD<br>High-frequency<br>NR TDD | IPsec (IPsec Tunnel Backup) | None            | <i>IPsec</i> |

### 7.3.2.6.3 Function Impacts

Active and standby routes cannot be configured for pre-encryption data in an IPv6 redundancy group.

## 7.3.3 Hardware

### Base Station Models

| RAT  | Base Station Model                                                                                                                                                                                                                                                            |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GSM  | 3900 and 5900 series base stations (macro base stations)                                                                                                                                                                                                                      |
| UMTS | <ul style="list-style-type: none"><li>3900 and 5900 series base stations (macro base stations)</li><li>DBS3900 LampSite and DBS5900 LampSite</li></ul>                                                                                                                        |
| LTE  | <ul style="list-style-type: none"><li>3900 and 5900 series base stations (macro base stations)</li><li>DBS3900 LampSite and DBS5900 LampSite</li></ul>                                                                                                                        |
| NR   | <ul style="list-style-type: none"><li>3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.</li><li>DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.</li></ul> |

### Boards

| NE    | Hardware Requirement                                                                                                                                                                             |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GBTS  | GTMUc+UMPT_L, with UMPT_L providing a co-transmission port to connect to the transport network                                                                                                   |
| eGBTS | UMPT_G/UMDU_G/MDUC_G, which provides a co-transmission port to connect to the transport network<br>UMPT_G+UTRPc, with UTRPc providing a co-transmission port to connect to the transport network |

| NE     | Hardware Requirement                                                                                                                                                                                           |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NodeB  | UMPT_U/UMDU_U/MDUC_U, which provides a co-transmission port to connect to the transport network<br>UMPT_U+UTRPc, with UTRPc providing a co-transmission port to connect to the transport network               |
| eNodeB | UMPT_L/UMPT_T/UMDU_L/UMDU_T, which provides a co-transmission port to connect to the transport network<br>UMPT_L/UMPT_T+UTRPc, with UTRPc providing a co-transmission port to connect to the transport network |
| gNodeB | UMPT_N, which provides a co-transmission port to connect to the transport network                                                                                                                              |
| MBTS   | UMPT/UTRPc/UMDU/MDUC, which provides a co-transmission port to connect to the transport network                                                                                                                |

## RF Modules

This function does not depend on RF modules.

### 7.3.4 Networking

In a network with IPsec Redundancy Among Multiple SeGWs:

- The base station uses one or multiple IP addresses to connect to the SeGWs.
- One primary SeGW and one or more secondary SeGWs are deployed.

SeGWs and the trusted domain must meet the following requirements:

- DPD is enabled on the SeGWs.
- If the status of IPsec SAs is normal, SeGWs can advertise the base station's downlink routing information to the trusted domain. If the status of IPsec SAs is abnormal, SeGWs can advertise the base station's downlink routing revocation information to the trusted domain.
- The trusted domain can learn the base station's downlink routing information advertised by SeGWs.

Currently, the E8000E and SeMG9811 meet the preceding deployment requirements.

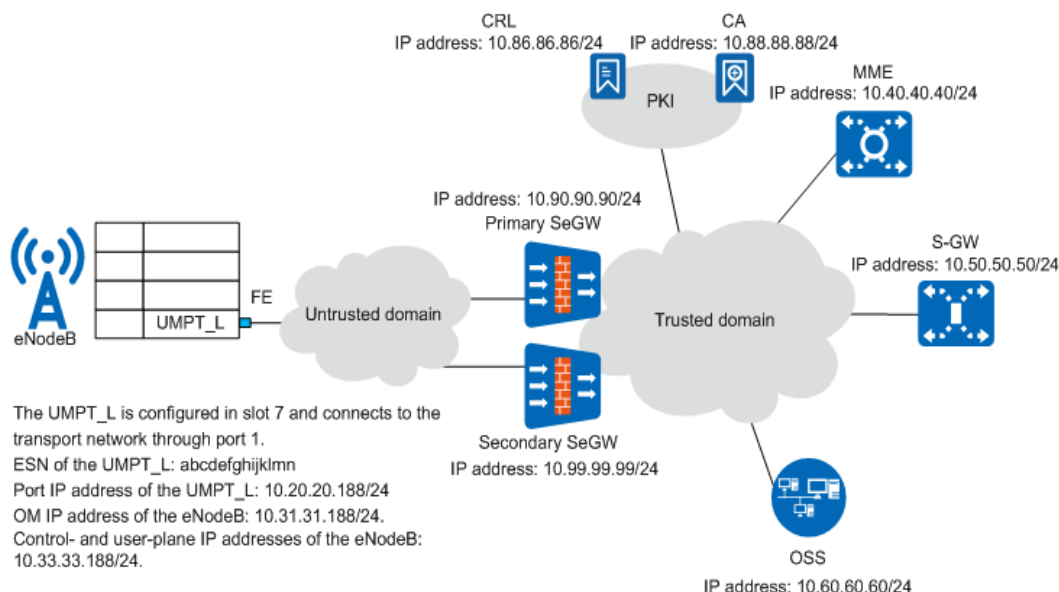
### 7.3.5 Others

None

## 7.4 Operation and Maintenance

This section uses the networking shown in [Figure 7-4](#) as an example to describe how to deploy the IPsec Redundancy Among Multiple SeGWs feature in on a PKI-based secure network. The configuration descriptions in this section apply to eGBTs, GBTs, NodeBs, gNodeBs, and eNodeBs.

**Figure 7-4** Example of deploying IPsec Redundancy Among Multiple SeGWs for an eNodeB on a PKI-based secure network



## 7.4.1 When to Use

After the IPsec feature is enabled, if the SeGW becomes faulty, base station services will be interrupted. To resume service rapidly, it is recommended that IPsec Redundancy Among Multiple SeGWs be enabled to improve IPsec tunnel reliability.

To implement IPsec Redundancy Among Multiple SeGWs, SeGWs must be deployed in different locations and meet certain capability requirements. For capability requirements of SeGWs, see [7.3.4 Networking](#).

In IPv6 IPsec redundancy among multiple SeGWs, a route to the active IKE peer must be configured. The route to the standby IKE peer is automatically generated and does not need to be configured. The generated standby route to the standby IKE peer has the same priority as the route configured for the active IKE peer. The total number of automatically generated routes and manually configured routes cannot exceed the upper limit for the base station.

### NOTE

A maximum of 32 routes are supported for the source IP address. When the maximum number of routes is required for IPv6 IPsec redundancy among multiple SeGWs (the ratio of active routes to standby routes is 1:7), only four routes to the active IKE peer can be configured.

If both packet filtering and IPsec Redundancy Among Multiple SeGWs are enabled, packet filtering rules need to be manually configured on the interfaces of the active and standby links. For detailed operations, see *Equipment Security*.

Before enabling IPsec Redundancy Among Multiple SeGWs, ensure that the active and standby IPsec tunnels are working properly. That is, the negotiation with the peer SeGW is normal and the uplink and downlink service data flows are normal.

## 7.4.2 Data Configuration

### 7.4.2.1 Data Preparation

- For details about IPv4 IPsec redundancy among multiple SeGWs, see [4.4.2.2 Data Preparation](#). The following table describes only different parameter configurations.
- For details about IPv6 IPsec redundancy among multiple SeGWs, see [5.4.1.1 Data Preparation](#). The following table describes only different parameter configurations.

**Table 7-1** Data to be prepared for an IKE peer

| Parameter Name              | Parameter ID                                          | Setting Notes                                                                                                                                                                                            |
|-----------------------------|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DPD Mode                    | <b>IKEPEER.DPD (LTE eNodeB, 5G gNodeB)</b>            | Enable the DPD function when the IPsec Redundancy Among Multiple SeGWs feature is enabled.                                                                                                               |
| DPD Idle Time               | <b>IKEPEER.DPDIDLETIME (LTE eNodeB, 5G gNodeB)</b>    |                                                                                                                                                                                                          |
| DPD Retransmission Interval | <b>IKEPEER.DPDRETRI (LTE eNodeB, 5G gNodeB)</b>       |                                                                                                                                                                                                          |
| DPD Retransmission Count    | <b>IKEPEER.DPDRETRN (LTE eNodeB, 5G gNodeB)</b>       |                                                                                                                                                                                                          |
| Redundancy Flag             | <b>IKEPEER.REDUNDANCYFLAG (LTE eNodeB, 5G gNodeB)</b> | When the IPsec Redundancy Among Multiple SeGWs feature is enabled, set this parameter to <b>MASTER</b> for the active IKE peer and set this parameter to <b>SLAVE</b> for the standby IKE peer.          |
| Master Peer Name            | <b>IKEPEER.MASTERPEERNAME (LTE eNodeB, 5G gNodeB)</b> | This parameter takes effect when the <b>IKEPEER.REDUNDANCYFLAG (LTE eNodeB, 5G gNodeB)</b> parameter is set to <b>SLAVE</b> .                                                                            |
| Priority                    | <b>IKEPEER.PRIORITY (LTE eNodeB, 5G gNodeB)</b>       | This parameter takes effect when the <b>IKEPEER.REDUNDANCYFLAG (LTE eNodeB, 5G gNodeB)</b> parameter is set to <b>SLAVE</b> . Different standby IKE peers must have different values for this parameter. |

**Table 7-2** Data to be prepared for basic IKE configurations

| Parameter Name                                 | Parameter ID                                                 | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IPSec Redundancy Switch Back                   | <b><i>IKECFG.IPSECSWITCHBACK (LTE eNodeB, 5G gNodeB)</i></b> | This parameter specifies whether the switchback to the active IPsec tunnel is allowed. Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                                                                                  |
| IPSec Redundancy Switch Back Wait Time         | <b><i>IKECFG.IPSECSBWAITTIME (LTE eNodeB, 5G gNodeB)</i></b> | Set this parameter based on the route convergence time of the trusted domain.                                                                                                                                                                                                                                                                                                                                                                                                         |
| IPSec Redundancy Switch Back Random Delay Time | <b><i>IKECFG.IPSECSBRANDTIME (LTE eNodeB, 5G gNodeB)</i></b> | Set this parameter based on the number of base stations connected to the SeGW (indicated by BTS_N) and the maximum number of IKE negotiations supported by the SeGW (indicated by IKE_M) in T seconds. The value of the <b><i>IKECFG.IPSECSBRANDTIME (LTE eNodeB, 5G gNodeB)</i></b> parameter should be greater than or equal to $BTS\_N \times T / IKE\_M$ (in unit of second).<br>If this parameter is set to <b>0</b> , random delay for IPsec redundancy switchback is disabled. |
| IPSec Redundancy Switchover Wait Time          | <b><i>IKECFG.IPSECSOWAITTIME (LTE eNodeB, 5G gNodeB)</i></b> | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| IPSec Redundancy Switchover Random Delay Time  | <b><i>IKECFG.IPSECSORANDTIME (LTE eNodeB, 5G gNodeB)</i></b> | If this parameter is set to <b>n</b> , random delay for IPsec redundancy switchover is enabled, and the maximum random delay is <i>n</i> (in unit of second).<br>If this parameter is set to <b>0</b> , random delay for IPsec redundancy switchover is disabled.                                                                                                                                                                                                                     |

| Parameter Name  | Parameter ID                                        | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IPSec Bind Mode | <b>IKECFG.IPSECBINDMODE (LTE eNodeB, 5G gNodeB)</b> | This parameter specifies the binding mode of an IPsec policy group, which can be global mode and single-interface mode. The two modes are mutually exclusive. This parameter takes effect only when an interface with the interface type being VLAN exists. In global mode, an IPsec policy group can be bound only to a normal interface and the IPsec policies take effect on the normal interface and all VLAN subinterfaces of the physical port on which the normal interface is located. In single-interface mode, the IPsec policies take effect only on the interface bound to the IPsec policy group. |

**Table 7-3** Parameters that must be set to configure an IPsec proposal

| Parameter Name     | Parameter ID                                           | Setting Notes                                                                                            |
|--------------------|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Encapsulation Mode | <b>IPSECPROPOSAL.ENCAPMODE (LTE eNodeB, 5G gNodeB)</b> | When the IPsec Redundancy Among Multiple SeGWs feature is enabled, set this parameter to <b>TUNNEL</b> . |

**NOTE**

The base station can use the same local IP address or different local IP addresses to communicate with the SeGWs. The **IKEPEER.LOCALIP (LTE eNodeB, 5G gNodeB)** parameter specifies the local IP address.

**7.4.2.2 Required Information**

For information to be collected, see [4.4.2.1 Required Information](#). For SeGWs, you need to collect information about the primary and secondary SeGWs.

**7.4.2.3 Using MML Commands**

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [7.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency,

and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

## Activation Command Examples

The following provides a configuration example in IPv4 networking:

Ethernet-interface-based configuration mode

```
//Setting the IPsec configuration mode
SET IKECFG: IPSECCFGMODE=ETHINTERFACE_BASED;
//Adding an IKE proposal
ADD IKEPROPOSAL: PROPID=10, AUTHMETH=IKE_CERT_SIG, DHGRP=DH_GROUP14, REAUTHSW=OFF;
//Adding a pair of active and standby IKE peers
ADD IKEPEER: PEERNAME="ikeMaster", PROPID=10, IKEVERSION=IKE_V2, IPVERSION=IPV4, IDTYPE=IPV4,
REMOTEP="10.90.90.90", REMOTENAME="segwmaster", DPD=PERIODIC, DPDIDLETIME=10, DPDRETRI=5,
DPDRETRN=5, LOCALIP="10.20.20.188", REDUNDANCYFLAG=MASTER, CERTSOURCE=APPCERT;
ADD IKEPEER: PEERNAME="ikeSlave", PROPID=10, IKEVERSION=IKE_V2, IPVERSION=IPV4, IDTYPE=IPV4,
REMOTEP="10.99.99.99", REMOTENAME="segwslave", DPD=PERIODIC, DPDIDLETIME=10, DPDRETRI=5,
DPDRETRN=5, LOCALIP="10.20.20.188", REDUNDANCYFLAG=SLAVE, MASTERPEERNAME="ikeMaster",
PRIORITY=1, CERTSOURCE=APPCERT;
//Adding an ACL
ADD ACL: ACLID=3000, ACLDESC="for IPsec";
//Adding ACL rules to the ACL
//Setting ACL rule 1 (RULEID = 1) for eNodeB signaling and service data flows
ADD ACLRULE: ACLID=3000, RULEID=1, PT=IP, SIP="10.33.33.188", SWC="0.0.0.0", DIP="0.0.0.0",
DWC="255.255.255.255", MDSCP=NO;
//Setting ACL rule 2 (RULEID = 2) for OM data flows and certificate management-related data flows
(including those generated during the interaction with the digital certificate and CRL database) of the
eNodeB when OM channels are protected by IPsec. If the OM channel does not need to be protected by
IPsec, the source IP address for certificate management-related data flows must be different from the OM
IP address. You are advised to configure a new internal-network IP address (for example, 10.45.45.45) as
the source IP address for certificate management-related data flows. Therefore, the SIP in the following ACL
rule needs to be changed to 10.45.45.45.
ADD ACLRULE: ACLID=3000, RULEID=2, PT=IP, SIP="10.31.31.188", SWC="0.0.0.0", DIP="0.0.0.0",
DWC="255.255.255.255", MDSCP=NO;
//Adding an IPsec proposal
ADD IPSECPROPOSAL: PROPNAME="prop0", TRANSMODE=ESP;
//Adding an IPsec policy
ADD IPSECPOLICY: SPGN="Policy0", SPSN=1, ACLID=3000, PROPNAME="prop0", PEERNAME="ikeMaster",
LTCFG=LOCAL, LTS=86400;
ADD IPSECPOLICY: SPGN="Policy0", SPSN=2, ACLID=3000, PROPNAME="prop0", PEERNAME="ikeSlave",
LTCFG=LOCAL, LTS=86400;
```

- Scenario 1:

```
//When GTRANSPARA.TRANSFCFGMODE is set to OLD
//Binding the IPsec policy to an outbound port
ADD IPSECBIND: SN=7, CN=0, SRN=0, SBT=BASE_BOARD, PT=ETH, PN=1, SPGN="Policy0";
```

- Scenario 2:

```
//When GTRANSPARA.TRANSFCFGMODE is set to NEW
//Binding the IPsec policy to an outbound port
ADD IPSECBINDITF: IPSECBINDITFID=0, ITFID=0, SPGN="Policy0";
```

```
//Setting basic IKE information
SET IKECFG: IPSECSWITCHBACK=ON, IPSECSBWAITTIME=2, IPSECSBRANDTIME=30, IPSECSOWAITTIME=15,
IPSECSORANDTIME=30;
//((Optional) Setting the IPsec replay alarm switch and threshold
SET IPGUARD: IPSECREPLAYCHKSW=ENABLE, IPSECREPLAYALMTHD=10;
```

Tunnel-interface-based configuration mode

```
//Setting the transmission configuration mode
SET GTRANSPARA: TRANSFCFGMODE=NEW;
//Setting the IPsec configuration mode
SET IKECFG: IPSECCFGMODE=TUNNELITF_BASED;
//Adding an IKE proposal
ADD IKEPROPOSAL: PROPID=10, AUTHMETH=IKE_CERT_SIG, DHGRP=DH_GROUP14, REAUTHSW=OFF;
```

```
//Adding a pair of active and standby IKE peers
ADD IKEPEER: PEERNAME="ikeMaster", PROPID=10, IKEVERSION=IKE_V2, IPVERSION=IPV4_TUNNELITF,
IDTYPE=IPV4, REMOTENAME="segwmaster", DPD=PERIODIC, DPDIDLETIME=10, DPDRETRI=5,
DPDRETRN=5, REDUNDANCYFLAG=MASTER, CERTSOURCE=APPCERT, IKEV2FRGSW=OFF;
ADD IKEPEER: PEERNAME="ikeSlave", PROPID=10, IKEVERSION=IKE_V2, IPVERSION=IPV4_TUNNELITF,
IDTYPE=IPV4, REMOTENAME="segwslave", DPD=PERIODIC, DPDIDLETIME=10, DPDRETRI=5, DPDRETRN=5,
REDUNDANCYFLAG=SLAVE, MASTERPEERNAME="ikeMaster", PRIORITY=1, CERTSOURCE=APPCERT,
IKEV2FRGSW=OFF;
//Adding an ACL
ADD ACL: ACLID=3000, ACLDESC="for IPsec";
//Adding ACL rules to the ACL
//Setting ACL rule 1 (RULEID = 1) for eNodeB signaling and service data flows
ADD ACLRULE: ACLID=3000, RULEID=1, PT=IP, SIP="10.33.33.188", SWC="0.0.0.0", DIP="0.0.0.0",
DWC="255.255.255.255", MDSCP=NO;
//Setting ACL rule 2 (RULEID = 2) for OM data flows and certificate management-related data flows
(including those generated during the interaction with the digital certificate and CRL database) of the
eNodeB when OM channels are protected by IPsec. If the OM channel does not need to be protected by
IPsec, the source IP address for certificate management-related data flows must be different from the OM
IP address. You are advised to configure a new internal-network IP address (for example, 10.45.45.45) as
the source IP address for certificate management-related data flows. Therefore, the SIP in the following ACL
rule needs to be changed to 10.45.45.45.
ADD ACLRULE: ACLID=3000, RULEID=2, PT=IP, SIP="10.31.31.188", SWC="0.0.0.0", DIP="0.0.0.0",
DWC="255.255.255.255", MDSCP=NO;
//Adding an IPsec proposal
ADD IPSECPROPOSAL:PROPNAME="prop0",TRANMODE=ESP;
//Adding an IPsec policy
ADD IPSECPOLICY: SPGN="Policy0", SPSN=1, ACLID=3000, PROPNAME="prop0", PEERNAME="ikeMaster",
LTCFG=LOCAL, LTS=86400;
ADD IPSECPOLICY: SPGN="Policy1", SPSN=2, ACLID=3000, PROPNAME="prop0", PEERNAME="ikeSlave",
LTCFG=LOCAL, LTS=86400;
//Configuring IP addresses for the IPsec tunnel
ADD TUNNELITF: PORTID=7, MODE=IPSEC_IPV4, LOCALIPREFMETHOD=IPADDR, LOCALIP4="10.20.20.188",
REMOTEIP4="10.90.90.90";
ADD TUNNELITF: PORTID=8, MODE=IPSEC_IPV4, LOCALIPREFMETHOD=IPADDR, LOCALIP4="10.20.20.188",
REMOTEIP4="10.99.99.99";
//Configuring outbound interfaces for the IPsec policy
ADD INTERFACE: ITFID=7, VRFID=2, ITFTYPE=NORMAL, PT=TUNNELITF, PORTID=7;
ADD INTERFACE: ITFID=8, VRFID=2, ITFTYPE=NORMAL, PT=TUNNELITF, PORTID=8;
//Binding the IPsec policy to outbound interfaces
ADD IPSECBINDITF: IPSECBINDITFID=1, SPGN="Policy0", ITFID=7;
ADD IPSECBINDITF: IPSECBINDITFID=2, SPGN="Policy1", ITFID=8;
//Setting basic IKE information
SET IKECFG: IPSECSWITCHBACK=ON, IPSECSBWAITTIME=2, IPSECSBRANDTIME=30, IPSECSOWAITTIME=15,
IPSECSORANDTIME=30;
//((Optional) Setting the IPsec replay alarm switch and threshold
SET IPGUARD: IPSECREPLAYCHKSW=ENABLE, IPSECREPLAYALMTHD=10;
```

### The following provides a configuration example in IPv6 networking:

```
//Adding an IKE proposal
ADD IKEPROPOSAL: PROPID=10, AUTHMETH=IKE_CERT_SIG, DHGRP=DH_GROUP14, REAUTHSW=OFF;
//Adding a pair of active and standby IKE peers
ADD IKEPEER: PEERNAME="ikeMaster", PROPID=10, IKEVERSION=IKE_V2, IPVERSION=IPV6, IDTYPE=IPV6,
REMOTENAME="segwmaster", DPD=PERIODIC, DPDIDLETIME=10, DPDRETRI=5, DPDRETRN=5,
REDUNDANCYFLAG=MASTER, CERTSOURCE=APPCERT;
ADD IKEPEER: PEERNAME="ikeSlave", PROPID=10, IKEVERSION=IKE_V2, IPVERSION=IPV6, IDTYPE=IPV6,
REMOTENAME="segwslave", DPD=PERIODIC, DPDIDLETIME=10, DPDRETRI=5, DPDRETRN=5,
REDUNDANCYFLAG=SLAVE, MASTERPEERNAME="ikeMaster", PRIORITY=1, CERTSOURCE=APPCERT;
//Adding an ACL
ADD ACL6: ACLID=3000, ACLDESC="for IPsec";
//Adding ACL rules to the ACL
ADD ACLRULE6: ACLID=3000, RULEID=10, PT=IP6, SIP="2001:DB8::1", SPFXLEN=32, DIP="2001:DB8::2",
DPFXLEN=32, MDSCP=YES, DSCP=25;
//Adding an IPsec proposal
ADD IPSECPROPOSAL:PROPNAME="prop0",TRANMODE=ESP;
//Adding an IPsec policy
ADD IPSECPOLICY: SPGN="Policy0", SPSN=1, ACLID=3000, PROPNAME="prop0", PEERNAME="ikeMaster",
LTCFG=LOCAL, LTS=86400;
ADD IPSECPOLICY: SPGN="Policy1", SPSN=2, ACLID=3000, PROPNAME="prop0", PEERNAME="ikeSlave",
LTCFG=LOCAL, LTS=86400;
```



```
//Configuring IP addresses for the IPsec tunnel
ADD TUNNELITF: PORTID=7, MODE=IPSEC_IPV6, LOCALIPREFMETHOD=IPADDR, VRFID=1,
LOCALIP6="2002::12:0:1", REMOTEIP6="2002::12:0:2";
ADD TUNNELITF: PORTID=8, MODE=IPSEC_IPV6, LOCALIPREFMETHOD=IPADDR, VRFID=1,
LOCALIP6="2002::12:1:1", REMOTEIP6="2002::12:1:2";
//Configuring outbound interfaces for the IPsec policy
ADD INTERFACE: ITFID=7, VRFID=2, ITFTYPE=NORMAL, PT=TUNNELITF, PORTID=7;
ADD INTERFACE: ITFID=8, VRFID=2, ITFTYPE=NORMAL, PT=TUNNELITF, PORTID=8;
//Binding the IPsec policy to outbound interfaces
ADD IPSECBINDITF: IPSECBINDITFID=1, SPGN="Policy0", ITFID=7;
ADD IPSECBINDITF: IPSECBINDITFID=2, SPGN="Policy1", ITFID=8;
//Setting basic IKE information
SET IKECFG: IPSECSWITCHBACK=ON, IPSECSBWAITTIME=2, IPSECSBRANDTIME=30, IPSECSOWAITTIME=15,
IPSECSORANDTIME=30;
//((Optional) Setting the IPsec replay alarm switch and threshold
SET IPGUARD: IPSECREPLAYCHKSW=ENABLE, IPSECREPLAYALMTHD=10;
```

## Deactivation Command Examples

For details, see [4.4.2.9 Deactivation](#).

### 7.4.2.4 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

#### NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

## 7.4.3 Activation Verification

After IPsec tunnels are enabled (For details on how to check whether IPsec tunnels are enabled, see [4.4.3 Activation Verification](#)), check whether IPsec tunnels are switched over if the active IPsec tunnel becomes faulty:

- Step 1** (Optional) Perform the following steps to check whether random delay for IPsec tunnel switchover has taken effect:
1. Record the time when the base station reports ALM-25891 IKE Negotiation Failure.
  2. On the LMT of the base station, start an IKE tracing task to obtain the time when the base station sends the first IKE negotiation message to a secondary SeGW.
  3. Calculate the difference between the two moments. If the difference is greater than or equal to the value of the **IPSec Redundancy Switchover Wait Time** parameter, this function has taken effect.
- Step 2** Run the **DSP IKESA** command to check the status of IPsec tunnels. If the value of the **Peer Address** parameter in the command output is the IP address of the secondary SeGW and the value of the **SA Flag** parameter is **Ready|StayAlive** or **Ready**, IPsec tunnels are successfully switched over.
- Step 3** Check whether services protected by IPsec are running properly by performing the following operations. If services are running properly, services have been successfully switched over to the standby IPsec tunnel.
- Initiate voice and data services and then check whether services are running properly.

- Check whether the corresponding base station is online on the MAE topology view.

----End

## 7.4.4 Network Monitoring

For details, see [4.4.4 Network Monitoring](#).

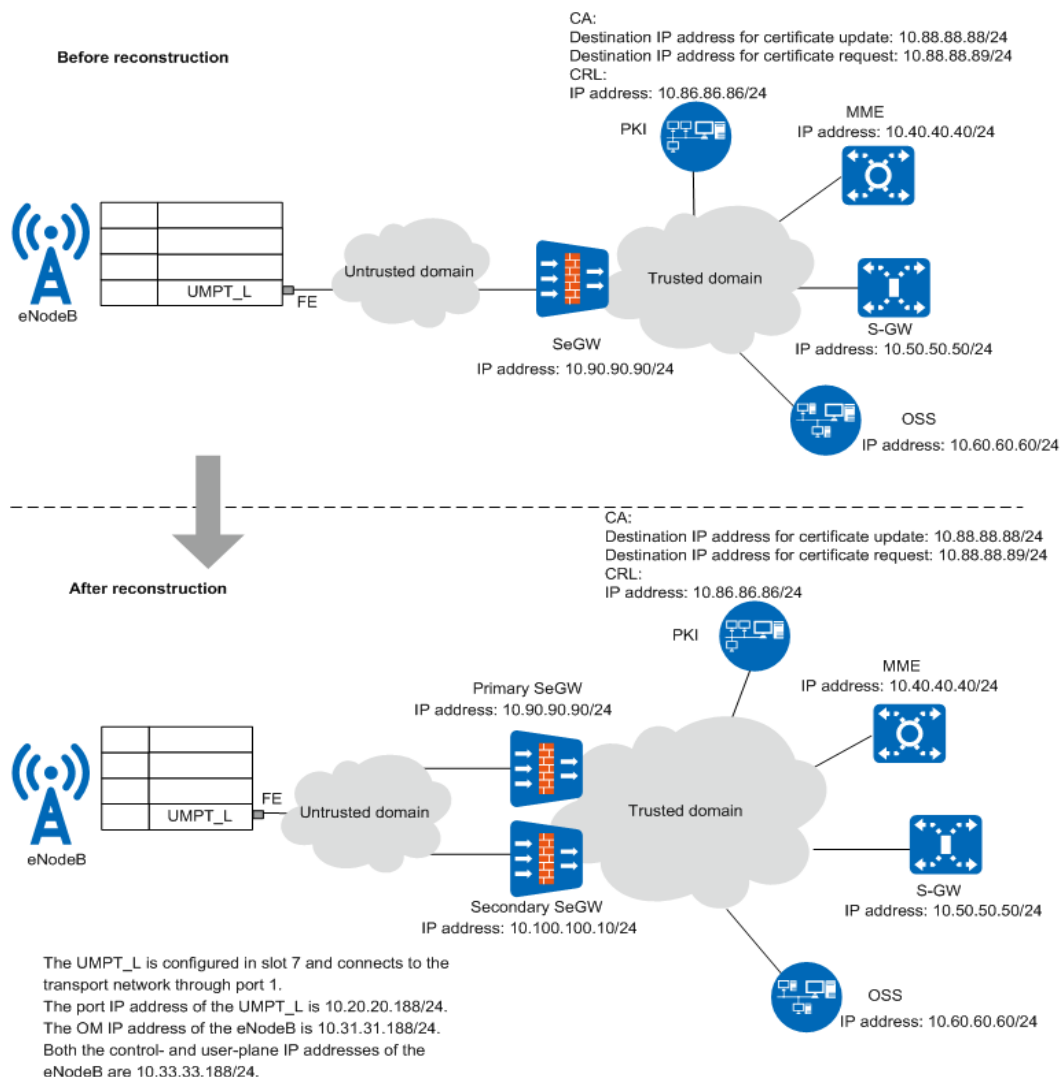
## 7.4.5 Reconstruction from a PKI/PSK-based Secure Network to a PKI/PSK-based IPsec Redundancy Network

The procedure for reconstructing from a PKI-based secure network to a PKI-based IPsec redundancy network is the same as that for reconstructing a PSK-based secure network to a PSK-based IPsec redundancy network. This section uses the reconstruction from a PKI-based secure network to a PKI-based IPsec redundancy network as an example to describe the reconstruction requirements and procedure. The reconstruction requirements and procedure are the same for the eGBTS, NodeB, eNodeB, gNodeB, and multimode base station.

This section uses an eNodeB as an example to describe the reconstruction.

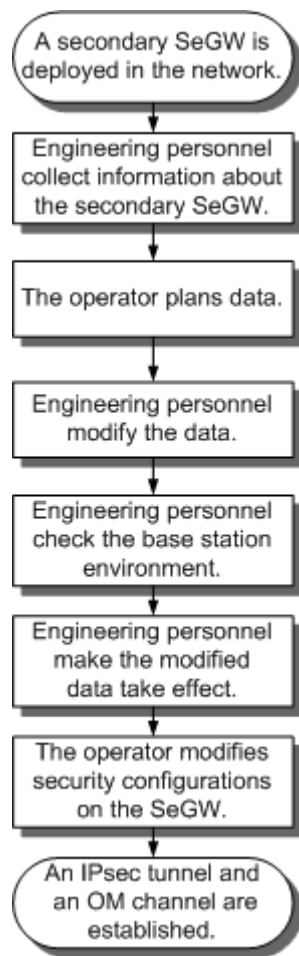
[Figure 7-5](#) shows the network topologies before and after the reconstruction.

**Figure 7-5** Example of reconstructing a PKI-based secure network into a PKI-based IPsec redundancy network



After the reconstruction, one active IPsec tunnel corresponds to one standby IPsec tunnel, and the value of the **IKEPEER.LOCALIP (LTE eNodeB, 5G gNodeB)** parameter for the standby IPsec tunnel is the same as the value of the **IKEPEER.LOCALIP (LTE eNodeB, 5G gNodeB)** parameter for the active IPsec tunnel. If the customer requires that the active and standby IPsec tunnels use different IP addresses, a new port IP address needs to be planned for the standby IPsec tunnel. The IP addresses for other data flows remain unchanged before and after the reconstruction.

## General Procedure



## Network Deployment and Information Collection

- Secondary SeGWs have been deployed on the network and meet the requirements described in [7.3.4 Networking](#).
- Engineering personnel have collected information about the secondary SeGWs. For details, see [4.4.2.1 Required Information](#).

## Security Data Planning

For details, see [7.4.2.1 Data Preparation](#).

## Preparing the Incremental Script

An incremental script is generated based on data of existing base stations and includes configuration modifications.

For details about how to modify IPsec configurations, see [4.4.2.10 Using the MAE-Deployment](#). ACL rules are configured according to the network plan.

## Checking the Base Station Environment

1. The base station meets the hardware requirements described in [7.3.3 Hardware](#).

2. The license for the IPsec feature has been activated on the base station.

## Downloading the Incremental Script

For details, see [Downloading the Incremental Script](#) in [4.5.1 Reconstruction from an Insecure Network to a PKI-based Secure Network](#).

## Modifying SeGW Configurations

Modify security-related parameter settings on the primary SeGW so that the following functions are supported:

- DPD is enabled.
- If the status of IPsec SAs is normal, SeGWs can advertise the base station's downlink routing information to the trusted domain.
- If the status of IPsec SAs is abnormal, SeGWs can advertise the base station's downlink routing revocation information to the trusted domain.

The trusted domain must be capable of learning the base station's downlink routing information advertised by SeGWs. The trusted domain must be modified if it cannot learn the base station's downlink routing information.

## Establishing IPsec Tunnels

The base station and the SeGW perform IKE negotiation to establish an IPsec tunnel. When the OM channel between the base station and the MAE is successfully established, network reconstruction succeeds.

## Activation Verification

For details, see [7.4.3 Activation Verification](#).

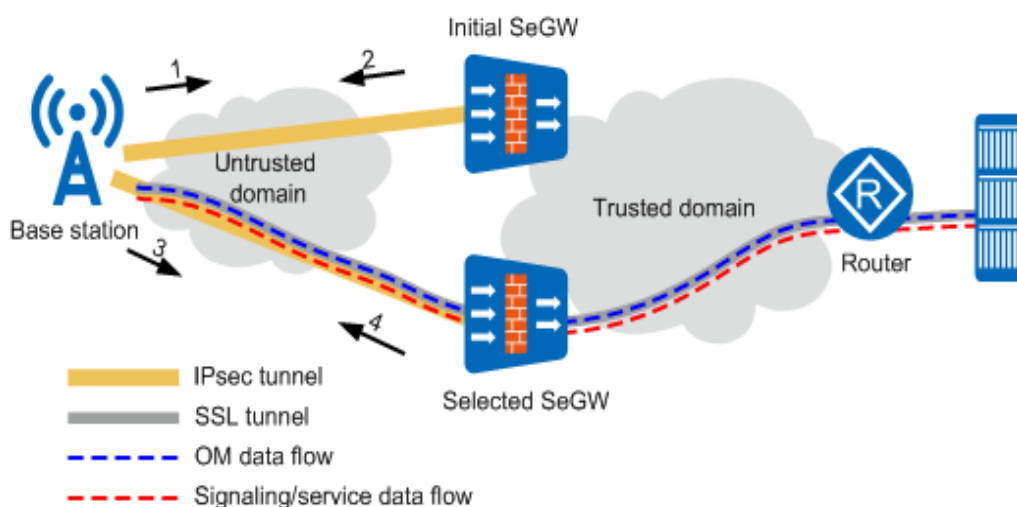
# 8 NE Supporting IPsec Redirection

## 8.1 Principles

The NE Supporting IPsec Redirection feature complies with IPsec redirection initiated by a base station as described in RFC 5685. When **IKEPEER.REDIRECTSW (5G gNodeB, LTE eNodeB)** is set to **ON**, the IKE\_SA\_INIT message sent by the base station contains the redirection support capability field. The SeGW decides whether to continue to provide services for the base station or initiate a redirection to the base station according to the redirection policy, thereby achieving load balancing between SeGWs and simplifying SeGW network planning. IPsec and IPv4 IPsec for IPv6 do not support this function in tunnel-interface-based configuration mode.

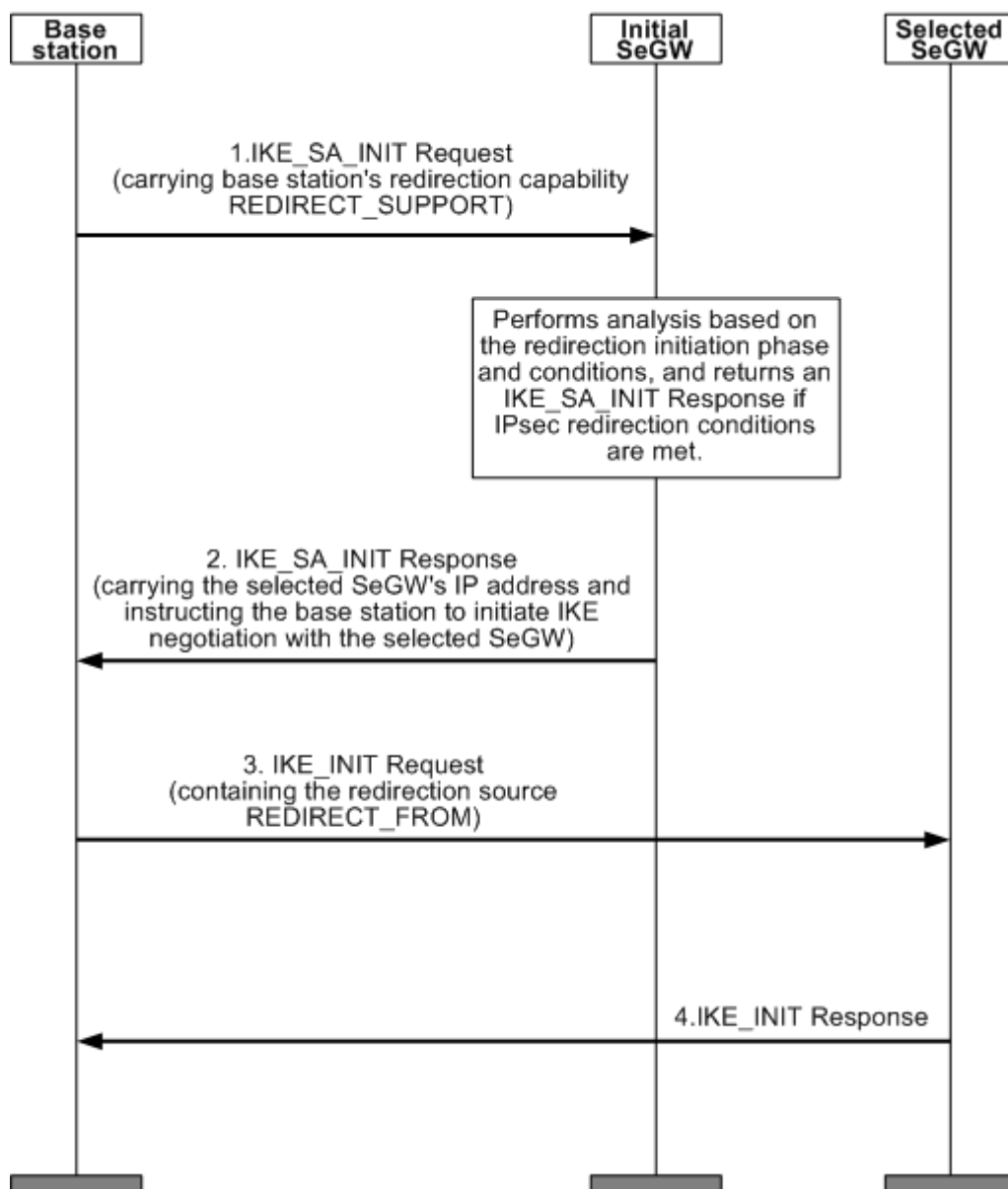
As shown in [Figure 8-1](#), the initial SeGW must be the SeGW configured on the base station, and the selected SeGW may be any one in the SeGW resource pool except the initial SeGW.

**Figure 8-1** IPsec redirection



[Figure 8-2](#) illustrates the message exchanges between the base station and the SeGW during the redirection procedure in [Figure 8-1](#).

**Figure 8-2** IPsec redirection procedure (using the INIT phase as an example)



**NOTE**

After the IPsec redirection, if the IKE negotiation between the base station and the selected SeGW fails, the base station initiates an IKE negotiation with the initial SeGW.

## 8.2 Network Analysis

### 8.2.1 Benefits

The IPsec redirection feature allows an SeGW connected to a base station to send a redirection request to the base station in case the SeGW is overloaded or faulty. After receiving the redirection request, the base station initiates an IKE negotiation

with a new SeGW. IPsec redirection helps balance the load between SeGWs and improves IPsec tunnel reliability.

## 8.2.2 Impacts

The impacts between this function and other functions are described in [8.3.2.1.3 Function Impacts](#), [8.3.2.2.3 Function Impacts](#), [8.3.2.3.3 Function Impacts](#), [8.3.2.4.3 Function Impacts](#), [8.3.2.5.3 Function Impacts](#), and [8.3.2.6.3 Function Impacts](#).

If the base station is redirected to a new SeGW, the ongoing services between the base station and the source SeGW will be affected. For example, ongoing voice calls will be dropped and data transmission rates will get extremely low.

## 8.3 Requirements

### 8.3.1 Licenses

| RAT     | Feature ID    | Feature Name                                | Model        | Sales Unit |
|---------|---------------|---------------------------------------------|--------------|------------|
| GSM     | GBFD-171206   | BTS Supporting IPsec Redirection            | LGB3IPSECRE  | Per BTS    |
| UMTS    | WRFD-171221   | NodeB Supporting IPsec Redirection          | LQW9IPSRD01  | Per NodeB  |
| LTE FDD | LOFD-081281   | eNodeB Supporting IPsec Redirection         | LT1SIPSECRE0 | Per eNodeB |
| NB-IoT  | MLOFD-081281  | eNodeB Supporting IPsec Redirection         | ML1SIPSECRE0 | Per eNodeB |
| LTE TDD | TDLOFD-081211 | eNodeB Supporting IPsec Redirection         | LT1STIPSR000 | Per eNodeB |
| NR      | FOFD-010080   | IPsec (gNodeB Supporting IPsec Redirection) | NR0S0IPSEC00 | Per gNodeB |

#### NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.



## 8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

### 8.3.2.1 GBFD-171206 BTS Supporting IPsec Redirection

#### 8.3.2.1.1 Prerequisite Functions

| RAT | Function Name        | Function Switch | Reference    |
|-----|----------------------|-----------------|--------------|
| GSM | BTS Integrated IPsec | None            | <i>IPsec</i> |

#### 8.3.2.1.2 Mutually Exclusive Functions

| RAT | Function Name       | Function Switch | Reference    |
|-----|---------------------|-----------------|--------------|
| GSM | IPsec Tunnel Backup | None            | <i>IPsec</i> |

#### 8.3.2.1.3 Function Impacts

None

### 8.3.2.2 WRFD-171221 NodeB Supporting IPsec Redirection

#### 8.3.2.2.1 Prerequisite Functions

| RAT  | Function Name          | Function Switch | Reference    |
|------|------------------------|-----------------|--------------|
| UMTS | NodeB Integrated IPsec | None            | <i>IPsec</i> |

#### 8.3.2.2.2 Mutually Exclusive Functions

| RAT  | Function Name       | Function Switch | Reference    |
|------|---------------------|-----------------|--------------|
| UMTS | IPsec Tunnel Backup | None            | <i>IPsec</i> |

### 8.3.2.2.3 Function Impacts

None

### 8.3.2.3 LOFD-081281 eNodeB Supporting IPsec Redirection

#### 8.3.2.3.1 Prerequisite Functions

| RAT     | Function Name | Function Switch | Reference    |
|---------|---------------|-----------------|--------------|
| LTE FDD | IPsec         | None            | <i>IPsec</i> |

#### 8.3.2.3.2 Mutually Exclusive Functions

| RAT     | Function Name       | Function Switch | Reference    |
|---------|---------------------|-----------------|--------------|
| LTE FDD | IPsec Tunnel Backup | None            | <i>IPsec</i> |

### 8.3.2.3.3 Function Impacts

None

### 8.3.2.4 MLOFD-081281 eNodeB Supporting IPsec Redirection

#### 8.3.2.4.1 Prerequisite Functions

| RAT    | Function Name | Function Switch | Reference    |
|--------|---------------|-----------------|--------------|
| NB-IoT | IPsec         | None            | <i>IPsec</i> |

#### 8.3.2.4.2 Mutually Exclusive Functions

| RAT    | Function Name       | Function Switch | Reference    |
|--------|---------------------|-----------------|--------------|
| NB-IoT | IPsec Tunnel Backup | None            | <i>IPsec</i> |

### 8.3.2.4.3 Function Impacts

None

### 8.3.2.5 TDLOFD-081211 eNodeB Supporting IPsec Redirection

### 8.3.2.5.1 Prerequisite Functions

| RAT     | Function Name | Function Switch | Reference    |
|---------|---------------|-----------------|--------------|
| LTE TDD | IPsec         | None            | <i>IPsec</i> |

### 8.3.2.5.2 Mutually Exclusive Functions

| RAT     | Function Name       | Function Switch | Reference    |
|---------|---------------------|-----------------|--------------|
| LTE TDD | IPsec Tunnel Backup | None            | <i>IPsec</i> |

### 8.3.2.5.3 Function Impacts

None

## 8.3.2.6 FOFD-010080 IPsec (IPsec Redirection)

### 8.3.2.6.1 Prerequisite Functions

| RAT                                                           | Function Name | Function Switch | Reference    |
|---------------------------------------------------------------|---------------|-----------------|--------------|
| NR FDD<br>Low-frequency<br>NR TDD<br>High-frequency<br>NR TDD | IPsec         | None            | <i>IPsec</i> |

### 8.3.2.6.2 Mutually Exclusive Functions

| RAT                                                           | Function Name | Function Switch | Reference    |
|---------------------------------------------------------------|---------------|-----------------|--------------|
| NR FDD<br>Low-frequency<br>NR TDD<br>High-frequency<br>NR TDD | IPsec         | None            | <i>IPsec</i> |

### 8.3.2.6.3 Function Impacts

None

## 8.3.3 Hardware

### Base Station Models

| RAT  | Base Station Model                                                                                                                                                                                                                                                            |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GSM  | 3900 and 5900 series base stations (macro base stations)                                                                                                                                                                                                                      |
| UMTS | <ul style="list-style-type: none"><li>3900 and 5900 series base stations (macro base stations)</li><li>DBS3900 LampSite and DBS5900 LampSite</li></ul>                                                                                                                        |
| LTE  | <ul style="list-style-type: none"><li>3900 and 5900 series base stations (macro base stations)</li><li>DBS3900 LampSite and DBS5900 LampSite</li></ul>                                                                                                                        |
| NR   | <ul style="list-style-type: none"><li>3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.</li><li>DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.</li></ul> |

### Boards

| NE     | Hardware Requirement                                                                                                                                                                                           |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GBTS   | GTMUc+UMPT_L, with UMPT_L providing a co-transmission port to connect to the transport network                                                                                                                 |
| eGBTS  | UMPT_G/UMDU_G/MDUC_G, which provides a co-transmission port to connect to the transport network<br>UMPT_G+UTRPc, with UTRPc providing a co-transmission port to connect to the transport network               |
| NodeB  | UMPT_U/UMDU_U/MDUC_U, which provides a co-transmission port to connect to the transport network<br>UMPT_U+UTRPc, with UTRPc providing a co-transmission port to connect to the transport network               |
| eNodeB | UMPT_L/UMPT_T/UMDU_L/UMDU_T, which provides a co-transmission port to connect to the transport network<br>UMPT_L/UMPT_T+UTRPc, with UTRPc providing a co-transmission port to connect to the transport network |
| gNodeB | UMPT_N, which provides a co-transmission port to connect to the transport network                                                                                                                              |
| MBTS   | UMPT/UTRPc/UMDU/MDUC, which provides a co-transmission port to connect to the transport network                                                                                                                |

## RF Modules

This function does not depend on RF modules.

### 8.3.4 Networking

On a network supporting IPsec redirection:

- The base station uses one IP address to connect multiple SeGWs.
- One initial SeGW and one or more selected SeGWs must be planned, and the SeGWs can advertise the base station's downlink routing information based on the status of IPsec SAs.

### 8.3.5 Others

To support this feature, the SeGWs must meet the following requirements:

- The SeGWs must support the RFC 5685 IKEv2 Redirect function.
- The SeGWs can generate internal dynamic routing based on IPsec SAs.
- The initial SeGW can establish IPsec tunnels with the base station so that base station deployment by plug and play (PnP) can be used.

### 8.3.6 Precautions

The application restrictions of IPsec redirection are as follows:

- The base station supports only IPv4-based IKEv2 redirection and can only serve as an initiator.
- IPsec redirection is not supported during base station deployment by PnP. In addition, the SeGW must support the establishment of IPsec tunnels.
- When IPsec redirection is enabled, ACL rules cannot be automatically established for selected SeGWs.

## 8.4 Operation and Maintenance

### 8.4.1 When to Use

If multiple SeGWs supporting IPsec redirection are deployed on the network, it is recommended that the Base Station Supporting IPsec Redirection feature be enabled on the network to implement load balancing among SeGWs and simplify network planning.

### 8.4.2 Data Configuration

#### 8.4.2.1 Data Preparation

(Optional) If ACL-based packet filtering is enabled on the base station, prepare ACL rule data (**ACLRULE** MO) for filtering packets between the base station and selected SeGW. The setting notes of the associated parameters are the same as those for ACL-based packet filtering. For details, see *Equipment Security*.

Prepare data for an IKE peer (**IKEPEER MO**). [Table 8-1](#) lists only new parameters compared with the parameters for IPsec. For other parameters to be prepared, see [4.4.2.2 Data Preparation](#).

**Table 8-1** Data to be prepared for an IKE peer

| Parameter Name        | Parameter ID                                      | Setting Notes                     |
|-----------------------|---------------------------------------------------|-----------------------------------|
| IKEv2 Redirect Switch | <b>IKEPEER.REDIRECTSW (5G gNodeB, LTE eNodeB)</b> | Set this parameter to <b>ON</b> . |

(Optional) Prepare data for IPsec redirection attack prevention (**IPGUARD MO**), as listed in [Table 8-2](#).

**Table 8-2** Data to be prepared for IPsec redirection attack prevention

| Parameter Name                  | Parameter ID                                            | Setting Notes                                                                                                                                                                                                                                                                                                 |
|---------------------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IKEv2 Redirect DOS Check Switch | <b>IPGUARD.REDIRECTDOSCHKSW (5G gNodeB, LTE eNodeB)</b> | <ul style="list-style-type: none"><li>In IPsec networking, the recommended value of this parameter is <b>ENABLE</b>.</li><li>In non-IPsec networking, the recommended value of this parameter is <b>DISABLE</b>.</li></ul>                                                                                    |
| Max Redirects Number            | <b>IPGUARD.MAXREDIRECTS (5G gNodeB, LTE eNodeB)</b>     | The default value is recommended.<br>If the SeGW's capacity is low or IPsec tunnels can be easily switched over, it is recommended that this parameter be set to a larger value. Note that the value of this parameter must not exceed the number of SeGWs. Otherwise, cyclic IPsec tunnel switchovers occur. |

### 8.4.2.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [8.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

- Step 1** (Optional) If ACL-based packet filtering is enabled on the base station, run the **ADD ACLRULE** command to add ACL rules for each selected SeGW.

 **NOTE**

ACL rules can be automatically established only for the initial SeGW and must be manually configured for selected SeGWs. IKE packets transmitted between the base station and selected SeGWs are blocked if ACL rules are not configured for selected SeGWs, causing IPsec redirections to fail. For details about ACL rules, see *Equipment Security*.

- Step 2** Run the **MOD IKEPEER** command to set **IKEv2 Redirect Switch**.

- Step 3** (Optional) Run the **SET IPGUARD** command to set **IKEv2 Redirect DOS Check Switch** and **Max Redirects Number**.

----End

### 8.4.2.3 MML Command Examples

IPsec redirection depends on the IPsec feature. Enable IPsec before enabling this feature. For details about how to enable IPsec, see [4.4.2.3.2 Using MML Commands](#).

The following example assumes that ACL-based packet filtering has been enabled on the base station. Rule 1 (RULEID = 1) and Rule 2 (RULEID = 2) are configured for selected SeGW 1 and selected SeGW 2, respectively.

```
//(Optional) Adding ACL rules to an ACL
ADD ACLRULE: ACLID=3000, RULEID=1, PT=IP, SIP="10.33.33.188", SWC="0.0.0.0", DIP="0.0.0.0",
DWC="255.255.255.255", MDSCP=NO;
ADD ACLRULE: ACLID=3000, RULEID=2, PT=IP, SIP="10.31.31.188", SWC="0.0.0.0", DIP="0.0.0.0",
DWC="255.255.255.255", MDSCP=NO;
//Setting the IPsec redirection switch to ON for the IKE peer
MOD IKEPEER: PEERNAME="ike", IKEVERSION=IKE_V2, IPVERSION=IPV4, IDTYPE=IPV4, REDIRECTSW=ON;
//(Optional) Configuring IPsec redirection protection
SET IPGUARD: REDIRECTDOSCHKSW=ENABLE, MAXREDIRECTS=4;
```

### 8.4.2.4 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 **NOTE**

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

## 8.4.3 Deactivation

### 8.4.3.1 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [8.3 Requirements](#) and complete the parameter configurations for related functions based on the dependency and mutually exclusive relationships between the functions, as well as the actual

network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

- Step 1** Run the **MOD IKEPEER** command to set **IKEv2 Redirect Switch** to **OFF**.
- Step 2** (Optional) Run the **RMV ACLRULE** command to remove ACL rules of each selected SeGW.
- Step 3** (Optional) Run the **SET IPGUARD** command to set **IKEv2 Redirect DOS Check Switch** to **DISABLE**.

----End

### 8.4.3.2 MML Command Examples

```
//Modifying the IKE peer
MOD IKEPEER: PEERNAME="ike", IKEVERSION=IKE_V2, IPVERSION=IPV4, IDTYPE=IPV4, REDIRECTSW=OFF;
//(Optional) Removing ACL rules of each selected SeGW
RMV ACLRULE: ACLID=3000, RULEID=1;
RMV ACLRULE: ACLID=3000, RULEID=2;
//(Optional) Disabling IPsec redirection protection against DoS attacks
SET IPGUARD: REDIRECTDOSCHSW=DISABLE;
```

### 8.4.4 Activation Verification

1. Run the **DSP IKEPEER** command. If the value of **IKEv2 Redirect Switch** in the command output is **ON**, IPsec redirection has been enabled on the base station.
2. Run the **DSP IKESA** command to query the status of IPsec tunnels. If the value of **IPv4 Peer Address** in the command output is the IP address of the selected SeGW, IPsec tunnels have been successfully switched over.
3. Perform the following operations to check whether services protected by IPsec of selected SeGWs are running properly. If the services are running properly and the base station is online on the topology view, IPsec redirection is running properly.
  - Initiate voice and data services and then check whether services are running properly.
  - Check whether the corresponding base station is online on the MAE topology view.

### 8.4.5 Network Monitoring

None



# 9 Parameters

---

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

## NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

### **FAQ 1: How do I find the parameters related to a certain feature from parameter reference?**

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

### **FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?**

- Step 1** Open the EXCEL file of the used reserved parameter list.

**Step 2** On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

# 10 Counters

---

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

## NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

## **FAQ: How do I find the counters related to a certain feature from performance counter reference?**

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

# 11 Glossary

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For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

# 12 Reference Documents

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- IETF RFC 2401, 2403, 2404, and 2409
- IETF RFC 4301, 4302, 4303, 4304, 4306, and 7296
- IETF RFC 3706
- IETF RFC 7383
- IETF RFC 4806
- *SRAN Networking and Evolution Overview*
- *PKI*
- *Automatic OMCH Establishment*
- *IPv4 Transmission*
- *Equipment Security*
- *BSS Local Switching in GBSS feature documentation*
- *S1 and X2 Self-Management in eRAN feature documentation*
- *NG and Xn Self-Management and eXn Self-Management in 5G RAN feature documentation*
- *X2 and S1 Self-Management in NSA Networking in eRAN feature documentation or 5G RAN feature documentation*
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